Ro.76A | 完整时钟局部载频电流负号障碍

A 在实际冻结 collar primitive 与完整时钟上构造 exact q=2 unresolved high-carrier cluster，证明局部载频电流及 correction row 严格为负。它只排除凭单边 offset spectrum 丢弃该行或宣称其非负的证明步骤；不反驳 W 的两模态通量估计，也不排除 perturbative、nonlocal signed 或 joint density/carrier cancellation。 **NOT CLAY.**

0 < DELTA_0 < DELTA | ACTUAL COLLAR PRIMITIVE | COMPLETE CLOCK | Q=2 UNRESOLVED CLUSTER | LOCAL CURRENT STRICTLY NEGATIVE | CORRECTION ROW STRICTLY NEGATIVE | CARRIER DENSITY RETAINED | SIGN-DROPPING CLOSED | W TWO-MODE PAYMENT INTACT | PERTURBATIVE ROUTE OPEN | NONLOCAL SIGNED ROUTE OPEN | JOINT CANCELLATION OPEN | VERSION-M CONDITIONAL | NO FIGURE / NO DNS | NOT CLAY

状态 · RO.76A STEP 52

PROFILE: 0 < DELTA_0 < DELTA

CARRIER: N=CEIL(16/ELL)

CLUSTER: (N,N+1), Q=2

CLOCK: 0 <= S <= 4

PRIMITIVE: XI_A >= 0, POSITIVE MASS

CURRENT: J < 0 LOCALLY

CORRECTION: |Z_Z|^2+2 ALPHA J < 0

POSITIVE DENSITY ROW: RETAINED

OBSTRUCTION: SIGN-DROPPING ONLY

GENERAL CLUSTER PAYMENT: OPEN

FORMAL FIGURE: NOT APPLICABLE

01 / 完整正文

0. 这一步得到什么

Ro.74S Steps 1--10 已把一侧球完成、terminal Abel 恒等式、四通道 circular recombination、未加权 genealogy 的抽象标量 no-go、低 Rayleigh 支付、no-exception exact-family no-go、canonical last exit 以及 paid/residual 六类分区逐项封存。Step 11 不再重复分区，而是精确回答两个 residual mechanisms 怎样共享一个 best-N budget，并分别把 short branch 与 scalar-excess branch 推到各自第一个仍缺失的 PDE 门槛。

本步先把局部耗散 clock 精确拆成黏性耗散与反常缺陷，再按优先顺序分成三类：反常缺陷至少承担终端 clock 的八分之一；高 Rayleigh 黏性耗散至少承担八分之一；否则低 Rayleigh 黏性耗散承担超过四分之一。第三类由抛物归一化动能时间质量、Jensen 不等式和继承的 padded-shell 三次付款同时给出

$$\sum_{k\in\mathcal{I}_{\rm lo}(\tau)} K_{k,R}(\tau) \le C\,\mathscr{L}(\boldsymbol{\lambda})^{1/3}(P_R^M)^{2/3}.$$

这里不需要例外壳层，也不需要新的 signed cancellation。这个结论是全部壳层同时成立的严格二次支付，但只覆盖低 Rayleigh 耗散支；它不推出低 Rayleigh 时间集具有统一正测度。

Step 8 引入标量 excess x 与 Jordan envelope X，证明 selected defect/high-Rayleigh residual 是既有 stopped-work ledger 的子账本。更关键的是，S.197--S.198 证明 no-exception stopped-work supremum $\mathfrak{W}_{\rm up}$ 与 full terminal clock、full positive cumulative flux 只差已由二次 ledger 支付的 B_Q。Ro.74O/P 的平滑精确族随后给出

$$\frac{\mathfrak{W}^{M,*}_{{\rm up},R_j}}{(P^{M,*}_{R_j})^{2/3}} \longrightarrow \infty,$$

从而严格否定普适 no-exception 二次界。S.38 仍是正确的条件蕴含；被反证的是它若要作为无条件定理所需的 antecedent。下一条可行路线必须回到固定 best-N、随终端变化的例外集合，并用 $\sqrt{N}, Y^{\mathrm{sf}}_{2,R}$ 支付尾部。

此前的 **PROVED ABSTRACT SCALAR NO-GO** 继续只排除 scalar completed-clock algebra 与未加权 genealogy；Step 8 的 no-go 由继承的真实平滑 NSE exact family 给出；Step 9 的 no-gain 说明 canonical stops 本身没有压缩。Step 10 证明四个 paid classes 合计只使用一个 $6B_Q$ ledger 和一个 C_5 cubic ledger，余项正好是 $\mathcal{R}_{\mathrm{sh}} \cup \mathcal{R}_x$ 上的共享 best-N residual gate。Step 11 进一步证明共享 budget 的离散 infimal convolution；short branch 得到 inverse-duration 与 nested-tent 控制，却仍缺 depth zero 的 terminal trace；scalar-excess branch 与 residual best-N 以字面常数 $1/5$ 与 3 等价。固定解的 tail tightness 不能替代与解和尺度无关的固定 N_0。现有 multi-packet exact families 也没有否定任意固定正 N。S.261、S.269、S.272、S.243、Q.12、Q.1、尺度收缩、正则性与奇点形成仍为 **OPEN / NOT CLAIMED**。**NOT CLAY.**

本节没有数值仿真、DNS 或 DGX。

1. 两个紧支撑一侧球 cutoff

保留 $r_m = 2^m R$、$\delta = R/8$ 与冻结过渡函数 ϑ，定义

$$\chi^-_{m,R}(y) = 1 - \vartheta\left(\frac{|y| - r_m}{\delta}\right), \qquad \chi^+_{m,R}(y) = \vartheta\left(\frac{r_m - |y|}{\delta}\right).$$

两者都光滑、径向、紧支撑且取值于 $[0,1]$。逐一检查 $r_m - \delta, r_m, r_m + \delta$ 划分的四个径向区间，得到

$$\boxed{0 \le \chi^-_{m,R} \le \chi^+_{m,R} \le 1, \qquad \beta^R_m = \chi^+_{m,R} - \chi^-_{m,R}, \qquad \psi^R_m = \chi^+_{m+1,R} - \chi^-_{m,R}.}$$

令 $\mathsf{B}^{\pm}_{m,R}$ 为它们的周期化。两条径向梯度都带负号，因此对 R0.74S Step 3 的 work vector 有

$$\boxed{\int_{\mathbb{T}^3} \mathcal{W}^M_R \cdot \nabla \mathsf{B}^-_{m,R} = -J^-_{m,R}, \qquad \int_{\mathbb{T}^3} \mathcal{W}^M_R \cdot \nabla \mathsf{B}^+_{m,R} = -J^+_{m,R}.}$$

这个符号决定了后面 root 与 outer 两行保留的是不同端点；不能在这里先取绝对值。

2. completed ball-clock tower

对任意非负紧支撑 lifted cutoff ϕ 及其周期化 Φ，定义端点能量、耗散、二次源项与通量四行

$$\begin{aligned}
\mathscr{E}_R[\Phi](t) &= \frac{\eta_R(t)}{2R} \int_{\mathbb{T}^3} \Phi |v_R|^2, \\
\mathscr{D}_R[\Phi](t) &= \frac{1}{R} \int_{(s_R,t)\times\mathbb{T}^3} \eta_R \Phi \, d\boldsymbol{\mu}, \\
\mathscr{Q}_R[\Phi](t) &= \frac{1}{2R} \int_{s_R}^t \int_{\mathbb{T}^3} [\eta_R' \Phi + \eta_R \Delta\Phi] |v_R|^2, \\
\mathscr{F}_R[\Phi](t) &= \frac{1}{R} \int_{s_R}^t \int_{\mathbb{T}^3} \eta_R \mathcal{W}^M_R \cdot \nabla\Phi.
\end{aligned}$$

写 $\mathscr{K}_R = \mathscr{Q}_R + \mathscr{F}_R$。R0.74P 的 suitable-weak 局部能量计算给出规范绝对连续代表

$$\boxed{\mathscr{K}_R[\Phi]=\mathscr{E}_R[\Phi]+\mathscr{D}_R[\Phi]\ge 0,\qquad \mathscr{K}_R[\Phi](s_R)=0.}$$

线性与上面的 cutoff 差分恒等式给出

$$\boxed{\begin{aligned}\mathscr{K}^+_{m,R}-\mathscr{K}^-_{m,R}&=\gamma_m^{-1}K^\partial_{m,R},\\ \mathscr{K}^+_{m+1,R}-\mathscr{K}^-_{m,R}&=\gamma_m^{-1}K_{m,R}.\end{aligned}}$$

于是

$$\boxed{\mathscr{K}^+_{m+1,R}-\mathscr{K}^+_{m,R}=\gamma_m^{-1}(K_{m,R}-K^\partial_{m,R})\ge 0.}$$

这是一条真正的单调 ball tower；但单调性本身只说明 residual 非负，并不把它从 ℓ^1 变成 ℓ^2。

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3. 从 collar 到 ball 仍是二次付款

令

$$d_m=\gamma_{m-1}-\gamma_m>0,\qquad m\ge 2.$$

冻结权重满足 $\sum_{k\ge j}\gamma_k\le(35/3)\gamma_j$。把 lifted ball 分成中心球、硬边界 collar 与 padded-shell 内部后，得到点态 packing

$$\begin{aligned}&\sum_{k\ge1}\gamma_k\chi^-_{k,R}+\sum_{k\ge1}\gamma_k\chi^+_{k+1,R}+\sum_{m\ge2}d_m\chi^+_{m,R}\\ &\quad\le C\left[\mathbf{1}_{\{|y|<4R\}}+\sum_{j\ge1}\gamma_j\mathbf{1}_{\operatorname{supp}\psi_j^R}(y)\right].\end{aligned}$$

对应绝对 Laplacian 的和乘以 R^2 后也满足同一控制。危险的 outer collar C_j^+ 上出现 γ_{j-1}，它由 $\operatorname{supp}\psi^R_{j-1}$ 支付；不能错误地用 γ_j 代替。中心球另由

$$R^{-3}\int_{I_{2R}}\int_{B_{4R}}|v_R|^2\le 32\,\mathcal{E}^{M,R}(z_0,8R)\le 32A_R$$

支付。合并后

$$\boxed{\sum_{k\ge1}\gamma_k\,\mathrm{TV}\,\mathscr{Q}^-_{k,R}+\sum_{k\ge1}\gamma_k\,\mathrm{TV}\,\mathscr{Q}^+_{k+1,R}+\sum_{m\ge2}d_m\,\mathrm{TV}\,\mathscr{Q}^+_{m,R}\le CA_R.}$$

因此 ball completion 没有制造新的低阶损失；真正的问题是保留下来的 terminal $\mathscr{K}$-clock。

4. 三条 signed channel 的时间方向

固定 Step 2 的 stopped family $(\tau, I, \boldsymbol{\sigma})$。对 $k \in I$，令 ρ_k 为前驱 merge/终端时刻，λ_k 为后继 merge/终端时刻；内部边界从 $\widehat{\sigma}_m = \max(\sigma_{m-1}, \sigma_m)$ 开始激活。

精确积分得到：

$$\frac{1}{R}\int_{s_R}^{\tau} \eta_R \mathcal{R}_R = -\sum_{k \in I_{\mathrm{rt}}} \gamma_k [\mathscr{F}^-_{k,R}(\rho_k) - \mathscr{F}^-_{k,R}(\sigma_k)],$$

$$-\frac{1}{R}\int_{s_R}^{\tau} \eta_R \mathcal{L}_R = \sum_{k \in I_{\mathrm{out}}} \gamma_k [\mathscr{F}^+_{k+1,R}(\lambda_k) - \mathscr{F}^+_{k+1,R}(\sigma_k)],$$

$$\frac{1}{R}\int_{s_R}^{\tau} \eta_R \mathcal{G}_R = \sum_{m \in I^{\partial}} d_m [\mathscr{F}^+_{m,R}(\tau) - \mathscr{F}^+_{m,R}(\widehat{\sigma}_m)].$$

用 $\mathscr{F} = \mathscr{K} - \mathscr{Q}$、$\mathscr{K} \geq 0$ 与二次付款，只能分别留下

$$\begin{aligned}
\left[R^{-1}\int \eta_R \mathcal{R}_R\right]_+ &\leq \sum_{k \in I_{\mathrm{rt}}} \gamma_k \mathscr{K}^-_{k,R}(\sigma_k) + CA_R, \\
\left[-R^{-1}\int \eta_R \mathcal{L}_R\right]_+ &\leq \sum_{k \in I_{\mathrm{out}}} \gamma_k \mathscr{K}^+_{k+1,R}(\lambda_k) + CA_R, \\
\left[R^{-1}\int \eta_R \mathcal{G}_R\right]_+ &\leq \sum_{m \in I^{\partial}} d_m \mathscr{K}^+_{m,R}(\tau) + CA_R.
\end{aligned}$$

不对称性是精确的：root 留在起始 stop，outer 留在 merge，weight-drop 留在终端。正性不会自动抹掉这些值。

5. terminal weight-drop 的 Abel 恒等式

在好时刻 t 写 $B_m = \mathscr{K}^+_{m,R}(t)$。有限求和分部给出

$$\sum_{m=2}^{M} d_m B_m = \gamma_1 B_2 + \sum_{m=2}^{M-1} \gamma_m (B_{m+1} - B_m) - \gamma_M B_M.$$

由 ball tower，中间项正好等于

$$\sum_{m=2}^{M-1}[K_{m,R}(t)-K^{\partial}_{m,R}(t)].$$

固定 R,t 时，周期化 ball clock 至多按 $1+2^{3M}$ 增长；而 $\gamma_M=\exp(-4^{M-1}/32)$，所以 $\gamma_M B_M\to 0$。取极限并补回 $m=1$ 得到

$$\boxed{\sum_{m\ge 2} d_m\mathscr{K}^{+}_{m,R}(t)=\gamma_1\mathscr{K}^{+}_{1,R}(t)+\sum_{m\ge 1}[K_{m,R}(t)-K^{\partial}_{m,R}(t)].}$$

每一项都非负。因此对任意有限 $H\subset\{2,3,\dots\}$，

$$\sum_{m\in H} d_m\mathscr{K}^{+}_{m,R}(t)\le \gamma_1\mathscr{K}^{+}_{1,R}(t)+Y^{\mathrm{clk}}_{1,R}.$$

这是正确且有限的 ℓ^1 估计，却没有任何平方函数压缩。把它代回 weight-drop 行，只会返回已经知道的大付款 ledger。

6. 抽象时钟塔严格饱和这个损失

取任意 $N\ge 1$ 与光滑单调函数 h，满足 $h(s_R)=0$、$h(\tau)=1$。规定

$$K_{m,R}(t)=\begin{cases}h(t), & 1\le m\le N,\\ 0, & m>N,\end{cases}\qquad K^{\partial}_{m,R}=0,\qquad \mathscr{K}^{+}_{1,R}=0,\qquad \mathscr{K}^{-}_{m,R}=\mathscr{K}^{+}_{m,R}.$$

其余 ball tower 由单调差分递归定义。再在纯标量层面取 $\mathscr{E}=\mathscr{K}$、$\mathscr{D}=\mathscr{Q}=0$、$\mathscr{F}=\mathscr{K}$，所有 completed-clock 与线性 tower 恒等式都逐项成立。

在终端时刻，前 N 个 shell positive variations 都等于 1，于是

$$\boxed{Y^{\mathrm{sf}}_{2,R}=\sqrt{N},\qquad \sum_{m\ge 2} d_m\mathscr{K}^{+}_{m,R}(\tau)=N.}$$

所以不存在仅由上述标量代数推出的普适常数 C，使

$$\sum_{m\ge 2} d_m\mathscr{K}^{+}_{m,R}(\tau)\le C\,Y^{\mathrm{sf}}_{2,R}.$$

这就是本节的 no-go：**positive completion + linear tower + ℓ^1 summation** 不能单独关闭匹配平方函数。这个见证没有空间算子或 PDE 实现，因此不排除真正利用 Navier--Stokes 方程、跨通道符号或有限 genealogy 的定理。

7. 四通道 signed recombination 精确返回原问题

对 $X\in\{E,D,Q,F,K\}$，把 root、outer、weight-drop 与 mismatch 四行按 stopped 时间方向组合成 $\mathfrak{C}_X$。有限块分解与 cutoff 线性给出

$$\boxed{\mathfrak{C}_X=\sum_{k\in I}[X_{k,R}(\tau)-X_{k,R}(\sigma_k)].}$$

特别地，$X=F$ 时 $\mathfrak{C}_F=W_R^M$，而 $F=K-Q$ 给出

$$\boxed{W_R^M=\sum_{k\in I}\Delta_{\sigma_k}^{\tau}K_{k,R}-\sum_{k\in I}\Delta_{\sigma_k}^{\tau}Q_{k,R}.}$$

二次 Q 行已由 CA_R 支付，但终端 upcrossing 恰好满足

$$\mathfrak{C}_K=\sum_{k\in I}\Delta_{\sigma_k}^{\tau}K_{k,R}>\frac{1}{4}\sum_{k\in I}K_{k,R}(\tau).$$

因此完整 signed recombination 没有把困难项压小，而是精确重建了要控制的对象。这是 **CIRCULAR ROUTE / PROVED**，不是 PDE 反例。

8. 拆出 mismatch 后的三通道正结果

令 Ω_A^R 为由 padded shells 减去内部 boundary bumps 得到的 genealogy cutoff。精确支撑几何证明 $\Omega_A^R\geq 0$，且插入新 shell 时 cutoff 单调增加。对三通道 stopped work $W_{R,3}^M$，所有起始与 merge 时钟相消，得到

$$\boxed{[W_{R,3}^M]_+\leq\Phi_I(\tau)+CA_R.}$$

这里 $\Phi_I=\mathscr{K}_R[\Omega_I^R]$。若最终块为 $[a,b]_{\mathbb{Z}}$，写 $r_m=K_{m,R}-K_{m,R}^{\partial}\geq 0$，则终端量有精确非负分解

$$\boxed{\Phi_I(t)=\sum_{[a,b]\in\mathrm{Comp}(I)}\left[K_{a,R}^{\partial}(t)+\sum_{m=a}^{b}r_m(t)\right].}$$

这是本节保留的正结果：三通道重组消除了 temporal genealogy debt；剩余障碍被定位到每个最终块的 root-boundary clock 与完整 ℓ^1 residual，而不是停止时刻本身。

9. 单块标量族排除未加权 genealogy 压缩

取 $I_N=\{1,\dots,N\}$，所有 shell 在同一时刻激活，边界 clocks 为零，令 $K_{k,R}=F_{k,R}=h$、$Q=D=0$，并用 tower identity 递归构造 ball clocks。所有标量 completed-clock 恒等式逐项成立，且

$$\boxed{W_N^{\mathrm{sc}} = N, \qquad Y_{2,R}^{\mathrm{sf}} = \sqrt{N}.}$$

同时

$$\sup_t \#\,\mathrm{Comp}(I(t)) = 1, \qquad \#\{\text{activation epochs}\} = 1, \qquad \#\{\text{block mergers}\} = 0.$$

因此 component、epoch 或 merger 数单独不能完成 $\ell^1 \to \ell^2$ 压缩。有限 genealogy 的精确计数为

$$\boxed{|I_{\mathrm{rt}}| + |I_{\mathrm{out}}| + |I^{\partial}| = 2|I| - e_{\mathrm{tie}},}$$

仍是 $O(|I|)$，不是无维数损失的平方函数界。这个结论只排除 **scalar completed-clock algebra + unweighted genealogy**；PDE-weighted block length、耗散支付与跨通道动力学符号仍可能有效。

10. 局部耗散的精确拆分与 Rayleigh 时间集

保留 R0.74P 的 suitable-weak 总局部耗散测度

$$\boldsymbol{\mu} = |\nabla u|^2\,dx\,dt + \boldsymbol{D}, \qquad \boldsymbol{D} \geq 0.$$

对每个壳层定义动能与黏性密度

$$e_{k,R}(t) = \frac{\gamma_k \eta_R(t)}{2R} \int_{\mathbb{T}^3} \Psi_k^R |v_R|^2, \qquad g_{k,R}(t) = \frac{\gamma_k \eta_R(t)}{R} \int_{\mathbb{T}^3} \Psi_k^R |\nabla v_R|^2.$$

把共同零测集上的代表取为零后，两行均可测。对好终端时刻 τ，耗散 clock 精确分解为

$$D_{k,R}(\tau) = \int_{s_R}^{\tau} g_{k,R}(t)\,dt + m_{k,R}(\tau), \qquad m_{k,R}(\tau) \geq 0,$$

其中 $m_{k,R}$ 是反常缺陷部分。给定正序列 $\boldsymbol{\lambda} = (\lambda_k)$，直接定义

$$L_{k,R} = \left\{ g_{k,R} \leq \frac{2\lambda_k}{R^2} e_{k,R} \right\}, \qquad H_{k,R} = (s_R, \tau) \setminus L_{k,R}.$$

当分母与 η_R 为正时，这等价于 cutoff-weighted ratio

$$\rho_{k,R} = R^2 \frac{\int \Psi_k^R |\nabla v_R|^2}{\int \Psi_k^R |v_R|^2} \le \lambda_k.$$

原定义不做除法；分母为零时两行同时为零，因此没有 $0/0$ 约定。

11. 八分之一、八分之一、四分之一三分法

对耗散主导壳层写 $T_k = K_{k,R}(\tau) > 0$ 且 $D_{k,R}(\tau) \ge T_k/2$。按优先顺序定义 defect、high 与 low 三类：

$$\begin{aligned} \mathcal{I}_{\mathrm{def}} &= \{m_{k,R}(\tau) \ge T_k/8\}, \\ \mathcal{I}_{\mathrm{hi}} &= \left\{k \notin \mathcal{I}_{\mathrm{def}} : \int_{H_{k,R}} g_{k,R} \ge T_k/8\right\}, \\ \mathcal{I}_{\mathrm{lo}} &= \mathcal{I}_D \setminus (\mathcal{I}_{\mathrm{def}} \cup \mathcal{I}_{\mathrm{hi}}). \end{aligned}$$

剩余的低 Rayleigh 类满足精确严格不等式

$$\int_{L_{k,R}} g_{k,R} > \frac{1}{4} T_k, \qquad \frac{1}{R^2} \int_{L_{k,R}} e_{k,R} > \frac{T_k}{8\lambda_k}.$$

令 $\delta_{k,R} = |L_{k,R}|/R^2$。冻结时间窗只给 $0 < \delta_{k,R} \le 4$，但这已经足够使 Jensen 给出

$$\frac{1}{R^2} \int_{L_{k,R}} e_{k,R}^{3/2} \ge \frac{1}{2} \left(\frac{T_k}{8\lambda_k} \right)^{3/2}.$$

时间集变薄不会破坏这一步：在固定动能时间质量下，它反而增大 $L_t^{3/2}$ 行。这里证明的是积分质量，不是统一时间厚度。

12. 全壳层低 Rayleigh 二次支付

把 R0.74R 的 padded-shell 三次付款限制到每个壳层自己的 $L_{k,R}$：

$$p_{k,R}^{\mathrm{lo}} = R^{-2} \gamma_k \int_{L_{k,R}} \eta_R^{3/2} \int_{\operatorname{supp} \psi_k^R} |\tilde{v}_R|^3.$$

空间 Hölder 与上一节合并，先得到逐壳层估计

$$T_k \le C_2 \lambda_k 2^k \gamma_k^{1/3} (p_{k,R}^{\mathrm{lo}})^{2/3}.$$

定义

$$\mathscr{L}(\boldsymbol{\lambda})=\sum_{k\ge 1}2^{3k}\gamma_k\lambda_k^3.$$

只要 $\mathscr{L}(\boldsymbol{\lambda})<\infty$，跨壳层 Hölder 与继承的非负付款给出

$$\boxed{\sum_{k\in\mathcal{I}_{\mathrm{lo}}(\tau)}K_{k,R}(\tau)\le C_3\mathscr{L}(\boldsymbol{\lambda})^{1/3}(P_R^M)^{2/3}.}$$

常数序列 $\lambda_k=1$ 可用；$\lambda_k=\gamma_k^{-\alpha}$ 在 $0\le\alpha<1/3$ 时可用；近临界序列

$$\lambda_k^{(\varepsilon)}=2^{-(1+\varepsilon)k}\gamma_k^{-1/3}$$

给出 $\mathscr{L}=2^{-3\varepsilon}/(1-2^{-3\varepsilon})$。临界 $\varepsilon=0$ 时每个 summand 等于 1，级数发散。这只是本论证的序列空间边界，不是任意解自动满足的 Rayleigh profile。

13. 未关闭 residual 与条件 finite-exception 接口

耗散主导族的完整剩余账本是

$$\sum_{k\in\mathcal{I}_D(\tau)}T_k\le C_3\mathscr{L}(\boldsymbol{\lambda})^{1/3}A_R+8\sum_{k\in\mathcal{I}_{\mathrm{def}}(\tau)}m_{k,R}(\tau)+8\sum_{k\in\mathcal{I}_{\mathrm{hi}}(\tau)}\int_{H_{k,R}}g_{k,R},\qquad A_R=(P_R^M)^{2/3}.$$

这条不等式没有隐藏新的 ℓ^1 clock remainder，但也没有控制最后两项。若未来 PDE 定理证明

$$\#(\mathcal{I}_{\mathrm{def}}(\tau)\cup\mathcal{I}_{\mathrm{hi}}(\tau))\le N_D,$$

则 Cauchy--Schwarz 立即给出剩余 clock 至多为 $\sqrt{N_D}Y_{2,R}^{\mathrm{sf}}$。这是 **PROVED CONDITIONAL IMPLICATION ONLY**；本步没有证明一致有限例外，更没有由此推出 (Q.1)。

14. 高频剪切诊断与严格边界

继承的光滑周期剪切

$$u_N(t,x)=Ae^{-N^2(t-t_-)}\sin(Nx_2)e_1,\qquad p_N=0,$$

满足固定 cutoff 上 $\rho_{k,R}^{(N)}/(R^2N^2)\to 1$。因此对固定阈值，充分高频时 high-Rayleigh 时间集可以在全部活跃时间出现；不能直接删除这一 residual。

但该剪切的 flux clock 满足 $F_{k,R}=0$，因而 $K_{k,R}=Q_{k,R}$，其 completed clocks 已由继承的 Q-variation ledger 支付。它只说明 high-Rayleigh 时间集非空，不是低 Rayleigh 定理或 completed-clock 估计的反例，也不把壳层自动放入优先类 $\mathcal{I}_{\mathrm{hi}}$。

Step 7 严格证明低 Rayleigh 耗散支的同时二次支付、精确 residual ledger 与条件 finite-exception implication。高 Rayleigh 支、反常缺陷支、stopped-work depletion、任意时钟 extraction、(Q.1)、尺度收缩与正则性仍为 **OPEN**。**NOT CLAY.**

15. 证书与独立审计

Step 5 最终确定性证书通过：

- 5/5 个精确 ledger 行；
- 7/7 个有限检查；
- 55/55 个结构检查；
- 4/4 个负向符号突变检查。

有限覆盖包括 312 个有理 cutoff 值、228 个导数样本、1024 个含并列 stop 的 stopped configuration、82432 个布尔激活比较、$M=2,\dots,8$ 的全部有限 Abel 端点，以及 $N=1,\dots,24$ 在五个有理时刻的抽象 tower。两个临时目录独立重算得到逐字节一致的 JSON 与 Markdown 报告。

Step 6 主证书另通过 4/4 exact、8/8 finite、58/58 structural、10/10 mutation；独立 Ruby 实现通过 9/9 independent 与 8/8 mutation，且与 producer 交叉一致。九类 forged JSON 全部被拒绝。

Step 7 主证书通过 16/16 exact、8/8 finite、52/52 structural、9/9 negative mutations。独立 Ruby 链通过 6/6 groups、31/31 structural 与 9/9 adversarial mutations，并与主 producer 交叉一致。主文 SHA-256 为 `e835a104f4a6f4d2281bef877dd6bfeb73f1c2396f6bd28203bb0812f7f8e3d3`。

这些是 **FINITE** 证书，只验证公式实现和符号哨兵；它们不机器证明 cutoff 光滑性、周期化/unfolding、suitable local-energy 计算、无限支撑估计或抽象见证的 PDE 实现。解析证明和有限证书必须继续分开。

16. 决定与下一门槛

本节已经 **PROVED**：

- 一侧 ball cutoff 与 flux 符号恒等式；
- completed ball-clock tower；
- 三族二次 $\mathcal{Q}$ 付款；
- root、outer、weight-drop 的精确 stopped 时间方向；
- terminal weight-drop Abel 恒等式；
- 标量 positive-clock 的 ℓ^1/ℓ^2 obstruction。
- 四通道 signed recombination 精确重建 stopped increments，证明该路线 circular；
- 三通道 genealogy cutoff 非负、插入符号有利、终端块分解精确；
- 单块标量族与有限 genealogy 计数的 abstract scalar no-go。
- 黏性/缺陷耗散的精确拆分与可测 low/high-Rayleigh 时间集；
- 八分之一、八分之一、四分之一优先三分法；
- 低 Rayleigh 动能时间质量、Jensen 转换与全壳层 $(P_R^M)^{2/3}$ 支付；
- admissible/critical Rayleigh profile 边界、精确 residual ledger 与条件 finite-exception implication。

本节关闭两条相邻的独立代数路线：分开估计所有 positive completions 只得到 ℓ^1；完整保留符号再线性重组则精确返回原未知量。未加权 component/epoch/merger 计数也不能修复差距。

Step 7 已经关闭耗散主导族的 low-Rayleigh 部分。下一步应只检验 high-Rayleigh 黏性 residual 与 anomalous-defect residual 能否由 PDE 结构支付，或是否能证明一致 finite-exception theorem；不能把条件接口写成已完成定理。

root/outer/weight-drop 动力学控制、high-Rayleigh/defect residual、R0.74R persistence hypotheses、无条件固定尺度不等式 (Q.1)、尺度收缩、正则性、奇点形成和 Clay 问题全部保持 **OPEN / NOT CLAIMED**。

LOW-RAYLEIGH BRANCH PAID. ABSTRACT SCALAR NO-GO RETAINED IN ITS ORIGINAL SCOPE. NOT CLAY.

17. 继承边界

- actual collar traces 与四通道 split：继承自 R0.74S Step 3，PROVED；
- stopped-family activation：继承自 R0.74S Step 2，PROVED；
- thin boundary clock 与 $K_m^\partial \le K_m$：继承自 R0.74S Step 4，PROVED；
- suitable-weak completed-clock operator：继承自 R0.74P，PROVED；
- weighted S_2 与 doubled-radius support ledger：继承自 R0.74H，PROVED；
- frozen adjacent-weight tail：继承自 R0.74S Step 1，PROVED。
- suitable-weak 耗散测度与 completed shell clock：继承自 R0.74P，PROVED；
- shell-dependent cubic payment 与 padded-shell Hölder：继承自 R0.74R，PROVED；
- 高频光滑剪切诊断：继承自 R0.73Y 与 R0.74B，在其原范围内 PROVED。

不声称新颖性、优先权、正则性或千禧年问题结论。

18. 三个时间测度与开放终端约定

固定 R0.74P--R0.74R 的 periodic suitable-weak Version-M 几何：黏性系数一、尺度 R、终端锚定路径 X_R、非降时间 cutoff η_R、padded shell cutoff Ψ_k^R 与权重 γ_k。写

$$\mathcal{T}_R = (s_R, t_0), \qquad J_\tau = (s_R, \tau), \qquad s_R = t_0 - 4R^2.$$

对 Borel 集 $A \subset \mathcal{T}_R$ 定义

$$\boxed{\begin{aligned} &\sigma_{k,R}(A) = R^{-2}\int_A e_{k,R}(t)\,dt,\\ &\nu_{k,R}(A) = \gamma_k R^{-1}\int_{A\times\mathbb{T}^3} \eta_R(t)\Psi_k^R(x - X_R(t))\,d\boldsymbol{\mu}(t,x),\\ &\beta_{k,R}(A) = \int_A |\dot{Q}_{k,R}(t)|\,dt. \end{aligned}}$$

σ 是抛物归一化动能测度，ν 是黏性加反常缺陷的总耗散测度，β 是 canonical Q-primitive 的 total-variation measure。区间 J_τ 在终端开口，与 inherited completed-clock 的 (s_R, τ) 约定一致，因此不会额外计入 $t=\tau$ 的耗散原子。冻结 cutoff 与其导数在 s_R 的公共邻域内为零，左端也没有质量逃逸。

若 $\boldsymbol{\delta}_{k,R}$ 是反常耗散的加权时间推前，则

$$d\nu_{k,R} = g_{k,R}\,dt + d\boldsymbol{\delta}_{k,R}, \qquad \boldsymbol{\delta}_{k,R}(J_\tau) = m_{k,R}(\tau).$$

在 inherited local-energy good terminal time 上，$\nu_{k,R}(J_\tau) = D_{k,R}(\tau)$，而 $\beta_{k,R}(J_\tau) = \mathrm{TV}_{J_\tau} Q_{k,R}$。这些是 S.163--S.166 的精确测度身份。

19. 标量 excess 与 Jordan envelope

固定正的确定性 profile $\boldsymbol{\lambda} = (\lambda_k)$，令

$$\alpha^{\boldsymbol{\lambda}}_{k,R} = \nu_{k,R} - \beta_{k,R} - 2\lambda_k\sigma_{k,R}.$$

两个 excess 层级分别是

$$\boxed{x^{\boldsymbol{\lambda}}_{k,R}(\tau) = \left[\alpha^{\boldsymbol{\lambda}}_{k,R}(J_\tau)\right]_+, \qquad X^{\boldsymbol{\lambda}}_{k,R}(\tau) = (\alpha^{\boldsymbol{\lambda}}_{k,R})^+(J_\tau).}$$

x 先计算整个终端区间的 signed mass，再取标量正部；X 是 Jordan decomposition 的正测度质量，会保留不同时间区间之间被净额相消的局部正 excess。Hahn decomposition、Radon measure 正则性与 Urysohn 逼近给出

$$0 \le [\alpha(J_\tau)]_+ \le \alpha^+(J_\tau) = \sup_{A\in\mathcal{B}(J_\tau)} \alpha(A) = \sup_{\substack{\phi\in C_c(J_\tau)\\ 0\le\phi\le 1}} \int_{J_\tau} \phi\, d\alpha.$$

因此 $0 \le x \le X$，并且

$$\nu(J_\tau) \le \beta(J_\tau) + 2\lambda\sigma(J_\tau) + x \le \beta(J_\tau) + 2\lambda\sigma(J_\tau) + X.$$

终端三分法使用较小的 x；X 用于局部化与弱极限稳定性，不是更小的终端上界。

20. 六分之一优先三分法与两个已付分支

对 dissipation-dominated shell 写 $T_k = K_{k,R}(\tau) > 0$、$D_{k,R}(\tau) \ge T_k/2$。按优先顺序检查

$$\beta_{k,R}(J_\tau) \ge \frac{1}{6}T_k, \qquad \sigma_{k,R}(J_\tau) > \frac{T_k}{12\lambda_k}.$$

若两项都失败，则

$$\alpha(J_\tau) > \frac{1}{2}T_k - \frac{1}{6}T_k - \frac{1}{6}T_k = \frac{1}{6}T_k,$$

所以 $x_k > T_k/6$。这给出 S.170--S.171 的 literal trichotomy：每个耗散主导正 clock 要么由 β 支付至少六分之一，要么有足够 kinetic time mass，要么进入 selected excess 类。

β-branch 由 inherited quadratic Q-variation ledger 支付。对 kinetic branch，令 $\delta_\tau = |J_\tau|/R^2 < 4$，Jensen 给出

$$R^{-2}\int_{J_\tau} e^{3/2}_{k,R} \ge \delta_\tau^{-1/2}\sigma_{k,R}(J_\tau)^{3/2} > \frac{1}{2}\left(\frac{T_k}{12\lambda_k}\right)^{3/2}.$$

与 inherited padded-shell cubic estimate 合并可得

$$T_k \le C_4 \lambda_k 2^k \gamma_k^{1/3} (p_{k,R}^{\tau})^{2/3}.$$

定义 $\mathscr{L}(\boldsymbol{\lambda}) = \sum_k 2^{3k} \gamma_k \lambda_k^3$。若 $\mathscr{L} < \infty$，有限壳层 Hölder 与 monotone convergence 给出

$$\boxed{\sum_{k \in \mathcal{I}_\sigma(\tau)} T_k \le C_5 \mathscr{L}(\boldsymbol{\lambda})^{1/3} (P_R^M)^{2/3}.}$$

21. selected/global excess ledger 与 Step 7 比较

β-branch 满足 $\sum_{\mathcal{I}_\beta} T_k \le C_\beta (P_R^M)^{2/3}$，selected excess branch 满足 $T_k \le 6x_k$。因此

$$\boxed{\sum_{k \in \mathcal{I}_D(\tau)} K_{k,R}(\tau) \le C_6 (1 + \mathscr{L}^{1/3}) (P_R^M)^{2/3} + 6 \sum_{k \in \mathcal{I}_x(\tau)} x_{k,R}^{\boldsymbol{\lambda}}(\tau).}$$

定义全壳层接口

$$\mathfrak{x}_{1,R}^{\boldsymbol{\lambda}} = \sum_k x_{k,R}^{\boldsymbol{\lambda}}, \qquad \mathcal{X}_{1,R}^{\boldsymbol{\lambda}} = \sum_k X_{k,R}^{\boldsymbol{\lambda}}, \qquad \mathfrak{x}_{1,R} \le \mathcal{X}_{1,R}.$$

selected inequality 因而可放宽为加上 $6\mathfrak{x}_{1,R}$，再放宽为加上 $6\mathcal{X}_{1,R}$。global sums 会覆盖已由 β 支付的壳层；X 还会保留被终端 signed mass 抵消的局部正 excess。

对 Step 7 同一个 high/low-Rayleigh 分割，逐壳层有

$$\boxed{x_{k,R} \le X_{k,R} \le m_{k,R}(\tau) + \int_{H_{k,R}} g_{k,R}(t)\,dt.}$$

这是对 old raw residual 的逐壳层支配，但由于两步 priority partition 不同，不能说 Step 8 的 global sum 数值上严格小于 Step 7 的 prioritized residual。

22. exact shear、lower semicontinuity 与光滑公式边界

对 inherited heat shear，Step 7 已证明 $F_k = 0$，故 $K_k = Q_k$。若 $T_k = K_k(\tau) > 0$，则

$$T_k = Q_k(\tau) \le \mathrm{TV}_{J_\tau} Q_k = \beta_k(J_\tau), \qquad x_k(\tau) = 0.$$

所以它进入 β-priority branch，不是 excess theorem 的反例；这里不声称 Jordan envelope X_k 为零。

在 R0.74P 的 fixed-scale Version-M topology 下，若 $u_n \to u$ strongly in L^3、$\nabla u_n \rightharpoonup \nabla u$ in L^2、$p_n \rightharpoonup p$ in $L^{3/2}$，则每个 fixed shell 上 $\nu_n \rightharpoonup^* \nu$ locally，$\sigma_n \to \sigma$ 与 $\beta_n \to \beta$ in total variation。开放集 Portmanteau 与 Jordan continuous-test formula 分别给出

$$x_k[u,p](\tau) \le \liminf_n x_k[u_n,p_n](\tau), \qquad X_k[u,p](\tau) \le \liminf_n X_k[u_n,p_n](\tau).$$

finite-shell Fatou 再接 monotone convergence，得到 $\mathfrak{x}_{1,R}$ 与 $\mathcal{X}_{1,R}$ 的全壳层 lower semicontinuity。这个结论只在固定 R 的 inherited topology 中成立，不提供 cross-scale compactness。

若解本身光滑，则 defect measure 为零，且

$$x_k = \left[\int_{s_R}^{\tau} \left(g_k - |\dot{Q}_k| - 2\lambda_k R^{-2} e_k\right) dt\right]_+, \qquad X_k = \int_{s_R}^{\tau} \left[g_k - |\dot{Q}_k| - 2\lambda_k R^{-2} e_k\right]_+ dt.$$

只有在另行提供满足上述 topology 的 smooth periodic NSE sequence 时，才能对这些光滑公式取 lim inf。本节不声称任意 suitable weak solution 都存在这样的 smooth approximants。

23. fixed-scale finiteness 与 terminal flux domination

由 signed-measure order $\alpha \le \nu$ 得 $\alpha^+ \le \nu$。Tonelli、$\Theta_R = \sum_k \gamma_k \Psi_k^R$ 的 inherited C^2-convergence 与 local finiteness 给出

$$\boxed{0 \le x_k \le X_k \le \nu_k(J_\tau), \qquad \mathcal{X}_{1,R}(\tau) \le \sum_k \nu_k(J_\tau) < \infty.}$$

这是 fixed-scale total-dissipation finiteness，不是关于 R 的一致二次界。

canonical primitives 在 s_R 为零，故 $\beta_k(J_\tau) \ge |Q_k(\tau)|$。completed-clock identity 给出

$$\alpha_k(J_\tau) = Q_k(\tau) + F_k(\tau) - E_k(\tau) - \beta_k(J_\tau) - 2\lambda_k \sigma_k(J_\tau) \le F_k(\tau).$$

所以

$$\boxed{x_k(\tau) \le [F_k(\tau)]_+, \qquad \mathfrak{x}_{1,R}(\tau) \le \mathfrak{L}^M_{\mathrm{abs},R} \le C P_R^M.}$$

线性 CP_R^M bound 在 $P_R^M > 1$ 时不是二次 $(P_R^M)^{2/3}$ bound。

24. selected excess 归入 Step 2 gate

在 $\eta_R = \eta_R' = 0$ 的共同初始区间内取 common good time σ_0。每个壳层都满足 $K_k(\sigma_0) = Q_k(\sigma_0) = F_k(\sigma_0) = 0$。若 $x_k(\tau) > 0$，则 $K_k(\tau) > 0$，所以这个共同零起点满足 Step 2 的 strict upcrossing condition。对任意有限非空 $G \subset \{k : x_k(\tau) > 0\}$，

$$W_R^M(\tau; G, (\sigma_0)_{k\in G}) = \sum_{k\in G} F_k(\tau) \geq \sum_{k\in G} x_k(\tau) > 0.$$

取 finite-family supremum 得

$$\boxed{\mathfrak{x}_{1,R}(\tau) \leq \mathfrak{W}^M_{\mathrm{up},R} \leq \mathfrak{L}^M_{\mathrm{abs},R} \leq CP_R^M.}$$

在 priority-selected excess class 上，$\beta_k < T_k/6$ 且 $|Q_k(\tau)| \leq \beta_k$，故

$$F_k(\tau) = T_k - Q_k(\tau) \geq T_k - |Q_k(\tau)| > \frac{5}{6}T_k, \qquad T_k < \frac{6}{5}F_k(\tau).$$

结合 β 与 kinetic 两个已付分支，S.196 给出

$$\boxed{\sum_{k\in\mathcal{I}_D(\tau)} K_{k,R}(\tau) \leq C_6(1+\mathscr{L}(\boldsymbol{\lambda})^{1/3})(P_R^M)^{2/3} + \frac{6}{5}\mathfrak{W}^M_{\mathrm{up},R}.}$$

这一步证明 selected defect/high-Rayleigh scalar residual 不是新的独立 obstruction，而是既有 stopped-work ledger 的子账本；但 Step 2 的 gate 本来就允许共同 zero start，所以它没有缩窄 no-exception supremum。

25. no-exception stopped work 与 full terminal flux 的等价

定义已付的 Q-variation、full terminal clock supremum 与 full positive cumulative flux：

$$\boxed{B^M_{Q,R} = \sum_k \mathrm{TV}_{[s_R,t_0)}\, Q_{k,R} \leq C_Q(P_R^M)^{2/3}, \quad \mathcal{K}_R^M = \sup_{\tau\in\mathcal{G}_R} \sum_k K_{k,R}(\tau), \quad \mathfrak{C}^M_{\mathrm{full},R} = \sup_{s_R<\tau<t_0} \left[\sum_k F_{k,R}(\tau)\right]_+.}$$

对 S.37 任意 admissible stopped family，把 work 与同一终端的 full flux 直接相减；start 与 omitted-shell 两部分中的 K_k 都非负，余下的 Q-差总计至多 B_Q。反向则在 common good terminal time 上，对 $K_k(\tau) > 0$ 且 $F_k(\tau) > 0$ 的有限壳层集合使用共同 zero stop，再取 monotone limit；若某个 omitted shell 满足 $K_k = 0 < F_k$，则 $F_k = -Q_k$，总量仍至多 B_Q。因此 S.197--S.198 得到

$$\boxed{\mathcal{K}_R^M - B^M_{Q,R} \leq \mathfrak{W}^M_{\mathrm{up},R} \leq \mathcal{K}_R^M + B^M_{Q,R}, \qquad |\mathfrak{W}^M_{\mathrm{up},R} - \mathfrak{C}^M_{\mathrm{full},R}| \leq B^M_{Q,R}.}$$

第二条的系数一已经被 single-shell scalar stress $K = 0, Q = -B, F = B$ 证明 sharp。故 Step 2 no-exception observable 只是在已付二次误差 B_Q 内等价于 full-cutoff positive cumulative flux；它不是一个更小的 signed-depletion quantity。

26. smooth exact-family no-go

使用 inherited R0.74O/P smooth periodic exact family，已有

$$\mathfrak{C}^{M,*}_{R_j} \asymp T_*, \qquad \mathfrak{C}^{M,*}_{\mathrm{full},R_j} \geq \mathfrak{C}^{M,*}_{R_j}, \qquad (P^{M,*}_{R_j})^{2/3} \asymp \frac{T_*}{K_*}, \qquad K_* \to \infty.$$

由于 S.198 的 additive Q-error 仅为 $O((P^{M,*}_{R_j})^{2/3})$，S.199 推出

$$\boxed{\frac{\mathfrak{W}^{M,*}_{\mathrm{up},R_j}}{(P^{M,*}_{R_j})^{2/3}} \longrightarrow \infty.}$$

也可只选 exact-family target shell：其 terminal clock 为 $\gtrsim T_*$，full Q-variation 为 $O(T_*/K_*)$，共同 zero stop 给 stopped work $\gtrsim T_*$。这个 witness 是 smooth、periodic、mean-zero、unforced、pressure-free 的真实 NSE solution family。

因此普适 antecedent

$$\mathfrak{W}^{M}_{\mathrm{up},R} \lesssim (P^M_R)^{2/3}$$

被严格 **REFUTED / no-go**。这不反驳 S.38 的条件代数、S.196、带 terminal exceptions 且由 $Y^{\mathrm{sf}}_{2,R}$ 支付的估计，也不反驳 (Q.1)。

27. 压力测试、证书与下一门槛

本步还保留六类 stress tests：interior atom 被 Jordan formula 完整检测；$\nu = \beta$ 的 already-paid dissipation 被 excess 扣除；高频 divergence-free functional family 排除仅靠 incompressibility/cutoff/Hölder 得到 x 或 X 的 cubic bound；符号密度的时间相消证明 X 可严格大于 x；质量向硬终端逃逸说明只能主张 lower semicontinuity；$Q_n = n^{-1}\sin(nt)$ 说明 primitive uniform convergence 不能替代 $\dot{Q}_n$ 的 strong L^1 convergence。这些 stress tests 的 PDE 身份边界逐项保留。

最终 Step 8 主证书通过 16/16 exact、19/19 finite、75/75 structural、20/20 negative mutations。独立 Ruby 审计通过 14/14 groups、22/22 exact rows、61/61 structural、14/14 source mutations、10/10 artifact mutations 与 6/6 report checks。主文 SHA-256 为 `0a79f2c5bb59644eca710b3d9341776853ceb4d1f65a36869c2465073f8c08ab`。有限证书支持实现可复现性，不替代 measure/PDE 解析证明。

Step 8 **PROVED**：S.163--S.196 的 three-measure excess interface、one-sixth trichotomy、两个已付分支、lower semicontinuity、fixed-scale finiteness、terminal flux domination 与 stopped-work bridge；S.197--S.198 的 no-exception two-sided equivalence；以及 S.199 的 smooth exact-family refutation。

继续 **OPEN**：固定 best-N_0、terminal-dependent exception estimate，并以 $\sqrt{N_0}Y^{\mathrm{sf}}_{2,R}$ 支付例外尾；Jordan envelope 的二次/平方函数/finite-exception bound；smooth NSE approximation existence；R0.74R extraction hypotheses；unconditional fixed-scale (Q.1)、scale contraction、prescribed-centre packing 与 regularity。

明确 **NOT CLAIMED**：exact shear 的 $X = 0$；selected excess sum lower semicontinuity；$X \leq \mathfrak{W}_{\mathrm{up}}$；Step 8 缩窄 Step 2 gate；硬终端 mass convergence；新颖性、优先权、奇点或 Clay 结论。

下一步不再尝试 no-exception supremum。唯一冻结方向是回到 R0.74Q (Q.7)--(Q.12) 的 fixed best-N、terminal-dependent exceptions，并保留 $\sqrt{N}, Y^{\mathrm{sf}}_{2,R}$ 付款。

UNIVERSAL NO-EXCEPTION STOPPED-WORK QUADRATIC BOUND: REFUTED. CONDITIONAL S.38: RETAINED. NOT CLAY.

28. Step 9 的结果与终端域

Step 9 不再修补已被 Step 8 反驳的 no-exception gate，而是回到 R0.74Q 的 fixed best-N terminal tail。必须区分 plateau 终端域 I_R 与 full clock interval $\mathcal{T}_R=(s_R,t_0)$：

$$\mathfrak{C}_R^M(\mathcal{D}):=\sup_{\tau\in\mathcal{D}}\left[\sum_{k\ge1}F_{k,R}(\tau)\right]_+,\qquad \mathfrak{C}_R^M(I_R)\le\mathfrak{C}_R^M(\mathcal{T}_R). \tag{S.200}$$

只保留这个不等式，不声称两个终端域相等。记

$$A_R=(P_R^M)^{2/3},\qquad Z_R=Y_{2,R}^{\mathrm{sf}},\qquad B_{Q,R}^M=\sum_k \mathrm{TV}\,Q_{k,R}\le C_QA_R. \tag{S.201}$$

Q,F,K 连续、从零起步，$K\ge0$；继承的 variation 与 Step 8 有限性保证下文所有无穷和绝对收敛。

29. signed best-N tail 与正确量词

对 $x\in\ell^1(\mathbb{N};\mathbb{R})$ 及固定整数 $N\ge0$，定义

$$\mathcal{S}_N(x):=\inf_{S\subset\mathbb{N},\,\#S\le N}\left[\sum_{k\notin S}x_k\right]_+. \tag{S.202}$$

对 F 与 K 分别在终端域上取 $\sup_\tau$，得到 $\mathcal{S}_{N,R}^F(\mathcal{D})$ 与 $\mathcal{S}_{N,R}^K(\mathcal{D})$，即 (S.203)。删除的只能是最大的 N 个正坐标，因而

$$\mathcal{S}_N(x)=\left[\sum_k x_k-\sum_{m=1}^N x_m^{+*}\right]_+,\qquad |\mathcal{S}_N(x)-\mathcal{S}_N(y)|\le\|x-y\|_{\ell^1}.$$

这是 forced full signed tail：负坐标的 cancellation 不能被任意 finite-subset supremum 取代。正确量词是

$$\boxed{\sup_\tau\ \inf_{\#S_\tau\le N}},$$

其中 N 与终端、尺度、解无关，但最优例外集 S_τ 可随 τ 变化。它不是更强的 $\inf_S\sup_\tau$。

30. F-half-exit：精确二分之一

固定终端 τ，令 $f_k=F_{k,R}(\tau)$。当 $f_k\ne0$ 时，取最后一个满足

$$\operatorname{sgn}(f_k)F_{k,R}(t) \le |f_k|/2$$

的时刻 $\ell_k^F(\tau)$；当 $f_k = 0$ 时定义 $\ell_k^F(\tau) = \tau$，即 (S.204)。连续性立即给出

$$F_{k,R}(\ell_k^F(\tau)) = \frac{1}{2}F_{k,R}(\tau), \qquad F_{k,R}(\tau) - F_{k,R}(\ell_k^F(\tau)) = \frac{1}{2}F_{k,R}(\tau). \tag{S.205}$$

在完整非例外 signed tail 上先求和，再取正部、$\inf_S$ 与 $\sup_\tau$，得到

$$\boxed{\mathfrak{W}^F_{1/2,N,R}(\mathcal{D}) = \frac{1}{2}\mathcal{S}^F_{N,R}(\mathcal{D}).} \tag{S.207}$$

对 plateau 域，这给出

$$\mathfrak{C}^M_R \le B^M_{Q,R} + \sqrt{N}\,Z_R + 2\mathfrak{W}^F_{1/2,N,R}(I_R). \tag{S.208}$$

但 F-half-exit 一般不满足 Step 2 的严格 S.25 upcrossing。精确反例 $F(t) = t$、$K(t) = \min\{2t, 1\}$、$\tau = 1$ 中，$\ell^F = 1/2$，但 $K(1) - K(1/2) = 0$。因此 (S.209) 排除了“canonical half-exit 自动是 S.25 admissible stop”的推断。

31. K-theta last exit 与尖锐一个 B_Q

对 $0 < \theta < 1$ 及 $T_k = K_{k,R}(\tau) > 0$，令 $\ell^K_{k,\theta}(\tau)$ 为最后一个满足 $K_{k,R}(t) \le \theta T_k$ 的时刻；$T_k = 0$ 时取 $\ell = \tau$。由 $F = K - Q$，

$$L_{k,\theta}(\tau) := F_k(\tau) - F_k(\ell^K_{k,\theta}) = (1-\theta)T_k - \Delta Q_{k,\theta}(\tau). \tag{S.211}$$

完整非例外 tail 的 last-exit observable 与 best-N clock tail 满足

$$\boxed{(1-\theta)\mathcal{S}^K_{N,R}(\mathcal{D}) - B^M_{Q,R} \le \mathfrak{W}^K_{\theta,N,R}(\mathcal{D}) \le (1-\theta)\mathcal{S}^K_{N,R}(\mathcal{D}) + B^M_{Q,R}.} \tag{S.214}$$

误差只是一个 B_Q，不是两个；系数 1 在单标量连续 clock 上可达，所以是尖锐的。对 $0 < \theta < 3/4$，正终端壳层有

$$K_k(\tau) - K_k(\ell^K_{k,\theta}) = (1-\theta)T_k > \frac{1}{4}T_k. \tag{S.215}$$

因此在 good terminal 上，任意有限正终端壳层族可用共同稠密 good-time set 逼近，落入 S.37 的闭包。这只是有限族、正终端、good terminal 的 closure statement；它不说 last-exit selector 连续，也不允许把一个无穷且时间不连续的 cutoff 直接插入 local energy inequality。$\theta = 3/4$ 时严格余量消失，故不包含端点。

32. 与 R0.74Q 既有 gate 等价：no-gain

signed F-tail 与非负 K-tail 之差由一个 B_Q 控制：

$$\left|\mathcal{S}^F_{N,R}(\mathcal{D})-\mathcal{S}^K_{N,R}(\mathcal{D})\right| \le B^M_{Q,R}. \tag{S.217}$$

因此，对与 R 及解都无关的固定 N_0，

$$\mathcal{S}^F_{N_0,R}(\mathcal{D}) \lesssim A_R \iff \mathfrak{W}^F_{1/2,N_0,R}(\mathcal{D}) \lesssim A_R,$$

$$\mathcal{S}^K_{N_0,R}(\mathcal{D}) \lesssim A_R \iff \mathfrak{W}^K_{\theta,N_0,R}(\mathcal{D}) \lesssim A_R. \tag{S.218}$$

这是 Step 9 的精确 no-gain/no-go：证明 last-exit 二次界，就等价于证明已经开放的 best-N terminal-tail bound，canonical stop 本身没有提供更弱的过渡定理。当 $\mathcal{D}=\mathcal{T}_R$ 时正好是 Q.12；当 $\mathcal{D}=I_R$ 时只是更弱的 plateau restriction。

33. 量词、相消与 full-history 压力测试

(S.219) 用两个终端状态 $x(\tau_1)=(1,0)$、$x(\tau_2)=(0,1)$ 证明

$$\sup_{\tau}\inf_{\#S_\tau\le 1}\sum_{k\notin S_\tau}x_k=0,\qquad \inf_{\#S\le 1}\sup_{\tau}\sum_{k\notin S}x_k=1.$$

(S.220) 用 $F(\tau)=(1,-1)$、$N=0$ 证明 forced signed tail 为零，而任意 subset supremum 会挑出正坐标并得到 $1/2$；后者破坏了需要保留的 cancellation。

(S.221) 取 $M>N$ 个同步 completed clocks，$K_k=F_k=h$、$Q_k=0$。平台终端上

$$\mathcal{S}_N(F)=\mathcal{S}_N(K)=(M-N)H,\quad \mathfrak{W}^F_{1/2,N}=\frac{M-N}{2}H,\quad \mathfrak{W}^K_{\theta,N}=(1-\theta)(M-N)H.$$

这是抽象连续-clock stress test，不是 NSE solution；它严格显示 last exit 不会自动把 ℓ^1 tail 变成 ℓ^2 payment。另外，若所有 level exit 都发生在某个“recent window”之前，该窗口内根本没有可用 exit；因此 (S.210) 必须保留从 s_R 到 τ 的 full history，除非另有 PDE 定理支付早期段。

34. theta=2/3 的兼容性与下一 PDE residual

$\theta=2/3$ 与 Step 8 的 one-sixth rows 兼容：

$$\Delta K_{k,2/3}=\frac{1}{3}T_k,\qquad |\Delta Q_{k,2/3}|<\frac{1}{6}T_k \Longrightarrow \Delta F_{k,2/3}>\frac{1}{6}T_k. \tag{S.222}$$

这是为下一 PDE 分解选择的兼容参数，不是全局最优化定理。下一真正新的目标是：在删去 Step 7 low-Rayleigh 支、Step 8 $\mathcal{I}_\beta$ 与 $\mathcal{I}_\sigma$ 两个已付分支后，定义并审计 forced full PDE residual tail；只允许至多 N_0 个随终端变化的例外，且以 $\sqrt{N_0}Z_R$ 支付。该 residual 可能仍包含 anomalous defect 或 high-Rayleigh dissipation，本步没有预先定义成有利对象。

35. Step 9 主张边界与双路审计

Step 9 **PROVED**：(S.200)--(S.203) 的终端域分离与 best-N tails；(S.204)--(S.207) 的 signed half-exit 精确表示；(S.208) 的 plateau reduction；(S.209) 的 S.25 失败反例；(S.210)--(S.215) 的 K-last-exit、尖锐一个 B_Q 误差与 $\theta < 3/4$ 有限 good-stop closure；(S.216)--(S.218) 的 plateau reduction、F/K tail comparison 与 no-gain 等价；以及 (S.219)--(S.222) 的量词、相消、平台和 $\theta = 2/3$ 压力测试。

继承的 Step 8 no-exception gate 仍是 **REFUTED**，S.38 条件蕴含仍然 **RETAINED**。canonical last exits 不产生新二次压缩，是 completed-clock algebra 层面的严格 **no-gain/no-go**。

继续 **OPEN**：存在与解和尺度无关的固定 N_0，使 (S.218) 的任一 best-N_0 PDE tail 获得二次界；删去已付分支后的 residual full tail；(Q.1)、R0.74R extraction hypotheses、scale contraction、prescribed-centre packing 与 regularity。

明确 **NOT CLAIMED**：F-half-exit 满足 S.25；canonical selector 对终端连续；一个无穷 stopped cutoff 可直接作 local-energy test；arbitrary subset supremum 可取代 forced full tail；sup inf 可换成 inf sup；plateau 与 full domain 相等；单个 dominant packet 证明 $N_0 = 1$；标量 stress tests 是 PDE solutions；以及新颖性、优先权、奇点形成或 Clay 结论。

主证书通过 9/9 exact、8/8 finite、57/57 structural 与 18/18 mutations。独立 Ruby 审计通过 12/12 groups、91,396/91,396 finite cases、49/49 structural、21/21 source mutations、15/15 artifact mutations与 6/6 report checks。主文 SHA-256 为 `85003b3fdfdf28618a82a57d241e86c086704ea3ed3a9b192de223f3b8c3a4dd`。有限证书只支持实现可复现性，不替代 PDE 解析证明。

CANONICAL BEST-N LAST EXITS: EXACT REPRESENTATIONS, NO NEW QUADRATIC COMPRESSION. NEXT: PDE RESIDUAL TAIL AFTER PAID BRANCHES. NOT CLAY.

36. Step 10 的问题与结论

Step 9 已说明 canonical last exit 只表示 R0.74Q 的 best-N terminal tail，并不自行压缩它。Step 10 的问题更窄：先删除所有已经被二次 Q-variation ledger 或 velocity-cubic ledger 支付的壳层，canonical 2/3-last-exit tail 还剩什么？

答案是一个精确六分区。四个 paid classes 合计只需

$$6B^M_{Q,R} + C_5\mathscr{L}(\boldsymbol{\lambda})^{1/3}A_R, \qquad A_R = (P^M_R)^{2/3},$$

其中 Q 行只记一次 $6B_Q$，$\mathcal{P}_\sigma$ 与 $\mathcal{P}_{\mathrm{LE}}$ 先合并再做 Hölder，只记一次 C_5 cubic ledger。余下两个 residual mechanisms 是：short non-D、Q-small 的 $\mathcal{R}_{\mathrm{sh}}$，以及 Step 8 scalar-excess class $\mathcal{R}_x = \mathcal{I}_x$。二者共享同一个 best-$N$ exception budget，不能各自删除 N 个坐标。

本步证明的是 exact reduction。它没有证明存在与解、尺度无关的固定 N_0，使 residual tail 获得二次界；没有证明 Q.12、Q.1、正则性或 Clay 结论。**NOT CLAY.**

37. canonical 2/3-last exit 与固定 profile

保留 periodic suitable-weak Version-M、full clock interval $\mathcal{T}_R = (s_R, t_0)$、plateau interval I_R、good-time set $\mathcal{G}_R$，以及

$$A_R = (P_R^M)^{2/3}, \qquad B_{Q,R}^M = \sum_{k\ge 1} \mathrm{TV}_{[s_R,t_0)}\, Q_{k,R} \le C_Q A_R, \qquad Z_R = \left(\sum_{k\ge 1} v_{k,R}^2\right)^{1/2}.$$

固定 $\tau \in \mathcal{G}_R \cap \mathcal{T}_R$。若 $T_k = K_{k,R}(\tau) > 0$，取最后一个 $K \le 2T_k/3$ 的时刻 ℓ_k，并写

$$\boxed{\begin{gathered}\ell_k = \max\{t \in [s_R,\tau] : K_{k,R}(t) \le 2T_k/3\}, \quad J_k^{\mathrm{LE}} = (\ell_k,\tau), \quad d_k = \frac{\tau-\ell_k}{R^2}, \\ 0 < d_k < 4, \quad K_{k,R}(\ell_k) = \frac{2T_k}{3}, \quad K_{k,R}(t) > \frac{2T_k}{3}\ (t \in J_k^{\mathrm{LE}}), \\ \Delta Q_k = Q_k(\tau) - Q_k(\ell_k), \quad \Delta F_k = F_k(\tau) - F_k(\ell_k), \\ \Delta K_k = \frac{T_k}{3}, \qquad \Delta F_k = \frac{T_k}{3} - \Delta Q_k.\end{gathered}} \tag{S.223}$$

若 $T_k = 0$，令 $\ell_k = \tau$、$d_k = 0$、residual coordinate 为零。ℓ_k 不必是 good time；这里只用 K_k 连续。

固定正的 deterministic profile $\boldsymbol{\lambda}$，它与 R, τ 和解无关，并满足

$$\boxed{\mathscr{L}(\boldsymbol{\lambda}) = \sum_{k\ge 1} 2^{3k}\gamma_k \lambda_k^3 < \infty.} \tag{S.224}$$

在正终端壳层中，按 $d_k \gtreqless \lambda_k^{-3/2}$、$|\Delta Q_k| \gtreqless T_k/6$、$D_{k,R}(\tau) \gtreqless T_k/2$ 分出 long/short、$Q+/Q-$、D/non-D。等号分别放入 long、absolute-Q-large 与 D-dominated 一侧。

38. D-first 六分区与 genealogy

Step 8 的 full-history priority partition 保持不变：

$$\mathcal{I}_D = \mathcal{I}_\beta \,\dot\cup\, \mathcal{I}_\sigma \,\dot\cup\, \mathcal{I}_x.$$

这里 $\mathcal{I}_\beta, \mathcal{I}_\sigma, \mathcal{I}_x$ 仍在完整 $J_\tau = (s_R, \tau)$ 上定义，不能改写成 last-exit interval。按 D-first 优先顺序定义

$$\boxed{\begin{array}{ll} \mathcal{P}_\beta = \mathcal{I}_\beta, & \mathcal{P}_\sigma = \mathcal{I}_\sigma, \\ \mathcal{P}_{\mathrm{LE}} = \mathcal{I}_{\neg D} \cap \mathcal{I}_{\mathrm{long}}, & \mathcal{P}_Q = \mathcal{I}_{\neg D} \cap \mathcal{I}_{\mathrm{short}} \cap \mathcal{I}_{Q+}, \\ \mathcal{R}_{\mathrm{sh}} = \mathcal{I}_{\neg D} \cap \mathcal{I}_{\mathrm{short}} \cap \mathcal{I}_{Q-}, & \mathcal{R}_x = \mathcal{I}_x, \\ \{k : T_k > 0\} = \mathcal{P}_\beta \,\dot\cup\, \mathcal{P}_\sigma \,\dot\cup\, \mathcal{P}_{\mathrm{LE}} \,\dot\cup\, \mathcal{P}_Q \,\dot\cup\, \mathcal{R}_{\mathrm{sh}} \,\dot\cup\, \mathcal{R}_x. & \end{array}} \tag{S.225}$$

$\mathcal{P}_Q$ 表示 absolute-Q-large，不是 Q 符号为正。Step 7 low-Rayleigh 支不会再生成第七类：

$$\boxed{\mathcal{I}_{\mathrm{lo}} \subset \mathcal{I}_\beta \cup \mathcal{I}_\sigma, \qquad \mathcal{I}_{\mathrm{lo}} \setminus (\mathcal{I}_\beta \cup \mathcal{I}_\sigma) = \varnothing, \qquad \mathcal{I}_x = \mathcal{I}_x \cap (\mathcal{I}_{\mathrm{def}} \cup \mathcal{I}_{\mathrm{hi}}).} \tag{S.226}$$

这是 genealogy 与 no-double-charge 结论，不是对 $\mathcal{I}_x$ 的新支付；它仍可能来自 anomalous-defect 或 high-Rayleigh 支。

39. 一个 Q ledger 与一个 cubic ledger

对 $\mathcal{P}_\beta$，Step 8 给出 $T_k \le 6\beta_k(J_\tau)$；对 $\mathcal{P}_Q$，定义给出 $T_k \le 6|\Delta Q_k|$。两类位于互不相交的 D 与 non-D shell sets，因此先求和再扩大到完整 variation ledger：

$$\boxed{\sum_{k\in\mathcal{P}_\beta\cup\mathcal{P}_Q} T_k \le 6\left(\sum_{k\in\mathcal{P}_\beta}\beta_k(J_\tau)+\sum_{k\in\mathcal{P}_Q}|\Delta Q_k|\right)\le 6B^M_{Q,R}\le 6C_QA_R.} \tag{S.227}$$

正确系数是一个 $6B_Q$，不是两个 $6B_Q$。

若 $k\in\mathcal{P}_{\rm LE}$，由 $D_k(t)\le D_k(\tau)<T_k/2$ 与 $K_k(t)>2T_k/3$，对 last-exit interval 上几乎处处 good times 有

$$\boxed{e_{k,R}(t)>\frac{T_k}{6},\qquad \frac{1}{R^2}\int_{J_k^{\rm LE}} e_{k,R}(t)^{3/2}\,dt > d_k\left(\frac{T_k}{6}\right)^{3/2}\ge \lambda_k^{-3/2}\left(\frac{T_k}{6}\right)^{3/2}.} \tag{S.228}$$

这里没有在可能非 good 的 ℓ_k 取 E_k, D_k 的值。接上 inherited padded-shell estimate 得到

$$\boxed{T_k\le C_{\rm LE}\lambda_k 2^k\gamma_k^{1/3}\big(p^{u,\eta}_{k,R}(J_k^{\rm LE})\big)^{2/3},\quad C_{\rm LE}=6C_1^{2/3}<C_4,\quad C_4=12(2C_1)^{2/3}.} \tag{S.229}$$

对 $\mathcal{P}_\sigma$ 使用 J_τ，对 $\mathcal{P}_{\rm LE}$ 使用各自的 $J_k^{\rm LE}$，在并集上先做 finite-shell Hölder，再只调用一次允许 shell-dependent time sets 的 (R.211)，得到

$$\boxed{\sum_{k\in\mathcal{P}_\sigma\cup\mathcal{P}_{\rm LE}} T_k\le C_5\mathscr{L}(\boldsymbol{\lambda})^{1/3}A_R,\qquad C_5=C_4C_P^{2/3}.} \tag{S.230}$$

把两支各自估成完整全局 ledger 会多收一次 C_5，本步不这样做。

40. residual vector 与尖锐比较

记四个 paid classes 的并为 $\mathcal{I}_{\rm pay}$，两个 residual classes 的并为 $\mathcal{I}_{\rm res}=\mathcal{R}_{\rm sh}\cup\mathcal{R}_x$。已有

$$\boxed{\sum_{k\in\mathcal{I}_{\rm pay}(\tau)} T_k\le 6B^M_{Q,R}+C_5\mathscr{L}(\boldsymbol{\lambda})^{1/3}A_R\le C_{\rm pay}(\boldsymbol{\lambda})A_R,\quad C_{\rm pay}=6C_Q+C_5\mathscr{L}(\boldsymbol{\lambda})^{1/3}.} \tag{S.231}$$

定义 residual stopped-flux vector

$$\boxed{r^{\boldsymbol{\lambda}}_{k,R}(\tau)=1_{\mathcal{I}_{\rm res}(\tau)}(k)\big[F_{k,R}(\tau)-F_{k,R}(\ell_k)\big],\qquad r_k=0 \text{ if } T_k=0.} \tag{S.232}$$

在 $\mathcal{R}_{\rm sh}$ 上，$|\Delta Q_k| < T_k/6$ 来自定义；在 $\mathcal{R}_x = \mathcal{I}_x$ 上，失败的 Step 8 β-test 给出同样严格界。配合 $\Delta F = T/3 - \Delta Q$，两个 residual classes 都满足

$$\boxed{\frac{T_k}{6} < r^{\boldsymbol\lambda}_{k,R}(\tau) < \frac{T_k}{2}, \qquad 2r^{\boldsymbol\lambda}_{k,R}(\tau) < T_k < 6r^{\boldsymbol\lambda}_{k,R}(\tau).} \tag{S.233}$$

全局补零后，

$$\boxed{0 \le r^{\boldsymbol\lambda}_{k,R}(\tau) \le \frac{T_k}{2} \le \frac{v_{k,R}}{2}, \quad \|r^{\boldsymbol\lambda}_R(\tau)\|_{\ell^2} \le \frac{Z_R}{2}, \quad \sum_k r^{\boldsymbol\lambda}_{k,R}(\tau) \le C_F P^M_R.} \tag{S.234}$$

ℓ^2 上界本身不能推出 fixed-N 的 ℓ^1 tail bound；最后一项在大 P^M_R 区域也只有线性尺度。

41. paid-branch deletion 的 best-N reduction

对非负 ℓ^1 vector 定义

$$\mathcal{S}_N(x) = \inf_{S\subset\mathbb{N}, \#S\le N} \sum_{k\notin S} x_k.$$

对每一个同样的 exceptional set S，四个 paid classes 与 residual comparison 给出

$$\boxed{\sum_{k\notin S} T_k \le 6B^M_{Q,R} + C_5\mathscr{L}(\boldsymbol\lambda)^{1/3}A_R + 6\sum_{k\notin S} r^{\boldsymbol\lambda}_{k,R}(\tau).} \tag{S.235}$$

用逼近 residual infimum 的同一批集合取极限，得到 fixed-good-terminal theorem

$$\boxed{\mathcal{S}_N((K_{k,R}(\tau))_k) \le 6B^M_{Q,R} + C_5\mathscr{L}(\boldsymbol\lambda)^{1/3}A_R + 6\mathcal{S}_N((r^{\boldsymbol\lambda}_{k,R}(\tau))_k).} \tag{S.236}$$

$\mathcal{R}_x$ 与 $\mathcal{R}_{\rm sh}$ 必须先合并再只做一次 best-N infimum；分别允许 N 个例外会悄悄把总预算增至 $2N$。

对 $\mathcal{D} \in \{I_R, \mathcal{T}_R\}$ 定义 good-terminal residual gate

$$\boxed{\mathfrak{R}^{\boldsymbol\lambda}_{N,R}(\mathcal{D}) = \sup_{\tau\in\mathcal{D}\cap\mathcal{G}_R} \mathcal{S}_N((r^{\boldsymbol\lambda}_{k,R}(\tau))_k).} \tag{S.237}$$

只用 terminal K-vector 的 inherited ℓ^1-continuity，把左侧从 dense good times 延伸到所有 terminal times；不使用 residual path、selector 或 masks 的连续性：

$$\boxed{\mathcal{S}^K_{N,R}(\mathcal{D}) \le C_{\mathrm{pay}}(\boldsymbol{\lambda})A_R + 6\mathfrak{R}^{\boldsymbol{\lambda}}_{N,R}(\mathcal{D}).} \tag{S.238}$$

反向由 coordinatewise $r \le K/2$ 与同一 exceptional set 得到

$$\boxed{\mathfrak{R}^{\boldsymbol{\lambda}}_{N,R}(\mathcal{D}) \le \frac{1}{2}\mathcal{S}^K_{N,R}(\mathcal{D}).} \tag{S.239}$$

因此对固定、与尺度和解无关的 N_0，

$$\boxed{\mathfrak{R}^{\boldsymbol{\lambda}}_{N_0,R}(\mathcal{D}) \lesssim A_R \quad\Longleftrightarrow\quad \mathcal{S}^K_{N_0,R}(\mathcal{D}) \lesssim A_R.} \tag{S.240}$$

这个等价移除了已知付款，但没有把 residual gate 变成定理；它准确标出还需要新 PDE 信息的位置。

42. plateau corollary、full gate 与 fallback

把 inherited plateau reduction 与 (S.238) 合并：

$$\boxed{\mathfrak{C}^M_R \le \sqrt{N}\,Z_R + 7B^M_{Q,R} + C_5\mathscr{L}(\boldsymbol{\lambda})^{1/3}A_R + 6\mathfrak{R}^{\boldsymbol{\lambda}}_{N,R}(I_R).} \tag{S.241}$$

七个 B_Q units 中，六个来自 paid partition，一个来自 terminal K-to-flux reduction。该式只在 plateau 域，不是 full-terminal Q.12。

绝对 F-variation 给出线性 fallback：

$$\boxed{\mathfrak{R}^{\boldsymbol{\lambda}}_{N,R}(\mathcal{D}) \le C_F P^M_R, \qquad P^M_R \le 1 \Longrightarrow \mathfrak{R}^{\boldsymbol{\lambda}}_{N,R}(\mathcal{D}) \le C_F A_R.} \tag{S.242}$$

因此 small-payment regime 已经闭合；当 $P^M_R > 1$ 时，线性 fallback 与目标之间仍差 $(P^M_R)^{1/3}$。真正开放的 full-domain statement 是

$$\boxed{\text{OPEN: 存在固定 } N_0 < \infty, C_{\mathrm{res}} < \infty,\text{ 使 } \mathfrak{R}^{\boldsymbol{\lambda}}_{N_0,R}(\mathcal{T}_R) \le C_{\mathrm{res}}A_R \text{ 对所有 } R \text{ 与解一致成立.}} \tag{S.243}$$

若 (S.243) 成立，则 (S.238) 推出 full Q.12，再由 inherited Q.9 推出 Q.1。只在 I_R 证明 residual bound 能直接给 plateau Q.1，但不能升级成 full Q.12。

43. sharpness、exception budget 与 D-persistence no-go

在 abstract continuous-clock 层面，取 $T = 1$、$0 < \varepsilon < 1/6$，并令

$$\boxed{\Delta Q=\frac{1}{6}-\varepsilon \Longrightarrow r=\frac{1}{6}+\varepsilon, \qquad \Delta Q=-\frac{1}{6}+\varepsilon \Longrightarrow r=\frac{1}{2}-\varepsilon.} \tag{S.244}$$

因此仅凭 $|\Delta Q|<T/6$，系数六与上界二分之一都是 limiting-sharp；等号 $|\Delta Q|=T/6$ 已归入 paid $\mathcal{P}_Q$。这些是 clock-algebra tests，不是 NSE solutions。

一份共享 exception budget 的必要性由 $T_1=T_2=3$、$r_1=r_2=1$ 给出：

$$\boxed{\mathcal{S}_1((r_1,r_2))=1, \qquad \mathcal{S}_1((r_1,0))+\mathcal{S}_1((0,r_2))=0.} \tag{S.245}$$

右式违法之处是给两个 residual labels 各花一个例外。固定 N 也不能被 truncation-dependent budget 替代：

$$\boxed{T_k=2^{-k}: \quad \mathcal{S}_1((T_k)_{k\ge1})=\frac{1}{2}, \quad \mathcal{S}_1((T_k)_{1\le k\le M})=\frac{1}{2}-2^{-M}, \quad \mathcal{S}_M((T_k)_{1\le k\le M})=0.} \tag{S.246}$$

最后，terminal D-dominance 不能免费局部化成 last-exit persistence。显式 rational piecewise-linear clock 取 $R^2=1, s_R=0, \tau=2, T=1$，last exit $\ell=1/4$，并安排 $D(t)=3/5$ 在后段保持不变；则

$$\boxed{D(\tau)=\frac{3T}{5}\ge\frac{T}{2}, \qquad \Delta D|_{(\ell,\tau)}=0, \qquad E(1)=\frac{7T}{100}<\frac{T}{6}.} \tag{S.247}$$

这严格排除把 $\mathcal{I}_D$ 直接塞入 long non-D persistence proof；Step 8 full-history trichotomy 必须保留。该 witness 仍只是 continuous clock stress test。

44. 路线决定

paid-branch deletion 已达到它的自然终点。下一 PDE 阶段应分别研究两类 residual mechanism，但最终必须用一个共享 exception budget 重组：

- **short non-D、Q-small packing**：正 stopped flux 与 T_k 可比，却生成在短于 $R^2\lambda_k^{-3/2}$ 的 terminal interval。新定理必须使用 spatial crowding、overlap 或 Carleson-type constraint 等跨壳层 PDE 信息；普通 ℓ^2 sequence inequality 不足以闭合。
- **scalar-excess ancestry**：$\mathcal{I}_x$ 只可能来自 anomalous-defect 或 high-Rayleigh shells，且 full-history Q-variation 与 kinetic mass 都低于 Step 8 thresholds。需要真正打包剩余 defect/high-Rayleigh mass；terminal D-dominance 不能免费变成 last-exit interval 内的支付。

任何候选估计都必须通过 (S.245)--(S.247)，并保留 inherited R0.74O/P exact family 的边界：该精确族只 refute no-exception gate，不证明 $N_0=1$ 足够。

45. Step 10 主张边界与双路审计

Step 10 **PROVED**：(S.223)--(S.224) 的 canonical 2/3-last-exit 与固定 profile；(S.225) 的 exact six-class partition；(S.226) 的 Step 7/8 compatibility；(S.227)--(S.231) 的 single $6B_Q$ 与 single C_5 payments；(S.232)--(S.234) 的 positive residual 与 $T/6<r<T/2$；(S.235)--(S.240) 的 fixed-good-terminal/domain-safe best-N reductions；以及 (S.241)--(S.242) 的 plateau corollary 与 small-payment fallback。

Step 10 **INHERITED**：R0.74P canonical clocks、variation ledgers 与 ℓ^1-terminal continuity；R0.74Q fixed-N reduction；R0.74R shell-dependent cubic payment；Step 7 Rayleigh trichotomy；Step 8 full-history $\beta/\sigma/x$ partition；Step 9 last-exit identity 与 finite good-stop closure。

Step 10 **REFUTED / RULED OUT**：额外 low-Rayleigh residual class；必须重复收取完整 Q 或 cubic ledger；两个 residual mechanisms 分别拥有 N 个 exceptions；terminal D-dominance 自动局部化到 last-exit interval；以及 last-exit algebra 单独产生 fixed-N shell compression。

继续 **OPEN**：固定、与解和尺度无关的 N_0 residual estimate (S.243)；$\mathcal{R}_{\mathrm{sh}}$ 与 $\mathcal{R}_x$ 的 PDE packing；full Q.12、Q.1、R0.74R extraction hypotheses、scale contraction、prescribed-centre packing 与 regularity。

明确 **NOT CLAIMED**：last-exit selector、branch masks 或 residual path 对终端连续/可测/lower semicontinuous；ℓ_k 是 good time；无穷 last-exit family 是一个 admissible local-energy test；Step 8 classes 可在 J_k^{LE} 重定义；plateau 与 full domain 相等；$\boldsymbol{\lambda}$ 已优化；scalar fixtures 是 NSE solutions；以及新颖性、优先权、奇点形成、正则性或 Clay 结论。

主证书通过 12/12 exact、10/10 finite、79/79 structural 与 47/47 negative mutations。独立 Ruby 通过 9/9 groups、65,681 cases、21/21 contract mutations、13/13 report checks 与 15/15 audit bindings；deterministic stdout SHA-256 为 `4877dc3a0de2c2f605641736c7355672f0a7a68cb97a37849d4a7c28495e8bbd`。主文 SHA-256 为 `9eb5f2a794021b49894adfc167d350f58d93c266e6be319ce835c58db2e0d74c`。有限证书不替代 inherited local-energy/PDE analysis。

PAID BRANCHES: DELETED WITH ONE Q LEDGER AND ONE CUBIC LEDGER. TWO RESIDUAL MECHANISMS SHARE ONE BEST-N GATE. S.243 OPEN. NOT CLAY.

46. Step 11 的问题与四项结论

Step 10 已把 full-terminal clock estimate 归约为两个不交 residual mechanisms 上的一个 combined best-N tail。Step 11 不证明这个 tail estimate，而是确定两支怎样精确重组、各自现有估计能走到哪里，以及哪一条新的 PDE 陈述足以闭合 short branch。

- combined residual 是两支 best-N tails 的精确离散 infimal convolution。两支可以分别研究，但 exception counts 必须相加；两个固定有限 count 仍足以满足最终“存在某个固定 N_0”的目标。
- short non-D branch 得到尖锐 inverse-duration coefficient、nested-tent integral 与任意正 backward depth 的控制；depth zero 的 terminal trace 仍缺失，critical s^2 Carleson endpoint 有 logarithmic divergence。
- scalar-excess branch 上，stopped residual 与 Step 8 priority-selected excess 在 best-N 意义下以字面常数 $1/5$ 与 3 等价。现有理论只有 linear summability 与 fixed-solution tightness，没有 solution- and scale-independent count。
- 现有 smooth exact families 只 refute zero-exception route。当前 multi-packet designs 的 cubic cost 过大，不能 refute 任意固定正 exception count。

因此 short branch 的缺口是 terminal anti-concentration，不是 interval overlap；excess branch 的缺口是 uniform weighted packing，不是 ancestry classification。两条都 **OPEN**。无新颖性、优先权、奇点、正则性或千禧年问题结论。**NOT CLAY.**

47. 两个 residual vectors 与共享 budget 恒等式

保留 Step 10 的全部定义。固定 good terminal τ，令 $T_k = K_{k,R}(\tau)$、$\ell_k = \ell^K_{k,2/3}(\tau)$、$d_k = (\tau - \ell_k)/R^2$，并把 residual 拆成不交支撑的两向量

$$\boxed{r_k^{\mathrm{sh}}(\tau) := \mathbf{1}_{\mathcal{R}_{\mathrm{sh}}(\tau)} r_k(\tau), \qquad r_k^{x}(\tau) := \mathbf{1}_{\mathcal{R}_{x}(\tau)} r_k(\tau), \qquad r = r^{\mathrm{sh}} + r^{x}.} \tag{S.248}$$

两支上都有 $T_k/6 < r_k < T_k/2 \le v_{k,R}/2$。对 $z \in \ell^1_+$ 写 $\mathcal{S}_N(z) = \inf_{\#S \le N} \sum_{k \notin S} z_k$。若 a, b 支撑不交，则

$$\boxed{\mathcal{S}_N(a+b) = \min_{0 \le n \le N} [\mathcal{S}_n(a) + \mathcal{S}_{N-n}(b)].} \tag{S.249}$$

这是 pointwise equality。取 terminal supremum 后只有 domain-safe inequality：

$$\boxed{\mathfrak{R}^{\boldsymbol{\lambda}}_{N,R}(\mathcal{D}) \le \min_{0\le n\le N}\left[\mathfrak{R}^{\rm sh}_{n,R}(\mathcal{D}) + \mathfrak{R}^{x}_{N-n,R}(\mathcal{D})\right].} \tag{S.250}$$

这里通常不能写等号，因为 supremum 与 finite minimum 不交换；最优 budget split 与 top-N shells 可以依赖 τ。若两支分别以固定 $N_{\rm sh}, N_x$ 闭合，令 $N_0 = N_{\rm sh} + N_x$，则

$$\boxed{\begin{gathered}\mathfrak{R}^{\boldsymbol{\lambda}}_{N_0,R} \le (C_{\rm sh} + C_x)A_R,\\ \mathcal{S}^{K}_{N_0,R} \le \left[6C_Q + C_5\mathscr{L}(\boldsymbol{\lambda})^{1/3} + 6C_{\rm sh} + 6C_x\right]A_R.\end{gathered}} \tag{S.251}$$

两个 best-N branch theorems 给出 combined best-$2N$，不是 best-N。fixture $a = (M, 0), b = (0, M)$ 给出

$$\boxed{\mathcal{S}_1(a) = \mathcal{S}_1(b) = 0, \qquad \mathcal{S}_1(a+b) = M, \qquad \mathcal{S}_2(a+b) = 0.} \tag{S.252}$$

它精确说明 duplicate N-budgets 会偷偷把总预算加倍。

48. Short branch 的 inverse-duration ledger

令 $\mathcal{H}_\tau = \mathcal{R}_{\rm sh}(\tau)$、$a_k = 2^{3k}\gamma_k$、$p_k = p^{u,\eta}_{k,R}(J^{\rm LE}_k)$。non-$D$ persistence 与继承的 cubic estimate 给出

$$\boxed{d_k\left(\frac{T_k}{6}\right)^{3/2} < C_1 a_k^{1/2} p_k, \qquad r^{\rm sh}_k < 3C_1^{2/3}(a_k d_k^{-2})^{1/3} p_k^{2/3}.} \tag{S.253}$$

对同一个 exceptional set 做 finite-shell Hölder，再只调用一次 shell-dependent-time-set estimate，得到

$$\boxed{\mathcal{S}_N(r^{\rm sh}(\tau)) \le 3C_1^{2/3}C_P^{2/3}\left(\mathfrak{D}^{\rm sh}_N(\tau)\right)^{1/3}A_R, \quad \mathfrak{D}^{\rm sh}_N := \inf_{\#S\le N}\sum_{k\in\mathcal{H}_\tau\setminus S} a_k d_k^{-2}.} \tag{S.254}$$

这只是 sufficient interface，不是对 $\mathfrak{D}^{\rm sh}_N$ 的新估计；它把遗留债务精确定位为 inverse-square duration。

49. Normalized depth 与 critical Carleson log gap

在 short branch 定义 $h_k = d_k\lambda_k^{3/2} \in (0,1)$、$w_k = a_k\lambda_k^3$。令 $\mathcal{H}_j = \{2^{-j-1} \le h_k < 2^{-j}\}$、$W_j = \sum_{k\in\mathcal{H}_j} w_k$，则

$$\boxed{w_k h_k^{-2} = a_k d_k^{-2}, \qquad \sum_j 4^j W_j \le \sum_k w_k h_k^{-2} \le 4\sum_j 4^j W_j.} \tag{S.255}$$

对原子测度 $\mu_\tau = \sum_{k\in\mathcal{H}_\tau} w_k\delta_{h_k}$，Tonelli 给出精确 layer-cake：

$$\boxed{\sum_k w_k h_k^{-2} = \mu_\tau((0,1)) + 2\int_0^1 s^{-3}\mu_\tau((0,s])\,ds.} \tag{S.256}$$

因此 $\mu_\tau((0,s]) \lesssim s^{2+\varepsilon}$ 足够，但 critical exponent two 不足。取固定 profile $\lambda_k = 1$、$w_k = 2^{3k}\gamma_k$、$h_k = w_k^{1/2}$，可有 $\mu((0,s]) \le 2s^2$，同时

$$\boxed{\sum_{k\ge k_0} w_k h_k^{-2} = \sum_{k\ge k_0} 1 = \infty.} \tag{S.257}$$

这就是 critical s^2 Carleson endpoint 的 logarithmic obstruction。它是 coefficient/clock stress test，不是 **NSE solution**。

50. Nested tent 能控制什么、不能控制什么

以 backward time $s = (\tau - t)/R^2$ 定义 $M_I(s) = \sum_{k\in I, d_k > s} r_k^{\mathrm{sh}}$、$V_I(s) = \sum_{k\in I, d_k>s} a_k$。所有 last-exit intervals 共用 terminal endpoint，故其 indicator 精确为 $\mathbf{1}_{\{s<d_k\}}$。weighted Hölder、(R.214) 与 (R.211) 给出

$$\boxed{\int_0^4 \frac{M_I(s)^{3/2}}{V_I(s)^{1/2}}\,ds \le 3^{3/2} C_1 C_P P_R^M.} \tag{S.258}$$

令 $\mathscr{A}_0 = \sum_k 2^{3k}\gamma_k < \infty$。对任意 $0 < \delta < 4$，

$$\boxed{\sum_{k\in I,\, d_k>\delta} r_k^{\mathrm{sh}} \le 3C_1^{2/3} C_P^{2/3} \mathscr{A}_0^{1/3} \delta^{-2/3} A_R.} \tag{S.259}$$

所以任何持续到固定正 backward depth 的 residual 已经得到 quadratic control；未解决的全部质量可以集中到 $d_k \downarrow 0$。一个 $L^{3/2}$-in-time tent bound 没有 depth-zero terminal trace。

严格嵌套本身也无能为力：连续 clock/payment tower 可同时满足 nested intervals 与很小 cubic integral，却有

$$\boxed{S_N(r) = M - N, \qquad A_M \asymp M^{2/3}, \qquad Z_M = 3\sqrt{M}.} \tag{S.260}$$

这是 abstract continuous clock/payment witness，不是 **NSE counterexample**。

51. Short branch 的第一个新 PDE 门槛

比 raw inverse-duration moment 更自然的目标是 amplitude-sensitive terminal anti-concentration：寻找与解和尺度无关的固定 N_{sh}、$0 < \delta_* < 4$、$0 \le \theta_* < 1$、$C_{\mathrm{nc}} < \infty$，使每个 good terminal 都存在同一个 S_τ、$\#S_\tau \le N_{\mathrm{sh}}$，满足

$$\boxed{\sum_{k\in\mathcal{H}_\tau\setminus S_\tau,\,d_k\le\delta_*} r_k^{\rm sh} \le \theta_* \sum_{k\in\mathcal{H}_\tau\setminus S_\tau} r_k^{\rm sh} + C_{\rm nc}A_R. \quad \textbf{OPEN}} \tag{S.261}$$

(S.261) 与 (S.259) 的 implication 已证明：它将闭合 short branch。boxed hypothesis 本身没有证明；它要求 PDE 排除一个非二次比例的 tail 完全在最后 $\delta_* R^2$ 时间内生成。

52. Scalar excess 与 residual 的精确 best-N 等价

保留 Step 8 scalar excess $x_k = [D_{k,R}(\tau) - \beta_{k,R}(J_\tau) - 2\lambda_k\sigma_{k,R}(J_\tau)]_+$，并令 $x_k^{\rm sel} = \mathbf{1}_{\mathcal{I}_x} x_k$。priority failures 与 terminal clock identity 给出字面坐标比较

$$\boxed{\frac{1}{5}x_k^{\rm sel} < r_k^x < 3x_k^{\rm sel} \qquad (k\in\mathcal{I}_x).} \tag{S.262}$$

常数在 scalar constraints 层面尖锐。优化同一个 exceptional set 后，

$$\boxed{\frac{1}{5}\mathcal{S}_N(x^{\rm sel}(\tau)) \le \mathcal{S}_N(r^x(\tau)) \le 3\mathcal{S}_N(x^{\rm sel}(\tau)).} \tag{S.263}$$

因此 $\mathcal{R}_x$ gate 已精确归约，但没有闭合。

53. Fixed-solution tightness 不等于 uniform N

Step 8 ancestor vector $b_k = \mathbf{1}_{\mathcal{I}_x}[m_{k,R} + \int_{H_{k,R}} g_{k,R}]$ 满足

$$\boxed{r_k^x \le 3x_k^{\rm sel} \le 3b_k, \qquad \sum_k x_k^{\rm sel} \le CP_R^M, \qquad \sum_k b_k \le C(A_R + P_R^M).} \tag{S.264}$$

这些在 $P_R^M > 1$ 时只是 linear ledger，Markov counting 不能产生 universal quadratic best-N tail。另一方面，因 $r_k^x \le v_{k,R}/2$ 且 $v_R \in \ell^1$，对每个固定 solution、固定 R 和 $\varepsilon > 0$，可以选取依赖于它们的 $N = N(u, R, \varepsilon)$，使

$$\boxed{\sup_{\tau\in\mathcal{G}_R\cap\mathcal{D}} \mathcal{S}_N(r^x(\tau)) \le \frac{1}{2}\sum_{k>N} v_{k,R} < \varepsilon.} \tag{S.265}$$

这是真实的 nonuniform compactness，但缺失的恰恰是与 solution、scale 无关的 uniform N 与 $O(A_R)$ rate；两者不能混同。

54. Ancestry 不能倒推到 last-exit interval

两个 exact rational scalar clocks 表明禁止的 shortcut。pure-defect row 满足

$$\boxed{\ell=1,\quad r^x=\frac{1}{3},\quad \sigma=\frac{959}{12000}<\frac{1}{12},\quad x=\frac{2641}{6000}>\frac{1}{6},\quad D(2)-D(\ell)=0.} \tag{S.266}$$

pure high-Rayleigh row 满足

$$\boxed{\ell=1,\quad r^x=\frac{1}{3},\quad \sigma=\frac{983}{12000}<\frac{1}{12},\quad x=\frac{2617}{6000}>\frac{1}{6},\quad \int_{H\cap J^{\mathrm{LE}}} g=0.} \tag{S.267}$$

两者都说明 full-history ancestry 不能 retrospectively restricted to J^{LE}。重复 pure-defect row 得到

$$\boxed{\mathcal{S}_N(r^x)=\frac{(M-N)_+}{3},\qquad A_M\asymp M^{2/3},\qquad Z_M=\sqrt{M}.} \tag{S.268}$$

(S.266)--(S.268) 都是 **ABSTRACT STRESS TESTS, NOT NSE COUNTEREXAMPLES**；它们排除只靠 scalar ℓ^1/ℓ^2 ledgers 的推导，不否定 (S.243)。exact minimal target 是

$$\boxed{\textbf{OPEN:}\quad \exists N_x, C_x<\infty \text{ uniformly such that} \mathcal{S}_{N_x}(x^{\mathrm{sel}}(\tau))\le C_x A_R.} \tag{S.269}$$

由 (S.263)，这与 $\mathcal{R}_x$ residual gate 在字面常数范围内等价。

55. Exact-family falsification criterion

若要 refute 某个固定 N，新的 smooth exact family 必须提供 $N+1$ 个不同 target shells，且

$$\boxed{\min_{1\le i\le N+1}\frac{K_{k_i,R}(\tau)}{A_R}\longrightarrow\infty.} \tag{S.270}$$

Step 10 将迫使这些 target shells 最终全部落入 combined residual，于是 $\mathcal{S}_N(r)/A_R\to\infty$。现有 R0.740/P single-packet family 只对 $N=0$ 通过此检验：一个正 exception 可以删除唯一大坐标。

现有 common-shear multi-packet construction 虽能制造 distinct terminal lobes，却同时证明 exterior cubic lower bound

$$\boxed{\frac{A_R^{(N)}}{NT}\ge\frac{c}{N}R^{2/3}L_N^{-1/3}\exp\left(\frac{5}{6}c_\gamma L_N^2\right)\longrightarrow\infty.} \tag{S.271}$$

所以其 clock lower scale 被 nonnegative cubic payment 压倒；它没有建立 (S.270)，也没有 refute fixed positive exception count。这是对现有设计的 quantitative obstruction，不是对全部 multi-packet architectures 的 no-go。

56. 合并后的开放定理

下一 PDE 阶段保留两个工作包：short terminal trace 检验 (S.261)；selected-excess packing 检验 (S.269)，并分开 anomalous measure 与 high-Rayleigh viscous mass。任何 adversarial exact family 先通过 (S.270)，不能只展示多个 terminal lobes。

最终 combined theorem 写成

$$\boxed{\begin{gathered}\textbf{OPEN: find fixed } N_{\rm sh}, N_x \in \mathbb{N}_0 \textbf{ and } C_{\rm sh}, C_x < \infty \textbf{ such that}\\ \sup_{\tau\in\mathcal{T}_R\cap\mathcal{G}_R} \mathcal{S}_{N_{\rm sh}}(r^{\rm sh}(\tau)) \le C_{\rm sh} A_R,\\ \sup_{\tau\in\mathcal{T}_R\cap\mathcal{G}_R} \mathcal{S}_{N_x}(r^{x}(\tau)) \le C_x A_R.\end{gathered}} \tag{S.272}$$

若 (S.272) 成立，(S.251) 以 $N_0 = N_{\rm sh} + N_x$ 推出 Step 10 (S.243)，继而条件性推出 R0.74Q (Q.12) 与 fixed-scale (Q.1)。implication 已证明，antecedent 仍 **OPEN**。

57. Step 11 主张边界与双路审计

Step 11 **PROVED**：(S.249)--(S.251) shared-budget identity 与 domain consequence；(S.253)--(S.254) inverse-duration estimate；(S.255)--(S.256) normalized-depth/dyadic/layer-cake identities；(S.257) coefficient-level critical Carleson failure；(S.258)--(S.259) nested-tent 与 positive-depth estimates；(S.262)--(S.263) scalar-excess/residual equivalence；(S.264)--(S.265) ancestry、linear ledger 与 fixed-solution compactness；以及从 (S.261)、(S.269)、(S.270)、(S.272) 出发的 conditional implications。

Step 11 **ABSTRACT STRESS TESTS, NOT NSE COUNTEREXAMPLES**：(S.252) duplicate-budget fixture；(S.257)、(S.260) critical Carleson sequence 与 nested tower；(S.266)--(S.267) ancestry-localization witnesses；(S.268) flat selected-excess tower。

Step 11 **OPEN**：terminal anti-concentration (S.261)；uniform selected-excess packing (S.269)；(S.272) 的两支估计；固定 universal N_0；Step 10 (S.243)、Q.12、Q.1、scale contraction、prescribed-centre packing 与 regularity；以及能通过 (S.270) 且没有 prohibitive full payment 的新 exact multi-packet family。

明确 **NOT CLAIMED**：moving masks、last-exit selectors、top-N sets 或 adaptive budget split 的连续性/可测性；terminal supremum 与 branch-budget minimum 交换；CKN-type singular-set estimates 对当前 shell residuals 计数；terminal ancestry 在 last-exit interval 持续；bounded literature search 穷尽；新颖性、优先权；或 Navier--Stokes Millennium problem 的解。

主证书通过 14/14 exact、7/7 finite、34/34 structural 与 7/7 negative mutations。独立 Ruby audit 通过 7/7 groups、206,891 cases、6/6 artifact locks、7/7 dependency locks 与 59/59 note checks；canonical stdout SHA-256 为 `506440647a0a9b5be9d65ded24762b6eb6f6ce8cf054473a0ac04bf8835a1ffb`。这些有限证书只支持实现可复现性，不替代 inherited local-energy/PDE analysis。

SHARED BUDGET: EXACT. SHORT TERMINAL TRACE: OPEN. SELECTED-EXCESS PACKING: OPEN. EXISTING MULTI-PACKET FAMILIES DO NOT REFUTE FIXED POSITIVE N. NOT CLAY.

58. Step 12：终端公共窗口与 conditional Morrey packing

Step 11 把 full-terminal clock estimate 归约为两个 best-N tail：short non-dissipation residual $r^{\rm sh}$ 与 selected dissipation excess $x^{\rm sel}$（等价地 r^x）。Step 12 没有证明这两个 universal tail estimate，而是把它们改写成更清晰的 PDE 接口，并证明一个带附加统一假设的 conditional packing theorem。

本步得到六项结论：short residual 可由一个共同终端窗内的 absolute shell-flux variation 与已证明的 positive-depth cubic term 控制；该新窗口泛函对 terminal 连续，但固定解的模量不对解与尺度统一；best-N 的 exact layer-cake identity 把问题变成所有阈值上的 shell counting；selected excess 的 anomalous-defect 与 high-Rayleigh ancestors 必须诚实相加 exception budgets；若 total local dissipation 具有统一 critical Morrey coefficient，且 lifted mollified path 的长度以 R 为单位一致有界，则 moving-tube covering 给出 conditional Morrey packing；最后，单个 inherited passive packet 的统一加速受到 kinematic screen，并不能绕过 cubic obstruction。

OPEN：S.280、S.288、S.303、Step 11 的 S.272、Q.12、Q.1、scale contraction、regularity 与 singularity。conditional Morrey 与 mixed-norm 结论的常数依赖其额外统一上界；不声称 novelty 或 priority。无 DNS、无 floating-point asymptotics、无 DGX。**NOT CLAY.**

59. 冻结设置与公共终端窗

保留 R0.74S Steps 10--11 的全部定义：

$$\mathcal{T}_R = (s_R, t_0), \qquad |\mathcal{T}_R| = 4R^2, \qquad A_R = (P_R^M)^{2/3},$$

并对每个 $\tau \in \mathcal{G}_R \cap \mathcal{T}_R$ 保留 r_k^{sh}、$d_k = (\tau - \ell_k)/R^2$ 以及 inherited absolute flux ledger

$$\sum_{k\geq 1} \int_{\mathcal{T}_R} |\dot{F}_{k,R}(t)|\, dt \leq \mathfrak{L}^M_{\mathrm{abs},R} \leq C_F P_R^M.$$

把 $|\dot{F}_{k,R}|$ 在 $\mathcal{T}_R$ 外延为零。对 $0 < \delta < 4$，定义

$$\boxed{\begin{aligned} J_{\tau,\delta} &= (\max\{s_R, \tau - \delta R^2\}, \tau), \\ f_{k,R}(\tau,\delta) &= \int_{J_{\tau,\delta}} |\dot{F}_{k,R}(t)|\, dt, \\ \mathcal{V}^F_{N,R}(\tau,\delta) &= \mathcal{S}_N((f_{k,R}(\tau,\delta))_{k\geq 1}). \end{aligned}} \tag{S.273}$$

这里用一个共同 shell deletion set 处理整段 window；不能在积分内部随时间改变。

60. Exact terminal variation-window reduction

若 $k \in \mathcal{R}_{\mathrm{sh}}(\tau)$ 且 $d_k \leq \delta$，则 $J_k^{\mathrm{LE}} \subset J_{\tau,\delta}$。因此对每个 $S \subset \mathbb{N}$，

$$\boxed{\sum_{\substack{k\in\mathcal{R}_{\mathrm{sh}}(\tau)\setminus S \\ d_k\leq\delta}} r_k^{\mathrm{sh}}(\tau) \leq \sum_{k\notin S} f_{k,R}(\tau,\delta).} \tag{S.274}$$

其余 $d_k > \delta$ 的坐标由 Step 11 (S.259) 以同一个 S 控制，故

$$\boxed{\mathcal{S}_N(r^{\mathrm{sh}}(\tau)) \leq \mathcal{V}^F_{N,R}(\tau,\delta) + C_{\mathrm{deep}}\delta^{-2/3}A_R, \quad C_{\mathrm{deep}} = 3C_1^{2/3}C_P^{2/3}\mathscr{A}_0^{1/3}.} \tag{S.275}$$

这个结构性增益把 last-exit selectors 与 branch masks 从新项中移除，改为固定窗口上的连续泛函。

61. 终端连续性与缺失的 uniform modulus

令 $g_R(t)=\sum_k|\dot F_{k,R}(t)|\in L^1(\mathcal{T}_R)$。当 $\tau_n\to\tau$ 时，

$$
\begin{gathered}
\|f_R(\tau_n,\delta)-f_R(\tau,\delta)\|_{\ell^1}\le\int_{J_{\tau_n,\delta}\triangle J_{\tau,\delta}}g_R(t)\,dt\to 0,\\
|\mathcal{S}_N(a)-\mathcal{S}_N(b)|\le\|a-b\|_{\ell^1}.
\end{gathered}
\tag{S.276}
$$

所以 $\tau\mapsto\mathcal{V}^F_{N,R}(\tau,\delta)$ 连续，good terminals 上的 supremum 等于全 $\mathcal{T}_R$ 上的 supremum。对每个固定 (u,R)，Lebesgue integral 的 absolute continuity 还给出

$$
\boxed{\Omega_{u,R}(\delta):=\sup_{\tau\in\mathcal{T}_R}\sum_k f_{k,R}(\tau,\delta)\longrightarrow 0\qquad(\delta\downarrow 0).}
\tag{S.277}
$$

但这个 modulus 依赖 solution 与 scale，且 S.275 的 positive-depth term 同时按 $\delta^{-2/3}A_R$ 增长，因此不能据此得到 universal estimate。

62. Layer cake：精确的 all-threshold counting problem

对 $z\in\ell^1_+$ 定义 $n_z(t)=\#\{k:z_k>t\}$。删除最大的 N 个坐标并用 Tonelli，得到

$$
\boxed{\mathcal{S}_N(z)=\int_0^\infty(n_z(t)-N)_+\,dt.}
\tag{S.278}
$$

因此一个充分的 distributional theorem 是找到 fixed N_F 与一个对解、尺度、terminal 均统一的 $\Phi\in L^1(0,\infty)$，使

$$
\boxed{\#\{k:f_{k,R}(\tau,\delta_*)>sA_R\}\le N_F+\Phi(s)\Longrightarrow\mathcal{V}^F_{N_F,R}(\tau,\delta_*)\le A_R\|\Phi\|_{L^1}.}
\tag{S.279}
$$

只在一个 threshold 上计数不够；critical A_R/t count 在 layer-cake endpoint 仍产生 logarithmic divergence。clean short-branch target 为

$$
\boxed{\begin{gathered}
\textbf{OPEN: find fixed } N_F,\ 0<\delta_*<4,\ C_F^*<\infty\textbf{ such that}\\
\sup_{\tau\in\mathcal{T}_R}\mathcal{V}^F_{N_F,R}(\tau,\delta_*)\le C_F^*A_R.\\
\text{Then (S.275) proves the short gate with } N_{\rm sh}=N_F.
\end{gathered}}
\tag{S.280}
$$

S.280 是 S.261 的 sufficient replacement，不是 equivalent reformulation；它更强，但目标连续、窗口固定且无 moving selector。

63. Inherited L_t^1 ledger 的 method boundary

固定 N，令 $M=N+1$。在同一 terminal window 内放置 M 个同步 AC spikes，可得

$$\boxed{\sum_k \int_{\mathcal{T}_R} |\dot F_{k,H}| = MH, \quad \mathcal{S}_N\left(\left(\int_{J_{\tau_0,\delta}} |\dot F_{k,H}|\right)_k\right) = H, \quad \frac{H}{(MH)^{2/3}} = \frac{H^{1/3}}{M^{2/3}} \to \infty.} \tag{S.281}$$

这是 vector-valued translated-spike witness，不是 **Navier--Stokes solution**。它只证明 AC 加 linear total-mass ledger 不能推出 uniform fixed-N、$P^{2/3}$-scaled window estimate；这是一个明确的 abstract no-go。

Fubini 仍给出 averaged terminal statement：

$$\boxed{\int_{\mathcal{T}_R} \sum_k f_{k,R}(\tau,\delta)\,d\tau \le \delta R^2 \int_{\mathcal{T}_R} g_R(t)\,dt \le C_F \delta R^2 P_R^M.} \tag{S.282}$$

除去至多 ηR^2 的 terminal times 后，

$$\sum_k r_k^{\mathrm{sh}}(\tau) \le C\left(\eta^{-1}\delta P_R^M + \delta^{-2/3} A_R\right). \tag{S.283}$$

若内点 optimizer 落在 $(0,4)$，平衡两项只得到

$$\boxed{\sum_k r_k^{\mathrm{sh}}(\tau) \le C\eta^{-2/5} A_R^{3/5} (P_R^M)^{2/5} = C\eta^{-2/5} (P_R^M)^{4/5}.} \tag{S.284}$$

这弱于所需 $P^{2/3}$，且不控制 supremum terminal；它是该方法的精确 exponent boundary，不声称对 NSE sharp。

64. Excess branch 与 honest exception accounting

在 Step 8 的 priority-selected set $\mathcal{I}_x(\tau)$ 上定义

$$\boxed{\begin{aligned} d_k^{\mathrm{def}}(\tau) &= \mathbf{1}_{\mathcal{I}_x(\tau)} m_{k,R}(\tau), \\ h_k(\tau) &= \mathbf{1}_{\mathcal{I}_x(\tau)} \int_{H_{k,R}} g_{k,R}(t)\,dt, \\ b_k(\tau) &= d_k^{\mathrm{def}}(\tau) + h_k(\tau), \qquad 0 \le x_k^{\mathrm{sel}}(\tau) \le b_k(\tau). \end{aligned}} \tag{S.285}$$

两个 ancestor vectors 可重叠，但 deletion sets 的 union 给出

$$\boxed{\mathcal{S}_{N_D+N_H}(x^{\mathrm{sel}}) \le \mathcal{S}_{N_D}(d^{\mathrm{def}}) + \mathcal{S}_{N_H}(h).} \tag{S.286}$$

若存在一个共享 exceptional set E_τ、$\#E_\tau \le N_b$，且在其外 $b_k \le q_k + c_k p_k^{2/3}$、$\sum q_k \le C_q A_R$、$\sum p_k \le C_p P_R^M$、$\sum c_k^3 \le C_c$，则 shellwise Hölder 得到

$$\boxed{\mathcal{S}_{N_b}(b) \le (C_q + C_c^{1/3} C_p^{2/3}) A_R.} \tag{S.287}$$

PDE 必须构造这些对象并保留 full-history ancestry。bare minimal statement 仍为

$$\boxed{\textbf{OPEN:}\quad \exists N_b, C_b \text{ fixed such that } \mathcal{S}_{N_b}(b(\tau)) \le C_b A_R.} \tag{S.288}$$

65. Conditional moving-tube Morrey theorem

令 $\widetilde{\boldsymbol{\mu}}$ 为 total local dissipation measure 的 periodic lift，$\widetilde{X}_R$ 为 mollified path 的 continuous lift，并定义 full-history tube

$$\boxed{\mathcal{U}_{k,R}(\tau) = \{(t,y) : s_R < t < \tau,\ y - \widetilde{X}_R(t) \in \operatorname{supp} \psi_k^R\}.} \tag{S.289}$$

假设在所限制的整个 solution class、所有尺度与 terminals 上存在共同常数 $M, L < \infty$：

$$\boxed{\sup_{Q_R^-(z,s)} \frac{\widetilde{\boldsymbol{\mu}}(Q_R^-(z,s))}{R} \le M, \qquad \mathcal{L}_R(\tau) = \frac{1}{R}\int_{s_R}^{\tau} |\dot{\widetilde{X}}_R(t)|\, dt \le L.} \tag{S.290}$$

按 elapsed time $O(R^2)$ 或 accumulated path variation $O(2^k R)$ 贪心停时，可用至多

$$\boxed{C_\psi(2^{3k} + L2^{2k})} \tag{S.291}$$

个 backward R-cylinders 覆盖 $\mathcal{U}_{k,R}(\tau)$。arc-length stopping 不可用 endpoint displacement 代替。又因 $\boldsymbol{\mu} = |\nabla u|^2 dxdt + \boldsymbol{D}$ 正好一次分解，defect 与 restricted high-Rayleigh viscous parts 相加后只需支付 tube total mass 一次：

$$\boxed{b_k(\tau) \le \frac{\gamma_k}{R} \widetilde{\boldsymbol{\mu}}(\mathcal{U}_{k,R}(\tau)) \le C_\psi M \gamma_k (2^{3k} + L2^{2k}).} \tag{S.292}$$

令 $\mathscr{A}_m = \sum_{k\ge1} 2^{mk}\gamma_k < \infty$（$m = 2,3$），则

$$\boxed{\sum_k x_k^{\mathrm{sel}}(\tau) \le \sum_k b_k(\tau) \le B(M,L) := C_\psi M(\mathscr{A}_3 + L\mathscr{A}_2).} \tag{S.293}$$

与 inherited linear cap $\sum x_k^{\mathrm{sel}} \le C_0 P_R^M$ 合并，在 $P_R^M \le 1$ 与 $P_R^M \ge 1$ 两区分别选择较强上界，得到

$$\boxed{\mathcal{S}_0(x^{\mathrm{sel}}(\tau)) \le \max\{C_0, B(M,L)\}A_R.} \tag{S.294}$$

这是真正证明的 **conditional theorem**。允许 $M = M(u,R)$ 或 $L = L(u,R)$ 只会恢复 nonuniform fixed-solution finiteness，不能冒充 bare suitable-weak class 的 uniform packing。

66. Scale-critical mixed-norm benchmark

令 $q \in [3,\infty]$、$r \in [3,\infty)$，$\theta = 3/r + 2/q$，并在 restricted class 的所有 target scales 上假设

$$\boxed{\mathcal{U}_{q,r}(R) = R^{1-\theta}\|u\|_{L_t^q(I_{8R};L_x^r(\mathbb{T}^3))} \le M_*.} \tag{S.295}$$

mean-zero periodic pressure gauge 与 Calderón--Zygmund 给出

$$\boxed{\|p-\bar{p}(t)\|_{L_t^{q/2}L_x^{r/2}} \le C_r M_*^2 R^{2\theta-2}.} \tag{S.296}$$

把等于一的 smooth spacetime test 插入 $\boldsymbol{\mu}$ 的 distribution definition，得到

$$\begin{aligned}\boldsymbol{\mu}(Q_R^-) \le C\bigg[&R^{-2}\int_{Q_{2R}^-}|u|^2 + R^{-1}\int_{Q_{2R}^-}|u|^3 \\ &+ R^{-1}\int_{Q_{2R}^-}|p-\bar{p}(t)|\,|u|\bigg].\end{aligned} \tag{S.297}$$

mixed Hölder 后所有 R powers 精确抵消为一次 R:

$$\boxed{\sup_{Q_R^-}\frac{\boldsymbol{\mu}(Q_R^-)}{R} \le C_{q,r}(M_*^2+M_*^3) =: M_\mu.} \tag{S.298}$$

同时 $|\dot{X}_R(t)| \le CR^{-3/r}\|u(t)\|_{L^r}$，故

$$\boxed{\frac{1}{R}\int_{s_R}^{\tau}|\dot{X}_R(t)|\,dt \le CM_*R^{-1-3/r+2-2/q+\theta-1} = CM_*.} \tag{S.299}$$

代入 S.294 得

$$\boxed{\mathcal{S}_0(x^{\mathrm{sel}}(\tau)) \le C_{q,r,M_*}A_R.} \tag{S.300}$$

这只是 conditional sanity check：critical strong norm ball 已有更强 regularity theory；当 $\theta > 1$ 时，单个解的 global mixed norm finiteness 不推出 S.295 所需的 scale-Morrey decay。这里也不把 weak L^3 未经 Lorentz endpoint argument 直接代入 cubic row。

67. Partial regularity 为什么没有闭合 excess gate

CKN 给出 singular set 的 parabolic $\mathcal{H}^1$-measure 为零，这是 support-size conclusion，不是 $\boldsymbol{D}$ 的 mass-density upper bound。抽象测度

$$\boxed{\boldsymbol{D}_M = \sum_{k=1}^{M} a_k \delta_{z_k}, \qquad a_k > 0} \tag{S.301}$$

可支撑在有限点集上，却在 M 个 moving annular tubes 中各有非零 weighted mass。它只是反驳“dimension 自动推出 packing”的 measure countermodel，不是 **NSE defect measure**。

high-Rayleigh ancestor 属于 $|\nabla u|^2 dxdt$，可完全位于 regular set。于是

$$\boxed{\text{large high-Rayleigh mass} \;\not\Longrightarrow\; \text{a singular point detected by epsilon regularity.}} \tag{S.302}$$

epsilon regularity 的 converse 不把每个大的 regular viscous mass 变成 singular point。即使有一个 threshold 的 singular-cylinder count，也仍需提升为 all-threshold integrable mass distribution 才能推出 S.288。

68. Bounded primary-source collision audit

有限文献检索覆盖 Caffarelli--Kohn--Nirenberg 的 partial regularity、De Rosa--Drivas--Inversi 的 anomalous dissipation support、Seregin 的 critical Morrey estimates、Barker 在附加 weak-L^3 bound 下的 singular-point count，以及 Neustupa 的 singular-point L^3 concentration。没有找到与 S.280 或 S.288 量词完全相同的 theorem。

这些来源分别控制 singular-set size、附加 integrability 下的 density、假设性的 critical coefficient 或依赖额外 norm 的 singular-point count；都没有从 bare inherited ledger 推出 full-history high-Rayleigh annular best-N packing。检索只用于标记 literature boundary，不构成 novelty 或 priority proof。

69. Route decision 与 combined target

short branch 应直接攻连续公共窗口 gate S.280，最好通过 all-threshold count S.279；scalar temporal L^1、terminal averaging 或没有 endpoint gain 的 critical one-threshold count 已被证明不足。excess branch 应攻 shared ancestor charging S.287 或足以触发 S.294 的 uniform moving-tube coefficient；defect 与 high-Rayleigh budgets 必须按 S.286 相加。

combined Step 12 target 是

$$\boxed{\begin{array}{c}\textbf{OPEN: find fixed } N_F, N_b \textbf{ and constants such that}\\ \sup_{\tau\in\mathcal{T}_R} \mathcal{V}^F_{N_F,R}(\tau,\delta_*) \lesssim A_R, \qquad \sup_{\tau\in\mathcal{G}_R\cap\mathcal{T}_R} \mathcal{S}_{N_b}(b(\tau)) \lesssim A_R.\\ \text{Then Step 11 closes with } N_{\rm sh}=N_F,\ N_x=N_b, \text{ and total budget } N_F+N_b.\end{array}} \tag{S.303}$$

两个 antecedents 在 bare suitable-weak class 中都 **OPEN**；第二项目前只在 Section 65/66 的附加统一假设下证明。

70. 单包加速的 kinematic screen

inherited R0.74F packet centre 满足 $Q(t)=q_{\rm pre}+B\int_0^t\theta(s,h)ds$、$|\theta|\le 1$、$0<B\le(32R^2)^{-1}$，故

$$\boxed{\mathrm{Var}_{[0,65R^2]}Q\le\frac{65}{32}<2\pi,\qquad \mathrm{Var}_{I_{2R}}Q\le\frac{1}{8}.}\tag{S.304}$$

冻结 exact family 的 packet centre 不发生一次完整 torus winding；R0.74F 的 all-winding estimates 是 Brownian-bridge heat kernel 的 periodic copies，不是 packet-centre orbits。

若 hypothetical monotone extension 满足 $0<\beta B\le q'(t)\le B$，对 torus measurable set J，令 $D=q(T)-q(0)$、$m=\lfloor D/(2\pi)\rfloor$，则 change of variables 给出 exact occupation bound

$$\boxed{\frac{m|J|}{B}\le\tau_J\le\frac{(m+1)|J|}{\beta B}.}\tag{S.305}$$

many-winding regime 中 $m\asymp BT$，访问次数与每次 residence time 的 B 因子抵消，不能产生 outer dyadic shells 的 exponential preference。若 $z_\ell\le H2^{p\ell}\Gamma^{4^\ell}$，且 $q_N=2^p\Gamma^{3\cdot 4^N}<1$，则

$$\boxed{\mathcal{S}_N(z)\le\sum_{\ell\ge N}z_\ell\le\frac{H2^{pN}\Gamma^{4^N}}{1-q_N}.}\tag{S.306}$$

所以 uniform speed-up 只改变共同 prefactor，不改变 super-Gaussian shell ratio。这是 kinematic screen，不是 **universal PDE no-go**，也没有把 earlier packet deposit 识别成 complete ancestor vector b。

71. Step 12 主张边界与双路审计

Step 12 **PROVED**：S.274--S.275 terminal variation-window reduction；S.276--S.277 continuity 与 fixed-solution modulus；S.278--S.279 best-N layer cake 与 all-threshold implication；S.281--S.284 L^1_t-only abstract no-go 与 averaged $P^{4/5}$ boundary；S.285--S.287 exception-budget recombination 与 conditional charging；S.289--S.294 moving-tube cover 与 conditional critical-Morrey theorem；S.295--S.300 mixed-norm sufficient benchmark；S.304--S.305 literal no-winding 与 occupation lemma；S.306 abstract super-Gaussian filter。

Step 12 **ABSTRACT BOUNDARY TESTS, NOT NSE COUNTEREXAMPLES**：S.281 synchronized temporal spikes；S.301 finite atomic defect-support model。

Step 12 **OPEN**：universal terminal-window gate S.280；universal ancestor gate S.288；combined S.303；Step 11 S.272；Q.12、Q.1；从 frozen payment alone 推出 uniform critical Morrey/path estimate；把 earlier moving-packet deposit 与完整 b 识别；bare suitable weak class 的 universal shell count；scale contraction、regularity、singularity 与 Clay。

主证书通过 16/16 exact、12/12 finite、51/51 structural 与 11/11 mutations。独立 Ruby audit 通过 12/12 groups、153,237 exact cases、6/6 artifact locks、6/6 dependency locks、39/39 note checks；两条实现相互独立地封存 source 与证书边界。有限检查只支持公式实现与防篡改，不替代 analytic proof。

72. Step 13：时间可积性上限与组合 Morrey 阈值

Step 12 把 short last-exit residual 写成一个共同终端窗内的绝对 physical-flux variation，并在统一 moving-tube Morrey 与 path-length 假设下证明了 selected-excess estimate。Step 13 检查普通时间可积性能够把 short gate 推到哪里，以及允许 Morrey coefficient 随 payment 增长时，excess gate 的精确标量阈值是什么。

本步得到五项结论。第一，对每个固定的周期 suitable weak solution 和固定尺度，physical shell-flux density 具有可求和的 $\ell^1(L_t^{4/3})$ 包络；这来自 energy-class interpolation $u \in L_t^4 L_x^3$、周期 Calderon--Zygmund pressure estimate 与固定尺度的 mollified drift bound。第二，维数化长度为 δ 的共同终端窗因此获得 $\delta^{1/4}$ 增益；一般的 L_t^p shell-tail estimate 获得 $\delta^{1-1/p}$，但整个窗口只能删除同一个 shell set。第三，这个时间增益不能修复 payment exponent：即使额外假设相关的 $\ell^1(L_t^p)$ tail 线性依赖 P_R^M，与 positive-depth term 平衡后仍只有 $P^{2(2p-1)/(5p-3)}$；energy-class 值为 $P^{10/11}$，$p=\infty$ 也只有 $P^{4/5}$，都达不到 $P^{2/3}$。第四，smooth $(N+1)$-coordinate witness 证明：任意阶 scalar temporal regularity 连同 payment-linear amplitude 仍不能单独推出 fixed-window best-N gate；这是抽象向量见证，不是 Navier--Stokes solution。第五，Step 12 的 separate uniform Morrey/path assumptions 可以弱化为 combined cover coefficient 至多按 $1+(P_R^M)^{2/3}$ 增长；$2/3$ 对只使用 two scalar caps 的论证是精确阈值，但不声称它对其他 PDE 证明是必要条件。

Universal terminal-window gate S.280、universal ancestor gate S.288 与 combined target S.303 继续 **OPEN**。本步不证明 Q.12、Q.1、scale contraction、regularity、singularity formation 或 Millennium problem；没有使用 DNS 或 DGX。**NOT CLAY.**

73. 无量纲 flux density 与共同删除次序

保留 R0.74S Step 12 的全部设定。令

$$\vartheta = \frac{\tau - s_R}{R^2} \in (0,4), \qquad t(\sigma) = s_R + R^2\sigma,$$

并把每个 derivative 在 $\mathcal{T}_R$ 外延拓为零。定义

$$\boxed{h_{k,R}(\sigma) := R^2 |\dot{F}_{k,R}(t(\sigma))|, \qquad 0<\sigma<4.} \tag{S.307}$$

则 S.273 的共同窗口 coordinate 精确为

$$\boxed{f_{k,R}(\tau,\delta) = \int_{(\vartheta-\delta,\vartheta)\cap(0,4)} h_{k,R}(\sigma)\,d\sigma.} \tag{S.308}$$

对 $1 \le p \le \infty$ 与整数 $N \ge 0$，定义 common-deletion temporal tail

$$\boxed{\mathfrak{H}^F_{p,N,R} := \inf_{\#S\le N} \sum_{k\notin S} \|h_{k,R}\|_{L^p(0,4)}.} \tag{S.309}$$

S.309 的次序不可交换：shell set 在 time norm 之外选定，因此整个窗口使用同一个删除集合。逐时移动的 minimizing shell set 不能控制这里的 common-window functional。

74. 固定解的 $L_t^{4/3}$ 事实

对 frozen class 中每个固定的周期 suitable weak solution 与固定 $R>0$，令

$$E_I := \operatorname*{ess\,sup}_{t\in\mathcal{T}_R}\|v_R(t)\|_2^2, \qquad D_I := \int_{\mathcal{T}_R}\|\nabla v_R(t)\|_2^2\,dt, \qquad e_R := \frac{E_I}{R}, \quad d_R := \frac{D_I}{R}.$$

则 Proposition 2.1 证明

$$\boxed{\mathfrak{H}^F_{4/3,0,R} \le C\left([e_R(e_R+d_R)]^{3/4} + e_R^{3/2}\right) \le C(e_R+d_R)^{3/2} < \infty.} \tag{S.310}$$

从而对每个 terminal 与 $0<\delta<4$，

$$\boxed{\mathcal{V}^F_{N,R}(\tau,\delta) \le C\delta^{1/4}[e_R(e_R+d_R)]^{3/4} + C\delta e_R^{3/2}.} \tag{S.311}$$

S.310 的有限 coefficient 可以依赖 solution 与 R；本步没有证明它由 P_R^M 一致控制。

采用 mean-zero periodic pressure gauge。Shellwise gauge cancellation 允许在 $F_{k,R}$ 的 signed derivative 中使用这个 gauge。Energy class 与空间插值给出

$$\boxed{v_R \in L_t^4 L_x^3, \qquad \pi_R - \overline{\pi}_R(t) \in L_t^2 L_x^{3/2}, \qquad a_R \in L_t^\infty.} \tag{S.312}$$

其中 $L_t^\infty L_x^2 \cap L_t^2 H_x^1 \subset L_t^4 L_x^3$，而周期 Calderon--Zygmund 估计为

$$\|\pi_R - \overline{\pi}_R(t)\|_{L^{3/2}} \le C\|v_R(t)\|_{L^3}^2.$$

固定尺度 convolution 把 L_x^2 映到 L_x^∞，故 $|a_R(t)| \le C_R\|v_R(t)\|_2$。继承的 cutoff bound 与 super-Gaussian shell weights 给出 $\sum_{k\ge1}\gamma_k(1+2^{3k}R^3)<\infty$。在 pressure gauge 已经改变之后再取 absolute value，可由 R0.74P 的 (2.9) 得到

$$\boxed{h_{k,R}(\sigma) \le C\gamma_k(1+2^{3k}R^3)\left[\|v_R(t)\|_3^3 + \|\pi_R(t)-\overline{\pi}_R(t)\|_{3/2}\|v_R(t)\|_3 + |a_R(t)|\|v_R(t)\|_2^2\right].} \tag{S.313}$$

完整尺度计算使用 $\sum_k \gamma_k|\nabla\Psi_k^R| \le CR^{-1}$ 与周期插值不等式

$$\|v_R(t)\|_3^4 \le C\|v_R(t)\|_2^2\left(\|\nabla v_R(t)\|_2^2 + R^{-2}\|v_R(t)\|_2^2\right).$$

由于 $|\mathcal{T}_R| = 4R^2$，cubic/pressure 部分的 $L_t^{4/3}$ norm 至多为 $CR^{-1/2}[e_R(e_R+d_R)]^{3/4}$，drift 部分的 L_t^∞ norm 至多为 $CR^{-2}e_R^{3/2}$。在 S.307 的变量替换下，前者获得 $R^{1/2}$，后者获得 R^2，从而得到 S.310；在 S.308 上对固定删除集使用 Holder，得到 S.311。

令 S.313 的 bracket 在 $t = t(\sigma)$ 处为 $\mathcal{B}_R(\sigma)$，则实际使用的是

$$\sum_k \|h_{k,R}\|_{4/3} \le C\left[\sum_k \gamma_k(1+2^{3k}R^3)\right]\|\mathcal{B}_R\|_{4/3}.$$

因此这是 $\ell^1(L^{4/3})$ estimate，不是把 $L^{4/3}(\ell^1)$ 非法互换过来。指数 $4/3$ 是 direct energy-class interpolation 对空间 cubic integral 给出的 endpoint；不排除加入额外 PDE 假设后获得更高时间可积性。

Energy-admissible pairs 满足

$$\frac{2}{q}+\frac{3}{r}=\frac{3}{2}, \qquad 2<r\le 6, \qquad q(r)=\frac{4r}{3(r-2)},$$

而 $r=2$ 单独理解为 $q(2)=\infty$。对三个空间 Holder reciprocal 和为一的 admissible factors，有

$$\sum_{i=1}^{3}\frac{1}{q_i}=\frac{9}{4}-\frac{3}{2}\sum_{i=1}^{3}\frac{1}{r_i}=\frac{3}{4},$$

故时间指数仍是 $4/3$。Pressure 的对称选择先把 $v_R\otimes v_R$ 放入强型 Calderon--Zygmund 范围 $L_x^{3/2}$；这里不使用 L^1 weak endpoint 或 L^∞-to-BMO endpoint。

75. 一般时间指数与精确优化

定义

$$a_p := 1-\frac{1}{p} \quad (1\le p<\infty), \qquad a_\infty := 1. \tag{S.314}$$

对 $\ell^1(L^p(0,4))$ 中的非负 shell densities，同一证明给出

$$\boxed{\mathcal{V}_{N,R}^F(\tau,\delta) \le \delta^{a_p}\mathfrak{H}_{p,N,R}^F.} \tag{S.315}$$

只为检验此方法，额外假设某个固定 $p\in(1,\infty]$、N、$\beta>0$ 与 C_H 对 solution、scale 与 terminal 一致满足

$$\mathfrak{H}_{p,N,R}^F \le C_H(P_R^M)^\beta. \tag{S.316}$$

S.315 与 S.275 合并为

$$\boxed{\mathcal{S}_N(r^{\mathrm{sh}}(\tau)) \le C_H\delta^{a_p}P^{\beta} + C_{\mathrm{deep}}\delta^{-2/3}P^{2/3}, \qquad P := P_R^M.} \tag{S.317}$$

当 $P \ge 1$、$\beta > 2/3$ 且 optimizer 落在 $(0,4)$ 时，平衡尺度为

$$\boxed{\delta_{p,\beta} \asymp P^{-(\beta-2/3)/(a_p+2/3)}.} \tag{S.318}$$

两项共同得到的 payment power 是

$$\boxed{E_{p,\beta} = \frac{2}{3}\frac{a_p+\beta}{a_p+2/3}, \qquad E_{p,\beta} - \frac{2}{3} = \frac{2}{3}\frac{\beta-2/3}{a_p+2/3} > 0.} \tag{S.319}$$

因此只要 temporal coefficient 比 $P^{2/3}$ 增长更快，任何 p，包括 $p = \infty$，都不能消除 exponent loss。反过来，在这个 two-term argument 内，$\beta \le 2/3$ 对 large-payment regime 已经充分；small-payment regime 由 $P \le P^{2/3}$ 与继承的 linear ledger 控制。

乐观的 linear case $\beta = 1$ 给出

$$\boxed{\delta_p \asymp P^{-p/(5p-3)}, \qquad E_p = \frac{2(2p-1)}{5p-3}.} \tag{S.320}$$

特别地，

$$\boxed{p = \frac{4}{3}: \quad \delta \asymp P^{-4/11}, \quad E_p = \frac{10}{11}; \qquad p = \infty: \quad \delta \asymp P^{-1/5}, \quad E_p = \frac{4}{5}.} \tag{S.321}$$

$p = 1$ 没有正 window power，linear term 仍为 P。S.320 在 $p \to \infty$ 时单调趋于 $4/5$，仍高于目标 $2/3$。

对 Proposition 2.1 实际证明的 coefficient，直接优化得到 fixed-solution estimate

$$\boxed{\mathcal{S}_N(r^{\mathrm{sh}}(\tau)) \le C\left[A_R + (\mathfrak{H}^F_{4/3,N,R})^{8/11}A_R^{3/11}\right].} \tag{S.322}$$

这里 A_R 覆盖 formal optimizer 超过 allowed window length 的情况。如果再有尚未证明的 $\mathfrak{H}^F_{4/3,N,R} \lesssim P$，mixed term 才是 $P^{10/11}$。这是对该 upper-bound method 的 ceiling，不是每个 NSE solution 都达到的 lower bound。

76. Smooth all-p 抽象饱和见证

固定 $N \ge 0$，令 $M = N+1$。取 $0 < \delta_0 < 4$ 与非负 $\phi \in C_c^\infty((-\delta_0, 0))$，满足 $\int \phi = 1$，并选择 terminal ϑ_0 使 translated support 位于 $(0,4)$。对 $H > 0$，定义

$$\boxed{h_{k,H}(\sigma) = \frac{H}{M}\phi(\sigma - \vartheta_0) \quad (1 \le k \le M), \qquad h_{k,H} = 0 \quad (k > M).} \tag{S.323}$$

每个 primitive 都 smooth 且 increasing，并且对每个 $1 \le p \le \infty$，

$$\boxed{\sum_k \|h_{k,H}\|_{L^p} = H\|\phi\|_{L^p}, \qquad \mathcal{V}_N^F(\vartheta_0, \delta_0) = \frac{H}{M}.} \tag{S.324}$$

若 abstract payment 归一化为 $P_H \asymp H$，则

$$\boxed{\frac{\mathcal{V}_N^F(\vartheta_0, \delta_0)}{P_H^{2/3}} \asymp \frac{H^{1/3}}{N+1} \longrightarrow \infty.} \tag{S.325}$$

这个 witness 有 fixed smooth time profile，属于所有 temporal L^p spaces。它只证明 logical boundary：temporal regularity 加 scalar linear-amplitude bound 不包含 S.280。它不是 velocity field、pressure、suitable weak solution 或 NSE counterexample。

还有一个饱和 adaptive $p = 4/3$ balance 的 smooth abstract witness。取 $P \ge 1$，选择 $0 \le \rho \in C_c^\infty((-1,0))$、$\|\rho\|_{4/3} = 1$，并令

$$c_\rho := \int_{-1}^{0} \rho(s)\, ds > 0, \qquad d := P^{-4/11} \le 1.$$

选择 ϑ_0 使 $\vartheta_0 + d \operatorname{supp} \rho \subset (0,4)$，并定义

$$h_{k,P}(\sigma) = \frac{P}{Md^{3/4}} \rho\left(\frac{\sigma - \vartheta_0}{d}\right), \qquad 1 \le k \le M.$$

于是

$$\sum_k \|h_{k,P}\|_{4/3} = P, \qquad \sum_k \|h_{k,P}\|_1 = c_\rho P^{10/11} \le c_\rho P.$$

删除 $N = M - 1$ 个 coordinates 后精确剩下 $c_\rho P^{10/11}/M$。为每个 coordinate 指定 abstract depth $d_{k,P} = d$ 与 residual $r_{k,P} = c_\rho P^{10/11}/M$。当 $\delta \ge d$ 时，共同终端窗收取全部 residual；当 $0 < \delta < d$ 时，coordinate 属于 deep class，且

$$\sum_k r_{k,P} = c_\rho P^{10/11} \le c_\rho P^{2/3} \delta^{-2/3}.$$

在 balancing depth 上，

$$P^{2/3} d^{-2/3} = P^{10/11}, \qquad \frac{10}{11} - \frac{2}{3} = \frac{8}{33}.$$

因此在固定常数 c_ρ 意义下，$10/11$ 对 linear $\ell^1(L^{4/3})$ rate bound、inherited linear L^1 ledger、depth allowance 与 fixed deletion budget 的抽象组合是 sharp。它仍只是 method-level countermodel，不是 NSE realization。

77. Moving-tube 路线的精确标量阈值

允许 Step 12 的 quantities 依赖 solution、scale 与 terminal：

$$M_R(\tau) := \sup_{Q_R^- \text{ in the buffer}} \frac{\tilde{\boldsymbol{\mu}}(Q_R^-)}{R},$$
$$L_R(\tau) := \frac{1}{R}\int_{s_R}^{\tau} |\dot{\tilde{X}}_R(t)|\, dt.$$

定义 combined cover coefficient

$$\boxed{B_R(\tau) := C_\psi M_R(\tau)\big(\mathscr{A}_3 + L_R(\tau)\mathscr{A}_2\big).} \tag{S.326}$$

只要这些 quantities 有限，Step 12 的 S.291--S.293 对每个 (u, R, τ) 点态成立，并给出

$$\boxed{\sum_k x_k^{\mathrm{sel}}(\tau) \le \min\{C_0 P_R^M, B_R(\tau)\}.} \tag{S.327}$$

因此 Step 12 分别假设 $M_R \le M$、$L_R \le L$ 比 algebraic closure 所需更强。

Proposition 5.1 是 conditional payment-dependent Morrey envelope：若一个 universal C_B 对所有 solution 与 scale 满足

$$\boxed{\sup_{\tau \in \mathcal{G}_R \cap \mathcal{T}_R} B_R(\tau) \le C_B\big[1 + (P_R^M)^{2/3}\big],} \tag{S.328}$$

则

$$\boxed{\sup_{\tau \in \mathcal{G}_R \cap \mathcal{T}_R} \mathcal{S}_0(x^{\mathrm{sel}}(\tau)) \le C(C_0, C_B)(P_R^M)^{2/3}.} \tag{S.329}$$

证明只分两个 payment regimes：$P_R^M \le 1$ 时使用 S.327 的 linear side 与 $P_R^M \le (P_R^M)^{2/3}$；$P_R^M \ge 1$ 时使用 S.328 与 $1 \le (P_R^M)^{2/3}$。特别地，

$$\boxed{B_R(\tau) \le C_B[1 + (P_R^M)^\theta], \qquad 0 \le \theta \le \frac{2}{3},} \tag{S.330}$$

即可闭合 selected-excess gate。M_R 与 L_R 可以分别 nonuniform，只要加权 cover cost 的组合增长不超过 quadratic payment scale。

对只知道 S.327 两个 scalar caps 的论证，指数阈值是 sharp。若 $\theta > 2/3$，固定 N，令 $M = N + 1$，对 large P 取 M 个等坐标，总质量 $T_P = \min\{C_0 P, C_B P^\theta\}$，并设 $x_k^{\mathrm{sel}} = b_k = T_P/M$。则

$$\boxed{\sum_k b_k = T_P, \qquad \mathcal{S}_N(b) = \frac{T_P}{N+1}, \qquad \frac{\mathcal{S}_N(b)}{P^{2/3}} \longrightarrow \infty.} \tag{S.331}$$

这是 two-cap inference 的 abstract sequence countermodel，不是 dissipation measure 或 NSE solution。使用更多 PDE structure 的证明仍可能在 $\theta > 2/3$ 时成功。

78. Dynamic high frequency 不能单独攻击 flux gate

在 2π-periodic torus 上取 $A > 0$、$T > 0$ 与 integer $n \geq 1$：

$$\boxed{u^{(n)}(t,x) = Ae^{-n^2 t}\sin(nx_2)e_1, \qquad p^{(n)} = 0.} \tag{S.332}$$

这是 unforced smooth Navier--Stokes solution：divergence free，$(u^{(n)} \cdot \nabla)u^{(n)} = 0$，并满足 heat equation。Mollified path velocity 平行于 e_1，而 moving velocity 与 y_1 无关，所以对每个周期 shell cutoff，

$$\boxed{\dot{F}^{(n)}_{k,R}(t) = \frac{\gamma_k \eta_R(t)}{2R}\int_{\mathbb{T}^3} |v^{(n)}_R|^2 (v^{(n)}_{R,1} - a_{R,1})\, \partial_{y_1}\Psi^R_k \, dy = 0.} \tag{S.333}$$

最后一个等号来自 y_1 上的周期积分。因此这个 exact family 的每个 $f_{k,R}$ 与 $\mathcal{V}^F_{N,R}$ 都为零。同时在 $[0,T]$ 上，

$$\boxed{\begin{aligned} \int_0^T\int_{\mathbb{T}^3} |\nabla u^{(n)}|^2 &= 2\pi^3 A^2(1-e^{-2n^2 T}), \\ \int_0^T\int_{\mathbb{T}^3} |u^{(n)}|^3 &= \frac{32\pi^2 A^3}{9n^2}(1-e^{-3n^2 T}). \end{aligned}} \tag{S.334}$$

其 dissipation-to-cubic ratio 按 n^2/A 增长，但 canonical physical-flux primitive 为零，completed clock 满足 $K = Q$。因此 high Fourier frequency 与 high Rayleigh ratio 本身不会产生 short physical-shell flux tail；这里的 k 是 physical moving annulus index，不是 Fourier shell。这个 screen 不推广为任意动态高频场的零 flux theorem。

79. 有限主文献边界

我进行了有界检索，没有找到能以 Step 13 所需量词直接给出 uniform temporal tail 或 S.328 的 theorem。

- Lei--Ren, *Quantitative partial regularity of the Navier--Stokes equations and applications*, Adv. Math. 445 (2024), 109654：定量化 dissipation-energy pigeonhole 并给出 logarithmically improved partial regularity；其 annular levels 与 constants 依赖 natural local energies，不是 common-terminal fixed-physical-shell best-N flux estimate。
- Choe--Yang, *Local kinetic energy and singularities of the incompressible Navier--Stokes equations*, JDE 264 (2018), 1171--1191：在 uniformly bounded scaled local kinetic energy 下得到 reverse Holder improvement；这正是 bare payment ledger 没有的额外信息。
- Guevara--Phuc, *Local energy bounds and epsilon-regularity criteria for the 3D Navier--Stokes system*, Calc. Var. PDE 56 (2017), 68：从 scale-integrated inputs 得到 pressure-sensitive local-energy 与 epsilon-regularity criteria；不产生 S.316 或 S.328 的系数。
- Koch--Tataru, *Well-posedness for the Navier--Stokes equations*, Adv. Math. 157 (2001), 22--35：在 critical small-data BMO^{-1} class 中使用 Carleson-type spacetime norm；不是从 P^M_R 为每个 bare suitable weak solution 推出 Carleson estimate。

这只是 collision check，不是 novelty 或 priority claim。相邻文献的 quantitative time/Morrey gains 都带有额外 scale information、smallness 或 energy-dependent constants，不能充当 S.280、S.288 或 S.328 的证明。

80. Critical 八叉树抽象反例

接着检查 bounded-branching ancestor tree 连同现有 linear 与 square ledgers 是否强迫额外 packing。答案是在 critical tree exponent 上不能。

固定整数 $m \geq 1$，令 $L = m^3$，取深度 $0 \leq d \leq L-1$ 的 complete eight-ary tree。对深度 d 的每个 node v，定义

$$\boxed{b_v = \frac{1}{m^2 8^d}, \qquad s_v = \frac{5}{3m^2 8^d}, \qquad c_v = 2^{-d}, \qquad p_v = \frac{1}{m^3 8^d}.} \tag{S.335}$$

按 s_v 缩放 Step 11 S.267 的 pure high-Rayleigh scalar row：

$$\boxed{T_v = s_v, \quad d_v^{\mathrm{def}} = 0, \quad \int_{H_v} g_v = b_v = \frac{3}{5}s_v, \quad \beta_v = 0, \quad \sigma_v = \frac{983}{12000}s_v < \frac{T_v}{12}, \quad x_v = \frac{2617}{6000}s_v > \frac{T_v}{6}, \quad r_v^x = \frac{s_v}{3}.} \tag{S.336}$$

每个 node 都严格处于 abstract $\mathcal{I}_x$ branch，ancestor 是 pure high-Rayleigh。逐层求和得到

$$\boxed{b_v = c_v p_v^{2/3}, \qquad \sum_v p_v = 1, \qquad \sum_v b_v = m, \qquad \sum_v s_v = \frac{5m}{3} =: P_m.} \tag{S.337}$$

这些 scalar rows 与 inherited linear clock/variation ledgers 及 zero Q-variation 相容，并满足

$$\boxed{\sum_v s_v^2 < \frac{200}{63m^4}, \qquad \sum_{w \succeq v} b_w^2 \leq \frac{8}{7}b_v^2, \qquad \sum_{w \in \mathrm{child}(v)} c_w^3 = c_v^3 \quad (0 \leq d(v) \leq L-2).} \tag{S.338}$$

其中第二项是 strong square-Carleson bound；第三项只对 nonterminal nodes 断言，并处于 exact critical equality：八个 children 各取一半系数，coefficient cube 总和守恒。

每个 coordinate 至多为 m^{-2}。删除任意固定 N 个 coordinates 后仍有

$$\boxed{\mathcal{S}_N(b) \geq m - \frac{N}{m^2}, \qquad A_m = P_m^{2/3} = \left(\frac{5m}{3}\right)^{2/3}, \qquad \frac{\mathcal{S}_N(b)}{A_m} \longrightarrow \infty.} \tag{S.339}$$

所以 linear total ledger、vanishing global square ledger、bounded branching、square-Carleson subtree estimate 与 critical child decay 仍不推出 S.288。停止一个 ancestor 不能被算成删除一个 shell exception，同时免费删除其全部 descendants；best-N functional 删除的是 individual shell coordinates。

这棵树是 strict abstract ledger model。其 nodes 没有被同时实现为一个 solution 的 physical moving annuli；它不满足一个共同 velocity field 的 coupled Navier--Stokes dynamics、pressure、diffusion、cross-cubic payment、periodic incidence 或 $K = Q + F$。它不是 NSE counterexample。

81. Conditional incidence theorem 与 cubic Dini interface

存在一个精确的充分替代条件。若对每个 terminal 都有 shell set E_τ、$\#E_\tau \le N_b$，以及非负 q_k、tree payments p_ν、coefficients c_ν 和 incidences $\nu \rightsquigarrow k$，使得在 E_τ 外

$$b_k \le q_k + \sum_{\nu \rightsquigarrow k} c_\nu p_\nu^{2/3}, \qquad \sum_k q_k \le C_q A_R,$$

并且

$$\sum_{\text{incidences}} p_\nu \le B_{\text{inc}} C_p P_R^M, \qquad \sum_{\substack{\text{incidences} \\ k \notin E_\tau}} c_\nu^3 \le C_c,$$

则在 incidence set 上使用指数 3 与 3/2 的 Holder 不等式，得到

$$\boxed{\mathcal{S}_{N_b}(b) \le \left[C_q + C_c^{1/3}(B_{\text{inc}} C_p)^{2/3}\right] A_R.} \tag{S.340}$$

三次方是精确 dual exponent，不是方便选择。对每个有限非负 coefficient vector，

$$\boxed{\sup_{p_\nu \ge 0,\ \sum p_\nu \le 1} \sum_\nu c_\nu p_\nu^{2/3} = \left(\sum_\nu c_\nu^3\right)^{1/3}.} \tag{S.341}$$

当 denominator 非零时，$p_\nu = c_\nu^3 / \sum_\omega c_\omega^3$ 取等号。若要把 tree inequality 变成 S.340 的 incidence coefficient bound，还需要三个 uniform inputs：

- root family 满足 $\sum_{v \in \text{roots}} c_v^3 \le C_{\text{root}}$；
- 每个固定 node 被计入的 incidence multiplicity 至多为 M_{inc}；
- 非负 child factors 满足 uniform Dini product sum

$$0 \le \theta_d, \qquad \sum_{w \in \text{child}(v)} c_w^3 \le \theta_d c_v^3, \qquad \sup_{d_0 \ge 0} \sum_{n \ge 0} \prod_{j=0}^{n-1} \theta_{d_0+j} \le C_D < \infty.$$

三项合起来给出 $\sum_{\text{incidences}} c_\nu^3 \le M_{\text{inc}} C_{\text{root}} C_D$，从而提供 S.340 所需的 C_c。Uniform $\theta_d \le \theta < 1$ 是简单特例。S.335 的 critical tree 有 $\theta_d = 1$，其 finite-depth Dini constant 按 $L = m^3$ 增长，因而不能提供 uniform coefficient bound。

82. Step 13 路线决定与主张账本

Step 13 排除两条无效方向。第一，提高整体 flux density 的 scalar time regularity 不够；short branch 的下一充分输入必须是 shell selective：

$$\boxed{\exists p\in(1,\infty],N_F,C:\quad \mathfrak{H}^F_{p,N_F,R}\le C(P^M_R)^{2/3}.} \tag{S.342}$$

S.342 的 deletion set 在时间上固定；pointwise moving exceptional set 不充分。第二，excess branch 应攻击较弱的 combined envelope S.328，而不是分别要求 M_R 与 L_R universal bounded；只要 product 保持 quadratic payment scale，较长 path 可以由较小 cylinder-density coefficient 抵消。第三，pure high-frequency heat shear 不是 physical-window gate 的反例候选；后续 exact-family search 必须制造 physical annular separation，并同时通过 Q、cubic、pressure 与 drift ledgers。

Step 13 **PROVED**：S.307--S.309 的 dimensionless/common-deletion representation；S.310--S.313 的 fixed-solution $\ell^1(L_t^{4/3})$ finiteness 与 $\delta^{1/4}$ common-window bound；S.314--S.315 的 general Holder bound；在明确假设 S.316 下 S.317--S.322 的 exact optimization algebra；S.323--S.325 的 smooth all-p abstract saturation witness；在 explicit geometric envelope S.328 下 S.326--S.330 的 payment-dependent Morrey implication；S.331 对 two-cap inference 的 sharpness；S.332--S.334 的 exact heat-shear identities；S.335--S.339 的 critical eight-ary abstract countermodel；以及 S.340--S.341 的 conditional incidence-charging theorem、exact cubic duality 与 Dini-subcritical criterion。

Step 13 **ABSTRACT BOUNDARY TESTS, NOT NSE COUNTEREXAMPLES**：S.323--S.325 的 synchronized temporal family；其后的 adaptive smooth rate/depth family；S.331 的 equal-coordinate two-cap family；S.335--S.339 的 eight-ary critical ancestor tree。

Step 13 **OPEN**：$\mathfrak{H}^F_{4/3,N,R}$ 的 uniform payment bound；quadratic shell-selective estimate S.342；S.340 的 PDE incidence data 与 strict cubic Dini-Carleson gain；bare suitable-weak class 的 payment-dependent moving-tube estimate S.328；universal gates S.280、S.288、S.303，Step 11 S.272，Q.12 与 Q.1；以及 scale contraction、regularity、singularity formation 与 Navier--Stokes Millennium problem。

有限证书通过 31/31 exact、11/11 finite、4/4 dependency、22/22 structural 与 32/32 negative mutations；独立 Ruby verifier 通过 9/9 groups、72,027 exact cases、6/6 artifact locks、4/4 dependency locks、32/32 note checks、1/1 primary-artifact group 与 2/2 negative groups。它们检查 algebra、finite fixtures、hashes、structure 与 claim wording，不 machine-prove PDE estimates、open packing gates、Morrey hypothesis、abstract model 的 NSE realization、regularity 或 Millennium problem。

本步的 advance 是固定解的 algebraic terminal modulus、当前 two-term method 的 exact time-integrability exponent ceiling、excess branch 的较弱 combined Morrey interface，以及把 strict cubic Dini decay 指认为下一条件性 packing interface 的 critical-tree obstruction。**NOT CLAY.**

83. Step 14：外侧 collar 对齐与 jump--corona obstruction

Step 13 把通向 quadratic payment scale 的可能修复分成两支：short branch 需要共同删除后的时间尾部 $\mathfrak{H}^F_{p,N,R}\lesssim(P^M_R)^{2/3}$，而 excess branch 需要对 full-history ancestors 建立 strict cubic Dini--Carleson charge。Step 14 把这两个接口放回实际 cutoff 几何与 total local-dissipation measure 的尺度中检验。以下仍记

$$A_R:=(P^M_R)^{2/3}.$$

本步得到八项结论。Physical shell-flux derivative 有 local cubic、local pressure、shell-scale harmonic pressure 与 moving-frame drift 的精确四通道 signed split；逐通道取绝对值后，继承估计只给出线性尺度 CP^M_R 的 L^1_t payment。Shell k 的 outer derivative collar 位于带有同一 γ_k 权重的 doubled-radius payment annulus，super-Gaussian 比值只帮助 $k\ge 3$ 的 inner collar，对无穷多个 outer collar 没有任何增益。平滑 $(N+1)$-coordinate construction 因而证明：aligned weighted L^1 ledger 本身不能推出任意 $p>1$ 的 common-deletion tail；这是 **ABSTRACT METHOD OBSTRUCTION**，不是 Navier--Stokes counterexample。

Excess branch 的正确代数接口是 shell-incidence multiset 上的 cubic coefficient sum，payment 也必须按同一 incidence multiplicity 重复计数。Scale-invariant dissipation pullback 给出 32-child parabolic tree；first density crossings 虽然稀疏，但 level parameter 在 critical cubic Holder estimate 中精确抵消。First relative density jumps 有 strict Dini coefficient，然而 jumps 之间留下的 low-transition corona 尚没有 quadratic payment。Exact heat shear 可显示任意多层 critical spatial mass splitting，同时所有 physical shell flux 恒为零；它只是否定“raw critical tree 自动说明 flux packing”的窄筛查，不是对 open gate 的 NSE counterexample。最终剩余任务被写成一个 shell-selective jump--corona lemma；该引理 **OPEN**，只有它推出 ancestor gate 的代数箭头已经证明。

Step 13 的 S.342、ancestor gate S.288、combined gate S.303、Step 11 S.272、Q.12 与 Q.1 都保持 **OPEN**。本步不证明 scale contraction、regularity、singularity formation 或 Navier--Stokes Millennium problem。没有使用 DNS 或 DGX。**NOT CLAY.**

84. 精确四通道 flux decomposition

令

$$\rho_k := 2^k R, \qquad \gamma_k := \exp\left(-\frac{4^{k-1}}{32}\right).$$

Euclidean lift 上的 frozen cutoff 为

$$\psi_k^R(y) = \vartheta\left(\frac{|y| - \rho_k}{R/8}\right)\vartheta\left(\frac{2\rho_k - |y|}{R/8}\right).$$

其 gradient 只支撑在两个 collars：

$$\boxed{\begin{aligned} C_{k,R}^- &:= \{\rho_k - R/8 < |y| < \rho_k\}, \\ C_{k,R}^+ &:= \{2\rho_k < |y| < 2\rho_k + R/8\}, \\ \operatorname{supp} \nabla\psi_k^R &\subset C_{k,R}^- \cup C_{k,R}^+ \subset B_{3\rho_k}. \end{aligned}} \tag{S.343}$$

最后一个 inclusion 必须跟随 ρ_k。只在固定半径 $2R$ 构造的 pressure remainder 仅在 B_{6R} harmonic，不能覆盖所有 outer shells。取 $0 \le \zeta \in C_c^\infty(B_4)$、$\zeta = 1$ on B_3，令 $\zeta_{\rho_k}(y) = \zeta(y/\rho_k)$，保留 fixed frozen gauge $c_R(t) = c_{2R}^{M,R}(t)$，并定义

$$\boxed{\begin{aligned} p_{k,R}^{\mathrm{loc}} &:= \mathcal{R}_i\mathcal{R}_j(\zeta_{\rho_k}\tilde{v}_{R,i}\tilde{v}_{R,j}), \\ h_{k,R}^{\mathrm{pr}} &:= \widetilde{\pi}_R - p_{k,R}^{\mathrm{loc}}. \end{aligned}} \tag{S.344}$$

在 distributions 意义下，$h_{k,R}^{\mathrm{pr}}$ 与 $h_{k,R}^{\mathrm{pr}} - c_R$ 都在 $B_{3\rho_k}$ harmonic；Weyl lemma 给出光滑代表。Gauge constant 不改变 flux，因为 $\int c_R(t)\tilde{v}_R \cdot \nabla\psi_k^R = 0$。对几乎处处的 t，periodized cutoff 展开后有

$$\boxed{\dot{F}_{k,R} = \dot{F}_{k,R}^{\mathrm{cub}} + \dot{F}_{k,R}^{\mathrm{loc}} + \dot{F}_{k,R}^{\mathrm{har}} + \dot{F}_{k,R}^{\mathrm{dr}},} \tag{S.345}$$

其中

$$\boxed{\begin{aligned} \dot{F}_{k,R}^{\mathrm{cub}} &:= \frac{\gamma_k}{R}\eta_R \int_{\mathbb{R}^3} \frac{|\tilde{v}_R|^2}{2}\tilde{v}_R \cdot \nabla\psi_k^R, \\ \dot{F}_{k,R}^{\mathrm{loc}} &:= \frac{\gamma_k}{R}\eta_R \int_{\mathbb{R}^3} p_{k,R}^{\mathrm{loc}}\tilde{v}_R \cdot \nabla\psi_k^R, \\ \dot{F}_{k,R}^{\mathrm{har}} &:= \frac{\gamma_k}{R}\eta_R \int_{\mathbb{R}^3} (h_{k,R}^{\mathrm{pr}} - c_R)\tilde{v}_R \cdot \nabla\psi_k^R, \\ \dot{F}_{k,R}^{\mathrm{dr}} &:= -\frac{\gamma_k}{R}\eta_R \int_{\mathbb{R}^3} \frac{|\tilde{v}_R|^2}{2} a_R \cdot \nabla\psi_k^R. \end{aligned}} \tag{S.346}$$

令 $t(\sigma) = s_R + R^2\sigma$，并以 $\widehat{h}_{k,R}^{\alpha}$ 表示四个 vector integrands 的非负 dimensionless majorants。则

$$
\boxed{h_{k,R}(\sigma) = R^2|\dot F_{k,R}(t(\sigma))| \le \sum_\alpha \widehat h^\alpha_{k,R}(\sigma).} \tag{S.347}
$$

这里的 prefactor 是 $\gamma_k R$，只有使用 $R|\nabla\psi_k^R| \le C$ 后才成为 $C\gamma_k$。Calderon--Zygmund、Young、fixed-gauge pressure majorant、local cubic 与 Jensen--Young drift estimate 合起来给出

$$
\boxed{\sum_{k\ge1}\sum_\alpha \|\widehat h^\alpha_{k,R}\|_{L^1(0,4)} \le CP_R^M.} \tag{S.348}
$$

S.348 是 **PROVED / INHERITED** 的 L^1 statement；它既不提供统一的 $p>1$ 时间可积性，也不提供 P_R^M 的 sublinear power。另有一个只对固定 solution 与固定 scale 成立的 tail。令

$$
T_K(R) := \sum_{k>K} \gamma_k(1+2^{3k}R^3), \qquad \mathfrak{T}^F_{4/3,K,R} := \sum_{k>K} \|h_{k,R}\|_{L^{4/3}(0,4)}.
$$

则

$$
\boxed{\mathfrak{H}^F_{4/3,K,R} \le \mathfrak{T}^F_{4/3,K,R} \le CT_K(R)\left([e_R(e_R+d_R)]^{3/4} + e_R^{3/2}\right).} \tag{S.349}
$$

对固定 $R_*>0$，$\sup_{0<R\le R_*} T_K(R) \to 0$，但 energy bracket 没有可用的统一 bound。因此 S.349 不是 uniform-in-R payment theorem。

85. Outer face 与 payment 权重完全对齐

Doubled-radius exterior payment 的 annuli 为 $A_j(2R) = \{2^{j+1}R \le |y| < 2^{j+2}R\}$。Collars 满足

$$
\boxed{C^+_{k,R} \subset A_k(2R) \quad (k\ge1), \qquad C^-_{k,R} \subset A_{k-2}(2R) \quad (k\ge3).} \tag{S.350}
$$

前两个 inner collars 在 B_{8R} core 内。对 $k\ge3$，inner face 的 target-to-payment ratio 为

$$
\boxed{\frac{\gamma_k}{\gamma_{k-2}} = \exp\left(-\frac{15\,4^{k-3}}{32}\right) \longrightarrow 0.} \tag{S.351}
$$

Outer face 则没有类似 gain：

$$
\boxed{\frac{\text{target coefficient on } C^+_{k,R}}{\text{payment coefficient on } A_k(2R)} = \frac{\gamma_k}{\gamma_k} = 1.} \tag{S.352}
$$

S.350--S.352 是对“先在每个 outer collar 取绝对值、再与 nonnegative exterior payment 比较”这一具体方法的 **PROVED GEOMETRIC OBSTRUCTION**；它不声称 signed outer flux 实际很大。删除任意固定数量的 inner shells 后，仍有无穷多个 aligned outer faces。

固定 $p \in (1,\infty]$、整数 $N, K_0 \ge 0$ 与 $C_*, P > 0$。令 $M = N+1$，任取不同的 $k_1,\dots,k_M > K_0$，并令 ϕ_d 是支撑在一个 interior terminal 附近、积分为一的 smooth bump。设

$$\boxed{w_i = \alpha_i = \gamma_{k_i}, \qquad g_i(\sigma) = \frac{P}{Mw_i}\phi_d(\sigma), \qquad H_i(\sigma) := \alpha_i g_i(\sigma).} \tag{S.353}$$

Aligned weighted payment 精确等于 P。删除 $N = M-1$ 个 coordinates 后，仍留下一个相同 target coordinate：

$$\boxed{\inf_{\#S \le N} \sum_{i \notin S} \|H_i\|_{L^p} = \frac{P}{N+1}\|\phi\|_{L^p} d^{1/p-1}.} \tag{S.354}$$

当 $d \downarrow 0$ 时右边发散，因此可令

$$\boxed{\inf_{\#S \le N} \sum_{i \notin S} \|H_i\|_{L^p} > C_* P^{2/3}.} \tag{S.355}$$

这些 g_i 是 smooth nonnegative scalar rates，不是同一个 velocity/pressure 产生的 flux。S.355 是 **ABSTRACT METHOD OBSTRUCTION**，不反驳 PDE target S.342；它只证明不能从 S.348、outer-shell deletion 与 super-Gaussian coefficients 单独推出 S.342。

86. 精确 coefficient-cube interface

令 $E_\tau \subset \mathbb{N}$、$\#E_\tau \le N_b$，并令 $\mathscr{I}_\tau$ 是 (ν,k) 的 countable incidence multiset，其中 $k \notin E_\tau$。假设

$$\boxed{b_k \le q_k + \sum_{\nu:(\nu,k)\in\mathscr{I}_\tau} a_{\nu k}, \qquad k \notin E_\tau.} \tag{S.356}$$

为每次 node occurrence 配置 $p_\nu \ge 0$，重复 incidence 重复计数，并约定 $a_{\nu k}^3/p_\nu^2 = 0$ when both vanish、$+\infty$ when $p_\nu = 0 < a_{\nu k}$。若

$$\boxed{\begin{aligned} \sum_{k\notin E_\tau} q_k &\le C_q A_R, \\ \sum_{(\nu,k)\in\mathscr{I}_\tau} p_\nu &\le C_p P_R^M, \\ \sum_{(\nu,k)\in\mathscr{I}_\tau} \frac{a_{\nu k}^3}{p_\nu^2} &\le C_{\mathrm{cor}}, \end{aligned}} \tag{S.357}$$

则 incidence multiset 上的 Holder inequality 给出

$$\boxed{\mathcal{S}_{N_b}(b(\tau)) \le \left(C_q + C_{\mathrm{cor}}^{1/3} C_p^{2/3}\right) A_R.} \tag{S.358}$$

这是 **PROVED / CONDITIONAL** implication：代数已证，S.356--S.357 的 PDE construction 未证。令 $a_{\nu k} = c_{\nu k} p_\nu^{2/3}$，最后一项正是 cubic coefficient sum。其 exponent 精确，因为

$$\boxed{\sup_{p_i \ge 0,\ \sum_i p_i \le 1} \sum_i c_i p_i^{2/3} = \left(\sum_i c_i^3\right)^{1/3}.} \tag{S.359}$$

若同一 node incident to many shells，就不能只计算 distinct-node coefficient sum；必须统一控制 incidence multiplicity，或直接证明 S.357 中 repeated cubic sum 与 repeated payment。

87. Scale-invariant parabolic measure 与 density roots

令 $\widetilde{\boldsymbol{\mu}}$ 是 total local-dissipation measure 的 periodic lift，$\widetilde{X}_R$ 是 lifted mollified path。在 dimensionless comoving coordinates 定义

$$\boxed{\begin{aligned}\Phi_R(\sigma, z) &:= \left(s_R + R^2\sigma, \widetilde{X}_R(s_R + R^2\sigma) + Rz\right),\\ \nu_R(A) &:= R^{-1}\widetilde{\boldsymbol{\mu}}(\Phi_R(A)).\end{aligned}} \tag{S.360}$$

R^{-1} 由 Navier--Stokes scaling 强制决定，因为 $|\nabla u|^2 dxdt$ 的 length dimension 为一。Parabolic dyadic cell 的 radius 减半时，有八个 spatial children 与四个 temporal children：

$$\boxed{\#\,\mathrm{child}(Q) = 32, \qquad \rho_{Q'} = \frac{1}{2}\rho_Q \quad (Q' \in \mathrm{child}(Q)).} \tag{S.361}$$

采用 half-open convention 以保留 boundary atoms 的精确加法性。令 $m_Q = \nu_R(Q)$、$\Theta(Q) = m_Q/\rho_Q$、$\mathfrak{M}_R = \nu_R(Q_0)$。First λ-roots 是 density 第一次穿过 λ 的 maximal cells；它们构成 antichain，并满足

$$\boxed{\lambda\rho_Q < m_Q \le 2\lambda\rho_Q, \qquad \sum_{Q \in \mathscr{R}_\lambda} \rho_Q \le \frac{\mathfrak{M}_R}{\lambda}.} \tag{S.362}$$

Root mass 具有精确 critical factorization：

$$\boxed{a_Q := m_Q, \qquad c_Q := \rho_Q^{1/3}, \qquad p_Q := m_Q^{3/2}\rho_Q^{-1/2}, \qquad a_Q = c_Q p_Q^{2/3}.} \tag{S.363}$$

于是

$$\boxed{\sum_{\mathscr{R}_\lambda} c_Q^3 \le \frac{\mathfrak{M}_R}{\lambda}, \qquad \sum_{\mathscr{R}_\lambda} p_Q \le (2\lambda)^{1/2}\mathfrak{M}_R.} \tag{S.364}$$

但 level parameter 在 cubic Holder product 中精确抵消：

$$\boxed{\left(\frac{\mathfrak{M}_R}{\lambda}\right)^{1/3}\left((2\lambda)^{1/2}\mathfrak{M}_R\right)^{2/3} = 2^{1/3}\mathfrak{M}_R.} \tag{S.365}$$

S.365 是 **PROVED THRESHOLD NO-GAIN**：density pigeonholing 加上 critical cubic duality 只返回 total measure mass。若 $\mathfrak{M}_R$ 的现有 estimate 仅线性依赖 payment，优化 λ 仍然是线性的。

88. Density jumps 稀疏，但 low-transition corona 未支付

固定 $\kappa > 1$ 与 $m_S > 0$ 的 tree node S。令 $\mathscr{J}_\kappa(S)$ 为每条 branch 中第一次满足 $\Theta(Q) > \kappa\Theta(S)$ 的 proper descendants。First-jump cells 互不相交，因此

$$\boxed{\sum_{Q \in \mathscr{J}_\kappa(S)} \rho_Q \le \frac{\rho_S}{\kappa}.} \tag{S.366}$$

更一般地，若 $c_Q^3 = \rho_Q^\alpha$、$\alpha \ge 1$，则

$$\boxed{\sum_{Q \in \mathscr{J}_\kappa(S)} c_Q^3 \le \theta_\alpha c_S^3, \qquad \theta_\alpha := \frac{2^{1-\alpha}}{\kappa} < 1.} \tag{S.367}$$

只沿 first-jump descendants 迭代得到 uniform Dini sum：

$$\boxed{\sum_{n \ge 0} \theta_\alpha^n = \frac{1}{1-\theta_\alpha}.} \tag{S.368}$$

S.366--S.368 是 **PROVED** measure-tree facts，但只控制 jump skeleton。Jumps 之间的 low-transition corona 只有 $\Theta(Q) \le \kappa\Theta(S)$，没有继承的 local-energy inequality 把它的全部 shell contributions 放入 S.357 的 quadratic q-row。逐层 strict factor 也不够：

$$\boxed{\theta_d = \frac{d+1}{d+2} < 1, \qquad \prod_{j=0}^{n-1}\theta_{d_0+j} = \frac{d_0+1}{d_0+n+1}, \qquad \sum_{n \ge 0}\prod_{j=0}^{n-1}\theta_{d_0+j} = \infty.} \tag{S.369}$$

所需性质是 uniform Dini summability，而不是 pointwise strictness。Step 13 的 critical corona model 在 32-child tree 中保留一个 temporal child 与全部八个 spatial children；深度 d 的 retained nodes 取 $\rho_v = 2^{-d}\rho_0$、$m_v = 8^{-d}m_0$，density 沿 branch 严格下降，却有

$$\boxed{\sum_{Q\in\mathrm{child}_{\mathrm{spatial}}(S)} c_Q^3 = 8\left(\frac{c_S}{2}\right)^3 = c_S^3.} \tag{S.370}$$

这里的 c 是 Step 13 incidence coefficient，不是 S.363 的 root-factor $\rho^{1/3}$。该例是 **ABSTRACT METHOD OBSTRUCTION**；它没有把整棵树实现为同一个 Navier--Stokes solution 的 clocks。

89. Shell incidence 与缺失的 analytic charge

在 comoving coordinates 中，展开每个 nonnegative periodized integral 后，collar family 静止。Physical spatial diameter 不超过 $2R$ 的 unperiodized lifted cell 至多遇到两个 shell indices：

$$\boxed{\#\{k : R\,\mathrm{pr}_z\,Q \cap \mathrm{supp}\,\psi_k^R \neq \varnothing\} \leq 2.} \tag{S.371}$$

Shell k 与 $k+2$ 的 supports 径向至少相隔 $2\rho_k - R/4 \geq 15R/4$，所以可能的 double incidence 只在相邻 padded shells 的同一 hard boundary。S.371 只适用于展开后的单个 Euclidean support，不是 torus cell 与全部 periodized copies 的 incidence bound。较大 cells 必须先切分到这一分辨率。

这是有利的 geometry，但 transported local-energy test 仍产生 Version-M drift；它的 absolute estimate 只在线性 payment S.348 中。Finite shell incidence 本身不能把 drift 或 low-transition corona 放进 quadratic q-budget；covering/top decomposition 重复使用同一 cell 时，payment 也必须按 S.357 重复计数。

90. Exact heat shear：raw tree 的窄 no-go

在 2π-periodic torus 上，取 $A > 0$、整数 $L \geq 1$ 与 $n = 2^L$：

$$\boxed{u^{(n)}(t,x) = Ae^{-n^2 t}\sin(nx_2)e_1, \qquad p^{(n)} = 0, \qquad n = 2^L.} \tag{S.372}$$

在 standard dyadic spatial grid 上，只要 generation 严格高于 wavelength，每个 child x_2-interval 都包含整数个 $\cos^2(nx_2)$ periods；density 在 x_1, x_3 方向常数，因此

$$\boxed{\int_J\int_{Q'} |\nabla u^{(2^L)}|^2 = \frac{1}{8}\int_J\int_Q |\nabla u^{(2^L)}|^2, \qquad Q' \in \mathrm{child}_{\mathrm{spatial}}(Q), \quad d(Q) < L.} \tag{S.373}$$

然而 moving velocity 与 y_1 无关，path velocity 平行于 e_1，在 y_1 上周期积分得到

$$\boxed{\dot F_{k,R}^{(2^L)}(t) = 0 \quad \text{for every } k, R, t.} \tag{S.374}$$

因此 deep critical dissipation tree 不必产生任何 physical shell-flux tail。该 exact family 不反驳 S.342 或 S.375，也没有实现 S.370 的 abstract ancestor failure；它的 completed clocks 已由 quadratic local-energy channel 支付。

91. Shell-selective jump--corona lemma

候选 PDE construction 必须先把每个 nonnegative collar row 展开到 Euclidean lift，再使用一个来自 fixed finite family of shifted grids 的 countable locally finite comoving parabolic forest 覆盖全部 lifted shell supports。每个 top cell T 可选择 $\lambda_T > 0$，取 first crossing roots，再迭代 first relative jumps。记 $\nu \rightsquigarrow k$ 仅表示与一个 unperiodized lifted support 的 incidence。

下述陈述对 bare periodic suitable-weak class **OPEN**。需要 universal $\kappa > 1$、universal integer N_b、常数 $C_q, C_p, C_{\mathrm{cor}}$，以及一个 common shell set E_τ、$\#E_\tau \le N_b$，使每个 solution、scale 与 good terminal 都能构造 nonnegative top、corona、jump 与 payment rows 满足

$$\boxed{\begin{gathered} b_k(\tau) \le q_k^{\mathrm{top}} + q_k^{\mathrm{cor}} + \sum_{\nu:\nu\rightsquigarrow k} a_{\nu k} \quad (k \notin E_\tau), \\ \sum_{k\notin E_\tau} (q_k^{\mathrm{top}} + q_k^{\mathrm{cor}}) \le C_q A_R, \\ \sum_{\substack{(\nu,k):\nu\rightsquigarrow k \\ k\notin E_\tau}} p_\nu \le C_p P_R^M, \\ \sum_{\substack{(\nu,k):\nu\rightsquigarrow k \\ k\notin E_\tau}} \frac{a_{\nu k}^3}{p_\nu^2} \le C_{\mathrm{cor}}. \end{gathered}} \tag{S.375}$$

同一个 E_τ 必须同时用于 defect 与 high-Rayleigh ancestors；它不能在 tree levels 或 payment channels 之间移动。所有 constants 与 κ 都独立于 solution、R、τ、λ_T、top count 与 forest depth。Payment sum 遍历完整 incidence multiset，包括 periodic copies、forest overlaps 与 repeated shell uses。Top row 包含 first crossing 之前的 cells 与全部 top-boundary contributions；corona row 包含 moving-frame drift 与 jump skeleton 没有到达的所有 nodes。

若 S.375 成立，把 $q_k = q_k^{\mathrm{top}} + q_k^{\mathrm{cor}}$ 代入 S.356--S.358，得到

$$\boxed{(\text{S.375}) \quad \Longrightarrow \quad \mathcal{S}_{N_b}(b(\tau)) \le \left(C_q + C_{\mathrm{cor}}^{1/3} C_p^{2/3}\right) A_R.} \tag{S.376}$$

S.376 因而 conditional 地关闭 ancestor gate S.288。已证的是 Holder arrow；未证的新数学内容是：在保留 shell incidence 与 payment additivity 的同时，以 quadratic scale 支付 top 与 low-transition corona 的 PDE estimate。

92. Step 14 路线决定

Short branch 不再继续使用“在 S.345 中逐通道取绝对值，再仅应用现有 nonnegative payment”的估计。Outer face 的 coefficient 完全对齐，S.353--S.355 的 smooth spike 饱和了这组信息。下一项可接受 input 必须是 outer collars 上的 signed local-energy cancellation，或在一个 common finite shell deletion 后仍 uniform 的 PDE time anti-concentration theorem。Target S.342 保持 **OPEN**。

Excess branch 的 positive algebra 已在 S.358 完成。下一任务是 S.375 的 PDE content，首先处理 low-transition corona 与 top-boundary row；只有在 repeated incidence payments 完整记录之后，才可使用 density thresholds 与 jump sparsity。Exact-family screen 必须 shell selective；高 Fourier frequency、deep raw dissipation tree 或大 Rayleigh ratio 都不够，因为 physical flux 可能为零，或 completed clock 已由 Q 支付。

本次路线决定不改变 frozen target 或其 quantifiers。

93. 有限的 primary-source boundary

Step 12--13 的 bounded primary-source screens 与本步两个接口逐项比较后，没有找到直接具备 S.342 common-deletion flux-tail 或 S.375 shell-selective jump--corona quantifiers 的 cited theorem。

Caffarelli--Kohn--Nirenberg 的 suitable-weak local energy inequality、epsilon regularity 与 singular-set parabolic size 不给出 S.375 的 repeated mass、incidence 或 low-transition-corona budgets；high-Rayleigh ancestor 也可能位于 regular set。J. Yang 的 mollified-flow trajectories、skewed cylinders 与 covering/maximal-function estimates 提供邻近几何，但不是 shell payment 或 cubic incidence charge。Koch--Tataru 的 BMO^{-1} critical small-data class 含 Carleson-type spacetime control，却不是从每个 bare suitable weak solution 的 P_R^M 自动推出。Lei--Ren 的 quantitative partial regularity 使用 scale selection 与 pigeonholing，但没有给出 common terminal window、固定 physical shell deletion 或 S.375 corona decomposition。Guevara--Phuc 的 pressure-sensitive local-energy 与 epsilon-regularity estimates 也不产生 aligned outer-collar L^p tail 或 payment-additive corona charge。

这只是 collision boundary，不是 novelty 或 priority claim，也不是 exhaustive literature review。

94. Step 14 主张账本与有限证书

Step 14 在 frozen setting 中 **PROVED**：S.343--S.347 的 shell-scale pressure decomposition 与 four-channel signed flux identity；S.348--S.349 的 inherited componentwise L^1 payment 与 fixed-solution tail boundary；S.350--S.352 的 collar inclusions、inner-face gain 与 outer-face alignment；S.356--S.359 的 incidence Holder theorem 与 exact cubic duality，其中 conclusion 对显示的 budgets 条件成立；S.360--S.365 的 scale-invariant measure、32-child scaling、first-root bounds、critical factorization 与 threshold cancellation；S.366--S.368 的 first-jump sparsity 与 strict Dini coefficient；S.369 的非 uniform strict-factor failure；S.371 的 bounded fine-scale shell incidence；以及 S.372--S.374 的 heat-shear mass split 与 zero-flux identities。

Step 14 的 **ABSTRACT METHOD OBSTRUCTIONS, NOT NSE COUNTEREXAMPLES** 是 S.353--S.355 的 aligned smooth $(N+1)$-coordinate rates，以及继承自 Step 13 ledger model 的 S.370 critical eight-ary corona embedding。

Step 14 的 **CONDITIONAL** statements 是：S.358 依赖 S.356--S.357 的 exact incidence budgets；S.376 依赖 open shell-selective jump--corona lemma S.375。

继续 **OPEN**：common-deletion temporal estimate S.342，包括 uniform $p > 1$ outer-collar anti-concentration；PDE shell-selective jump--corona lemma S.375，尤其 top-boundary、low-transition-corona 与 moving-drift charge；ancestor gate S.288、combined gate S.303、Step 11 S.272、Q.12 与 Q.1；以及 scale contraction、regularity、singularity formation 与 Millennium problem。

冻结主文 SHA-256 为 `c843284d68c0d7d441214b0b3e67e97ca4c5ebda5f527a957eb6e9bdc07f55f9`。Python certificate 通过 12/12 exact、9/9 finite groups、74,287 finite rational cases、3/3 dependency、37/37 structural 与 49/49 negative mutations。Independent Ruby verifier 通过 7/7 groups、82,788 cases、6/6 artifact locks、2/2 dependency locks、68/68 note checks、1/1 primary-artifact group 与 2/2 negative groups。有限程序检查 exact rational algebra、fixtures、upstream hashes、equation numbering、selected formula bindings 与 claim wording；它们不 machine-prove analytic pressure decomposition、inherited PDE estimates、open packing gates、S.375、abstract fixture 的 NSE realization、regularity 或 Millennium problem。

本步的 advance 是严格的方法边界：super-Gaussian weights 不改善 aligned outer flux face，density threshold 不改善 critical cubic payment power，density jumps 还留下当前 PDE ledger 未控制的 corona；下一正向命题被精确写成 open lemma S.375。**NOT CLAY.**

95. Step 15：混合通量等价与终端 crown 强制性缺口

Step 14 把 short branch 的共同删除 flux tail S.342 与 excess branch 的 jump--corona input S.375 分成两条路线。Step 15 先证明一个此前未记录的精确归约：short last-exit residual 与 selected-excess residual 可以写进同一个非负 stopped physical-flux vector，并且在同一个 best-N 删除集合下与完整 residual 逐坐标等价。因而，若仍开放的 S.342 将来成立，同一个 N_F 会一次支付两条 residual 分支，不再需要另加 ancestor budget。

对独立的 crown 路线，本步把每个 terminal corona 压成只计一次的 disjoint crown，并证明其 cubic coefficient budget 与停止深度无关。这个组合结果没有提供缺失的 PDE coercivity：selected-crown nonlinear payment S.407 仍是 **OPEN PDE INPUT**。Periodic positive-measure tree 与 selected scalar clock 只分别通过几何和标量压力测试；它们没有被耦合成一个 completed-clock/measure fixture，更不是 Navier--Stokes 反例。

本步没有证明 S.342、S.375、S.407、Q.12、Q.1、scale contraction、regularity、singularity formation 或 Millennium problem，也不声称 novelty 或 priority。有限证书检查代数、组合、哈希、结构与边界措辞，不 machine-prove 开放的 PDE 输入。**NOT CLAY.**

96. Hybrid start 与同一物理通量向量

固定一个 Version-M suitable weak solution、尺度 R 与 good terminal τ。保留

$$T_k = K_{k,R}(\tau), \qquad \ell_k = \ell^K_{k,2/3}(\tau), \qquad \mathcal{I}_{\rm res}(\tau) = \mathcal{R}_{\rm sh}(\tau) \dot\cup \mathcal{I}_x(\tau).$$

选取 frozen initial interval 中共同的 local-energy good time σ_0，使所有 shell 都有 $K(\sigma_0) = Q(\sigma_0) = F(\sigma_0) = 0$。定义

$$\boxed{\sigma_k^{\rm hyb}(\tau) := \begin{cases} \ell_k(\tau), & k \in \mathcal{R}_{\rm sh}(\tau), \\ \sigma_0, & k \in \mathcal{I}_x(\tau), \\ \tau, & k \notin \mathcal{I}_{\rm res}(\tau), \end{cases} \qquad z^\lambda_{k,R}(\tau) := F_{k,R}(\tau) - F_{k,R}(\sigma_k^{\rm hyb}(\tau)).} \tag{S.377}$$

于是

$$\boxed{z_k = r_k^{\rm sh} = r_k \quad (k \in \mathcal{R}_{\rm sh}), \qquad z_k = F_{k,R}(\tau) = [F_{k,R}(\tau)]_+ \quad (k \in \mathcal{I}_x), \qquad z_k = r_k = 0 \quad (k \notin \mathcal{I}_{\rm res}).} \tag{S.378}$$

Selected excess 因此是该 physical-flux vector 的 subcoordinate：

$$\boxed{0 \le x_k^{\rm sel}(\tau) \le z_k(\tau) \qquad (k \ge 1).} \tag{S.379}$$

S.377--S.379 是表示与符号结论，不是对 z 的 PDE bound。

97. 同一个 Q-variation diamond 与 best-N 等价

对 $k \in \mathcal{I}_x(\tau)$，令

$$U_k = Q_{k,R}(\ell_k), \qquad V_k = Q_{k,R}(\tau) - Q_{k,R}(\ell_k).$$

两段 Q-variation 来自同一个 full-history variation measure，故不是两个可独立饱和的误差：

$$\boxed{|U_k| + |V_k| \le \mathrm{TV}_{J_\tau} Q_{k,R} < \frac{T_k}{6}.} \tag{S.380}$$

Terminal clock 与 last-exit identity 为

$$\boxed{z_k = T_k - U_k - V_k, \qquad r_k = \frac{T_k}{3} - V_k.} \tag{S.381}$$

在 S.380 的同一个菱形约束内直接优化，得到 sharp scalar constants

$$\boxed{\frac{1}{5} z_k < r_k < \frac{3}{7} z_k \qquad (k \in \mathcal{I}_x(\tau)).} \tag{S.382}$$

与 short branch 上的 equality 合并，得到

$$\boxed{\frac{1}{5} z_k(\tau) \le r_k(\tau) \le z_k(\tau), \qquad z(\tau) \in \ell^1_+.} \tag{S.383}$$

对同一个任意 shell set S、$\#S \le N$ 先求和再优化，得到

$$\boxed{\frac{1}{5} \mathcal{S}_N(z(\tau)) \le \mathcal{S}_N(r(\tau)) \le \mathcal{S}_N(z(\tau)).} \tag{S.384}$$

令 $\mathfrak{Z}^{\lambda}_{N,R}(\mathcal{D})$ 为 good terminals 上 $\mathcal{S}_N(z)$ 的 supremum，则

$$\boxed{\frac{1}{5} \mathfrak{Z}^{\lambda}_{N,R}(\mathcal{D}) \le \mathfrak{R}^{\lambda}_{N,R}(\mathcal{D}) \le \mathfrak{Z}^{\lambda}_{N,R}(\mathcal{D}).} \tag{S.385}$$

因此完整 hybrid-flux gate 与完整 combined-residual gate 在同一个 terminal-dependent deletion budget 下按常数 $1/5$ 与 1 等价。S.396 的 abstract scalar-ledger rows 分别把比值逼近 $1/5$ 与 $3/7$，证明 S.382 的常数在已用标量约束内 sharp；它们不是 NSE counterexamples。

98. 一个 common-deletion temporal tail 同时支付两条分支

保留 Step 13 的 dimensionless flux density $h_{k,R}$。Absolute continuity 与 S.377 给出

$$\boxed{0 \le z_k(\tau) \le \int_0^4 h_{k,R}(\sigma)\, d\sigma \le 4^{1-1/p} \|h_{k,R}\|_{L^p(0,4)}, \qquad 1 \le p \le \infty.} \tag{S.386}$$

在取 terminal 或 branch 之前，整个 time norm 只删除同一个 shell set，因此

$$\boxed{\mathcal{S}_N(r(\tau)) \le \mathcal{S}_N(z(\tau)) \le 4^{1-1/p} \mathfrak{H}^F_{p,N,R}.} \tag{S.387}$$

若把仍开放的 Step 13 estimate 明确作为 antecedent，

$$\boxed{\mathfrak{H}^F_{p,N_F,R} \le C_H A_R, \qquad A_R = (P^M_R)^{2/3},} \tag{S.388}$$

则同一个 N_F 给出

$$\boxed{\mathfrak{R}^{\boldsymbol{\lambda}}_{N_F,R}(\mathcal{T}_R) \le 4^{1-1/p} C_H A_R,} \tag{S.389}$$

$$\boxed{\mathcal{S}^K_{N_F,R}(\mathcal{T}_R) \le \left[C_{\mathrm{pay}}(\boldsymbol{\lambda}) + 6\,4^{1-1/p} C_H\right] A_R,} \tag{S.390}$$

并经 inherited terminal reduction 得到

$$\boxed{\mathfrak{C}^M_R \le C(\boldsymbol{\lambda}, p, N_F, C_H)\,[A_R + Y^{\mathrm{sf}}_{2,R}].} \tag{S.391}$$

S.388--S.391 证明的是 implication，不是 S.342 或 Q.1 本身。其新意义是 route-level：若 S.342 成立，它会同时关闭 short 与 selected-excess residual，不需要另加 N_b 或 ancestor coefficient。

99. Signed common window 的 start-clock debt

把 Step 14 的四通道 identity 积分在 hybrid active blocks 上，可写成

$$\boxed{\sum_{k\in G\cap\mathcal{I}_{\mathrm{res}}(\tau)} z_k(\tau) = \sum_{\alpha\in\{\mathrm{cub,loc,har,dr}\}} \sum_{k\in G\cap\mathcal{I}_{\mathrm{res}}(\tau)} \int_{\sigma_k^{\mathrm{hyb}}(\tau)}^{\tau} \dot{F}^{\alpha}_{k,R}(t)\,dt.} \tag{S.392}$$

这允许新的 PDE cancellation，但 algebraic regrouping 本身不产生 gain。对 common terminal window 起点 $a=\max\{s_R,\tau-\delta R^2\}$，若 shallow shell 满足 $a\le \ell_k$，则

$$\boxed{r_k^{\mathrm{sh}} = G_{k,\tau,\delta} + \left[K_{k,R}(a) - \frac{2T_k}{3}\right] + [Q_{k,R}(\ell_k) - Q_{k,R}(a)].} \tag{S.393}$$

定义 start-clock overshoot

$$\omega_{k,\tau,\delta} := \mathbf{1}_{\mathcal{R}^{\le\delta}_{\mathrm{sh}}(\tau)}(k)\left[K_{k,R}(a) - \frac{2T_k}{3}\right]_+.$$

同一个 deletion set 下的求和给出

$$\boxed{\sum_{k\in\mathcal{R}_{\rm sh}^{\le\delta}(\tau)\setminus S} r_k^{\rm sh} \le \left[\sum_{k\in\mathcal{R}_{\rm sh}^{\le\delta}(\tau)\setminus S} G_{k,\tau,\delta}\right]_+ + \sum_{k\notin S}\omega_{k,\tau,\delta} + C_Q A_R.} \tag{S.394}$$

于是 minimal signed common-window gate 为

$$\boxed{\begin{aligned}\mathcal{S}_N(r^{{\rm sh},\le\delta}(\tau)) \le C_Q A_R + \inf_{\#S\le N}\Bigg\{&\left[\sum_{k\in\mathcal{R}_{\rm sh}^{\le\delta}(\tau)\setminus S} G_{k,\tau,\delta}\right]_+ \\ &+ \sum_{k\notin S}\omega_{k,\tau,\delta}\Bigg\}.\end{aligned}} \tag{S.395}$$

Last-exit maximality只控制 $\ell_k < t \le \tau$，不能控制更早的共同起点 a。S.397 的 abstract clock 取 $K(a)=M$、$K(\ell)=2$、$K(\tau)=3$，得到

$$\boxed{r=1, \qquad G=3-M, \qquad \omega=M-2, \qquad r=G+\omega.} \tag{S.397}$$

所以 signed synchronization 将逐坐标 absolute increment 换成了两个同删集合任务：common-window cancellation 与 start-clock overshoot control。这个 debt 是精确代数事实，但见证仍只是 abstract clock check。

100. Shellwise ownership 与 terminal crowns

对 selected ancestor，定义 finite positive Borel submeasure $\alpha_{k,\tau}^{\rm anc}$。Frozen definition 与 cutoff domination 给出

$$\boxed{b_k(\tau)=\alpha_{k,\tau}^{\rm anc}(\mathbb{R}\times\mathbb{R}^3), \qquad 0\le d\alpha_{k,\tau}^{\rm anc} \le \gamma_k \mathbf{1}_{\widehat{\mathcal{U}}_{k,R}(\tau)}\, d\nu_R.} \tag{S.398}$$

对 countable locally finite half-open forest-top occurrences，分别按 shell 构造 Borel ownership：

$$\boxed{\widehat{\mathcal{U}}_{k,R}(\tau) = \dot{\bigcup_{T:(T,k)\in\mathscr{I}_{\rm top}}} \mathcal{O}_{Tk}, \qquad \mathcal{O}_{Tk}\subset T\cap\widehat{\mathcal{U}}_{k,R}(\tau).} \tag{S.399}$$

Forest overlap 因而不重复 ancestor mass；adjacent-shell incidence、shifted-grid occurrence、periodic copy 与 repeated top occurrence 仍各自保留。定义 incidence-weighted top content

$$\boxed{\mathscr{C}_{\rm top} := \sum_{(T,k)\in\mathscr{I}_{\rm top}} \gamma_k\rho_T < \infty.} \tag{S.400}$$

这里重复一个 top 会相应增加 $\mathscr{C}_{\rm top}$，不会被隐藏在几何常数中。

101. First roots、first jumps 与深度无关系数预算

取 canonical top level $\lambda_T = m_T/\rho_T$。First crossing roots 满足

$$\boxed{\sum_{S\in\mathscr{R}(T)} \rho_S \le \frac{m_T}{\lambda_T} = \rho_T.} \tag{S.401}$$

从每个 root 迭代 first proper κ-jump descendants，则

$$\boxed{\sum_{S\in\mathscr{J}_j(T)} \rho_S \le \kappa^{-j}\rho_T, \qquad \sum_{j\ge 0}\sum_{S\in\mathscr{J}_j(T)} \rho_S \le \frac{\kappa}{\kappa-1}\rho_T.} \tag{S.402}$$

在任意 finite stopping depth L，top crown、各非终端 jump crown 与 terminal-depth crowns 构成 exact half-open partition：

$$\boxed{T = \Omega_T \,\dot\cup \dot\bigcup_{0\le j\le L}\ \dot\bigcup_{S\in\mathscr{J}_j(T)} \Omega_S.} \tag{S.403}$$

每个 low-transition corona 只按一个 crown 计数，而不是在每一 dyadic generation 重复计数。Infinite-jump mass 保留在最后一个 finite-depth crown 中，不可在极限里丢弃。完整 crown--shell incidence multiset 因而满足

$$\boxed{\begin{aligned} \sum_{(\Omega_S,T,k)\in\mathscr{C}_L} \gamma_k\rho_S \le C_{\kappa,L}\mathscr{C}_{\mathrm{top}} \le C_\kappa \mathscr{C}_{\mathrm{top}}, \\ C_\kappa = 1+\frac{\kappa}{\kappa-1} = \frac{2\kappa-1}{\kappa-1}. \end{aligned}} \tag{S.404}$$

S.404 的常数与 depth、top count 和 top levels 无关。这关闭的是 cubic coefficient side，不是 payment estimate。

102. Open nonlinear crown payment 与 conditional closure

先选择一个 common shell exception set E_τ、$\#E_\tau \le N_b$，再对所有 tops、crowns、defect 与 high-Rayleigh channels 使用同一个集合。设 owned crown mass 分解为

$$\boxed{a_{Sk} = q_{Sk} + a_{Sk}^{\mathrm{pay}}, \qquad \sum_{\mathscr{C}_L(E_\tau)} q_{Sk} \le C_q A_R.} \tag{S.405}$$

对 paid part 定义 canonical crown payment

$$\boxed{p_{Sk}^{\mathrm{crown}} = \frac{(a_{Sk}^{\mathrm{pay}})^{3/2}}{(\gamma_k\rho_S)^{1/2}}, \qquad \frac{(a_{Sk}^{\mathrm{pay}})^{3}}{(p_{Sk}^{\mathrm{crown}})^2} = \gamma_k\rho_S \mathbf{1}_{\{a_{Sk}^{\mathrm{pay}}>0\}}.} \tag{S.406}$$

缺失的 PDE statement 是

$$\boxed{\sum_{\mathscr{C}_L(E_\tau)} p^{\mathrm{crown}}_{Sk} = \sum_{\mathscr{C}_L(E_\tau)} \frac{(a^{\mathrm{pay}}_{Sk})^{3/2}}{(\gamma_k\rho_S)^{1/2}} \le C_p P^M_R. \qquad \textbf{OPEN}} \tag{S.407}$$

S.407 要求 constant 对 solution、R、τ、forest、top count、levels 与 stopping depth 一致；同一份 frozen payment 若用于多个 occurrences，必须在 sum 中重复记录。它不是本步 local-energy inequality 的推论。

若 S.400、S.405、S.407 同时成立，则 Hölder 与 S.404--S.406 给出

$$\boxed{\mathcal{S}_{N_b}(b(\tau)) \le \left[C_q + (C_\kappa \mathscr{C}_{\mathrm{top}})^{1/3} C_p^{2/3}\right] A_R.} \tag{S.408}$$

Proposition 3.1 因而是 **PROVED CONDITIONAL**：top boundary 与所有 low-transition crowns 的 bookkeeping 已闭合，但 antecedent S.407 仍开放。

103. Converse Hölder 与 linear-payment 方法障碍

对非负 a_i, p_i，$A = \sum a_i$、$P = \sum p_i$，Hölder 的精确 converse 为

$$\boxed{\sum_i \frac{a_i^3}{p_i^2} \ge \frac{A^3}{P^2}, \qquad \inf_{p_i \ge 0,\, \sum p_i = P} \sum_i \frac{a_i^3}{p_i^2} = \frac{A^3}{P^2}.} \tag{S.409}$$

Equality 恰在 $p_i = (P/A)a_i$ 取得。固定任意 deletion budget N_b，取 $M = N_b + 1$ 个 distinct shell coordinates，并令

$$\boxed{b_{k_i} \ge H \quad (1 \le i \le M), \qquad P_H = C_M H, \qquad A_H = (C_M H)^{2/3}.} \tag{S.410}$$

允许 forest、grids、levels、depth 与 common exception set 在看到 data 后自适应选择，完整 repeated incidence multiset 仍满足：若 q-budget 是 $O(A_H)$、payment 是 $O(P_H)$，则对 large H

$$\boxed{\sum_i \frac{a_i^3}{p_i^2} \ge \frac{H}{8C_p^2 C_M^2}.} \tag{S.411}$$

等价地，若 cubic coefficient sum 被一个 fixed C_{cor} 控制，则

$$\boxed{\sum q_k \ge H - C_{\mathrm{cor}}^{1/3}(C_p C_M H)^{2/3}.} \tag{S.412}$$

所以只有 linear payment 时，normalized q-budget 或 cubic coefficient budget 至少一项必须发散。这个 conclusion 是对 formal nonnegative incidence data 的 **ABSTRACT METHOD OBSTRUCTION**，不是 S.407 的 PDE 反例。

104. 两个未耦合的压力测试

Periodic positive-measure tree 取

$$\boxed{d\nu_H(\sigma,z) := \sum_{n\in\mathbb{Z}^3}\sum_{i=1}^{M}\frac{H}{\gamma_{k_i}}\delta_{\sigma_*}(d\sigma)|Q_i^x|^{-1}\mathbf{1}_{Q_i^x+(2\pi/R)n}(z)\,dz.} \tag{S.413}$$

在 standard grid 中，每个 retained child 满足

$$\boxed{\rho_v = 2^{-d}\rho_0, \qquad m_v = 8^{-d}m_0, \qquad \Theta(v) = 4^{-d}\Theta(0).} \tag{S.414}$$

所以它没有 upward κ-jump，却保留任意深 low-transition coronas；所有 periodic copies 仍进入 finite constant C_M。独立地，把 Step 11 pure-defect scalar fixture 按 $s = 5H/3$ 缩放，得到

$$\boxed{T = \frac{5H}{3}, \quad b = m = H, \quad r^x = \frac{5H}{9}, \quad \sigma = \frac{959H}{7200} < \frac{T}{12}, \quad x = \frac{2641H}{3600} > \frac{T}{6}, \quad \beta = 0.} \tag{S.415}$$

取 $M = N_b + 1$ 个 scalar copies 后

$$\boxed{\frac{\mathcal{S}_{N_b}(b)}{A_H} \geq C_M^{-2/3}H^{1/3} \longrightarrow \infty.} \tag{S.416}$$

S.413--S.414 只测试 measure geometry；S.415--S.416 只测试 selected scalar-clock arithmetic。两者没有共享一个 completed-clock/measure identity，更没有共同 velocity-pressure Navier--Stokes realization。不得把它们的并列展示写成 coupled counterexample。

105. 路线决定、文献边界与证书

Step 15 把下一步可接受的输入压缩为两条独立路线。

1. 证明 S.342。由 S.387--S.391，这一个 common-deletion theorem 用同一个 N_F 关闭整个 combined residual，ancestor jump--corona route 随即不再需要。 2. 若保留 ancestor route，则 S.404 已给出所有 finite-depth terminal crowns 的 depth-independent coefficient budget；剩余的新 PDE burden 是 S.407 的 occurrence-level $3/2$-coercivity。它必须在同一个 shell deletion 后同时记录 top/stopping faces、pressure、moving drift、defect 与 infinite-jump remainder。

还有一条更直接但同样未证的接口：用 signed local-energy cancellation 证明 $\mathfrak{Z}^{\boldsymbol{\lambda}}_{N,R}(\mathcal{T}_R) \lesssim A_R$。由 S.385，它与 Step 10 residual gate 只差 literal factor 5。对 shallow short intervals，S.395 说明任何 signed common-window proof 还必须控制 start-clock overshoot。

Frozen note 的 bounded collision boundary 比较了 physical-space flux locality、skewed-cylinder maximal/covering estimates、reverse Holder under extra local kinetic control、anomalous-dissipation support results、quantitative partial regularity 与 Navier--Stokes inequality flexibility。它们都没有提供 S.342 或 S.407 的 frozen quantifiers。这只是有限 collision check，不是 exhaustive review，也不是 novelty 或 priority claim。

本步 **PROVED**：S.377--S.387 的 hybrid vector、sharp coordinate comparison 与 same-deletion best-N equivalence；S.388--S.391 的 conditional implication；S.392--S.395 的 signed-window debt identity；S.396--S.397 的 abstract sharpness checks；S.398--S.404 的 ancestor submeasure、ownership、terminal crowns 与 depth-independent coefficient content；S.405--S.408 的 exact factorization 与 conditional Hölder closure；S.409--S.412 的 converse-Hölder method obstruction；S.413--S.416 的两个 separate stress tests。

本步 **OPEN PDE INPUTS**：common-deletion temporal flux tail S.342；selected-crown nonlinear payment S.407；以及 inherited S.375。继续 **OPEN**：S.288、S.303、S.272、Q.12、Q.1、scale contraction、regularity、singularity formation 与 Navier--Stokes Millennium problem。

Frozen primary certificate 通过 9/9 finite groups、3,941 finite cases、5/5 dependency、45/45 structural 与 20/20 negative checks。Independent Ruby verifier 从头重建 vectors、deletion sets、trees、crowns 与 incidence rows，并锁定两篇主文、两份实现、certificate 与 report 的哈希；它检查 finite algebra 与 combinatorics，不证明 S.342、S.407 或任何 PDE realization。Primary analytic audit 与 independent audit 均在这个边界内 PASS。**FINITE ONLY. NOT CLAY.**

106. Step 16：光滑精确解否定所有 $p > 1$ 的二次时间尾估计

Step 15 说明，只要一个 common-deletion temporal flux tail 成立，同一个删除集就能支付 combined terminal residual 的两条分支。Step 16 检查 Step 13 候选 S.342 额外要求的 $p > 1$ 时间可积性。检验对象不是数值近似，也不是奇异解，而是 Taylor 1923 bi-periodic decaying vortex 的平移与振幅族。

结论是严格的：对每个 $p > 1$、每个有限删除预算 N 和每个拟议常数 C，都能选择 admissible R, z_0 与足够大的振幅 A，使 S.342 失败。因此 S.342 必须标记为 **FALSE**，不能继续列为 open candidate。

这个否定结果只针对 supercritical temporal-tail statement S.342。它不否定 direct hybrid terminal-flux gate，不否定临界 $p = 1$ 候选 S.444，也不证明或否定 S.407、S.375、Q.12、Q.1、scale contraction、regularity、singularity formation 或 Navier--Stokes Millennium problem。这里没有 singular solution。**NOT CLAY.**

107. Taylor 1923 二周期衰减涡是三维周期类中的光滑精确解

在 $\mathbb{T}^3 = (-\pi, \pi]^3$ 上定义

$$\boxed{W(x) = \big(\sin x_1 \cos x_2, -\cos x_1 \sin x_2, 0\big), \qquad p_W(x) = \frac{\cos 2x_1 + \cos 2x_2}{4}.} \tag{S.417}$$

直接微分得到

$$\boxed{\nabla \cdot W = 0, \qquad \Delta W = -2W, \qquad (W \cdot \nabla) W = -\nabla p_W.} \tag{S.418}$$

固定 $t_0 > 0$，令

$$b_A(t) := A e^{-2(t-t_0)}, \qquad u_A(t,x) := b_A(t) W(x), \qquad p_A(t,x) := b_A(t)^2 p_W(x). \tag{S.419}$$

于是

$$\partial_t u_A - \Delta u_A + (u_A \cdot \nabla) u_A + \nabla p_A = 0. \tag{S.420}$$

对每个 $A>0$，这都是光滑、精确、周期、mean-zero、无外力的三维不可压 NSE 解。它独立于 x_3，所以是嵌入三维解类的二维光滑 screen。这里能自由改变振幅，是因为这个特定 W 同时是 steady Euler field 与 Laplace eigenfield；这不是一般 NSE 解的振幅对称性。

108. 固定坐标 Bernoulli 通量精确抵消，Version-M 只留下移动漂移

冻结的 even radial mollifier 对 W 的所有 $\sqrt{2}$-频率模式给出同一个实乘子

$$\varphi_R^{\mathrm{per}} * W = \mu_R W, \qquad \mu_R := \int_{\mathbb{R}^3} \varphi(z) \cos\big(R(1,1,0)\cdot z\big)\, dz. \tag{S.421}$$

当 $R \downarrow 0$ 时 $\mu_R \to 1$。取 R 足够小，可令 $1/2 \le \mu_R \le 1$。以 $x_* = (\pi/4, 0, 0)$ 为终点，Version-M 轨迹和移动场满足

$$\boxed{\dot\xi = \mu_R b_A W(\xi), \quad v_R(t,y) = b_A(t) W(y+\xi(t)), \quad \pi_R(t,y) = b_A(t)^2 p_W(y+\xi(t)).} \tag{S.422}$$

唯一性给出 $\xi_2 = \xi_3 = 0$，并且

$$\dot\xi_1 = \mu_R b_A \sin \xi_1, \qquad \boxed{\tan\frac{\xi_1(t)}{2} = \tan\frac{\pi}{8} \exp\left(-\mu_R \int_t^{t_0} b_A(s)\, ds\right).} \tag{S.423}$$

对 shell cutoff Ψ_k^R，记

$$m_{k,R} := \int_{\mathbb{R}^3} \psi_k^R(y)\, dy, \qquad J_{k,R}(\xi) := \int_{\mathbb{T}^3} \Psi_k^R(y) |W(y+\xi)|^2\, dy. \tag{S.424}$$

Bernoulli 函数 $B_W = |W|^2/2 + p_W$ 满足 $\nabla \cdot (B_W W) = 0$。因此固定坐标中的 kinetic 与 physical-pressure shell flux 经过周期分部积分后精确抵消；time-dependent pressure gauge 也因不可压而积分为零。Version-M 的完整公式只剩 moving-cutoff drift：

$$\boxed{\dot F_{k,R}(t) = \frac{\gamma_k \mu_R \eta_R(t) b_A(t)^3}{2R} W(\xi(t)) \cdot \nabla_\xi J_{k,R}(\xi(t)).} \tag{S.425}$$

这是本步的关键区分。固定坐标 Bernoulli 通量为零，不代表冻结的移动观测量没有通量；沿非恒定局部速度移动的 cutoff 产生可计算漂移，而且这项不能从 Version-M flux 中删除。

109. 一个径向 Fourier 乘子同时激活任意 $N+1$ 个物理 annuli

Taylor 平面场满足

$$|W(x)|^2 = \frac{1 - \cos 2x_1 \cos 2x_2}{2}. \tag{S.426}$$

令 $q_+ = (2,2,0)$，并定义

$$c_{k,R} := \int_{\mathbb{R}^3} \psi_k^R(y)\cos(q_+\cdot y)\,dy. \tag{S.427}$$

径向对称和积化和差公式给出

$$\boxed{J_{k,R}(\xi) = \frac{m_{k,R}}{2} + c_{k,R}\left(|W(\xi)|^2 - \frac{1}{2}\right), \qquad \nabla J_{k,R} = c_{k,R}\nabla|W|^2.} \tag{S.428}$$

先固定任意有限删除预算 $N \geq 0$，令 $M = N+1$。再选择 R 使

$$\boxed{0 < R < \min\left\{\frac{\pi}{16}, \frac{\pi}{6\sqrt{2}(2^{M+1}+1/8)}\right\}, \qquad \mu_R \geq \frac{1}{2}, \qquad \bar{I}_{8R} \Subset (0,T).} \tag{S.429}$$

这样，对前 M 个物理 shell cutoff 的整个支撑都有 $|q_+\cdot y| \leq \pi/3$，从而

$$\boxed{c_{k,R} \geq \frac{1}{2}m_{k,R} > 0, \qquad 1 \leq k \leq M.} \tag{S.430}$$

这些是移动空间坐标中的 $N+1$ 个不同物理 annuli，不是 Fourier-shell index。沿 $\xi_2 = 0$ 有

$$W(\xi)\cdot\nabla|W(\xi)|^2 = \sin\xi_1 \sin 2\xi_1. \tag{S.431}$$

选一个固定小 $\delta > 0$。当 A 足够大时，终端物理时间块 $[t_0 - \delta/A, t_0)$ 位于 I_R，且 $\pi/8 \leq \xi_1(t) \leq \pi/4$。于是前 $N+1$ 个 shell 同时满足

$$\boxed{|\dot{F}_{k,R}(t)| = \dot{F}_{k,R}(t) \geq \frac{\gamma_k\mu_R c_{k,R} g_0}{2R}A^3, \quad 1 \leq k \leq M, \quad t_0 - \delta/A \leq t < t_0,} \tag{S.432}$$

其中 $g_0 = \sin(\pi/8)\sin(\pi/4) > 0$。

110. $p > 1$ 的 common-deletion temporal tail 逐量词失败

沿用 Step 13 的无量纲 density

$$h_{k,R}(\sigma) = R^2|\dot{F}_{k,R}(s_R + R^2\sigma)|, \qquad 0 < \sigma < 4.$$

S.432 的无量纲时间长度是 $\delta/(AR^2)$。因此，对 $1 < p < \infty$，

$$\boxed{\|h_{k,R}\|_{L^p(0,4)} \geq \frac{\gamma_k \mu_R c_{k,R} g_0}{2} \delta^{1/p} R^{\,1-2/p} A^{\,3-1/p}, \qquad 1 \leq k \leq M.} \tag{S.433}$$

对 $p=\infty$，

$$\boxed{\|h_{k,R}\|_{L^\infty(0,4)} \geq \frac{\gamma_k \mu_R c_{k,R} g_0 R}{2} A^3.} \tag{S.434}$$

删除至多 $N=M-1$ 个 index，前 M 个正坐标中至少留下一个。所以

$$\boxed{\mathfrak{H}^F_{p,N,R} \geq c_{p,N,R} A^{\,3-1/p}, \qquad p \in (1,\infty],} \tag{S.435}$$

其中 $c_{p,N,R}>0$ 与 A 无关。另一方面，完整 payment 的每个非负 row 都必须纳入比较。固定 R 后，平移不改变 pointwise amplitude；local energy、exterior velocity/pressure、fixed gauge 与 algebraic harmonic row 分别至多是 $O_R(A^2)$ 或 $O_R(A^3)$。因此

$$\boxed{P^M_R \leq C_R A^3.} \tag{S.436}$$

这里 exterior $\mathcal{G}$ 的 all-copy sum 由 super-Gaussian shell weights 收敛；harmonic $\mathcal{H}$ row 则使用冻结的 order-—4 algebraic kernel。两种收敛机制不能混写。

结合 S.435--S.436，

$$\frac{\mathfrak{H}^F_{p,N,R}}{(P^M_R)^{2/3}} \geq c'_{p,N,R} A^{\,1-1/p} \longrightarrow \infty \qquad (A \to \infty). \tag{S.437}$$

因此 S.342 的精确量词否定是

$$\boxed{\begin{array}{c}\text{对每个 } p \in (1,\infty],\ N \in \mathbb{N}_0,\ C>0, \\ \text{都存在 admissible } R, z_0 \text{ 和一个光滑、周期、无外力解，使得} \\ \mathfrak{H}^F_{p,N,R} > C(P^M_R)^{2/3}.\end{array}} \tag{S.438}$$

顺序不能交换：先由对手给定 p,N,C，再取 $M=N+1$，选择一个 admissible R，最后令 $A\to\infty$。R 可以依赖 N，因为 S.342 要求同一个有限 N 对所有 admissible scales 一致成立。

111. 更一般的付款幂与时间反集中指数也受到限制

同一族还给出更尖锐的指数边界。若某个固定 $p\in[1,\infty]$ 存在

$$\mathfrak{H}^F_{p,N,R} \leq C(P^M_R)^{\beta},$$

则必须有

$$\boxed{\beta \geq 1 - \frac{1}{3p}.} \tag{S.438a}$$

这里 $1/\infty = 0$。所以 $2/3$ 只在 $p = 1$ 与振幅标度相容；每个 $p > 1$ 都要求严格更大的 payment power，除非加入额外因子。

更一般地，在同一个无量纲终端块上，若

$$\int_I h_{k,R} \leq C(P_R^M)^\beta |I|^\alpha,$$

则 $|I| \asymp_R A^{-1}$、$\int_I h_{k,R} \asymp_R A^2$ 强制

$$\boxed{3\beta - \alpha \geq 2.} \tag{S.438b}$$

特别地，在 bare class 中，二次 payment power $\beta = 2/3$ 不允许任何正的 time anti-concentration exponent $\alpha > 0$。

112. 临界 $p = 1$ 只达到振幅饱和，新的 S.444 仍然 OPEN

对所有 k，S.428 仍成立且 $|c_{k,R}| \leq m_{k,R}$。沿特征线使用

$$d\xi_1 = \mu_R b_A \sin \xi_1 \, dt$$

作变量替换，μ_R 精确抵消，同时少掉一个振幅因子。于是

$$\int_{s_R}^{t_0} |\dot{F}_{k,R}(t)| \, dt \leq C_R \gamma_k m_{k,R} A^2. \tag{S.439}$$

由 $m_{k,R} \leq C 2^{3k} R^3$ 与 $\sum_k 2^{3k} \gamma_k < \infty$，

$$\mathfrak{H}_{1,N,R}^F \leq C_R A^2. \tag{S.440}$$

把 S.432 在长度 δ/A 的终端块上积分，并再次使用 $N+1$ 个 shell 的 pigeonhole，得到反向振幅下界

$$\mathfrak{H}_{1,N,R}^F \geq c_{N,R} A^2. \tag{S.441}$$

取趋向 t_0 的 local-energy good times，buffered local energy 的 endpoint 项还给出

$$P_R^M \geq c_R A^3. \tag{S.442}$$

所以在固定 (N, R) 下，

$$\boxed{\mathfrak{H}^F_{1,N,R} \asymp_{N,R} A^2, \qquad P_R^M \asymp_R A^3, \qquad \mathfrak{H}^F_{1,N,R} \asymp_{N,R} (P_R^M)^{2/3}.} \tag{S.443}$$

这只是振幅 exponent 的饱和。它既没有给出 universal N_1, C，也没有证明下面的新候选：

$$\boxed{\begin{gathered} \exists N_1 \in \mathbb{N}_0,\ C > 0 \text{ universal，使得} \\ \forall \text{ admissible Version-M solutions, } R, z_0 \text{ and terminal settings,} \\ \mathfrak{H}^F_{1,N_1,R} \leq C(P_R^M)^{2/3}. \end{gathered}} \tag{S.444}$$

S.444 仍是 **OPEN**。Step 15 的 S.386--S.387 包含 $p = 1$；若将 S.444 作为 antecedent，S.389--S.391 的同一推理仍能支付完整 hybrid residual。Step 13 的 $p > 1$ 时间窗增益不得继续使用。

113. ABC 精确族提供独立代数 screen，但不是第二个所需定理

等参数 ABC field

$$U = (\sin x_3 + \cos x_2,\ \sin x_1 + \cos x_3,\ \sin x_2 + \cos x_1)$$

满足 $\nabla \times U = U$ 与 $\Delta U = -U$。因此 $u = Ae^{-(t-t_0)}U$，连同 mean-zero pressure $-A^2 e^{-2(t-t_0)}(|U|^2 - 3)/2$，也是一个光滑周期精确解。在 $\xi_* = 0$ 有

$$U(0)\cdot\nabla|U|^2(0) = 6.$$

速度模式的频率长度为 1，$|U|^2$ 的非恒定模式长度为 $\sqrt{2}$。径向乘子、small-R shell positivity 与 $O(A^{-1})$ trajectory residence 再次产生 $A^{3-1/p}$ obstruction。

这只是对机制的独立 exact-family screen。Taylor 族的主证明不依赖它，也不需要把它提升为第二个 theorem。ABC 的历史语境见 Dombre et al., *Chaotic streamlines in the ABC flows* (1986)。

114. 原始来源只支持经典背景，不承担本项目的否定结论

S.417 的精确场采用 Taylor 1923 的 bi-periodic decaying vortex。现代数值文献有时把它称为 two-dimensional Taylor--Green vortex，但这里不把它与 Taylor--Green 1937 的 fully three-dimensional datum 混为一谈。

历史来源是 G. I. Taylor, *On the decay of vortices in a viscous fluid* (1923)。Taylor--Green 1937 只作为历史语境，见 *Mechanism of the production of small eddies from large ones*。现代 exact-flow 背景包括 Chai--Wu--Fang 的 *Single-scale two-dimensional-three-component generalized-Beltrami-flow solutions of incompressible Navier--Stokes equations* (2020) 与 Antuono 的 *Tri-periodic fully three-dimensional analytic solutions for the Navier--Stokes equations* (2020)。这些 classical fields、generalized-Beltrami mechanism 与 exponential decay 都不是 novelty claims。

附近的分析背景还包括 Caffarelli--Kohn--Nirenberg 的 suitable-weak partial regularity DOI，Wolf 的 local pressure decomposition，Dascaliuc--Grujić 的 physical-scale flux locality，Koch--Tataru 的 critical Carleson-type norm，以及 Yang 的 skewed-cylinder maximal estimates DOI。它们不提供本项目特定的 terminal mollified trajectory、periodized physical annuli、time norm 之前的 fixed shell deletion 与 P_R^M payment 的组合定理。

冻结审计只做了 bounded collision search。没有找到相同量词的 theorem 或 counterexample，不等于 exhaustiveness，也不构成 novelty 或 priority claim。S.438 的证明依赖页面展示的直接 substitution 与 payment comparison，不依赖文献搜索没有命中。

115. 主张账本、有限证书与下一条允许路线

在冻结的 Version-M setting 中，本步 **PROVED**：S.417--S.420 的精确光滑 NSE identities；S.421--S.423 的径向 mollifier 与 terminal path；S.424--S.425 的 fixed-frame Bernoulli cancellation 与 moving-drift identity；S.426--S.432 对任意 $N+1$ 个 physical shells 的 simultaneous positivity；S.433--S.437 的 temporal-tail lower bounds 与 complete-payment upper bound；S.438 对 S.342 的量词级否定；以及 S.439--S.443 在固定 (N,R) 下的临界 L_t^1 振幅饱和。

本步新写出的 S.444 仍 **OPEN**。继续 **OPEN AND UNCHANGED**：direct hybrid terminal-flux gate、selected-crown estimate S.407、S.375、S.288、S.303、S.272、Step 10 S.243、Q.12、Q.1、scale contraction、regularity、singularity formation 与 Millennium problem。

路线校正是明确的：以后不能再把 S.342 当作 proof antecedent，因为它在光滑周期精确解上已经为 false。short-flux 路线下一步只能分析临界 L_t^1 tail S.444，并保留 signed moving-drift cancellation 与 common deletion set。terminal-crown 路线仍可通过另外开放的 coercivity estimate S.407 继续。

主证书通过 7/7 finite groups、2,207 cases、7/7 structural groups 与 3/3 dependency locks。Independent Ruby verifier 通过 9/9 groups、2,839 independent cases，并锁定主文、两份审计、Python/Ruby 实现、主证书 JSON 与证书报告。另有 11 个外部负探针，全部按预期失败关闭。两套有限程序核对 exact Fourier identities、pigeonhole、representative support/path inequalities、amplitude exponents、hashes、structure 与 claim labels；它们不 machine-prove arbitrary-mollifier continuity、continuous payment estimate、S.444、S.407、regularity 或 Clay problem。
FINITE ONLY. NOT CLAY.

116. 路线修正：闭流线复现使绝对时间变差从 A^2 升到 A^3

Step 16 在同一 Taylor 1923 光滑精确解上选取 terminal centre $x_*=(\pi/4,0,0)$。该点位于非复现的 separatrix 路径：轨迹只穿过关键相位一次，因此绝对 flux variation 是 A^2。这个计算对该特殊 terminal setting 完全正确，却不能决定对所有 terminal settings 量化的 S.444。

Step 17 把终点移到同一解的一条 regular closed streamline。在固定物理时间窗中，Version-M 中心完成 $O_R(A)$ 次闭轨道返回；每次 flux density 的瞬时尺度为 A^3。有符号增量在一圈圈之间抵消，但绝对时间变差会逐圈累加。

最终结论是：对所有 $p\in[1,\infty]$、所有有限删除预算 N 以及所有 $\beta<1$，power-only absolute temporal-tail 都失败。尤其 $p=1,\beta=2/3$ 正是否定 S.444 的量词组合，所以 **S.444** 为 **FALSE**。

这不是 DNS，也不是数值 Navier--Stokes 仿真。图中轨道与曲线来自精确解公式的确定性解析可视化。例子全局光滑；S.472、direct hybrid gate、S.407、Q.12、Q.1、scale contraction 与 regularity 仍 OPEN。**NOT CLAY.**

117. 同一 Taylor 精确族上的 regular closed streamline

在 $\mathbb{T}^3=(-\pi,\pi]^3$ 上定义

$$\boxed{\psi=\sin x_1\sin x_2,\qquad W=(\sin x_1\cos x_2,-\cos x_1\sin x_2,0).}\tag{S.445}$$

令 $p_W=(\cos 2x_1+\cos 2x_2)/4$、$b_A(t)=Ae^{-2(t-t_0)}$、$u_A=b_AW$、$p_A=b_A^2p_W$。由 $\nabla\cdot W=0$、$\Delta W=-2W$ 和 $(W\cdot\nabla)W=-\nabla p_W$，它对每个 $A>0$ 都是 smooth、periodic、mean-zero、unforced 的精确三维 NSE 解。

取 $\Gamma=\{\sin x_1\sin x_2=1/2\}$ 在 $(0,\pi)^2$ 中的分支。它由上下两条显式图形拼成一个 compact regular oval，且 W 在其上无零点并沿切向流动。令 $\chi'=W(\chi)$、$\chi(0)=(\pi/4,\pi/4,0)$，则存在有限周期 $T_*>0$：

$$\boxed{\chi(s+T_*)=\chi(s),\qquad T_*=\int_\Gamma\frac{d\ell}{|W|}.}\tag{S.447}$$

记 $g(s)=|W(\chi(s))|^2$、$q=g'$。同一轨道上两点的 g 值分别为 $1/2$ 与 $3/4$，所以 $q\not\equiv 0$，且对所有 $p\geq 1$，一个足够长的 phase interval 都包含线性多份完整周期：

$$L\geq 2T_*\implies\int_a^{a+L}|q(s)|^p ds\geq\frac{V_p}{2T_*}L,\qquad V_p=\int_0^{T_*}|q|^p>0.\tag{S.449}$$

118. 任意 $N+1$ 个物理 annuli 同时激活，Version-M 中心线性多次回返

先固定任意有限删除预算 N，再令 $M=N+1$，最后选择足够小的 admissible R。Step 16 的 Fourier multiplier 与 support/cosine 论证给出前 M 个 physical shells 的系数

$$c_{k,R}\geq\frac{1}{2}m_{k,R}>0,\qquad 1\leq k\leq M.$$

Version-M 的 terminally anchored path 恰为

$$\boxed{\theta_A(t)=\mu_R\int_{t_0}^t b_A(r)dr,\qquad\xi_A(t)=\chi(\theta_A(t)),\qquad\theta_A'=\mu_Rb_A.}\tag{S.452}$$

在 $I_R=(t_0-R^2,t_0)$ 上，phase length 是

$$\boxed{L_A=\frac{\mu_RA}{2}(e^{2R^2}-1)\asymp_R A.}\tag{S.453}$$

因此随 $A\to\infty$，中心在固定物理时间窗里完成线性多次 closed-orbit recurrence。这一机制与 Step 16 的单次 separatrix passage 不同。

119. 绝对时间尾对每个 $p\geq 1$ 都是 A^3

固定框架的 kinetic 与 physical-pressure Bernoulli flux 仍精确抵消，pressure gauge 也逐壳层抵消；完整 Version-M 只留下 moving-cutoff drift：

$$\boxed{\dot{F}_{k,R}(t)=\frac{\gamma_k\mu_R c_{k,R}}{2R}\eta_R(t)b_A(t)^3 q(\theta_A(t)).} \tag{S.454}$$

在 I_R 上 $\eta_R=1$。用 $d\theta=\mu_R b_A\,dt$ 和周期平均后，所有有限 p 的 L^p_t norm 都获得 A^{3p} 的 p 次方下界；$p=\infty$ 也在一个完整周期中达到 RA^3 尺度。删除至多 $N=M-1$ 个 shell，前 M 个正坐标中至少留下一个。因此，对所有 $p\in[1,\infty]$，

$$\boxed{d_{p,N,R}A^3\le \mathfrak{H}^F_{p,N,R}\le D_{p,R}A^3,\qquad A\ge A_0(R).} \tag{S.459}$$

下界是 continuum analytic statement；有限证书只核对精确恒等式、计数、删除量词和 exponent bookkeeping。

120. 完整 Version-M payment 仍是 A^3，所以所有 $\beta<1$ 都失败

沿 compact orbit 平移不会改变 fixed smooth profiles 的幅值。local energy、exterior velocity/pressure、quadratic cutoff 与 harmonic rows 逐项给出至多 A^2 或 A^3；super-Gaussian all-copy sum 和 order-−4 harmonic sum分别保持可和。good times 趋近 t_0 时，buffered local-energy endpoint 给出正的 A^2 trace，其 $3/2$ 次幂给出 payment 下界。于是

$$\boxed{c_RA^3\le P^M_R\le C_RA^3.} \tag{S.461}$$

把它与 S.459 合并，若 $\beta\ge 0$ 使用 payment 上界，若 $\beta<0$ 使用 payment 下界，得到

$$\boxed{\frac{\mathfrak{H}^F_{p,N,R}}{(P^M_R)^\beta}\ge c_{p,N,R,\beta}A^{3(1-\beta)}\longrightarrow\infty,\qquad \beta<1.} \tag{S.462}$$

量词顺序不可交换：对手先固定 p,N,β,C，然后取 $M=N+1$、选择 admissible R,z_0，最后令 A 足够大。取 $p=1,\beta=2/3$ 就精确否定 S.444。对 absolute temporal-tail 的 power-only 形式，必要 payment exponent 至少为 1。

121. 有符号正向 excursion 仍只有 A^2

同一公式也说明 absolute value 丢失了什么。由于 $d_tg(\theta_A)=\mu_Rb_Aq(\theta_A)$，

$$\dot{F}_{k,R}=\frac{\gamma_k c_{k,R}}{2R}\eta_R b_A^2\frac{d}{dt}g(\theta_A). \tag{S.464}$$

分部积分后，端点项与 cutoff/damping 项都只有 quadratic amplitude scale，从而

$$\boxed{\sum_{k\ge1}\operatorname{osc}_{[s_R,t_0)}F_{k,R}\le C_RA^2.} \tag{S.466}$$

沿正确 orientation 选择同一周期内从 $g=1/2$ 到 $g=3/4$ 的有序两时刻，可得前 $N+1$ 个坐标均有正向 $c_{k,R}A^2$ excursion。于是 signed range 和 positive excursion 恰为 A^2，而 absolute variation 为 A^3。

122. BV/Jordan 分解精确识别 recurrent backtracking debt

在 $p=1$ 时 dimensionless normalization 完全抵消：

$$\mathfrak{H}^F_{1,N,R} = \inf_{\#S\le N} \sum_{k\notin S} \mathrm{TV}_{[s_R,t_0)} F_{k,R}. \tag{S.467}$$

令 $V^\pm_{k,R}=\int[\pm\dot F_{k,R}]_+dt$、$B_{k,R}=\min(V^+,V^-)$。由共同零起点和 Jordan decomposition，

$$\boxed{\mathrm{TV}\,F_{k,R} = |F_{k,R}(t_0^-)| + 2B_{k,R}.} \tag{S.468}$$

在 recurrent orbit 上，terminal endpoint 或 signed range 只有 A^2，但 backtracking debt $B_{k,R}$ 在每个激活 shell 上达到 A^3。S.444 实际要求为每次上下往返付费；signed terminal problem 并不需要这笔重复债务。

123. 正确的后继目标是 fixed-deletion positive excursion / simultaneous height

定义正向有序 excursion 与 common-deletion tail

$$\boxed{\mathfrak{O}^{F,+}_{N,R} := \inf_{\#S\le N} \sum_{k\notin S} \sup_{a<b}[F_{k,R}(b)-F_{k,R}(a)]_+.} \tag{S.469}$$

每个 Step 15 hybrid coordinate 都是同一 $F_{k,R}$ 的两时刻 increment，因此

$$\boxed{\mathfrak{Z}^\lambda_{N,R}(\mathcal{T}_R) \le \mathfrak{O}^{F,+}_{N,R} \le \mathfrak{H}^F_{1,N,R}.} \tag{S.470}$$

在当前 exact family 上，

$$\boxed{\mathfrak{O}^{F,+}_{N,R} \asymp A^2 \asymp (P^M_R)^{2/3}, \qquad \mathfrak{H}^F_{1,N,R} \asymp A^3.} \tag{S.471}$$

因此 S.472--即 universal fixed-deletion positive-excursion estimate--是与 signed endpoint problem 对齐的充分输入，但目前仍 **OPEN**。它比 direct Step 15 gate 在量词上更强；更弱的 direct hybrid terminal-flux gate 也仍 OPEN，未来证明不必一定先证明 S.472。

把完成的非负 clocks 写成 $K=F+Q$，且 $B_{Q,R}=\sum_k \mathrm{TV}\,Q_{k,R} \lesssim (P^M_R)^{2/3}$，则 S.473--S.475 证明：S.444 对应 positive-variation packing，而正确的后继只需 maximal simultaneous height 或 positive-excursion packing。

124. 三份审计、双语言证书与期刊级四联图

Primary analytic audit 对 S.445--S.475 的 closed-orbit topology、periodic averaging、dimensionless L_t^p normalization、complete payment、fixed deletion、signed integration by parts 与 completed-clock inequalities 逐项复核，结论为 PASS。Independent adversarial audit 保留并关闭五项修复记录，最终也为 PASS。Literature audit 逐条检查 Taylor、Yang、Dascaliuc--Grujić、Wolf 与 Duchon--Robert 的来源、适用范围与 non-implication 边界；有限检索不构成 novelty 或 priority claim。

Python 主证书通过 12/12 finite groups、4,325 cases、11/11 structural checks 与 2/2 dependency locks。独立 Ruby 实现不调用 Python：7/7 exact groups 共 294 assertions、4/4 artifact/commit locks、20/20 semantic checks、32/32 negative mutations、3/3 artifact-path substitutions 与 14/14 reproducibility assertions 全部 PASS。

正式四联图展示 regular closed streamline、一个 orbit period 的 g、四次 return 中 signed cancellation 对 absolute debt 的分离，以及 slope 2 / slope 3 的 amplitude classes。图、caption、source-data、plot/validate scripts、manifest 与 QA 均公开；它是 analytic exact-field visualization，不是 DNS 或数值 NSE simulation。

125. 主张账本与严格下一接口

本节 **PROVED**：regular closed streamline 与 recurrence lemma；任意 $N+1$ 个 physical shells 同时激活；exact Version-M recurrent path 与 flux identity；所有 $p \geq 1$ 的 A^3 absolute temporal-tail；完整 $P_R^M \asymp A^3$；所有 $\beta < 1$ 的 power-only absolute tail 失败，尤其 **S.444 为 FALSE**；signed A^2 range、BV/backtracking identity、positive-excursion implication 与 completed-clock comparison。

本节 **OPEN**：S.472 fixed-deletion positive excursion / simultaneous-height estimate。继续 **OPEN AND UNCHANGED**：direct hybrid gate、terminal-crown coercivity S.407、Q.12、Q.1、scale contraction 与 regularity。

路线决定不可回退：任何后续证明都不得再使用 S.342、S.444 或 $\mathfrak{H}_{p,N,R}^F \lesssim (P_R^M)^\beta$ with $\beta < 1$。下一 temporal task 只能研究 signed positive-excursion tail、fixed-deletion simultaneous height，或直接处理 Step 15 hybrid last-exit increments；terminal-crown 路线仍独立可用。**NOT CLAY.**

126. 结论与修正：真正缺失的是时间量词的中间层

Step 17 已证明 absolute temporal-variation tail 的所有次线性幂估计都为 FALSE，并保留逐坐标 positive excursion $\mathfrak{O}_{N,R}^{F,+}$ 作为一个充分的 signed 后继。但它对每个 shell 分别取时间上确界，再对 shell 求和；Step 15 的实际 residual gate 则先固定一个 terminal time。Step 18 补上两者之间缺失的 fixed-deletion functional，并把三种量词次序精确分开。

直接 Step 15 gate 是 $\sup_\tau \inf_{\#S \leq N} \sum_{k \notin S} z_k(\tau)$。若要求一个异常 shell 集合对所有 common good terminal times 同时有效，就得到更强但仍非 separable 的 $\inf_{\#S \leq N} \sup_\tau \sum_{k \notin S} z_k(\tau)$。它仍弱于逐坐标 positive excursion。

completed-clock simultaneous height 在一次支付已控制的 Q-variation 后控制 fixed-set hybrid tail；反向上，Step 10 paid-branch inequality 用 fixed hybrid tail 加已知 A_R-scale payments 控制 simultaneous height。因此两者只在目标尺度上等价，不是字面相等，也不等同于更弱的 moving-deletion gate。

互不相交的 triangular clocks 证明前两条层级不等式都可严格，并且逐坐标 maxima 可以任意倍数高估 simultaneous height。正确的严格例子需要 $M \geq N+2$，不是 $M = N+1$。同一抽象族还证明 inherited nonnegativity、Q-variation 与 linear absolute-flux ledger 不能纯代数推出所需 $2/3$ -power estimate。这是 ABSTRACT information-theoretic obstruction，不是 Navier--Stokes 反例。

Step 17 的 recurrent Taylor smooth family 不否定任何仍存活的 gate：固定 N 后按 S.451 固定 R，它只把 positive excursion 饱和在 quadratic scale，同时继续否定已经弃用的 absolute-variation route。本节是严格 route reduction 与 correction；不证明 fixed-deletion gate、direct hybrid gate、Q.12、Q.1、scale contraction 或 regularity。**NOT CLAY.**

127. 冻结设定与三种 deletion order

固定一个 Version-M suitable weak solution、一个 admissible scale R、一个 admissible profile $\boldsymbol{\lambda}$、一个 terminal domain $\mathcal{D} \in \{I_R, \mathcal{T}_R\}$ 以及一个 $N \in \mathbb{N}_0$。令 $I = [s_R, t_0)$、$\mathcal{D}_g = \mathcal{D} \cap \mathcal{G}_R$。继承对象满足

$$\boxed{K_{k,R} = F_{k,R} + Q_{k,R} \geq 0, \qquad F_{k,R}(s_R) = Q_{k,R}(s_R) = K_{k,R}(s_R) = 0.} \tag{S.476}$$

并且

$$B_{F,R} := \sum_{k\geq 1} \mathrm{TV}_I\, F_{k,R} \leq C_F P_R^M, \qquad B_{Q,R} := \sum_{k\geq 1} \mathrm{TV}_I\, Q_{k,R} \leq C_Q A_R, \qquad A_R = (P_R^M)^{2/3}.$$

保留 Step 15 的 nonnegative stopped-flux vector

$$z_k(\tau) = F_{k,R}(\tau) - F_{k,R}(\sigma_k^{\mathrm{hyb}}(\tau)), \qquad \tau \in \mathcal{D}_g,$$

其中 active coordinate 上 $\sigma_k^{\mathrm{hyb}}(\tau) \in [s_R, \tau]$，否则 $z_k(\tau) = 0$；Step 15 已证明 $z(\tau) \in \ell_+^1$。设 $\mathscr{S}_N = \{S \subset \mathbb{N} : \#S \leq N\}$，定义

$$\boxed{\begin{aligned} \mathfrak{H}_{N,R}^{\mathrm{hyb}}(\mathcal{D}) &:= \sup_{\tau\in\mathcal{D}_g} \inf_{S\in\mathscr{S}_N} \sum_{k\notin S} z_k(\tau) = \mathfrak{Z}_{N,R}^{\boldsymbol{\lambda}}(\mathcal{D}), \\ \mathfrak{H}_{N,R}^{\mathrm{fix}}(\mathcal{D}) &:= \inf_{S\in\mathscr{S}_N} \sup_{\tau\in\mathcal{D}_g} \sum_{k\notin S} z_k(\tau). \end{aligned}} \tag{S.477}$$

这里的 fix 只表示：对每个已经固定的 solution、scale、centre 与 terminal domain，同一个 shell set 要用于全部 common good terminal times；该集合仍可依赖这些固定数据。它不冻结 hybrid starts，后者仍依赖 terminal time。

为比较，定义

$$\boxed{o_{k,R}^F := \sup_{s_R\leq a<b<t_0} [F_{k,R}(b) - F_{k,R}(a)]_+, \qquad \mathfrak{O}_{N,R}^{F,+} := \inf_{S\in\mathscr{S}_N} \sum_{k\notin S} o_{k,R}^F.} \tag{S.478}$$

因为 $o_{k,R}^F \leq \mathrm{TV}_I\, F_{k,R}$，序列 $(o_{k,R}^F)_k \in \ell_+^1$，且对每个 k, τ 都有 $0 \leq z_k(\tau) \leq o_{k,R}^F$。以下所有级数都由同一可和序列控制，不交换任何无限 signed sum。

128. 精确层级与 layer-cake incidence

minimax inequality 与逐坐标支配直接给出

$$\boxed{\mathfrak{H}_{N,R}^{\mathrm{hyb}}(\mathcal{D}) \leq \mathfrak{H}_{N,R}^{\mathrm{fix}}(\mathcal{D}) \leq \mathfrak{O}_{N,R}^{F,+} \leq \mathfrak{H}_{1,N,R}^F.} \tag{S.479}$$

第一条只用 sup inf $\le$ inf sup，不假设 minimax equality 或 attainment。第二条来自 $z_k(\tau)\le o^F_{k,R}$，第三条来自 $o^F_{k,R}\le \mathrm{TV}\,F_{k,R}$；若 infimum 不取到，就用 ε-minimizer 再令 $\varepsilon\downarrow 0$。

对 $\lambda>0$，令 $A_\tau(\lambda)=\{k:z_k(\tau)>\lambda\}$、$A_o(\lambda)=\{k:o^F_{k,R}>\lambda\}$。layer-cake 与 best-$N$ rearrangement 给出

$$\boxed{\begin{aligned}\mathfrak{H}^{\mathrm{hyb}}_{N,R}(\mathcal{D})&=\sup_{\tau\in\mathcal{D}_g}\int_0^\infty(\#A_\tau(\lambda)-N)_+\,d\lambda,\\ \mathfrak{H}^{\mathrm{fix}}_{N,R}(\mathcal{D})&=\inf_{S\in\mathscr{S}_N}\sup_{\tau\in\mathcal{D}_g}\int_0^\infty\#(A_\tau(\lambda)\setminus S)\,d\lambda,\\ \mathfrak{D}^{F,+}_{N,R}&=\int_0^\infty(\#A_o(\lambda)-N)_+\,d\lambda.\end{aligned}}\tag{S.480}$$

对非负 ℓ^1 序列 x，所用恒等式为

$$\inf_{\#S\le N}\sum_{k\notin S}x_k=\int_0^\infty(\#\{k:x_k>\lambda\}-N)_+\,d\lambda.$$

先对有限截断排序，再用 monotone convergence 即得无限序列版本。S.480 中间一行仅在固定 S,τ 后使用 Tonelli；两个 optimization 都没有穿过积分号。

这正定位量词缺口：direct gate 可在知道每个 terminal time 后改变删除集；fixed gate 要求一个集合击中全部 common good terminal times；separable excursion gate 又把 $A_\tau(\lambda)$ 换成更大的 coordinatewise envelope $A_o(\lambda)$，因而忘记不同 shell peaks 是否只在互斥时刻出现。这里只断言 $\bigcup_\tau A_\tau(\lambda)\subseteq A_o(\lambda)$，不断言相等。

129. Completed-clock simultaneous height 与目标尺度等价

定义 terminal domain 上的 fixed-deletion simultaneous height

$$\boxed{\mathfrak{L}^K_{N,R}(\mathcal{D}):=\inf_{S\in\mathscr{S}_N}\sup_{t\in\mathcal{D}}\sum_{k\notin S}K_{k,R}(t).}\tag{S.481}$$

它不同于 Step 17 的 separable maximum $\mathfrak{M}^K_{N,R}=\inf_S\sum_{k\notin S}\sup_{t\in I}K_{k,R}(t)$：$\mathfrak{L}^K$ 的时间上确界位于 shell sum 外，但仍保留一个 deletion set。

对 active hybrid coordinate，$K=F+Q$、$K\ge 0$ 与 $\sigma^{\mathrm{hyb}}_k(\tau)\le\tau$ 给出

$$\boxed{\begin{aligned}z_k(\tau)&=K_k(\tau)-K_k(\sigma^{\mathrm{hyb}}_k(\tau))-Q_k(\tau)+Q_k(\sigma^{\mathrm{hyb}}_k(\tau))\\&\le K_k(\tau)+\mathrm{TV}_I\,Q_k.\end{aligned}}\tag{S.482}$$

在同一个固定 S 外求和，先取 terminal supremum，再优化，得到

$$\boxed{\mathfrak{H}^{\mathrm{fix}}_{N,R}(\mathcal{D})\le\mathfrak{L}^K_{N,R}(\mathcal{D})+B_{Q,R}\le\mathfrak{L}^K_{N,R}(\mathcal{D})+C_QA_R.}\tag{S.483}$$

所有 clock sums 都有限，因为共同零起点给出 $K_k(t)\le o^F_{k,R}+\mathrm{TV}_I\,Q_{k,R}$，故 $\sum_k\sup_t K_k(t)\le B_{F,R}+B_{Q,R}<\infty$。

令 $\Pi_R^{\boldsymbol{\lambda}} = 6B_{Q,R} + C_5\mathscr{L}(\boldsymbol{\lambda})^{1/3}A_R \le C_{\mathrm{pay}}(\boldsymbol{\lambda})A_R$。Step 10 S.235 对同一个固定 S 与每个 good terminal time 给出 $\sum_{k\notin S} K_{k,R}(\tau) \le \Pi_R^{\boldsymbol{\lambda}} + 6\sum_{k\notin S} z_k(\tau)$。利用 $t \mapsto (K_{k,R}(t))_k$ 在 ℓ^1 中连续、common good-time set 稠密，得到反向估计

$$\boxed{\mathfrak{L}_{N,R}^K(\mathcal{D}) \le \Pi_R^{\boldsymbol{\lambda}} + 6\mathfrak{H}_{N,R}^{\mathrm{fix}}(\mathcal{D}).} \tag{S.484}$$

所以对固定 admissible profile 和固定 universal N，$\mathfrak{H}^{\mathrm{fix}} \lesssim A_R$ 当且仅当 $\mathfrak{L}^K \lesssim A_R$。这是支付已知项后的 target-scale equivalence，不是 literal equality，也不把任一 functional 等同于更弱的 moving-deletion tail。

最后，固定删除集后，sum 的 supremum 不超过 coordinatewise suprema 之和；结合 Step 17 S.475，

$$\boxed{\mathfrak{L}_{N,R}^K(\mathcal{D}) \le \mathfrak{M}_{N,R}^K \le \mathfrak{D}_{N,R}^{F,+} + B_{Q,R}.} \tag{S.485}$$

不存在从 $\mathfrak{L}^K$ 到 $\mathfrak{M}^K$ 的 universal reverse comparison；后面的 abstract clocks 给出无界分离。

130. 修正后的 OPEN targets 与无条件线性 fallback

最弱且精确匹配 Step 15 route 的 direct target 仍是某个 universal finite N_0, C 下 $\mathfrak{H}_{N_0,R}^{\mathrm{hyb}}(\mathcal{T}_R) \le CA_R$；它与 full residual gate 相差 Step 15 S.385 的字面因子 5。

若坚持一个 exceptional shell set 对全部 common good terminal times 同时有效，route-minimal successor 是

$$\boxed{\begin{array}{l} \exists N_{\mathrm{fix}} \in \mathbb{N}_0,\ C_{\mathrm{fix}} < \infty \text{ for the fixed universal admissible profile, such that} \\ \mathfrak{H}_{N_{\mathrm{fix}},R}^{\mathrm{fix}}(\mathcal{T}_R) \le C_{\mathrm{fix}}(P_R^M)^{2/3} \quad \text{uniformly in the solution and admissible } R, z_0. \end{array}} \tag{S.486}$$

S.486 为 **OPEN**。由 S.479，它蕴含 direct hybrid gate，进而通过既证 reductions 蕴含 Step 10 S.243、Q.12 与 Q.1；它比 direct gate 更强，因为 direct exceptional set 可依赖 τ。

目标尺度等价的 completed-clock 表述是

$$\boxed{\begin{array}{l} \exists N_L \in \mathbb{N}_0,\ C_L < \infty \text{ for the fixed universal admissible profile, such that} \\ \left(\mathfrak{L}_{N_L,R}^K(\mathcal{T}_R)\right)^{3/2} \le C_L P_R^M \quad \text{uniformly in the solution and admissible } R, z_0. \end{array}} \tag{S.487}$$

S.487 同样为 **OPEN**。S.483--S.484 只证明它与 S.486 在目标尺度等价，常数仅依赖同一冻结 profile；二者都强于 direct moving-deletion gate。

继承的无条件信息只有

$$\boxed{\mathfrak{H}_{N,R}^{\mathrm{fix}}(\mathcal{D}) \le B_{F,R} \le C_F P_R^M, \qquad \mathfrak{L}_{N,R}^K(\mathcal{D}) \le B_{F,R} + B_{Q,R} \le C_F P_R^M + C_Q A_R.} \tag{S.488}$$

当 $P_R^M \le 1$ 时第一式已不超过 $C_F A_R$；当 $P_R^M > 1$ 时，它还差因子 $(P_R^M)^{1/3}$。只重排 displayed linear ledger 无法消掉这一因子。

131. Disjoint triangular clocks 的精确 ABSTRACT separation

固定整数 $M > N$、高度 $H > 0$ 与 $I = [0,1]$。对 $1 \le j \le M$，令

$$\boxed{\phi_j(t) = \left(1 - 2M\left|t - \frac{2j-1}{2M}\right|\right)_+, \qquad K_j(t) = F_j(t) = H\phi_j(t), \qquad Q_j(t) = 0,} \tag{S.489}$$

其 support interiors 两两不交，后续坐标为零，并取 common-zero-start increment $z_j(\tau) = H\phi_j(\tau)$。每个时刻至多一个坐标为正；任一至多删除 $N < M$ 项的固定集合都遗漏某个 peak。于是

$$\boxed{\begin{aligned} &\mathfrak{H}_N^{\mathrm{hyb}} = \begin{cases} H, & N = 0, \\ 0, & N \ge 1, \end{cases} && \mathfrak{H}_N^{\mathrm{fix}} = H, && \mathfrak{L}_N^K = H, \\ &\mathfrak{D}_N^{F,+} = \mathfrak{M}_N^K = \mathfrak{V}_N^K = (M-N)H, && \mathfrak{H}_{1,N}^F = 2(M-N)H, && \sum_j \mathrm{TV}\, F_j = 2MH. \end{aligned}} \tag{S.490}$$

特别地，当 $N \ge 1$ 且 $M \ge N+2$ 时，

$$\boxed{0 = \mathfrak{H}_N^{\mathrm{hyb}} < \mathfrak{H}_N^{\mathrm{fix}} = \mathfrak{L}_N^K < \mathfrak{D}_N^{F,+} = \mathfrak{M}_N^K.} \tag{S.491}$$

比值 $\mathfrak{D}_N^{F,+}/\mathfrak{H}_N^{\mathrm{fix}} = M - N$ 在固定 N、$M \to \infty$ 时无界。因此 abstract clock class 中不存在从 simultaneous functional 到 separable coordinatewise maximum 的 universal reverse comparison。先前诱人的 $M = N+1$ 只分开 moving 与 fixed deletion，不能使第二条不等式严格。

以 full absolute-flux ledger $\mathcal{P} = \sum_j \mathrm{TV}\, F_j = 2MH$ 归一化。固定 N 与任意 $M > N$，令 $H \to \infty$，则

$$\boxed{\frac{\mathfrak{H}_N^{\mathrm{fix}}}{\mathcal{P}^{2/3}} = \frac{\mathfrak{L}_N^K}{\mathcal{P}^{2/3}} = \frac{H^{1/3}}{(2M)^{2/3}} \longrightarrow \infty.} \tag{S.492}$$

所以 $K \ge 0$、$K = F + Q$、$B_Q \lesssim \mathcal{P}^{2/3}$ 与 $\sum_k \mathrm{TV}\, F_k \lesssim \mathcal{P}$ 这些 abstract assumptions，不能对任何预先指定的有限 N 推出 fixed-deletion quadratic bound。高度 H 是独立参数；只令 $M = N+1$、$H = M^3$ 改变的是对 N 的 uniformity，不能否定 fixed-N statement。

S.489--S.492 全部是 **ABSTRACT CLOCK STRESS TESTS**：它们不是 spatial fields，不实现 Version-M payment，也不是 Navier--Stokes solutions 或 counterexamples。

132. Recurrent Taylor family 通过所有仍存活的 gates

对 Step 17 的 exact smooth family，先固定有限 N，再按 S.451 选择并固定 R。对该量词次序以及 $A \ge A_0(N,R)$，

$$\mathfrak{D}_{N,R}^{F,+} \asymp_{N,R} A^2, \qquad B_{Q,R} = O_R(A^2), \qquad P_R^M \asymp_R A^3.$$

因此 S.479 与 S.485 给出

$$\boxed{\mathfrak{H}^{\mathrm{hyb}}_{N,R} \le \mathfrak{H}^{\mathrm{fix}}_{N,R} \lesssim_{N,R} A^2 \asymp_{N,R} (P_R^M)^{2/3}, \qquad \mathfrak{L}^K_{N,R}(\mathcal{T}_R) \lesssim_{N,R} A^2.} \tag{S.493}$$

这是 fixed-R screen，不是 universal S.486 或 S.487 的证明。它只说明摧毁 absolute temporal variation 的 smooth recurrent family 与全部存活的 quadratic gates 相容：recurrence 产生 $O(A)$ 次重复绕行，但 peaks 处于同一 phase geometry，signed excursion 仍为 $O(A^2)$。

133. 成功的 PDE theorem 必须补入什么

精确 reductions 留下三个嵌套研究目标：最弱的 direct hybrid target 是 $\mathfrak{H}^{\mathrm{hyb}} \lesssim A_R$，完全匹配 Step 15；fixed-deletion target 是 S.486，要求一个共同有限 shell set，但保留 simultaneous terminal incidence；completed-clock target 是 S.487，或更强的 Step 17 positive-excursion bound。simultaneous-height form 在已知 payment 后与 fixed hybrid gate 等价，而 separable positive excursion 仍要求额外 cross-time information。

triangular clocks 已证明新输入不能只有 nonnegativity 与 inherited linear ledgers。可行 theorem 至少要加入一个真正 PDE-specific mechanism，例如：

- simultaneous height-to-cubic-payment estimate；
- persistence 或 dwell-time，使高 clock aggregate 占据足够 parabolic time 并由 cubic payment 支付；
- deterministic stopping-time / Carleson charge，控制 S.480 的 time--shell incidence sets；
- 与 hybrid first-passage intervals 绑定的 signed entrance 或 collar-flux payment。

这些只是 mechanism classes，不是已证明 lemmas。

134. 一手来源 collision boundary

有界两轮一手来源检索没有找到同时具有 S.486 全部量词的 theorem：deterministic suitable weak solutions；对每个固定 solution、scale、centre 与 terminal domain，保留一个 finite shell deletion 覆盖全部 common good terminal times；对 forward stopped increments 作 infinite-shell ℓ^1 sum；删除预算 universal；并由 $(P_R^M)^{2/3}$ 支付。

- Dascaliuc--Grujić 的 physical-scale energy cascade / flux locality 证明 inertial-range 条件下的 signed time/ensemble-averaged flux estimates，不是带一个 shell deletion 的 terminal-time maximum。
- Yang 的 flow-generated skewed cylinders maximal functions 给出 space-time averages 的 weak-$(1,1)$ 与 strong-(p,p) bounds，不控制 simultaneous clock height 或 fixed best-N terminal functional。
- Yu 的 finite-chain CKN bad-scale counting 在预先指定的有限 scale chain 上用 nonnegative channel costs 得到至少一个 CKN-small scale，不给 infinite-shell、all-good-terminal 的 S.486。
- Yu 的 coarse-grained pressure-flux work depletion 给出 exact fixed-chain signed-work depletion，并明确保留 negative work / backscatter；它不声称 negative set smallness、uniform moving-window constants 或 chain length 趋于无穷时的 summability。

这些是相邻工具与可能 ingredients，不是 open gate 的证明。检索有界；未命中不构成 novelty、priority 或 exhaustiveness claim。

135. 主张账本、证书与严格后续边界

本节 **PROVED**：S.476--S.479 的 hierarchy 与 infinite-shell justification；S.480 的 exact layer-cake incidence formulas；S.481--S.485 的 completed-clock 双向 target-scale reduction；S.488 的 unconditional linear fallback；S.489--S.492 的 abstract triangular-clock exact values；S.493 的 fixed-R Taylor compatibility screen。

本节 **ABSTRACT ONLY**：S.491 的 strictness 与 unbounded reverse ratio；S.492 中 listed ledger assumptions 不能推出 2/3-power bound。二者都不是 NSE counterexample。

继续 **OPEN**：direct moving-deletion hybrid gate；route-minimal fixed-deletion gate S.486；target-scale-equivalent simultaneous-height gate S.487；Step 17 positive-excursion gate S.472；terminal-crown coercivity S.407；Q.12、Q.1、scale contraction 与 regularity。**NOT CLAY.**

Python 主证书通过 5/5 exact finite groups、283,157 rational cases、5/5 structural groups 与 5/5 hash locks。独立 Ruby verifier 通过 8/8 groups、72,144 assertions；Python/Ruby 分别拒绝 12/12 与 13/13 intentional mutations；三组 Python hash seed 和一次跨工作目录 Ruby 重放均 byte-identical。

正式四联图的 25 文件档案通过 39 checks 与 verify-only。Panel A 是 proved inequalities 与 known-payment links 的精确 schematic；Panels B--D 是 exact abstract clocks。彩色、灰度与独立 PDF render 均完成视觉 QA。它不是 PDE data，不是 DNS，不是 NSE simulation，也不证明 OPEN S.486--S.487 或 Clay 问题。

route decision 已精确固定：后续 fixed-deletion 工作应直接研究 $\mathfrak{H}^{\mathrm{fix}}$，不应自动升级到更强的 $\mathfrak{O}^{F,+}$；completed-clock 工作可研究在已支付项后等价的 $\mathfrak{L}^{K}$；最弱有效路线仍是 terminal-dependent deletion set 的 direct Step 15 hybrid gate。本站在此停于冻结 Step 18，等待下一份明确冻结包。

结果与精确范围：错开峰值不能消除普通驻留的付款

Step 18 把 fixed-deletion 路线归约到 completed-clock simultaneous height

$$\mathfrak{L}^{K}_{N,R}(D) := \inf_{\#S\le N} \sup_{t\in D} \sum_{k\notin S} K_{k,R}(t).$$

互不相交的 triangular clocks 说明，不同壳层原则上可以在不同时间达到峰值。Step 19 检查更窄的问题：只把两个 packet lobe 的目标时间错开，能否让 completed-clock witness 相对于 Version-M payment 变便宜？

答案只在明确的 lobe-floor 层面给出，而且是否定的：只要保留既有 outer-lobe 几何和不可忽略的驻留时间，外叶的正动能下界就会被非负 velocity-cubic payment 强制支付。这个结论不需要内叶与外叶同时出现。

本节证明四件事：持续 θR^3 的 outer lobe 产生精确 Hölder coercivity；任意甚至互不相交的两个正 K-clock floor 只产生 fixed-deletion completed-clock witness；既有相邻壳参数下，低付款要求归一化 dwell 指数坍缩；同一个 exact smooth periodic unforced common-shear Navier--Stokes 解确实能够实现两个互不相交的 R^3 lobe windows，但它的 payment-to-witness ratio 发散。

这里没有 full completed-clock upper bound，没有把 K 替换成 $\mathfrak{H}^{\mathrm{fix}}$，没有任意实时间 scheduling theorem，也没有正则性或奇性结论。NOT CLAY。

冻结的 Version-M 设置与 persistent outer-lobe floor

固定 $R>0$、终端时刻 t_0，以及 R0.74P Version-M 设置中的一个光滑、周期、无外力 Navier--Stokes 解。完整付款含有非负 exterior velocity row

$$P_R^M \ge \mathcal{G}_u, \qquad \mathcal{G}_u = (2R)^{-2} \int_{I_{2R}} \int_{\mathbb{T}^3} W_{2R}(x) |u(t,x)|^3 \, dx \, dt. \tag{T.1}$$

对 outer target shell k_2，记

$$L_2 = \lambda 2^{k_2}, \qquad \Gamma_2 = \gamma_{k_2} = e^{-c_\gamma L_2^2}. \tag{T.2}$$

R0.74Q 的物理 lobe 位于 $A_{k_2}(R) = A_{k_2-1}(2R)$，因此该区域上

$$W_{2R} \ge \gamma_{k_2-1} = \Gamma_2^{1/4}. \tag{T.4}$$

取可测集合 $J_2 \subset I_R \subset I_{2R}$，令 $|J_2| = \theta R^3$。对几乎处处的 $t \in J_2$，假设 moving lobe $\Omega_2(t)$ 联合可测，并满足

$$\Omega_2(t) \subset A_{k_2}(R), \qquad |\Omega_2(t)| = \frac{1}{16} L_2 R^3, \qquad \Psi_{k_2}^R = 1 \text{ on } \Omega_2(t). \tag{T.6}$$

定义 persistent normalized lobe kinetic floor

$$h_2 := \operatorname*{ess\,inf}_{t \in J_2} \frac{\Gamma_2}{2R} \int_{\Omega_2(t)} |u(t,x)|^2 \, dx > 0. \tag{T.7}$$

由于 defect-completed clock 是非负端点动能项与累积耗散项之和，$J_2 \subset I_R$ 上的时间 cutoff 等于一，所以 $K_{k_2,R}(t) \geq h_2$ 几乎处处成立。

Schedule-invariant outer-lobe Hölder coercivity

空间 Hölder 不等式给出

$$\int_{\Omega_2(t)} |u|^3 \geq |\Omega_2(t)|^{-1/2} \left(\int_{\Omega_2(t)} |u|^2 \right)^{3/2}. \tag{T.11}$$

代入冻结的 lobe volume 与 kinetic normalization，得到

$$\int_{\Omega_2(t)} |u|^3 \geq 2^{7/2} h_2^{3/2} \Gamma_2^{-3/2} L_2^{-1/2}. \tag{T.12}$$

把 T.1 的非负积分限制到 moving lobe，再使用 weight floor 与 $|J_2| = \theta R^3$，得到 Step 19 的核心估计

$$\boxed{P_R^M \geq 2\sqrt{2}\,\theta h_2^{3/2} R \Gamma_2^{-5/4} L_2^{-1/2}.} \tag{T.9}$$

等价地，令

$$\Lambda_2 := \theta^{2/3} R^{2/3} \Gamma_2^{-5/6} L_2^{-1/3},$$

则

$$\boxed{(P_R^M)^{2/3} \geq 2\Lambda_2 h_2.} \tag{T.10}$$

证明中没有出现内 packet、内叶目标时间或两个目标区间的交叠。因此，只要 outer-lobe hypotheses 保持成立，估计对相对 schedule 不变。

若只知道 $W_{2R} \geq c_W \Gamma_2^{1/4}$ 和 $|\Omega_2(t)| \leq C_\Omega L_2 R^3$，同一证明给出

$$P_R^M \ge 2^{-1/2} c_W C_\Omega^{-1/2} \theta h_2^{3/2} R \Gamma_2^{-5/4} L_2^{-1/2}. \tag{T.15}$$

取 $c_W = 1$、$C_\Omega = 1/16$ 就精确恢复 T.9。

两枚可能不同时的 K-clock floor：只能推出 one-deletion witness

令 $k_1 \ne k_2$，并在任意可测正测度集合 $J_i \subset D$ 上假设

$$K_{k_i,R}(t) \ge h_i > 0 \quad \text{a.e. on } J_i, \qquad i = 1, 2. \tag{T.16}$$

不要求 J_1 与 J_2 相交。任意 $\#S \le 1$ 的删除集合至少留下 k_1, k_2 中的一枚；在该坐标自己的目标时间集合取值并利用全部 clock 的非负性，便得到

$$\boxed{\mathfrak{L}_{1,R}^K(D) \ge h_* := \min(h_1, h_2).} \tag{T.17}$$

如果 outer floor 满足 $h_2 \ge h_*$，T.10 进一步给出

$$\boxed{(P_R^M)^{2/3} \ge 2\Lambda_2 h_*.} \tag{T.18}$$

这两式只能作单向解释。h_* 是 $\mathfrak{L}_{1,R}^K$ 的一个下界 witness；其他时间、壳层或累积耗散可能让 full functional 更大。因此不能把 T.18 改写成 $(P_R^M)^{2/3} \gtrsim \mathfrak{L}_{1,R}^K(D)$。

同样不能把 T.17 中的 completed clock K_k 替换成 Step 18 的 stopped forward-flux functional $\mathfrak{H}^{\mathrm{fix}}$。两者之间已证明的桥必须保留 Step 18 的 payment terms 和原方向。

指数 dwell threshold

沿 $L_{1,n} \to \infty$ 的序列，采用既有相邻壳参数

$$L_2 = 2L_1, \qquad \Gamma_2 = e^{-c_\gamma L_2^2}, \qquad S = \log(1/R),$$

其中

$$c_\gamma = \frac{8}{3969}, \qquad a_S = \frac{75}{22528}. \tag{T.21}$$

R0.74F 的 inherited sufficient bridge-survival window 是 $S - a_S L_1^2 \to -\infty$。它只是该证明的充分条件，不是所有 packet 的必要条件。记

$$d_L := a_S L_1^2 - S \to +\infty. \tag{T.23}$$

代入 Λ_2 后得到精确恒等式

$$\boxed{\log \Lambda_2 = \frac{2}{3}\left[\log \theta + (5c_\gamma - a_S)L_1^2 + d_L - \frac{1}{2}\log L_2\right].} \tag{T.24}$$

指数余量严格为正：

$$\boxed{5c_\gamma - a_S = \frac{603445}{89413632} > 0.} \tag{T.25}$$

所以 $\theta = 1$ 时 $\Lambda_2 \to \infty$。更一般地，如果要让 $(P_{R_n}^M)^{2/3}/h_{*,n}$ 保持有界，T.18 首先强迫 $\Lambda_{2,n} = O(1)$，进而得到必要的 dwell ceiling

$$\boxed{\theta_n \le C L_{2,n}^{1/2} e^{-(5c_\gamma - a_S)L_{1,n}^2 - d_{L,n}}.} \tag{T.28}$$

等价的对数形式为

$$\boxed{\log \theta \le -(5c_\gamma - a_S)L_1^2 - d_L + \frac{1}{2}\log L_2 + O(1).} \tag{T.29}$$

既有 R0.74F/R0.74Q lobe interval 的 $\theta = 1$。任意移动内叶都不会进入 T.24，也不能改变结论。真正的逃逸必须让 maximal comparable-floor persistence 本身指数缩短，而不是只在分析中截取一个早已长时间存在的 lobe。

Packet-amplitude 恢复与 abstract sharpness

在 exact common-shear packet family 中，all-lobe dominance 给出 $|u| \ge c_0 \mathfrak{a}_2$，因此

$$h_2 \ge c\Gamma_2 \mathfrak{a}_2^2 L_2 R^2. \tag{T.30}$$

代入 T.9，恢复任意 normalized dwell 下的 amplitude formula

$$P_R^M \ge c\theta \mathfrak{a}_2^3 \Gamma_2^{1/4} L_2 R^4. \tag{T.31}$$

$\theta = 1$ 时这正是 R0.74Q (Q.168)。关键澄清是：coercive step 只需要 outer lobe，不需要它与其他目标同时出现。

T.9 的各个幂次在纯测度论假设下是 sharp 的。取一个时空矩形，在其上令 lobe 体积、时间长度、shell weight 固定，并让向量场为常数，空间 Hölder 取等号，便精确得到

$$\theta^1 h_2^{3/2} R^1 \Gamma_2^{-5/4} L_2^{-1/2}. \tag{T.33}$$

固定瞬时高度而令 $\theta \downarrow 0$，cubic payment 随 θ 线性消失，说明 peak-only estimate 不能替代 persistence input。这些矩形只是 ABSTRACT SHARPNESS TESTS：不是周期散度自由解，不实现 Version-M ledger，也不是 Navier--Stokes counterexample。

同一个 exact common-shear 解中的 asynchronous lobe construction

沿上一节序列，另行保留中央 chart 与 common-shear platform 条件：$L_2 = 2L_1$、$R \le 1/32$、$L_2 R \le 5/144$、$L_1 \ge 9216$，以及 $R^{-1} e^{-a_S L_1^2} \to 0$ 。

在 terminal slab $I_R = (64R^2, 65R^2)$ 内任选 $0 < \theta_i \le 1$ 与 terminal times τ_i，使

$$J_i = (\tau_i - \theta_i R^3, \tau_i) \subset I_R. \tag{T.35}$$

定义相应水平预移位

$$q_{\mathrm{pre},i} := -B \int_0^{\tau_i} \theta_R(s, y_i^\circ)\, ds, \qquad Q_i(t) = q_{\mathrm{pre},i} + B \int_0^t \theta_R(s, y_i^\circ)\, ds. \tag{T.36}$$

于是 $Q_i(\tau_i) = 0$，并且在 J_i 上有 $|Q_i(t)| \le R/64$。水平平移与 common scalar advection-diffusion equation 对易，inversion partner 保持完全奇对称，因此有限和

$$u = (\mathfrak{a}_1 G_1 + \mathfrak{a}_2 G_2, b, 0), \qquad p = 0 \tag{T.39}$$

仍是同一个 exact smooth periodic mean-zero unforced Navier--Stokes solution。这里没有把在不同 shear 下演化的两个独立解相加。

既有 shear error、heat-age reserve、inversion suppression、vertical cross-tail 与 annular margins 对两枚 admissible target times 都统一成立。特别地，每个 J_i 都携带目标 lobe，且

$$K_{k_i,R}(t) \ge c A_*^2 R^2 = cT, \qquad \mathfrak{L}^K_{1,R}(I_R) \ge cT. \tag{T.41}$$

这个结论只允许在 stated slab 内选择相对 schedule；它不是已演化解的独立时间平移，也不是任意实目标时间定理。

两个严格分离的 unit-dwell windows 仍然昂贵

当 $R < 1/3$ 时取

$$\theta_1 = \theta_2 = 1, \qquad \tau_1 = 64R^2 + 2R^3, \qquad \tau_2 = 65R^2. \tag{T.42}$$

则

$$J_1 = (64R^2 + R^3, 64R^2 + 2R^3), \qquad J_2 = (65R^2 - R^3, 65R^2)$$

都位于 I_R，且两者间隔精确为 $R^2 - 3R^3 > 0$。它们是同一 common-shear solution 中两个真正互不相交的 R^3-long lobe windows。

然而 outer interval 仍有 $\theta_2 = 1$，所以 T.24-T.26 给出

$$\frac{(P_R^M)^{2/3}}{T} \longrightarrow \infty. \tag{T.43}$$

这个 construction 证明 asynchronous clock floors 可以实现，却不能把这些 floors 变成 low-payment witness。它排除的只是“普通 R^3 驻留加错峰”这一条明确机制，不是所有可能的 asynchronous PDE 构造。

一手文献的有限未命中边界

Hölder 本身与把非负积分限制到一个可测集合，都是经典测度论事实，不带 novelty claim。2D3C/passive-component reduction、parallel Kelvin waves、prescribed periodic shear 下的 scalar dispersion、forced passive-scalar mixing blocks、alternating-shear time schedules，以及 physical-shell flux locality，也都必须作为既有机制归属到相应文献。

本次两轮有界检索只核对六组一手来源：Singh-Sridhar；Biferale-Buzzicotti-Linkmann；Jiménez-Urias-Haine；Bruè-De Lellis；Bruè-Colombo-Crippa-De Lellis-Sorella；Dascaliuc-Grujić。

检索没有找到同时包含以下六项的 theorem：同一 exact common-shear/unforced solution 内的多个 passive packets；stated slab 内可独立选择的目标窗口；统一的 total-field physical lobe floor；对非负 weighted exterior $|u|^3$ payment 的转换；互不相交时间窗口仍给 completed-clock fixed-deletion floor；由既有 shell weight 与 survival window 导出的指数 dwell threshold。

这个结论只能写成 six-source bounded non-hit。它不证明 novelty、priority、correctness、optimality 或 publishability，也没有穷尽 MathSciNet、zbMATH、完整引用图、学位论文、非英语来源和未公开材料。LITERATURE BOUNDARY。

Claim ledger、证书与下一边界

- PROVED：T.9-T.10 的 exact outer-lobe Hölder coercivity，以及在 lobe hypotheses 保持时对相对 target-time schedule 的不变性。
- PROVED：同一个 exact common-shear solution 中两个 disjoint admissible lobe windows，T.34-T.43。
- PROVED：T.17 的 two-clock completed-height witness；对象严格是 K-clock，不是 stopped flux。
- PROVED：T.24-T.29 的 logarithmic dwell identity 与必要 collapse threshold；并且 measure-theoretic class 内各幂次 sharp。
- INHERITED：Version-M payment、shell weights、exact common-shear finite-packet solution、R0.74F/R0.74Q lobe placement 与 sufficient bridge-survival proof window。
- FINITE COMPUTATION：Python 31/31 groups、18,933 cases；independent Ruby 11/11 groups、9,201 assertions；Python/Ruby 分别拒绝 26/26 与 27/27 mutations；三组 PYTHONHASHSEED 与独立 Ruby regeneration 均逐字节一致。
- FIGURE：25 文件、47 checks、18/18 deterministic-core hashes；ANALYTIC SCHEMATIC / DERIVED ANALYTIC VALUES / NOT PDE DATA / NOT DNS / NOT CLAY。
- OPEN：stated slab 外的 scheduling；指数短 maximal comparable-floor persistence 的 PDE 构造；full $\mathfrak{L}^K_{1,R}(D)$ 的 payment-scale upper bound；off-target clocks 与 accumulated dissipation；无 Step 18 payment terms 的 K-to-$\mathfrak{H}^{\mathrm{fix}}$ bridge；fixed-deletion、direct hybrid、Q.12、Q.1、scale contraction、regularity 与 singularity。

下一条可检验路线必须真正构造 maximal outer comparable-floor persistence 指数缩短、且更长区间上没有可比 floor 的 packet；或者放弃一个既有 shell-weight、survival 或 lobe-floor 假设。只改变两个普通 R^3-long lobes 的先后顺序，或在分析中截短它们，不再是一条可行逃逸路线。

本节是 local coercivity theorem 与 route reduction，不是 full-clock theorem，更不是 Millennium problem 的解答。NOT CLAY。

本节结论：先分清两个时间集合

Step 20 对冻结的 canonical common-shear packet architecture 证明一个明确的运动学事实：包中心以 `R^-2` 量级穿过水平空间，而物理 annulus 给它 `L_i R` 量级的中心余量，因此显式认证几何走廊 `R_i^cert` 的时间尺度是 `L_i R^3`。这是对几何走廊的双边结论。completed-clock 的 `K_{k_i,R}` 超水平集可能更大，本节对它仅证明包含关系与下测度界；没有 converse，也没有上测度界。

冻结架构与完整终端时间片

对象仍是同一个 exact smooth periodic、mean-zero、unforced common-shear Navier–Stokes 解，压力为零；两个 inversion-paired derivative-heat packets 共用一个 shear coefficient。对 `i=1,2`，令 `L_2=2L_1`、`L_1>=9216`，并在终端片 `I_R=(64R^2,65R^2)` 上追踪包中心 `Q_i(t)`。平台校准给出严格单调速度区间

$$\frac{1-\varepsilon_i}{128R^2} \le Q_i'(t) \le \frac{1}{128(1-\varepsilon_1)R^2}, \qquad \varepsilon_i = 4e^{-a_D L_i^2} < \frac{1}{4}.$$

这里没有把先前较短的 `R^3` 窗口悄然放大；完整时间片上的 packet survival、inversion suppression、cross-packet tail 与 periodic remainder 分别由冻结主文列出的全时间输入重新核对。

物理 annulus 的精确中心余量

选定移动 lobe box `Omega_i(t)`，其体积恰为 `L_iR^3/16`。显式中心余量

$$A(L) = \sqrt{\left(\frac{2}{\lambda}\right)^2 - \frac{1}{256} - \left(c_h + \frac{1}{L}\right)^2} - \frac{b_2}{L}$$

在 `L>=9216` 时满足 `3/8<A(L)<1`。因此，只要 `|Q_i(t)|<A(L_i)L_iR`，整个 lobe 都位于选定的物理空间 annulus `A_{k_i}(R)` 内，并且 cutoff `Psi_{k_i}^R` 在 lobe 上恒为一。这是 physical-space shell 陈述，不是 Fourier frequency shell，也不是对所有可能中心位置的最优刻画。

认证几何走廊的双边尺度

定义完整的显式充分条件预像

$$\mathscr{R}_i^{\mathrm{cert}} = \{t \in I_R : \ |Q_i(t)| < A(L_i)L_iR\}.$$

中心余量除以速度给出 `A(L_i)L_iR^3` 量级；即使零点落在时间片端点，至少一侧仍保留足够时间。精确的 slab truncation 与单调预像估计得到

$$\frac{72}{5}L_iR^3 \le |\mathscr{R}_i^{\mathrm{cert}}| \le \min\left\{R^2, \frac{256A(L_i)}{1-\varepsilon_i}L_iR^3\right\} < \frac{1024}{3}L_iR^3.$$

所以 `|R_i^cert|=Theta(L_iR^3)`。上式右端只约束这个认证几何走廊；它不是 maximal physical residence，更不是完整 `K`-超水平集的上界。

从总场 lobe floor 到 completed-clock：只有下包含

在认证走廊内，direct packet、inversion partner、另一 packet 与 periodic copies 的比较都在总场中完成，得到 `|u(t,x)|>=c a_i` 的 lobe floor。时间 cutoff 与空间 cutoff 在 lobe 上等于一，而 completed clock 的其余项非负，故

$$
\mathscr{R}_i^{\mathrm{cert}} \subset \{t \in I_R : K_{k_i,R}(t) \geq c_K T\}, \qquad \left|\{K_{k_i,R} \geq c_K T\} \cap I_R\right| \geq \frac{72}{5} L_i R^3.
$$

这是本页最重要的量词边界：只证明 `corridor subset K-superlevel` 和 full `K`-superlevel 的 lower measure bound。accumulated dissipation、off-target endpoint rows、另一 packet 或 common shear 都可能让 `K` 在 lobe 离开后继续保持高值，因而不能反向包含，也不能把上一节的 `1024/3` 上界转移给 `K`。

认证驻留进入 cubic payment

令外包的 normalized certified dwell 为 `theta_cert,2=|R_2^cert|/R^3`，则

$$
\theta_{\mathrm{cert},2} \geq \frac{72}{5} L_2.
$$

R0.74T 的 measurable-lobe Hölder coercivity 可用于整个认证走廊。把 lobe floor `h_2>=cT` 与上述驻留下界代入，得到

$$
\frac{(P_R^M)^{2/3}}{T} \geq c R^{2/3} \Gamma_2^{-5/6} L_2^{1/3}.
$$

这里使用的是空间 Hölder 与非负积分对可测时间集的限制，二者都是经典工具；本节的新工作边界在冻结 PDE 架构、物理 lobe、总场比较、completed-clock 与 payment 量词的组合，而不是 Hölder 本身。

指数短驻留逃逸为何冲突

沿冻结尺度关系写 `S=log(1/R)`、`d_L=a_SL_1^2-S -> +infinity`。若反设 normalized payment 保持有界，R0.74T 的必要条件会强迫

$$
\theta_{\mathrm{cert},2} \leq C L_2^{1/2} \exp\left[-(5c_\gamma - a_S) L_1^2 - d_L\right], \qquad 5c_\gamma - a_S = \frac{603445}{89413632} > 0.
$$

这不是对真实走廊的无条件上界，而是“付款若有界”推出的必要上界。它与已证明的 `theta_cert,2>=(72/5)L_2` 指数级不相容。因此，frozen canonical common-shear architecture 中的 exponentially short-dwell escape 被关闭；结论没有推广到任意 shear、任意 packet 或任意 clock。

显式相位还可改善常数

对冻结显式相位 `tau_1=64R^2+2R^3`、`tau_2=65R^2`，一侧 slab room 可以直接加强：inner forward corridor 至少为 `(96/5)L_1R^3`，outer one-sided corridor 至少为 `(144/5)L_2R^3`。这些是认证几何走廊的改进 lower constants，不改变完整 `K`-超水平集仍然只有下界的结论。

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有限证书、图包与可复现边界

冻结 Python certificate 为 PASS：31/31 checks、869 个 exact finite cases；独立 Ruby audit 为 PASS：9/9 groups、1,651 个 Rational assertions。Python/Ruby mutation suites 分别拒绝 23/23 与 24/24 个故意错误，Python hash seeds 0、1、42 输出 byte-identical。期刊图包通过 47/47 validation checks，deterministic core 为 18/18，SVG、one-page PDF 与 600 dpi PNG 均绑定到冻结来源。

这些结果是 exact arithmetic、kinematic、structural、dependency 与 hash QA；图是 analytic schematic / derived analytic values，不是 PDE data，也不是 DNS。有限计算不替代 continuum PDE proof，图包 seal 也不认证数学正确性。

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文献近碰撞与停止线

有限一手来源筛查没有找到同时陈述六部分冻结组合的来源；这只是 bounded-search non-hit，不是 novelty、priority、correctness、nonexistence 或 publishability 证明。最接近的名称碰撞是 Inage（2026）的 coherent same-scale Fourier–helical triads “residence-time compression”：它对 low phase-drift set 给出 scale-decaying upper temporal estimate；这里则对 physical-space annulus 中的 canonical packet lobe 证明 lower residence，并只向 completed-clock 超水平集传递下界。对象、shell、假设与不等式方向都不同。

仍 OPEN：完整 `K`-超水平集上测度界；包含 off-target endpoint rows、viscous accumulation、cross terms 与 shear baseline 的 full completed-clock upper ledger；arbitrary-clock lobe extraction；high-Rayleigh / anomalous-defect；fixed deletion；direct hybrid；Q.12；Q.1；scale contraction；regularity 与 singularity formation。本节不是 suitable-weak 任意解定理，不是 Navier–Stokes counterexample，**NOT CLAY**。

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结论先行：这是路线备忘录，不是上界定理

R0.74V 把 completed-clock upper 的证明义务拆成可核查的账本。已建立的是精确分解、粗尺度预算、条件代数和失败条件；目标坐标上界与全壳层上界均未建立。R0.74U 的认证几何走廊仍只给 `K` 超水平集的单向包含和下测度界，不能反向使用。

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完整时钟的三行非负账本

在 local-energy good times，完整时钟由端点动能、累积普通黏性和累积异常缺陷三行组成。对一般 suitable weak solution，hard time 必须使用 canonical absolutely continuous representative；不能把原始端点表达式强行解释为处处成立。冻结光滑族的异常缺陷为零，但推广时仍须单独保留。

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shear、packet 与交叉项

对精确 common-shear 解，速度分量正交给出

$$K_{k,R} = K^b_{k,R} + K^G_{k,R}.$$

展开两个 packet 后，端点与黏性交叉项有符号，但 Cauchy-Schwarz/Young 将它们安全吸收到两条对角行：

$$K_{k,R}(t) \le K^b_{k,R}(t) + 2\sum_{m=1}^{2}\big(E^m_k(t) + D^m_k(t)\big).$$

这解决了点态上界中的交叉项，却不自动控制 positive variation。

lifted multiplicity 不能被 torus 长度截断

周期化 cutoff 是欧氏 lift 的和，不是投影后的 `0-1` 指示函数。令 `s_k=(2^{k+1}+1/8)R`，正确的一维弦长预算是

$$\ell_k = s_k + s_k^3.$$

同理，体积使用精确平铺恒等式

$$\int_{\mathbb{T}^3} \Psi^R_k = \int_{\mathbb{R}^3} \psi^R_k,$$

而不是用一个 torus 体积上限抹去 lifted multiplicity。

已有的粗尺度预算

全局 packet 能量与 lifted chord 给出 `H_{k<-m}=gamma_k a_m^2 ell_k R` 级别的端点加黏性上界。common-shear 则形成持续基线；目标壳层上其归一化尺度为 `Xi_i^sh ~ Gamma_i B^2 L_i^3/A_*^2`。若这条基线没有严格低于 `kappa T`，时长上界可能直接失败。全壳层 shear 求和还要求 `B^2/A_*^2` 受控，因此不能对 R0.74U 允许的任意 `A_*>0` 给出统一结论。

V.47-V.50：有限表 occupation 仍是开放输入

当前只提出六个 central-chart pair 上的 whole-annulus moving-centre `L^2/H^1` occupation 估计。它必须同时处理 inversion partners、periodic copies、weighted remainders、累积黏性和 derivative collar。现有 near-lobe 比较只覆盖主叶附近，不能替代这组全 annulus 估计。V.47-V.50 仍为 OPEN；任何 all-k 使用还须另证 lifted-copy summation。

V.56 只有条件代数，没有解析闭合

若 common-shear、两条累积黏性与异常缺陷组成的 persistent baseline 小于 `kappa T/2`，并且 V.47-V.50 的 weighted remainder gates 全部通过，则 union bound 把目标坐标超水平集约化到两个 packet endpoint 集合，形式上得到

$$|\{t \in I_R : K_{k_i,R}(t) \geq \kappa T\}| \leq C_\kappa L_i R^3.$$

这就是 V.56 的条件路线。因为 occupation inputs 尚未证明，V.56 本身仍为 OPEN。

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adjacent-inward free comparator 给出正指数

在相邻内壳层，权重增益与自由热核尾部代价组合成

$$\chi(65) = \frac{12191}{132088320} > 0.$$

这是 exact finite arithmetic，说明 free comparator 预测一个指数放大的 inward tail。但它不是 common-shear solution 的下界。还必须证明 remote strip 上的相对 bridge comparison、inversion control 和另一 packet 的 noncancellation；因此不能据此宣布全壳层反例。

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七类失败条件

路线会在以下任一处失败：common-shear floor 过高；黏性或异常缺陷形成永久基线；加权尾部余项不可积；广义尺度过慢；把 terminal plateau 错当成完整 cutoff interval；混淆 height、duration 与 variation；或在大壳层错误复用单一 central-lift distance。每一项都必须在定理陈述中成为显式假设或已证估计。

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停止线与下一项研究

下一项最小命题是 common-shear remote adjacent-inward comparison；之后才是六对有限表的 weighted annular occupation。以下全部保持 OPEN：V.47-V.50、V.56、所有 all-k lifted-copy extension、common-shear remote/adjacent-inward comparison、all-shell matching upper、fixed deletion、arbitrary-clock extraction、scale contraction、regularity、singularity 与 Navier-Stokes Millennium problem。本发布没有文献审计，不作 novelty、priority 或 publishability 判断；没有 DNS、仿真或 PDE 数据。正式图件：NOT APPLICABLE。本节纯解析，没有 Navier--Stokes 数值仿真、DNS、DGX 或正式图件。**NOT CLAY.**

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结论先行：严格阈值，而不是近叶估计的直接复用

本节在冻结的 two-packet exact smooth common-shear family 内，完成 R0.74V 的第一个 remote endpoint comparison。结论是相对于 free derivative-heat comparator 的 survival/sweeping 尺度二分；它不是绝对 (o(1)) 代换，也不是任意 suitable weak solution 的结论。

$$p = \frac{1}{\lambda} = \frac{32}{63}, \qquad q(\ell) = \frac{p^2}{4\ell}, \qquad q_{64} = \frac{4}{3969}, \qquad q_{65} = \frac{256}{257985} < q_{64}.$$

在 central conditional bridge 概率下，若 (t=\tau_m=\ell R^2)、(64\le\ell\le65)，则

$$\mathbb{P}^{\rm br}_{0,y}\left\{e^{-(q(\ell)+\eta)L^2}\le\mathfrak{S}_t\le e^{-(q(\ell)-\eta)L^2}\right\}\longrightarrow 1\qquad(\eta>0).$$

这是概率意义的 logarithmic asymptotic，不是 deterministic prefactor asymptotic。位移 deficit 使用 age (t=\ell R^2)；free heat comparator 的总 age 是 (T=(\ell+1)R^2)，两者不得混同。

冻结 family 与显式 remote strip

沿用

$$\lambda=\frac{63}{32},\quad c_h=\frac{15}{16},\quad d=c_h-p=\frac{433}{1008},\quad L_2=2L_1,\quad h_m=c_hL_mR,$$

以及 (L_1\ge9216)、(L_2R\le5/144) 和 (R^{-1}e^{-a_SL_1^2}\to0)，其中 (a_S=75/22528)。在 (t=\tau_m) 时定义

$$\mathcal{S}_m=\left\{x:\ |x_1|<\frac14\sqrt{pL_m}R,\ \frac54R<x_2<\frac32R,\ pL_mR-R<x_3<pL_mR-\frac12R\right\}.$$

直接几何计算给出

$$\mathcal{S}_m\subset A_{k_m-1}(R),\qquad \Psi^R_{k_m-1}=1\text{ on }\mathcal{S}_m,\qquad |\mathcal{S}_m|=\frac1{16}\sqrt{pL_m}\,R^3.$$

令 (z=x_2)、(y=x_3-h_m=-(dL+\delta)R) 且 (1/2<\delta<1)。free comparator 为

$$H_m(t,z,y)=R^3\partial_zK^{\rm per}_{R^2+t}(z)K^{\rm per}_{R^2+t}(y),\qquad |H_m|\asymp e^{-(dL+\delta)^2/[4(1+\ell)]}.$$

精确 all-winding 条件桥表示

对 direct positive packet，时间反向 Feynman--Kac 表示保留每一个 vertical winding：

$$G^+_m(t,Q_m(t)+z,h_m+y)=R^3\sum_{n\in\mathbb{Z}}w_n\,\mathbb{E}^{\rm br}_{n,y}\left[\partial_zK^{\rm per}_T(z+\mathfrak{S}_t)\right],$$

$$w_n=k_T(2\pi n-y),\qquad T=t+R^2,\qquad \mathfrak{S}_t=B\int_0^t[\theta_R(t-s,h_m)-\theta_R(t-s,h_m+Y_s)]ds.$$

相应 free comparator 是

$$H_m=R^3\partial_z K_T^{\mathrm{per}}(z)\sum_{n\in\mathbb{Z}}w_n.$$

因此比较必须除以完整 winding sum。非中心质量没有被删除，而是满足

$$\omega_{\mathrm{per}}:=\frac{\sum_{n\neq 0}w_n}{w_0}\le C\exp\left[-\frac{1}{11R^2}\right]\le Ce^{-75L^2}=o(1).$$

remote saturation deficit 与短时层

令 (A(r,x)=1-\theta_R(r,x)\ge0)。上、下周期 Gaussian 界在 (x=pLR+O(R))、(r=\ell R^2) 给出同一指数：

$$-\frac{1}{L^2}\log A(\ell R^2,pLR+O(R))\longrightarrow q(\ell)=\frac{p^2}{4\ell}.$$

central bridge 的半群恒等式为

$$\mathbb{E}^{\mathrm{br}}_{0,y}A(t-s,h_m+Y_s)=A(t-s+v_s,h_m+\mu_s),\quad \mu_s=\frac{T-s}{T}y,\quad v_s=\frac{s(T-s)}{T}.$$

指数差把积分局部化到 (s\asymp R^2/L^2) 的短时层，因而

$$\mathbb{E}^{\mathrm{br}}_{0,y}|\mathfrak{S}_t|\le CL^{-2}e^{-q(\ell)L^2+CL}+Ce^{-c_h^2L^2/260+CL}.$$

反向的高概率下界来自同一短时层；对足够小的固定 (epsilon>0)，

$$\mathfrak{S}_t\ge cL^{-2}\exp\left[-\frac{(p+\epsilon)^2}{4\ell}L^2-CL\right]>0$$

以 (1-Ce^{-c\epsilon^2L^4}) 的 central-bridge 概率成立。这里没有假设冻结 saturation profile 单调。

relative survival、uniform slab 与 sweeping

若

$$\frac{1}{RL^2}e^{-q(\ell)L^2+C_0L}\longrightarrow 0,$$

则在 remote strip 上

$$\sup_{x\in\mathcal{S}_m}\left|\frac{G_m^+(t,x_2,x_3)}{H_m(t,x_2,x_3-h_m)}-1\right|\longrightarrow 0.$$

特别地，uniform slab survival 的充分条件是

$$\limsup\frac{\log(1/R)}{L^2}<q_{65}.$$

反之，若 (\ell_j\to\ell_\infty\) 且

$$\liminf\frac{\log(1/R_j)}{L_j^2}>q(\ell_\infty),$$

则 (G_m^+/H_m\to0)；在全 slab 上统一成立的充分条件是

$$\liminf\frac{\log(1/R)}{L^2}>q_{64}.$$

窄带 (q_{65}\le\log(1/R)/L^2\le q_{64}) 不能整体称作"未分类"：固定极限 (\ell) 时，与 (q(\ell)) 的严格比较仍决定一侧；只有 equality 及其 critical law 保持 OPEN。

inversion、另一 packet 与 amplitudes 之后的非抵消

物理第一分量是

$$U=\mathfrak{a}_1(G_1^++G_1^-)+\mathfrak{a}_2(G_2^++G_2^-),\qquad \frac{\mathfrak{a}_2}{\mathfrak{a}_1}=2^{-1/2}e^{3q_{64}L_1^2}.$$

inversion partner 与 cross-packet 项必须在插入实际 amplitudes 后比较。冻结精确 margins 为

$$\frac{5}{693}>0,\qquad \delta_{1\leftarrow2}=\frac{100043}{29804544}>0,\qquad \delta_{2\leftarrow1}=\frac{3667}{17611776}>0.$$

连同 periodic-copy reserve，它们给出

$$\frac{\mathfrak{a}_{3-m}|G_{3-m}| + \mathfrak{a}_m|G_m^-|}{\mathfrak{a}_m|H_m|} \longrightarrow 0.$$

这是一条 amplitude-weighted relative noncancellation statement，不是未加权的 absolute (o(1))。

冻结尺度上两 packet 的相反结果

对 outer packet (m=2)，

$$4q_{65} - a_S = \frac{3719797}{5811886080} > 0$$

把 R0.74U reserve 转成全 slab survival，故

$$\sup_{x\in\mathcal{S}_2}\left|\frac{U(t,x)}{\mathfrak{a}_2H_2(t,x_2,x_3-h_2)} - 1\right| \longrightarrow 0.$$

inner packet 在一般冻结假设下没有统一结果。对原始尺度 (R=e^{-L_1^2/320})，精确 margins

$$\frac{1}{320} - q_{64} = \frac{2689}{1270080} > 0, \qquad q_{65} - \frac{1}{1280} = \frac{13939}{66044160} > 0$$

给出 packet 1 swept、packet 2 survives：

$$\frac{U}{\mathfrak{a}_1H_1} \longrightarrow 0 \text{ on } \mathcal{S}_1, \qquad \frac{U}{\mathfrak{a}_2H_2} \longrightarrow 1 \text{ on } \mathcal{S}_2.$$

加权端点发散与 fixed-deletion 边界

相邻内壳权重满足

$$\frac{\gamma_{k_m-1}}{\Gamma_m} = e^{(3/4)c_\gamma L_m^2}.$$

packet 2 的 relative survival、remote strip 体积和 completed-clock 非负 endpoint row 合并为

$$\boxed{\frac{K_{k_2-1,R}(\tau_2)}{T_*} \geq cL_2^{-1/2}e^{\chi(65)L_2^2-CL_2} \longrightarrow \infty, \qquad \chi(65)=\frac{12191}{132088320}>0.}$$

因此 matching all-shell (O(T_*)) upper bound 对这个 frozen placement 为 FALSE。但唯一发散坐标是 (k_2-1=k_1)，one-shell fixed deletion 可以恰好删去它；所以 fixed-deletion obstruction、whole-shell upper、time occupation、accumulated viscosity 与 target-coordinate duration 都没有被证明。

四联图、证书与文献边界

冻结四联图只展示 derived analytic values：Panel A 是 physical-shell schematic；Panel B 是 (q(\ell)) 阈值；Panel C 是 exact all-winding bridge proof map；Panel D 是 leading analytic endpoint factor。它没有 sampled path、PDE data、DNS 或 finite-(L_2) numerical lower certificate。

- primary analytic audit：PASS，0 blockers；
- Python certificate：33/33 checks，33 exact cases；
- independent Ruby：6/6 groups，56 assertions；
- negative mutations：Python 23/23、Ruby 24/24 rejected；
- seeds 0、1、42 与 independent regeneration：byte-identical；
- literature：截至 2026-09-03 的 bounded primary-source screen 仅给 finite non-hit，不证明 novelty、priority、correctness、nonexistence 或 publishability。

停止线与开放问题

本节只对一个 frozen exact smooth common-shear family 建立 remote adjacent-inward relative comparison，并由 packet 2 导出一个 frozen-placement all-shell upper obstruction。

- PROVED：exact all-winding disintegration；central-bridge logarithmic rate；严格阈值两侧的 relative survival/sweeping；amplitude 后的 inversion/cross-packet noncancellation；packet-2 加权 endpoint divergence。
- OPEN：critical equality law；fixed deletion；whole-shell (H^1) occupation；time occupation；positive-variation upper；accumulated viscosity；payment normalization；arbitrary-clock extraction；scale contraction；一般 suitable weak solutions；regularity 与 singularity。
- PUBLICATION BOUNDARY：ANALYTIC SCHEMATIC | DERIVED ANALYTIC VALUES | NOT PDE DATA | NOT DNS | LOCAL DIRECT TRANSLATION | NO DGX | NOT CLAY。

这不是 Navier--Stokes Millennium problem 的解决，也不声称构造了一般解的反例。**NOT CLAY.**

结论先行：两坐标 endpoint obstruction 已证，actual gate 反例未证

本节只讨论一个 frozen three-packet exact smooth periodic common-shear family。它把 R0.74W 的一个发散坐标扩展为两个不同的发散坐标，但随后证明这两个 audited W-strip witnesses 仍不足以击败 actual cubic-payment normalization。

$$\boxed{\begin{array}{c}\textbf{TWO-COORDINATE ENDPOINT OBSTRUCTION RELATIVE TO } T_*\textbf{: PROVED,}\\ \textbf{ACTUAL } (P_R^M)^{2/3}\textbf{-NORMALIZED COUNTEREXAMPLE: NOT PROVED,}\\ \textbf{EQUAL-TARGET W-STRIP ROUTE: NO-GO BY CUBIC PAYMENT.}\end{array}}$$

这不是 whole-shell clock upper/lower theorem，不控制 accumulated dissipation，也不是任意 suitable weak solution、regularity 或 singularity 结论。**NOT CLAY.**

三 packet 精确 family 与共同归一化

取

$$k_2=k_1+1,\qquad k_3=k_1+2,\qquad L_2=2L_1,\qquad L_3=4L_1,$$

并在同一 heat-evolved shear (b) 下重演化三个 inversion-paired derivative-heat packets：

$$U_3=\sum_{m=1}^{3}\mathfrak{a}_mG_m,\qquad u^{(3)}=(U_3,b,0),\qquad p^{(3)}=0,$$

$$\Gamma_m=e^{-c_\gamma L_m^2},\qquad \mathfrak{a}_m=A_*(\Gamma_mL_m)^{-1/2},\qquad \Gamma_m\mathfrak{a}_m^2L_mR^2=A_*^2R^2=:T_*.$$

这是 exact smooth periodic unforced Navier--Stokes solution。equal target-clock normalization 不等于 equal raw-energy normalization；改变共同振幅 (A_*) 不能修复后文的 payment ratio。

remote strips、relative survival 与完整 cross audit

对 (m=2,3)，使用 R0.74W 的 adjacent-inward strip

$$\mathcal{S}_m=\left\{x:\ |x_1|<\frac14\sqrt{pL_m}R,\ \frac54R<x_2<\frac32R,\ pL_mR-R<x_3<pL_mR-\frac12R\right\}.$$

outermost chart 条件 (L_3R\le5/144) 与 inherited U-reserve 给出

$$4q_{65}-a_S=\frac{3719797}{5811886080}>0,\qquad 16q_{65}-a_S=\frac{72925813}{5811886080}>0.$$

因此 packets 2、3 都在 full closed slab 上 relatively survive。插入真实 amplitudes 后，四个正 cross margins 为

$$\delta_{2\leftarrow1}=\frac{3667}{70447104},\quad \delta_{2\leftarrow3}=\frac{100043}{29804544},\quad \delta_{3\leftarrow2}=\frac{3667}{70447104},\quad \delta_{3\leftarrow1}=\frac{147359}{281788416}>0.$$

inversion margin (5/693>0)，noncentral winding reserve (123450676/1091475>0)。所以 inversion partners、相邻与非相邻 cross packets、全部 periodic windings 都已按 amplitude-weighted 比较控制；这里没有 diagonal-only 假设。

两个不同坐标的 T-star endpoint 发散

adjacent-shell weight、strip volume、free kernel 与共同 (T_*) 归一化给出

$$K_{k_m-1,R}(\tau_m) \geq cT_*L_m^{-1/2}e^{\chi(65)L_m^2-CL_m}, \qquad m=2,3,$$

$$\chi(65)=\frac{12191}{132088320}>0.$$

因为 (k_2-1=k_1)、(k_3-1=k_2)，得到两个 distinct coordinates：

$$\frac{K_{k_1,R}(\tau_2)}{T_*}\longrightarrow\infty, \qquad \frac{K_{k_2,R}(\tau_3)}{T_*}\longrightarrow\infty.$$

这些是 explicit strip lower witnesses。它们不提供任一 whole-shell upper bound。

fixed deletion 的量词次序：时间可以不同

实际 budget-one deletion functional 是

$$\boxed{\mathfrak{L}^K_{1,R}(\mathcal{D})=\inf_{\substack{S\subset\mathbb{N}\\ \#S\leq 1}}\sup_{t\in\mathcal{D}}\sum_{k\notin S}K_{k,R}(t).}$$

删除集 (S) 必须先固定，之后才取时间上确界。于是无论删除哪个坐标，另一个 witness 都可在自己的时间被选中：

$$\mathfrak{L}^K_{1,R}(\mathcal{D})\geq\min\{K_{k_1,R}(\tau_2),K_{k_2,R}(\tau_3)\}.$$

令 (\mathcal D=\mathcal T_R)，便有

$$\boxed{\frac{\mathfrak{L}^K_{1,R}(\mathcal{T}_R)}{T_*}\longrightarrow\infty.}$$

(\tau_2) 与 (\tau_3) 可以不同；simultaneity 不是证明所需。可选地令二者相等，也可得到同一 smooth time 的 simultaneous vector-height statement。若 deletion set 被错误地允许依赖时间，上述 pigeonhole 会失效。

actual gate 使用 cubic-payment normalization

真正待检验的不等式不是 (O(T_*))，而是

$$\mathfrak{L}^K_{1,R}(\mathcal{T}_R) \le C_L^{2/3}(P_R^M)^{2/3}.$$

packet 3 的 target lobe 在 payment radius (2R) 上满足

$$A_{k_3}(R) = A_{k_3-1}(2R), \qquad \gamma_{k_3-1} = \Gamma_3^{1/4}.$$

三 packet amplitude-weighted target-lobe dominance 与 nonnegative exterior velocity-cubic row 强制

$$P_R^M \ge c\mathfrak{a}_3^3\Gamma_3^{1/4}L_3R^4 = cA_*^3R^4\Gamma_3^{-5/4}L_3^{-1/2},$$

所以

$$\boxed{\frac{(P_R^M)^{2/3}}{T_*} \ge cR^{2/3}L_3^{-1/3}e^{(5/6)c_\gamma L_3^2}.}$$

该比值中的 (A_*) 精确抵消，不能通过共同振幅调节消除。

payment rate 严格压过两个 audited W-strip rates

写 (\rho_R=\log(1/R)/L_1^2)。U-reserve 给出 (\rho_R\le a_S+o(1))，从而 payment lower rate 是

$$\frac{40}{3}c_\gamma - \frac{2}{3}a_S = \frac{3306805}{134120448} > 0.$$

最大的 audited W-strip exponent 是

$$16\chi(66) = \frac{244208}{134120448}, \qquad \chi(66) = \frac{15263}{134120448}.$$

二者严格 gap 为

$$\boxed{\frac{3306805 - 244208}{134120448} = \frac{3062597}{134120448} > 0.}$$

因此对两个实际 strip integrals，

$$\boxed{\frac{E_2^{\rm strip}+E_3^{\rm strip}}{(P_R^M)^{2/3}} \longrightarrow 0.}$$

这只是 two-strip upper comparison，不是 whole-shell clock upper bound，也不排除尚未证明的 whole-shell 或 accumulated-dissipation effect。

precise no-go 与 next proposition X.52

本节精确排除的是 equal-target three-packet W-strip architecture：两个 endpoints 相对 (T_*) 发散，但外 packet 的 cubic payment 具有更大指数，因此这两个 witnesses 不可能反驳 actual normalized gate。

下一构造目标必须直接满足

$$\boxed{\frac{\min\{K_{r,R}(t_r),K_{s,R}(t_s)\}}{(P_R^M)^{2/3}} \longrightarrow \infty, \qquad r\neq s, \quad t_r,t_s\in\mathcal{T}_R.}$$

这就是 X.52。它不要求 (t_r=t_s)，但必须把两个 undeletable clock heights 与 outer exterior cubic payment 解耦。改变 amplitude law、shell placement、weight interaction 或 packet geometry 都需要新的 exact normalization、survival proof 与 all-cross-packet audit。

冻结四联图、证书与 bounded literature screen

四联图只编码 analytic scale index、different-time fixed-deletion pigeonhole、exact exponent comparison 与 claim hierarchy；没有 sampled trajectories、PDE data、DNS 或 simulation。

- primary analytic audit：PASS，0 blockers；
- Python certificate：31/31 checks，231 exact cases/assertions；
- independent Ruby：5/5 groups，36 assertions；
- negative mutations：Python 24/24、Ruby 25/25 rejected；
- seeds 0、1、42 与 independent regeneration：byte-identical；
- figure archive：25 files、3,096,940 bytes，deterministic 18/18，visual QA PASS；
- literature：截至 2026-09-03 的 bounded primary-source screen 仅给 finite non-hit，不证明 novelty、priority、correctness、nonexistence 或 publishability。

项目术语 fixed deletion、common-shear packet、simultaneous height、remote adjacent-inward 不能作为标准文献术语呈现。

停止线与开放边界

- PROVED：frozen exact three-packet smooth NSE family；packets 2、3 的 relative survival；两个 distinct (T_*)-normalized endpoint divergences；fixed-set/different-time pigeonhole；(\mathfrak L^K_{1,R}/T_*\to\infty)；两个 audited strip integrals 相对 actual cubic payment 可忽略。
- NOT PROVED：actual ((P_R^M)^{2/3})-normalized fixed-deletion counterexample；whole-shell clock upper/lower；accumulated-dissipation enhancement。
- NO-GO：equal-target W-strip route 被 outer exterior velocity-cubic payment 阻断。
- OPEN：payment-compatible two-coordinate construction X.52；positive-variation upper；accumulated viscosity；arbitrary-clock extraction；scale contraction；一般 suitable weak solutions；regularity 与 singularity。

- PUBLICATION BOUNDARY: ANALYTIC SCHEMATIC | DERIVED ANALYTIC VALUES | NOT PDE DATA | NOT DNS | LOCAL DIRECT TRANSLATION | NO DGX | NOT CLAY。

这不是 Navier--Stokes Millennium problem 的解决，也不声称构造了一般解的反例。**NOT CLAY.**

结论先行：冻结几何 no-go 已证，取消构造未证

本节检验的目标是

$$\frac{\min\{K_{r,R}(t_r),K_{s,R}(t_s)\}}{(P_R^M)^{2/3}}\longrightarrow\infty,\qquad r\neq s.$$

四条路线的结论必须分级读取：非等振幅本身在 frozen common-shear heat-packet geometry 中严格失败；非相邻 dyadic placement 更差；没有指数级 target-field cancellation 的几何修改无法让两个不同 W endpoints 同时通过；真正的指数级 field cancellation 只留下一个形式必要指数窗口，尚未构造。accumulated viscosity 分支只有 dimensional screen，没有严格的 (H^1)/occupation upper。

$$\boxed{\begin{array}{c}\textbf{FROZEN-GEOMETRY NO-GO: PROVED,}\\ \textbf{FORMAL CANCELLATION WINDOW: NECESSARY ONLY,}\\ \textbf{ACCUMULATED-VISCOSITY BRANCH: OPEN / NOT CERTIFIED.}\end{array}}$$

这是 route screen，不是 cancellation-cell theorem、payment-compatible counterexample、regularity 或 singularity 结论。**NOT CLAY.**

fixed-deletion functional 与不同 witness times

固定删除量是

$$\mathfrak{L}^K_{1,R}(\mathcal{D})=\inf_{\substack{S\subset\mathbb{N}\\ \#S\le 1}}\sup_{t\in\mathcal{D}}\sum_{k\notin S}K_{k,R}(t).$$

一个删除集必须在时间上确界之前选定。因此，两个不同坐标在两个可能不同的时间有下界时，已有

$$\mathfrak{L}^K_{1,R}(\mathcal{D})\geq\min\{K_{r,R}(t_r),K_{s,R}(t_s)\}.$$

不需要同一时间的向量下界。对 frozen exact smooth family，(K_{k,R}=K^b_{k,R}+K^G_{k,R}\geo)，anomalous-defect row 为零。后文所有 strip 结论都不被提升为 whole-shell estimate。

universal endpoint-versus-self-payment ledger

令一个 packet 的尺度为 (L)，高度为 (h=(p+d)LR)，其中 (p=32/63)。在 adjacent inward strip 上，W-type endpoint 的指数尺度是

$$E^{\mathrm{adj}} \asymp \mathfrak{a}^2 R^2 \Gamma^{1/4} \exp\left[-\frac{d^2}{2\mathsf{a}} L^2\right], \qquad \Gamma = e^{-c_\gamma L^2}, \quad c_\gamma = \frac{8}{3969}.$$

若该 packet 的 target lobe 没有被 actual field cancellation 消掉，nonnegative exterior velocity-cubic row 强制

$$P_R^M \gtrsim |\mathfrak{a}|^3 R^4 \Gamma^{1/4}, \qquad (P_R^M)^{2/3} \gtrsim \mathfrak{a}^2 R^{8/3} \Gamma^{1/6}.$$

于是振幅平方精确抵消，必要条件成为

$$\rho_L > \frac{3}{2}\Theta(d,\mathsf{a}), \qquad \Theta(d,\mathsf{a}) = \frac{c_\gamma}{12} + \frac{d^2}{2\mathsf{a}},$$

而 W bridge survival 要求 $(\rho_L<q(\ell)=p^2/(4\ell))$。关键是 bridge deficit age 为 (ℓ)，free heat age 为 $(\ell+1)$，不得合并分母。冻结高度下

$$\Xi_{\mathrm{fr}}(\ell) = \frac{p^2}{6\ell} - \frac{c_\gamma}{12} - \frac{d^2}{2(\ell+1)}$$

在 $([64,65])$ 上递增，并满足

$$\boxed{\max_{64\le\ell\le65} \Xi_{\mathrm{fr}}(\ell) = \Xi_{\mathrm{fr}}(65) = -\frac{875993}{968647680} < 0.}$$

所以 frozen geometry 中任何依赖同一 packet 的 W-type adjacent witness、同时支付该 packet target lobe 的方案，都不可能 payment-compatible。

Route A：非等振幅本身严格失败

把 packet (i) 的振幅写成 $(\log|\mathfrak a_i|=\alpha_iL_i^2+o(L_i^2))$。endpoint 与其 self-payment 的 two-thirds 指数分别含同一个 $(2\alpha_i)$，两者之差恒为

$$\kappa_i - \pi_i = \frac{2}{3}\rho_i - \Theta(d_i,\mathsf{a}_i).$$

因此 amplitude tilt 可以均衡两个 clocks，也可以改善 cross-packet dominance，却不能创造缺失的 self-payment gap。结合 frozen survival ceiling，结论是

$$\boxed{\text{NON-EQUAL AMPLITUDES ALONE: STRICT NO-GO.}}$$

此结论只覆盖 target lobe 未被实际场消去的 frozen geometry；它不是对所有可能 cancellation architecture 的否定。

Route B：非相邻 dyadic placement 更差

若 $L_{\rm out}=rL_{\rm in}$ 且 $r=2^j\ge2$，则 $\rho_{\rm out}=\rho/r^2$。outer packet 的必要 payment compatibility 要求

$$\rho > r^2\frac{3}{2}\Theta(d_{\rm out},\mathsf{a}_{\rm out}),$$

但 inner survival 要求 $\rho<q_{64}$。即使令 $d=0$，也有 $\Theta\ge c_\gamma/12$，在最有利的相邻 dyadic 情形 $r=2$ 正好得到边界等式

$$4\cdot\frac{3}{2}\cdot\frac{c_\gamma}{12}=\frac{c_\gamma}{2}=q_{64}.$$

两侧所需都是严格不等式，因此 $r=2$ 已失败，$r>2$ 只会更差。

$$\boxed{\text{NON-ADJACENT PLACEMENT: STRICT NO-GO.}}$$

失败来源是 outer packet 自己的 cubic payment，不是 large-packet cancellation。

Route C：只有 target field 的指数级取消未被排除

仅缩小几何间距 d 不能改变 $A_k(R)=A_{k-1}(2R)$ 与不可约的 $\Gamma^{1/4}$ payment weight；把 chord、volume 或 residence time 只改成 L 的多项式也不能改变指数符号。payment row 含 $|u|^3$，相反的 signed flux 不能抵消它；必须在整个 target spacetime box 上取消 actual field。

若 corrector 让 target-lobe residual field 减少 $e^{-\zeta_iL_i^2}$，并让 spacetime volume 再减少 $e^{-\omega_iL_i^2}$，必要 self-compatibility 变成

$$\boxed{\frac{2}{3}\rho_i-\Theta(d_i,\mathsf{a}_i)+2\zeta_i+\frac{2}{3}\omega_i>0.}$$

现有 heat-packet width 与 speed 只产生多项式变化，所以当前 architecture 中 $\omega_i=0$。screen 后唯一未被排除的机制是真正的指数级 field cancellation。它在 exact PDE algebra 上原则上可由同一 shear 下有限个 inversion-paired passive correctors 表述，但本节没有构造这种 corrector。

changed geometry 的形式必要指数窗口

形式 ledger 取

$$d=\frac{7}{32},\qquad \mathfrak{a}_0=\frac{131}{2},\qquad \rho=\frac{9}{10000},\qquad \sigma=\frac{203461}{473260032},\qquad \zeta=\frac{1}{5000}.$$

在这一组 rational data 下，formal W survival reserve、U-reserve、reference-height deficit separation 与两个 remote cross margins 都严格为正。outer amplitude tilt 后，两个 formal adjacent endpoints 具有相同指数；tilt 削弱的 cross margin仍为

$$\delta_{2\leftarrow 1}-\sigma=\frac{1893805}{8518680576}>0.$$

postulated outer target-lobe cancellation 超过最低必要值的余量为

$$\zeta-\left(\frac{\Theta_0}{2}-\frac{\rho}{12}\right)=\frac{16723709}{1996565760000}>0,$$

modeled outer-lobe payment 相对共同 endpoint height 的最后 gap 为

$$\frac{16723709}{249570720000}>0.$$

这些正分数只证明 necessary exponent inequalities 有非空窗口。尚缺 changed-height common-shear platform、central-reference comparison、all-winding survival、full-box cancellation cell、remote-strip negligibility、corrector 自身 payment control 和完整 (P_R^M) upper。这里没有 sufficient feasibility 或 constructed-family claim。

accumulated viscosity：dimensional screen，不是 no-go theorem

如果未来能证明 all-winding derivative 与 occupation uppers，moving remote packet 在 fixed (R)-width strip 中的 residence counting 会形式上给出

$$D_i^{\mathrm{adj}}\lesssim \mathrm{poly}(L_i)\,\mathfrak{a}_i^2R^3\Gamma_i^{1/4}e^{-d_i^2L_i^2/(2\mathfrak{a}_i)}.$$

与 cubic payment 比较的形式 rate 是

$$\frac{1}{L_i^2}\log\frac{D_i^{\mathrm{adj}}}{(P_R^M)^{2/3}}\le -\frac{1}{3}\rho_i-\Theta(d_i,\mathfrak{a}_i)+o(1)<0.$$

target shell 的同类 dimensional rate 也为负。这说明当前 counting 预测 time integration 多出一个 (R) 因子，monotonicity 本身不能修复 exponent；但 frozen sources 没有给出所需的 (H^1)/occupation upper，所以不能把这些公式写成严格结论。

更高频的 passive profile 属于新 architecture，需要重新审计 heat damping、initial energy、complete payment 与 survival。

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certificate 与 bounded literature boundary

- independent primary analytic audit：PASS，0 blockers；
- Python certificate：24/24 checks，244 cases；
- independent Ruby audit：21 assertions；
- negative mutations：Python 22/22、Ruby 23/23 rejected；
- (PYTHONHASHSEED=0,1,42)：byte-identical；
- certificate scope：finite exact arithmetic、source structure、hashes 与 claim boundaries，不是 continuous PDE proof；
- literature：截至 2026-09-03 的 bounded primary-source screen 只得到 finite non-hit，不证明 novelty、priority、nonexistence、correctness、sharpness 或 publishability。

文献中最接近的是 exact Navier--Stokes shearing waves、shear-flow passive-scalar pathwise dissipation、heat observability/control cost 与 propagation of smallness。它们提示指数级局部取消可能伴随指数级全局代价，但没有完成这里的 corrector/payment construction。

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下一 proposition Y.57 与停止线

最小下一目标 Y.57 是：在同一 common shear 下构造两个 adjacent inversion-paired primaries 与有限 corrector family，使 outer target spacetime box 上发生指数级 field cancellation，同时 correctors 在两个 remote inward strips 上可忽略，并证明

$$(P_R^M)^{2/3} = o(\min\{K_{k_1-1,R}(t_1), K_{k_2-1,R}(t_2)\}).$$

本节不证明 Y.57。

- PROVED：frozen geometry 下同一 packet 的 W-type adjacent endpoint 不能击败其 mandatory target-lobe cubic payment；amplitude powers 精确抵消；non-adjacent dyadic placement 更差；(\Xi_{\rm fr}(65)=-875993/968647680<0)。
- FORMAL NECESSARY ONLY：changed geometry 的 rational exponent window；没有 platform、corrector 或 complete payment theorem。
- OPEN / NOT CERTIFIED：accumulated-viscosity (H^1)/occupation upper、Y.57、whole-shell clock、arbitrary suitable weak solutions、scale contraction、regularity 与 singularity。
- PUBLICATION BOUNDARY：正式图件：NOT APPLICABLE。本节纯解析，没有 Navier--Stokes 数值仿真、DNS、DGX 或正式图件。ROUTE SCREEN ONLY | NO FORMAL FIGURE | NO PDE DATA | NO DNS | LOCAL DIRECT TRANSLATION | NO DGX | NOT CLAY。

这不是 Navier--Stokes Millennium problem 的解决，也不提出一般解反例。**NOT CLAY.**

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结论先行：持续 tube 会支付，endpoint-only 尚未关闭

R0.74Y 留下的目标是在 outer target spacetime box 上以指数精度消去 field，同时保留同一 packet 的 adjacent-inward remote tail 作为第二 clock coordinate。R0.74Z 对这个 cancellation-cell continuation 给出两级结论。

若 remote weighted kinetic floor (h) 在长度 (|J|=\theta_LR^3) 的 spacetime tube 上持续，则它自己强制 exterior cubic payment；严格低于临界 rate 的 persistence sequence 不能形成 W-kinetic payment escape。若只在 endpoint 保留 remote field，则 smoothness 只给某个 persistence interval，不能自动给 uniform (R^3) residence，critical 和更短时间集中仍开放。

**PERSISTENT-TUBE KINETIC COERCIVITY: PROVED,
TIME-TAME ENDPOINT ROUTE: CONDITIONALLY BLOCKED,
FULL-CLOCK Y.57 CANCELLATION CELL: OPEN.**

没有 payment-compatible cell 被构造，也没有 whole-shell、regularity 或 singularity 结论。**NOT CLAY.**

exact common-shear algebra 与禁止的平移

令

$$\mathcal{L}_b F = (\partial_t + b(t,x_3)\partial_2 - \Delta_{23})F.$$

同一 odd shear (b) 下任意有限个 smooth periodic solutions (C_j^+) 与 inversion partners

$$C_j^-(t,x_2,x_3) = -C_j^+(t,-x_2,-x_3)$$

的实系数线性组合仍解同一 passive equation。与 primary field 合并后，(u=(U_{\rm primary}+C,b,0)) 是 exact smooth periodic mean-zero unforced Navier--Stokes solution，pressure 可取零，negative coefficients 允许。

但 exact algebra 不允许把已演化 packet 任意垂直或时间平移：

$$[\partial_2,\mathcal{L}_b]=0, \qquad [\partial_3,\mathcal{L}_b]=(\partial_3 b)\partial_2,$$

$$\mathcal{L}_b[C_0(t-\tau)] = [b(t,x_3)-b(t-\tau,x_3)]\partial_2 C_0(t-\tau,x).$$

因此 vertical centers、time offsets 或 Hermite cells 必须从 initial data 在实际 (b(t,x_3)) 下重新演化；不能静默删去 periodic windings。

两次 fourth-root weight shift 与 tube coercivity

outer packet 的 adjacent-inward shell 是 (k=k_2-1)。写

$$\omega = \gamma_{k_2-1} = \Gamma^{1/4}.$$

同一物理 annulus 在 doubled radius 上满足

$$A_{k_2-1}(R) = A_{k_2-2}(2R), \qquad \gamma_{k_2-2} = \omega^{1/4} = \Gamma^{1/16}.$$

这是第二次 fourth-root shift；不能再次使用 (\Gamma^{1/4}) 作为 doubled-radius payment weight。

若 (\Omega(t)\subset A_{k_2-1}(R))、(|\Omega(t)|\le C_\Omega L^\nu R^3)，且对 (t\in J)、(|J|=\theta_LR^3) 有

$$\frac{\omega}{2R}\int_{\Omega(t)}|u(t,x)|^2\,dx\ge h,$$

则 spatial Hölder 和 nonnegative exterior velocity row 给出 exact deterministic coercivity：

$$P_R^M\ge cC_\Omega^{-1/2}\theta_Lh^{3/2}R\omega^{-5/4}L^{-\nu/2},$$

$$\boxed{(P_R^M)^{2/3}\ge cC_\Omega^{-1/3}\theta_L^{2/3}hR^{2/3}\omega^{-5/6}L^{-\nu/3}.}$$

该下界使用 total field，而不是某个 summand，所以已包含 primaries、correctors、inversion partners 与全部 periodic copies 的 interference。

strict persistence threshold 与临界层

令

$$\theta_L=e^{-\kappa L^2+o(L^2)}.$$

R0.74Y 的 frozen rational parameters 给出

$$\Delta_{\rm rem}=\frac{5}{24}c_\gamma-\frac{\rho}{6}=\frac{64279}{238140000}>0,\qquad \kappa_*=\frac{3}{2}\Delta_{\rm rem}=\frac{64279}{158760000}.$$

于是

$$\liminf_{L\to\infty}\frac{1}{L^2}\log\frac{(P_R^M)^{2/3}}{h}\ge\Delta_{\rm rem}-\frac{2}{3}\kappa.$$

因此每个满足

$$\boxed{\limsup_{L\to\infty}\frac{-\log\theta_L}{L^2}<\kappa_*}$$

的 persistence sequence 都被严格排除为 W-kinetic payment escape。等号层 (-L^{-2}\log\theta_L=\kappa_*+o(1)) 仍是 **OPEN**：poly(L) 和 (o(L^2)) 因子可能决定符号。

time-tame endpoint-to-tube 只是一条条件路线

remote free comparator 的 field scale 为

$$\mathcal{A}_{\rm rem} = |\mathfrak{a}_2| e^{-\beta L^2 + O(L)}, \qquad \beta = \frac{49}{268288},$$

并有 exact reserve

$$\frac{\rho}{4} - \beta = \frac{7103}{167680000} > 0.$$

定义 moving derivative

$$\mathscr{D}_2 = \partial_t + Q_2'(t)\partial_2.$$

如果 total corrector 在固定 normalized remote neighborhood 上满足 time-tame envelope

$$R^2 |\mathscr{D}_2 C| \le |\mathfrak{a}_2| e^{o(L^2)},$$

endpoint preservation 又在略放大的 remote strip 上成立，并且 W comparison 在移动 strip、全部 windings 与 (O(R)) center/heat-age perturbation 下 uniform，则 endpoint field 可在 (R^3) 时间 tube 上持续。结合上一节的 exact coercivity，W-kinetic route 被阻断。

第一步依赖 displayed envelope、endpoint condition 和 moving-strip all-winding uniformity，是 **CONDITIONAL LEMMA**；第二步才是 exact payment theorem。不得把 (Z.22) 单独写成 unconditional persistence theorem。

Gaussian、multi-center 与 Hermite cells 的边界

加入 identical negative copy 会在 target 与 remote strip 上同时把 primary 完全消去，因而也毁掉所需第二 coordinate。一个 displaced restoring Gaussian 可以在两个选定 points 做插值，其 target suppression rate 为

$$\zeta_{2c} = \frac{49}{134144}, \qquad \zeta_{2c} - \frac{1}{5000} = \frac{13857}{83840000} > 0.$$

但其 remote ratio 的 logarithmic slope 是 (-\Theta(L))，跨固定宽度 strip 会变化 (e^{\Theta(L)})，不能给 uniform (1+o(1)) restoration。即使只恢复一点，普通 (R)-width packet 的 remote field 也会在 (R^3) interval 上持续并触发 tube coercivity。

更多 centers 或高阶 finite differences 可以 flatten 更多 derivatives，却不能在 family size、coefficient condition number 和 separations 任意依赖 (R) 时推出 uniform no-go。fixed 或 coefficient-tame subexponential families 若满足 time-tame envelope，则落入条件阻断；arbitrary exponentially ill-conditioned finite network 仍开放。qualitative analyticity 也不能在没有 frequency/global-norm bound 时传播 (e^{-\zeta L^2}) smallness。

endpoint-focused escape 的必要复杂度

令 (\mathcal N_L) 表示 (Z.22) 中 normalized time derivative 与 conditioning factor。endpoint preservation 所保证的 residence rate 形式上满足

$$\theta_L \gtrsim \min\left\{1, \frac{\exp[(\rho/4-\beta)L^2+o(L^2)]}{\mathcal{N}_L}\right\}.$$

在该 derivative/conditioning model 内，想避开 strict subcritical persistence no-go，必须至少有

$$\boxed{\log \mathcal{N}_L \geq \left(\frac{476239}{1064835072}+o(1)\right)L^2.}$$

这是 **necessary leading rate**，不是 sufficient construction；等号仍开放。fixed (M)、polynomial (M) 与 polynomial Hermite order 只有在 total conditioning 和 derivative envelope 也 subexponential 时才被排除。极窄 packet 的 backward heat amplification 对 isolated modes 给出强烈代价，但推广到 arbitrary cancelling finite sums 仍需要 spectral/observability estimate。

W-kinetic 结论不能提升为 full-clock Y.57

tube coercivity 比较的是 payment 与 chosen region 上的 kinetic floor (h)。endpoint clock 只知 (K_{k_2-1,R}(t_2)\ge h)，而没有证明 (K\le e^{o(L^2)}h)。accumulated ordinary-viscosity row 可能比 (h) 大；若要关闭它，必须证明该 row 自身也强制 comparable central-energy 或 exterior payment。

因此已证的是：strict subcritical residence 下，time-tame corrector 不能把 persistent W-type remote kinetic strip witness 单独变成 clock-over-payment counterexample。未证的是：任何 time-tame cell 都不可能满足完整 Y.57 ratio。

下一 proposition (Z.39) 是 cancellation-robust remote endpoint persistence/payment dichotomy：要么 field 在 rate 严格低于 (\kappa_*) 的 tube 上持续并由本节关闭，要么 critical/shorter concentration 必须在 complete Version-M ledger 中支付。第二分支、critical layer、full completed clock 与 arbitrary ill-conditioned finite family 均保持开放。

certificate、literature 与冻结四联图

- independent primary analytic audit：PASS，0 blockers；
- Python certificate：10/10 checks；
- independent Ruby audit：11/11 assertions；
- negative mutations：Python 22/22、Ruby 23/23 rejected；
- (PYTHONHASHSEED=0,1,42)：byte-identical；
- figure archive：25 files、3,032,354 bytes，18/18 deterministic rerender，SVG/PNG/PDF 与 greyscale QA PASS；
- certificate scope：finite exact arithmetic、source structure、hashes 与 claim boundaries，不是 continuous PDE proof；
- literature：截至 2026-09-03 的 bounded primary-source screen 仅为 finite non-hit，不证明 novelty、priority、nonexistence、correctness 或 publishability。

四联图编码 exact weight ladder、strict persistence threshold、conditional time-tame route 与 full-clock claim hierarchy。它是 analytic schematic 和 derived analytic values，不是 PDE simulation、DNS、sampled trajectory 或 empirical fit。

发布停止线

- **PROVED**：finite same-(b) inversion-paired superposition 的 exact NSE closure；two fourth-root weight shifts；persistent remote tube 的 Hölder coercivity；strict (\limsup(-L^{-2}\log\theta_L)<\kappa_*) W-kinetic no-go。
- **CONDITIONAL**：endpoint preservation 加 (Z.22) 与 moving-strip all-winding uniformity 推出 (R^3) persistence；在该 model 内的 complexity lower rate。
- **OPEN / NOT CERTIFIED**：critical layer、endpoint-only branch、arbitrary exponentially ill-conditioned finite family、accumulated clock rows、full-clock Y.57、complete payment upper、whole-shell estimates、fixed deletion、general suitable weak solutions、regularity 与 singularity。
- **PUBLICATION BOUNDARY**：没有构造 payment-compatible cancellation cell；finite literature non-hit 不是 novelty evidence。ANALYTIC SCHEMATIC | DERIVED ANALYTIC VALUES | NOT PDE DATA | NOT DNS | NO NOVELTY CLAIM | LOCAL DIRECT TRANSLATION | NO DGX | NOT CLAY。

本节不解决 Navier--Stokes Millennium problem，也不提出一般解反例。**NOT CLAY.**

结论先行：任意短的 endpoint focusing 也必须付款

在冻结的光滑周期 common-shear family 中，W-remote 正体积 endpoint core 不能靠把时间集中压到任意短来逃避 cubic payment。对完整总场 F 直接使用 moving-cutoff 恒等式后，只有两个穷尽分支：局部能量在长度 c R^3 的回看窗内保持，或者它的快速上升本身由同一个扩大 moving strip 内的质量支付。两支都给出同一个 spacetime lower bound；这关闭了 R0.74Z 留下的 critical 与 arbitrarily shorter smooth focusing 逃逸，但没有控制完整时钟 K。

精确 common-shear 闭合与移动坐标

速度 u=(F,b,0) 由同一奇剪切 b 与任意有限个被动 packet、inversion partner 的总和构成，仍是光滑、周期、零均值、无外力且压力为零的精确 Navier--Stokes 解。沿参考高度的水平平移 z=x_2-Q_2(t) 后，总场满足带残余剪切 c(t,x_3) 的线性漂移扩散方程。证明始终作用于总场，因此已经包含 packet、corrector、inversion partner 与全部周期 winding 的相消或叠加。

A.18：moving-cutoff 局部能量恒等式

取固定在移动坐标中的非负 cutoff phi，使其在 endpoint core 上为一并支撑于扩大 remote strip。精确积分分部给出

$$\frac{1}{2}E'(t)+\int\phi|\nabla_{z3}\widetilde{F}|^2=\frac{1}{2}\int\big(c\,\partial_z\phi+\Delta_{z3}\phi\big)|\widetilde{F}|^2.$$

冻结几何满足 $|c|\lesssim R^{-2}$、$|\partial_z\phi|\lesssim R^{-1}$、$|\Delta\phi|\lesssim R^{-2}$；在 $R\le 1$ 时，误差统一由 $K_\phi R^{-3}$ 乘以 enlarged-strip mass 控制。这里的 transport sign 与 R^{-3} 次数已由双实现证书锁定。

A.26：persistence 与 rapid-rise 两支穷尽

令 E_* 为 endpoint core 的总场 L^2 能量，并回看 $J=[t_2-c_0R^3,t_2]$。若 $E(t)$ 在 J 上始终不少于 $E_*/2$，直接积分得到 $X=\int_J M(t)\,dt\ge c_1E_*R^3$。若不然，存在较早时刻能量跌到 $E_*/2$ 以下；积分 A.18 便把至少 $E_*/2$ 的上升充入同一 $M(t)$，仍得到相同 lower bound。不存在第三分支，persistent、critical 与任意更短的光滑 endpoint focusing 均已覆盖。

从局部 L2 质量到 Version-M cubic payment

扩大 tube 的 spacetime 体积至多为 $CL^{1/2}R^6$。Hölder 因而把 A.26 转成 $\int |F|^3 \gtrsim E_*^{3/2}R^{3/2}L^{-1/4}$。该 tube 位于 scale $2R$ 的 exterior row，且权重至少为 $\omega^{1/4}$，所以

$$P_R^M \gtrsim \omega^{1/4}E_*^{3/2}R^{-1/2}L^{-1/4}.$$

再代入 endpoint lower $E_* \geq (2R/\omega)h_{\mathrm{rem}}$，得到冻结主结论

$$(P_R^M)^{2/3} \gtrsim h_{\mathrm{rem}}R^{2/3}\omega^{-5/6}L^{-1/6}.$$

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精确指数余量

冻结参数 omega=Gamma^(1/4)；从 remote clock 到 doubled-radius payment 的可见权重是 omega^(1/4)=Gamma^(1/16)，不能重复使用 Gamma^(1/4)。全部 R、L、omega 次数代入后，严格正余量为

$$\frac{5}{24}\frac{8}{3969} - \frac{1}{6}\frac{9}{10000} = \frac{64279}{238140000} > 0.$$

这个结论是精确有理数账本，不是数值拟合。证书同时拒绝 reciprocal p、错误 transport sign、R^-2 或 R^-4 cutoff、错误 omega 权重、遗漏 critical/shorter 分支以及 full-clock promotion。

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Fourier 账本的正确用途与限制

水平 Fourier 模态满足精确能量衰减；向前高频衰减与向后放大因子可以逐模态写出。但水平 band 不是完整 generator control，全局 modal energy 也不是自动的局部 strip payment。若改走一般 observability，常数会依赖缩小几何、频率与 conditioning；本 lemma 不需要把这种常数静默设成统一常数，因为 moving-cutoff 直接对总场工作。

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文献边界：最近先例不是 novelty 证据

Wang--Wang--Zhang--Zhang, arXiv:1711.04279, Section 3.2 的纯热方程 nested-cutoff inner-endpoint / outer-spacetime estimate 是最近的方法先例；在 T about R^3、r-r' about R 时也呈现主导 R^-3 尺度。R0.75A 增加的是残余剪切、移动周期各向异性 strip、shell weight 与 Version-M cubic conversion。七篇一手来源的 bounded screen 没有发现直接覆盖全部链条的定理，但这种有限未命中不证明 novelty、priority、nonexistence、correctness 或 publishability。

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证据等级与冻结复核

解析主文与 primary audit 的结论等级是 theorem/lemma；Python 证书 14/14、独立 Ruby 17/17，并对 8 个 targeted mutations fail-closed，它们只复核精确有理数、哈希、公式 sentinels、指数和边界，不替代连续 PDE 证明。正式图档共 25 个文件、2,588,462 bytes，SVG/PNG/PDF 主件

与 manifest、validation、QA 均按冻结哈希发布；图是 analytic schematic，不是 PDE simulation、DNS、sampled trajectory 或 empirical fit。

停止线与下一命题 A.63

PROVED 仅限精确光滑有限 common-shear family 的 W-remote endpoint persistence/payment dichotomy，以及相应 horizontal modal identities。完整 completed clock K、accumulated/off-target rows、whole-shell upper、fixed deletion、任意 suitable weak solution 延拓、scale contraction、regularity 与 singularity 均仍 OPEN。下一命题 A.63 是 remote complete-clock extraction：必须同时控制 endpoint、accumulated 与 off-target rows，且不得把 strip lower 改写成 whole-shell upper。NO NOVELTY CLAIM. NOT CLAY.

结论先行：完整时钟只剩 outer accumulated dissipation

在冻结的 exact smooth periodic inversion-paired common-shear family 中，safe complete subclock、inner padding 与完整 endpoint row 已由 Version-M cubic payment 支付。唯一尚未关闭的项是 outer transition collar 上的 full-time accumulated physical dissipation。完整 $K_{k,R}$、fixed deletion、任意 suitable-weak extension 与 regularity 仍为 OPEN。

> **SAFE SUBCLOCK: PAID;　FULL ENDPOINT: PAID;　OUTER ACCUMULATED DISSIPATION: OPEN.**

冻结精确族与尺度

令 $k=k_2-1$、$L=L_2$、$\omega=\Gamma^{1/4}$、$pL=2^{k_2}$。速度场保持

$$u=(F,b,0),\qquad (\partial_t+b\partial_2-\Delta_{23})F=0,\qquad \partial_t b-\partial_3^2 b=0.$$

结论只对 inversion-paired exact family 成立；此时 Version-M mollified trajectory 恒为零。它不是任意 suitable weak solution 的定理。

safe cutoff 与 outer collar 必须分开

冻结 shell cutoff 被分成 safe cutoff χ_k^R 与 outer-collar cutoff ξ_k^R。safe support 在 doubled-radius shell 上保有 $W_{2R}\geq\omega^{1/4}$，outer half 只能统一使用 $W_{2R}\geq\omega$。这一 index shift 决定了 safe rate 与 outer accumulated rate 的符号差异。

$$A_k(R)=A_{k-1}(2R),\qquad \gamma_{k-1}=\omega^{1/4},\qquad \gamma_k=\omega.$$

time-cutoff Caccioppoli 账本

对空间 cutoff χ 与冻结时间 cutoff η_R 做精确分部积分，同时保留 F 与 shear b 的耗散，得到

$$K^{\chi}_{k,R}(t_2) \le C\omega R^{-4} \int_{I_{2R}} \int_{\operatorname{supp}\chi} |u|^2.$$

transport 项在右侧的符号为正；初始项因 η_R 在 s_R 附近为零而消失。证书锁定了该符号、R^{-4} prefactor 与全部 numbered references。

safe complete subclock 已支付

safe support 的 spacetime volume 至多为 CL^3R^5。Hölder 与 doubled-radius payment row 给出

$$K^{\mathrm{safe}}_{k,R}(t_2) \le CLR^{-1}\omega^{5/6}(P^M_R)^{2/3}.$$

冻结参数下的 exact rate 为

$$\frac{\rho}{4} - \frac{5c_\gamma}{24} = -\frac{92837}{476280000} < 0.$$

因此 safe region 的 endpoint 与 full accumulated smooth dissipation 两行同时关闭。

outer-collar endpoint 也已支付

对 terminal R^3 回看窗使用与 Step 26 相同的穷尽二分：outer localized energy 要么持续，要么 rapid rise 强制同一 collar 的 spacetime mass。由较小的 collar volume 得到

$$(P^M_R)^{2/3} \ge cH^{\mathrm{out}}_{k,R}(t_2)R^{2/3}\omega^{-1/3}L^{-2/3}, \qquad \frac{c_\gamma}{12} - \frac{\rho}{6} = \frac{4279}{238140000} > 0.$$

safe endpoint 与 outer endpoint 合并后，完整 endpoint row 已支付。

adverse full-window rate 不是反例

outer-collar accumulated dissipation 的 coarse full-window estimate 只有

$$D^{\mathrm{out}}_{k,R} \le CL^{2/3}R^{-1}\omega^{1/3}(P^M_R)^{2/3}, \qquad \frac{\rho}{4} - \frac{c_\gamma}{12} = \frac{27163}{476280000} > 0.$$

正系数只说明这一 upper-bound method 不能被冻结指数统一吸收；没有构造 saturating exact solution，因此不是 complete-clock counterexample。

精确剩余量是 temporal packing

把 I_{2R} 分成 R^3 blocks，并以各 block payment p_m 定义

$$N_{\rm eff}=\frac{(\sum_m p_m^{2/3})^3}{(\sum_m p_m)^2},\qquad 1\le N_{\rm eff}\le N.$$

则

$$D^{\rm out}_{k,R}\le CL^{2/3}R^{-2/3}\omega^{1/3}N_{\rm eff}^{1/3}(P_R^M)^{2/3}.$$

充分条件精确化为

$$\limsup_{L\to\infty}\frac{\log N_{\rm eff}}{L^2}<\frac{4279}{79380000}.$$

证据等级与文献边界

Primary analytic audit 为 PASS、0 blocker。Python certificate 为 8/8，独立 Ruby 为 9/9；20/20 与 21/21 定向 mutations 均被拒绝，三个 hash seeds 字节一致，47 个 equation tags 与 references 完整解析。它们只验证 exact arithmetic、source binding 与 structure，不替代连续 PDE proof。

bounded primary-source screen 确认 local-energy/Caccioppoli 是既有方法，但没有在三篇筛选来源中发现冻结 weighted ledger 与 $N_{\rm eff}$ threshold 的直接陈述。该 finite non-hit 不构成 novelty 或 priority 判断。

停止线与下一命题

下一命题必须二选一：证明 outer-dissipation packing condition，或构造 accounting 完整的 exact smooth counterexample。两者均未完成。

$$D^{\rm out}_{k,R}\le C(P_R^M)^{2/3}\quad \text{仍为 OPEN.}$$

不得把 strip lower 改写为 whole-shell upper。正式图件：NOT APPLICABLE。本节纯解析，没有 Navier--Stokes 数值仿真、DNS、DGX 或正式图件。NO NOVELTY CLAIM. NOT CLAY.

结论先行：total-cubic packing 是背景剪切假阳性

在冻结的 exact smooth periodic saturation-shear family 中，outer collar 的 total-velocity cubic block masses 在全部 (N\asymp R^{-1}) 个短块上可比，因此 (N_{\rm eff}^{\rm sh}\asymp R^{-1})。这严格否定了 B.44 threshold 必须普适成立的命题，但不否定 B.44 作为充分条件的正确性。

这不是 Navier--Stokes counterexample；只是对一个 auxiliary universal packing condition 的 exact-family counterexample。完整 (K)、fixed deletion、任意 suitable-weak extension 与 regularity 仍为 OPEN。NOT CLAY。

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comparable blocks 与精确 threshold gap

对 shear-only field (u^{\rm sh}=(0,b,0))，固定正 cap 在整个 (I_{2R}=(61R^2,65R^2)) 上满足 (|b|\ge B/2)。每个扩大短块的 payment 都有

$$p_m^{\rm sh} \asymp \omega L^2 R^{-2}, \qquad cR^{-1} \le N \le CR^{-1}.$$

所以

$$N_{\rm eff}^{\rm sh} \asymp N \asymp R^{-1}, \qquad \lim_{L\to\infty} \frac{\log N_{\rm eff}^{\rm sh}}{L^2} = \frac{\rho}{4} = \frac{9}{40000}.$$

相对 B.44 sufficient threshold 的精确超额为

$$\frac{9}{40000} - \frac{4279}{79380000} = \frac{27163}{158760000} > 0.$$

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被计数的 background shear dissipation 仍已支付

球面 outer collar 的固定 (x_3)-slice 面积至多为 (CLR^2)。冻结 saturation datum 的一维 BV 范数一致有界；periodic heat kernel 与 Young 不等式给出

$$\int_{61R^2}^{65R^2} \|\partial_3\theta_R(t)\|_2^2\,dt \le CR.$$

结合 scale-(2R) exterior velocity payment lower，得到

$$D_{k,R}^{{\rm out},b} \le C\omega^{1/3}L^{-1/3}(P_R^M)^{2/3} = o\Big((P_R^M)^{2/3}\Big).$$

这一结论在加入任意 passive component (F) 后仍成立，因为 ((F^2+b^2)^{3/2}\ge |b|^3)。large (N_{\rm eff}) 只记录低频背景在时间块上的持久性，并未记录其梯度代价。

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corrected gate：只剩 passive dissipation

对一般 exact common-shear field (u=(F,b,0))，outer dissipation 精确分解为

$$D^{\mathrm{out}}_{k,R} = D^{\mathrm{out},F}_{k,R} + D^{\mathrm{out},b}_{k,R}.$$

shear row 已支付，最小未解命题因此是

$$D^{\mathrm{out},F}_{k,R} \overset{?}{\le} C(P^M_R)^{2/3}.$$

替代 observable 必须看到 passive gradient 或其 frequency scale，不能只依赖 total (|u|^3) block masses。direct outer-dissipation estimate B.45 在本节仍是 NEITHER PROVED NOR DISPROVED。

冻结证据与主张边界

Primary analytic audit 为 PASS、0 mathematical blocker、0 release blocker。Python certificate 为 8/8，独立 Ruby 为 9/9；18/18 与 19/19 定向 mutations 均被拒绝，三个 hash seeds 字节一致，36 个 equation tags 与 references 完整解析。它们只验证 finite exact arithmetic、source binding 与 structure，不替代连续 PDE proof。

本次冻结白名单没有新增 literature-collision artifact；handoff 只允许 bounded finite non-hit 表述。它不构成 novelty、priority、nonexistence、correctness 或 publishability 判断。正式图件：NOT APPLICABLE；本节纯解析，无 simulation、DNS 或 DGX。

停止线与下一命题

R0.75C 在 corrected passive row 停止。下一步只能证明一个 frequency-sensitive passive block estimate，或构造 accounting 完整的 exact forward passive family；二者均未完成。

$$\boxed{\textbf{TOTAL-CUBIC PACKING REJECTED; PASSIVE DISSIPATION OPEN; NOT CLAY.}}$$

不得把 auxiliary B.44 false positive 写成 Navier--Stokes counterexample，不得把 B.45 写成已否定，也不得把 shear-row payment 写成 passive-row closure。R0.75D/E/F/G 与其他后续工作未读取、未公开。

结论先行：passive row 首次得到严格的两区间 fallback

对剩余的被动外 collar 耗散 $D^{\mathrm{out},F}_{k,R}$，本节证明无条件估计

$$\boxed{D^{\mathrm{out},F}_{k,R} \le C L^{2/3}\omega^{1/3}(P^M_R)^{2/3} + C P^M_R.}$$

因此 $P_R^M \le 1$ 时，目标 $D_{k,R}^{\mathrm{out},F} \lesssim (P_R^M)^{2/3}$ 已闭合；但冻结 common-shear 分支属于 large-payment regime，不能由这一推论覆盖。这里证明的是 absolute Hölder/Young 路线的精确能力边界，不是完整 B.45，也没有构造反例。

无条件 Caccioppoli ledger 与精确 mixed homogeneity

令

$$p_F := R^{-2}\omega \int_{I_{2R}} \int_{\operatorname{supp} \xi_k^R} |F|^3, \qquad p_b := R^{-2}\omega \int_{I_{2R}} \int_{\operatorname{supp} \xi_k^R} |b|^3.$$

scale-$2R$ exterior velocity row 给出 $p_F + p_b \le C P_R^M$。time/Laplacian cutoff row 与 drift row 分别满足

$$\omega R^{-3} \int_{I_{2R}} \int_{\mathrm{out}} |F|^2 \le C L^{2/3} \omega^{1/3} p_F^{2/3},$$

$$\omega R^{-2} \int_{I_{2R}} \int_{\mathrm{out}} |b||F|^2 \le p_b^{1/3} p_F^{2/3}.$$

第二式保留了 drift factor；重复使用绝对值 Hölder/Young 不能把线性 payment homogeneity 改成纯 2/3 次。

small-payment regime 已支付

当 $P_R^M \le 1$ 时，$P_R^M \le (P_R^M)^{2/3}$，且 $L^{2/3}\omega^{1/3} \to 0$。所以

$$P_R^M \le 1 \quad \Longrightarrow \quad D_{k,R}^{\mathrm{out},F} \le C(P_R^M)^{2/3}.$$

这是 passive outer-padding 的严格小支付结论，但不是任意 suitable weak solution 的 regularity statement，也不提供 complete-clock extraction。

低频只在 localization 与 cubic comparability 条件下支付

若所选分量 G 在扩大 outer collar 上满足 localized Rayleigh bound

$$\int |\nabla_{23} G|^2 \le K^2 \int |G|^2,$$

并另有该分量的 cubic comparability，则 $K \le K_{\mathrm{low}}$ 时可支付，其中

$$K_{\text{low}} = cR^{-1}L^{-1/3}\omega^{-1/6}, \qquad L^{-2}\log K_{\text{low}} = \frac{147163}{476280000} + o(1).$$

这不是无条件 Littlewood--Paley lemma：总 $|F|^3$ 不逐点控制投影分量。只分 horizontal modes 也不够，因为零 horizontal mode $F_m = e^{-m^2 t}\sin(mx_3)$ 可以有任意大的 vertical frequency。

冻结 common-shear branch 是严格 large payment

冻结背景 cubic atom 满足

$$p_b \asymp L^2\omega R^{-3}, \qquad \lim_{L\to\infty} L^{-2}\log p_b = \frac{27163}{158760000} > 0.$$

所以 $P_R^M \geq cp_b \to \infty$，small-payment implication 在该分支不可用。线性项不能被当前推论吸收，只说明 absolute Hölder/Young treatment 的限制；不能写成目标估计失败，也不能写成 exact counterexample。

精确 interaction gate 与仍未闭合的频带

mixed drift term 达到目标二次尺度，精确需要

$$\boxed{p_b p_F^2 \leq C(P_R^M)^2.}$$

该 interaction gate 尚未证明。短块强 damping 阈值为 $K \gg R^{-3/2}$，而 low threshold 与它之间仍有

$$K_{\text{low}} \ll K \lesssim R^{-3/2}, \qquad \frac{3\rho}{8} - \left(\frac{\rho}{4} + \frac{c_\gamma}{24}\right) = \frac{27163}{952560000} > 0.$$

signed transport improvement、high-frequency local capture、intermediate band、$b_3\partial_2 F$ commutator、cutoff/projection leakage、periodic weights 与 complete clock 全部保持 OPEN。

冻结证据与文献边界

Primary analytic audit 为 PASS，mathematical blockers 0、release blockers 0。Python certificate 为 20/20，独立 Ruby 为 23/23；双方各拒绝 41/41 定向 mutations，unknown mutation fail closed，三个 hash seeds 字节一致，D.1--D.23 与 23/23 displays 完整解析。

冻结 primary-source screen 只支持三项方法背景：localized divergence-free transport 保留 cutoff flux；定量 local estimate 通常保留 drift norm/profile；shear-specific localization 可行但现有 enhanced-dissipation theorem 不直接给出本站 weighted physical-collar estimate。有限 non-hit 不构成 literature completeness、novelty、priority、nonexistence、correctness 或 publishability 判断。

停止线与下一命题

R0.75D 在 large-payment interaction gate 停止。下一步必须证明 $p_b p_F^2 \lesssim (P_R^M)^2$，或用 signed transport / localized parabolic frequency dichotomy 替代它，或构造 accounting 完整的 exact forward counterexample；三者均未完成。

$$\boxed{\textbf{SMALL PAYMENT PAID; LARGE-PAYMENT INTERACTION OPEN; NOT CLAY.}}$$

不得把 $P^{2/3}+P$ fallback 写成完整 B.45，不得把 low-frequency 条件引理写成无条件分解，也不得把 linear-term non-absorption 写成反例。R0.75E/F/G/H 与其他后续工作未读取、未公开。本节无正式图、simulation、DNS 或 DGX。

结论先行：大背景不等于大通量，实零模已获得全支付闭合

对 frozen common-shear equation

$$(\partial_t + b(t,x_3)\partial_2 - \Delta_{23})F = 0,$$

固定 physical collar cutoff 产生的 transport flux 只含不同 horizontal modes 之间的差频耦合。所有 diagonal terms $n=m$ 精确消失。因此，对任意 admissible real horizontal zero mode，

$$\boxed{\partial_2 F = 0 \Longrightarrow D_{k,R}^{\mathrm{out},F} \le C(P_R^M)^{2/3} \quad (L \ge L_0).}$$

该结论不要求 $P_R^M \le 1$，也允许任意高 vertical frequency。R0.75D 的高 vertical-frequency zero mode 只阻止 horizontal-to-full Rayleigh inference，并不阻止目标耗散估计。

带终端项和正确符号的 localized energy identity

令 $\xi = \xi_k^R(x_1,x_2,x_3)$，并保留 frozen time cutoff η_R。乘以 $\eta_R \xi \overline{F}$、积分并取实部，得到

$$\begin{aligned}\frac{1}{2}\int_{\mathbb{T}^3} \xi |F(t_2)|^2 + \int_{s_R}^{t_2}\int_{\mathbb{T}^3} \eta_R \xi |\nabla_{23}F|^2 \\ = \frac{1}{2}\int_{s_R}^{t_2}\int_{\mathbb{T}^3} [\eta_R'\xi + \eta_R \Delta_{23}\xi]|F|^2 + \mathcal{T}_\xi(F,b),\end{aligned}$$

其中

$$\mathcal{T}_\xi(F,b) := \frac{1}{2}\int_{s_R}^{t_2}\int_{\mathbb{T}^3} \eta_R b\, \partial_2 \xi\, |F|^2.$$

右侧 transport sign 为正；终端 endpoint 非负，只能在上界中丢弃，不能暗中等同于 payment row。

精确的 horizontal difference-frequency identity

因为 $b = b(t, x_3)$ 与 x_2 无关，每个 horizontal mode 独立演化：

$$\partial_t f_n - \partial_3^2 f_n + (n^2 + inb) f_n = 0.$$

以 $\Xi_\ell(x_3) = \int_{-\pi}^{\pi} \hat{\xi}_\ell(x_1, x_3)\, dx_1$ 表示 cutoff 的 x_1-平均，transport flux 精确写成

$$\boxed{\mathcal{T}_\xi(F, b) = \pi \operatorname{Re} \sum_{n,m\in\mathbb{Z}} i(m-n) \int \eta_R b\, \Xi_{m-n} f_n \overline{f_m}.}$$

因 multiplier $i(m-n)$ 在 $n = m$ 时为零，通量是纯 off-diagonal difference-frequency quantity。大 background cubic atom p_b 本身不能推出大 localized transport flux。

zero-flux spectral sectors 与实零模

horizontal support S 由 modal equation 保持。若

$$\Xi_{m-n} = 0 \qquad (n, m \in S,\ n \neq m),$$

则所有 off-diagonal summands 消失，$\mathcal{T}_\xi(F, b) = 0$。这是充分的 spectral-orthogonality condition，不声称 generic radial collar 对非平凡实支撑都满足它。

特别地，$S = \{0\}$ 是真实可容许的不变子空间；此时 $|F|^2$ 与 x_2 无关，周期积分直接消去 $\partial_2 \xi$。这包括任意高 vertical sine modes。

零通量扇区的 pure two-thirds payment

保留 local cubic atom

$$p_F := R^{-2}\omega \int_{I_{2R}} \int_{\operatorname{supp}\xi} |F|^3, \qquad p_F \le C P_R^M.$$

当 transport flux 为零时，time/Laplacian cutoff rows 与 spacetime Hölder 给出

$$\begin{aligned} D_{k,R}^{\mathrm{out},F} &\le C\omega R^{-3} \int |F|^2 \\ &\le C L^{2/3} \omega^{1/3} p_F^{2/3} \\ &\le C L^{2/3} \omega^{1/3} (P_R^M)^{2/3}. \end{aligned}$$

又因 $L^{2/3}\omega^{1/3}=L^{2/3}\exp[-(c_\gamma/12)L^2]\to 0$，故充分大的 L 上得到全支付 $P^{2/3}$ bound；无需 R0.75D 的 interaction hypothesis。

现实性边界：complex singleton 不是物理实场

nonzero complex singleton 也有零 flux，但它只是 complexified scalar diagnostic，不能提升为 physical real Navier--Stokes velocity。真实场满足 $f_{-n}=\overline{f_n}$，所以 nonzero real harmonic 具有成对支撑 $\{n,-n\}$，其 $2n$ difference frequency 一般会耦合 cutoff。

冻结有限 witness

$$\Xi(x)=2+\cos(2x)+\sin(2x),\qquad F(x)=2\cos x+\sin x$$

同时由直接 Laurent multiplication 与 ordered off-diagonal sum 得到

$$\boxed{\mathcal{T}_\xi/\pi=-\frac{1}{2}\neq 0.}$$

该 witness 只验证 signed convolution 的代数归一化；它不是完整 spacetime trajectory，也不是 geometric collar 的有限模型。

任意实场的精确剩余 gate

定义目标归一化下的正 signed cross-mode flux

$$\mathfrak{X}_{\xi,R}(F,b):=\frac{\pi\omega}{R}\left[\operatorname{Re}\sum_{n\neq m} i(m-n)\int \eta_R b\,\Xi_{m-n} f_n\overline{f_m}\right]_+ .$$

则 exact reduction 为

$$\boxed{D^{\mathrm{out},F}_{k,R}\le CL^{2/3}\omega^{1/3}(P^M_R)^{2/3}+\mathfrak{X}_{\xi,R}(F,b).}$$

任意实场所需的下一命题是

$$\boxed{\mathfrak{X}_{\xi,R}(F,b)\le C(P^M_R)^{2/3}.}$$

这一 signed phase-mixing / difference-frequency gate 尚未证明。general real $\pm n$ pairs、cross-mode aggregation、cutoff Fourier tails 与 localized observability 都保持 OPEN。

冻结证据、文献边界与停止线

Primary analytic audit 为 PASS，mathematical blockers 0、release blockers 0。Python certificate 为 13/13，独立 Ruby 为 16/16；双方各拒绝 39/39 定向 mutations，unknown mutations fail closed，三个 hash seeds 字节一致，E.1--E.24 与 24/24 displays 完整解析。

bounded primary-source screen 只支持相邻机制：shear 保持 streamwise modes；streamline average 解一维 diffusion；physical localization 保留 drift flux。没有检索到本站 spherical-collar convolution 与 Version-M payment 的组合定理；finite non-hit 不构成 completeness、novelty、priority、correctness 或 publishability 判断。

REAL ZERO MODE PAID FOR ALL PAYMENT; GENERAL CROSS-MODE GATE OPEN; NOT CLAY.

R0.75E 在任意实场的 signed cross-mode gate 停止。complete clock、fixed deletion、suitable-weak transfer、regularity 与 singularity 均未闭合。R0.75F/G/H 与其他后续工作未读取、未公开。本节无正式图、simulation、DNS 或 DGX。

结论先行：相位积分精确回收离对角账本，但不给出新估计

对 $g_{nm} = f_n\overline{f_m}$ 与 $\ell = m - n$，把 modal-product equation 代回 R0.75E 的 signed collar flux，并在时间与 x_3 上分部积分，精确得到

$$\boxed{\mathcal{T}_\xi = \mathcal{E}_{\rm off} - \mathcal{A}_{\rm off} + \mathcal{D}_{\rm off}.}$$

这三项正是原 localized energy identity 的 off-diagonal endpoint、cutoff 与 dissipation rows。代回后全部相消，只留下 diagonal modal identity。因而 direct modal phase integration 是原能量恒等式的离对角投影，不产生独立符号、小因子或 observability bound。

精确 modal-product equation：无须除以剪切或差频

两个模式方程分别含 $-inbf_n$ 与 $+imb\overline{f_m}$。乘积求导并使用 vertical product rule，得到

$$\boxed{i\ell b g_{nm} = \partial_t g_{nm} - g''_{nm} + 2f'_n\overline{f_m}' + (n^2+m^2)g_{nm}.}$$

这里 $\ell = m - n$，cross-gradient coefficient 必须是 2。等式没有除以 b 或 ℓ，因此在 shear zeros 处仍成立；它本身只是代数恒等式，不是 oscillatory estimate。

diagonal/off-diagonal forms 与归一化

在 frozen $1/(2\pi)$ Fourier convention 下，endpoint 与 cutoff rows 来自 half-energy，系数为 π；dissipation row 使用整周期因子 2π。horizontal-gradient product 为

$$(in)(-im)g_{nm} = nm\, g_{nm}.$$

完整分解为

$$\mathcal{E}_{\mathrm{diag}}+\mathcal{E}_{\mathrm{off}}+\mathcal{D}_{\mathrm{diag}}+\mathcal{D}_{\mathrm{off}}=\mathcal{A}_{\mathrm{diag}}+\mathcal{A}_{\mathrm{off}}+\mathcal{T}_{\xi}.$$

任何遗漏 endpoint、改动 transport sign，或把 dissipation 的 2π 改成 π，都会破坏冻结证书。

两次分部积分与完全相消

因为 $\eta_R(s_R)=0$、$\eta_R(t_2)=1$，时间导数给出 terminal off-diagonal endpoint 减去 η_R' cutoff row。周期 x_3 分部积分给出

$$-\int \eta_R\Xi_\ell g_{nm}''=-\int \eta_R\Xi_\ell'' g_{nm}.$$

再用

$$n^2+m^2=(m-n)^2+2nm=\ell^2+2nm,$$

所有项无余项地组装成 $\mathcal{E}_{\mathrm{off}}-\mathcal{A}_{\mathrm{off}}+\mathcal{D}_{\mathrm{off}}$。代回完整 identity 后得到

$$\boxed{\mathcal{E}_{\mathrm{diag}}+\mathcal{D}_{\mathrm{diag}}=\mathcal{A}_{\mathrm{diag}}.}$$

为什么这是一条 proof-route no-go

最后的 diagonal identity 也可以逐个模式直接乘以 $\eta_R\Xi_0\overline{f_n}$、取实部得到。因此 circular phase substitution 只重写了同一账本，没有添加 resolvent、hypocoercivity、trajectory separation、residence time 或 payment sensitivity。

冻结 two-mode closed solution 先从 $i\ell bg$ 独立计算 transport，再分别检查 time/vertical integration by parts 以及两个 energy identities；其 F.12、F.17 与 F.18 residual 均精确为零。这排除了“证书先假定待证 identity”的循环。

cutoff 正性也不能单独控制 localized form

对奇数 $N=2M+1$，取 real Dirichlet/Fejer family

$$D_N=\sum_{k=-M}^{M}e^{ikx},\qquad a_N=\frac{D_N}{\sqrt{N}},\qquad X_N=\frac{|D_N|^2}{N^2}.$$

它满足 $0\le X_N\le 1$、$\langle X_N\rangle=1/N$、$\langle |a_N|^2\rangle=1$，但 ordered difference counts 给出

$$\langle |D_N|^4\rangle = \frac{2N^3+N}{3}, \qquad \frac{\langle X_N|a_N|^2\rangle}{\langle X_N\rangle\langle |a_N|^2\rangle} = \frac{2N+N^{-1}}{3} \longrightarrow \infty.$$

精确有限值为 $N=3,5,7$ 时依次 $19/9$、$17/5$、$33/7$。这只否定 positivity-only diagonal comparison；该 family 不是 frozen geometric collar，也不是 E.24 的 counterexample。

下一条成功估计必须加入真正的新信息

- quantitative uncertainty：把 x_2-thin collar concentration、horizontal frequency 与 heat damping 联结起来；
- resolvent 或 hypocoercive estimate：利用整个时间窗内的 shear phase，而非代数替换；
- pathwise residence-time bound：控制轨道在 fixed collar 中的停留；
- payment-sensitive positive Toeplitz bound：直接控制正 signed off-diagonal form。

这些方向仍可行，但本节均未证明。任意实场 E.24、complete-clock extraction、fixed deletion、suitable-weak transfer、regularity 与 singularity 保持 OPEN。

冻结证据、文献边界与停止线

Primary analytic audit 为 PASS，mathematical blockers 0、release blockers 0。Python certificate 为 16/16，独立 Ruby 为 20/20；双方各拒绝 43/43 定向 mutations，unknown mutations fail closed，三个 hash seeds 与 regeneration 均字节稳定，F.1--F.23 与 23/23 displays 完整解析。冻结 ledger 为 12/12，并显式包含两套验证器直接依赖的 fixtures 与 expected JSON。

bounded primary-source screen 只确认：实际 enhanced dissipation 使用 resolvent/semigroup 等额外信息；pathwise 方法加入 trajectory information；physical localization 保留 drift flux。有限 non-hit 不构成 completeness、novelty、priority、correctness 或 publishability 判断。

$$\boxed{\text{PHASE SUBSTITUTION TAUTOLOGICAL; POSITIVITY-ONLY DIAGONAL CONTROL FALSE; E.24 OPEN;}}$$

R0.75F 只裁剪两条证明路线，不构造 frozen-collar counterexample。complete clock、fixed deletion、suitable-weak transfer、regularity 与 singularity 均未闭合。后续工作未读取、未公开。本节无正式图、simulation、DNS 或 DGX。

结论先行：独立增益的精确充分阈值

R0.75F 已证明 direct modal phase substitution 只会重建同一个 localized energy ledger；R0.75G 因而把剩余问题改写成一个可证伪的定量门槛。若独立的动力学论证能够证明

$$\mathfrak{X}_{\xi,R}(F,b) \le CR^{\alpha} p_b^{1/3} p_F^{2/3}, \tag{G.1}$$

则精确充分阈值为

$$\boxed{\alpha > \alpha_* := 1 - \frac{c_\gamma}{3\rho} = \frac{27163}{107163} \approx 0.2534736803.} \tag{G.2}$$

这只是一条针对 (G.1) 的 conditional sufficient threshold：本节没有证明任何正的 R^{α} gain，也没有证明这是所有可能路线的必要条件，更没有在阈值处或阈值以下构造 counterexample。

冻结输入、local atoms 与 background 大小

保持冻结尺度与常数

$$R=\exp\Big(-\frac{\rho}{4}L^2\Big),\qquad \omega=\exp\Big(-\frac{c_\gamma}{4}L^2\Big),\qquad \rho=\frac{9}{10000},\quad c_\gamma=\frac{8}{3969}. \tag{G.5}$$

在 outer collar cylinder 上定义

$$\begin{aligned} p_b &:= R^{-2}\omega\int_{I_{2R}}\int_{\operatorname{supp}\xi}|b|^3,\\ p_F &:= R^{-2}\omega\int_{I_{2R}}\int_{\operatorname{supp}\xi}|F|^3. \end{aligned} \tag{G.6}$$

非负的 scale-$2R$ exterior velocity row 给出

$$p_b+p_F\le CP_R^M, \tag{G.7}$$

而 R0.75D 的 absolute Hölder estimate 是零增益情形

$$\mathfrak{X}_{\xi,R}\le Cp_b^{1/3}p_F^{2/3}. \tag{G.8}$$

时间窗长为 $O(R^2)$，collar 体积为 $O(L^2R^3)$，且 $|b|\le CR^{-2}$，所以

$$\begin{aligned} p_b &\le CR^{-2}\omega(R^2)(L^2R^3)(R^{-6})\\ &\le CL^2\omega R^{-3}. \end{aligned} \tag{G.9}$$

因此

$$p_b^{1/3}\le CL^{2/3}\omega^{1/3}R^{-1}. \tag{G.10}$$

R0.75C 的 matching lower bound 不是这条充分蕴含所需的输入；这里只使用 (G.9) 的上界。

阈值推导与 strictness

假设 (G.1)，由 (G.7) 与 (G.10) 得

$$\begin{aligned}\mathfrak{X}_{\xi,R} &\le CR^{\alpha}p_b^{1/3}p_F^{2/3}\\ &\le CL^{2/3}\omega^{1/3}R^{\alpha-1}(P_R^M)^{2/3}.\end{aligned}\tag{G.11}$$

其系数的指数率为

$$\lim_{L\to\infty}\frac{1}{L^2}\log\left(L^{2/3}\omega^{1/3}R^{\alpha-1}\right)=\frac{(1-\alpha)\rho}{4}-\frac{c_\gamma}{12}.\tag{G.12}$$

严格负当且仅当

$$3(1-\alpha)\rho<c_\gamma,\qquad \alpha>1-\frac{c_\gamma}{3\rho}.\tag{G.13}$$

精确有理数计算为

$$1-\frac{(8/3969)}{3(9/10000)}=1-\frac{80000}{107163}=\frac{27163}{107163}.\tag{G.14}$$

在等号处指数率为零，但 $L^{2/3}$ 仍增长；因此对这个未精炼估计，(G.2) 必须保持 strict。与 R0.75E (E.22) 合并只得到条件蕴含

$$\boxed{\text{(G.1) for some } \alpha>\alpha_* \quad\Longrightarrow\quad D_{k,R}^{\mathrm{out},F}\le C(P_R^M)^{2/3}.}\tag{G.15}$$

一组三分之一足够；四分之一不够这条路线

代入 $\alpha=1/3$ 的严格指数 margin 是

$$\frac{\rho}{6}-\frac{c_\gamma}{12}=-\frac{4279}{238140000}<0.\tag{G.3}$$

所以 $R^{1/3}$ gain 在条件上足够。代入 $\alpha=1/4$ 则为

$$\frac{3\rho}{16}-\frac{c_\gamma}{12}=\frac{1489}{1905120000}>0.\tag{G.4}$$

所以 $R^{1/4}$ 对这条 reduction 不充分；这不是反例，也不排除其他可能的证明机制。

amplitude scaling 不会创造增益

固定 shear，对任意 $A>0$ 以 AF 替换 F，则

$$\mathfrak{X}_{\xi,R}(AF,b)=A^2\mathfrak{X}_{\xi,R}(F,b),\qquad p_{AF}^{2/3}=A^2p_F^{2/3}. \tag{G.16}$$

因此当分母非零时，dimensionless correlation ratio

$$\mathscr{C}_R(F,b):=\frac{\mathfrak{X}_{\xi,R}(F,b)}{p_b^{1/3}p_F^{2/3}} \tag{G.17}$$

在 passive-field amplitude scaling 下不变；signed numerator 为零时约定 $\mathscr{C}_R=0$。缺失的小因子必须来自 sign、phase、dynamics 或 geometry，不能来自 passive amplitude 的重新归一化。R0.75E 的 horizontal zero sector 恰有 $\mathscr{C}_R=0$。

residence-time 解释与精确 exponent

若未来论证能用 nonnegative interaction atom 替代完整 background atom，并证明

$$\mathfrak{X}_{\xi,R}\le C(p_b^{\mathrm{int}})^{1/3}p_F^{2/3},\qquad p_b^{\mathrm{int}}\le CR^{\beta}p_b, \tag{G.18}$$

则 $\alpha=\beta/3$，阈值等价于

$$\boxed{\beta>\beta_*:=3\alpha_*=\frac{27163}{35721}\approx 0.7604210408.} \tag{G.19}$$

calibrated plateau speed 与 R^{-2} 同阶。对一个 unwrapped monotone real lift，一次穿过宽度 $O(R)$ 的区间时有纯运动学 occupation bound

$$|\{t:q(t)\in J_R\}|\le\frac{|J_R|}{\inf|q'|}\le CR^3. \tag{G.20}$$

相对于完整 $O(R^2)$ window，这只是 $O(R)$ fraction，形式上对应 $\beta=1$、$\alpha=1/3$。它解释 (G.3) 的 favorable margin，却不证明 arbitrary diffusing and interfering passive field 满足 (G.18)。

pure-transport benchmark 与 diffusion obstruction

令 H 解一维 pure transport equation

$$\partial_t H+b(t)\partial_2 H=0. \tag{G.21}$$

对 spatially constant drift 与固定 smooth cutoff ξ，直接积分给出

$$\frac{1}{2}\frac{d}{dt}\int \xi|H|^2=\frac{1}{2}\int b(t)\partial_2\xi|H|^2, \tag{G.22}$$

从而

$$\frac{1}{2}\int_s^t\int b\partial_2\xi|H|^2=\frac{1}{2}\int \xi|H(t)|^2-\frac{1}{2}\int \xi|H(s)|^2. \tag{G.23}$$

冻结 exact fixture 给出的 positive flux 及 endpoint half-energy difference 都是 1/32。full-window absolute Hölder 丢失了这项 exact crossing cancellation；但恢复 diffusion 后，localized identity 同时包含正在估计的 dissipation，把 flux 从该 identity 解出来只会重复 R0.75F 的 circularity。因此 (G.23) 是机制 benchmark，不是 passive advection-diffusion 的证明。

最小下一命题与五道 falsification gates

数值上有余量的目标现在可以精确写成

$$\boxed{\mathfrak{X}_{\xi,R}(F,b)\le CR^{1/3}p_b^{1/3}p_F^{2/3}.} \tag{G.24}$$

- Dynamic gate：gain 必须对 total passive solution 成立，而不是只对预选 packet 或 static trigonometric family 成立。
- Diffusion gate：Brownian 或 heat recrossing 与 vertical diffusion 必须纳入，不能把 unknown dissipation 移到另一边。
- Geometry gate：spherical collar、x_1 averaging、全部 periodic copies，以及 radial normal 几乎横切 drift 的区域都必须保留。
- Transition gate：b 很小或变号的 bands 必须由更小 geometry 或独立 shear estimate 支付。
- Payment gate：(G.18) 的 interaction atom 必须由 Version-M 已有 rows 支付，不能假设 E.24 或目标 dissipation bound。

允许的下一结果只有三类：证明 (G.24)，证明某个 $\alpha>\alpha_*$ 的较弱 (G.1)，或构造 exact frozen-family sequence 使 $R^{-\alpha_*}\mathscr{C}_R$ 无界。本节三者都没有建立。

冻结证据、文献边界与停止线

Primary analytic audit 为 PASS、零 mathematical blockers、零 release blockers。Python certificate 为 16/16，独立 Ruby 为 18/18；双方拒绝全部 57/57 定向 mutations，unknown mutations fail closed，3 个 Python hash seeds 字节稳定，G.1--G.24 与 24/24 displays 完整解析。完整 frozen dependency ledger 为 12/12，并显式包含两套验证器直接读取的 fixtures 与 expected JSON。

bounded primary-source screen 只确认：邻近 shear-flow 工作会使用 resolvent 或 semigroup coercivity、pathwise trajectory information 与 local shear，physical localization 仍保留 drift flux。没有一个 inspected source 给出 (G.1)、(G.24) 或 Version-M spherical-collar theorem；有限 non-hit 不构成 literature completeness、novelty、priority、nonexistence、correctness 或 publishability 判断。

本节已证明 (G.2)--(G.4)、(G.9)--(G.19) 的阈值和缩放推导，以及 (G.22)--(G.23) 的 pure-transport benchmark。每个 arbitrary-real positive gain、interaction atom (G.18)、G.24、E.24、complete clock、fixed deletion、suitable-weak transfer、regularity 与 singularity 仍 OPEN。

EXACT SUFFICIENT THRESHOLD ONLY; POSITIVE GAIN OPEN; NOT CLAY.

R0.75G 只量化一条 conditional route，不完成任意实场 closure。后续工作未授权、未读取、未公开。本节无正式图、simulation、numerical fit、DNS 或 DGX。

Result and exact boundary

R0.75G identifies `R^(1/3)` as a sufficient gain for the frozen signed collar flux. The present note proves that gain in the exact pure-transport benchmark, with the frozen nondecreasing time cutoff and a terminal tube of the same collar volume.

Let `H` solve

$$\partial_t H + q'(t)\partial_2 H = 0 \tag{H.1}$$

on the periodic spatial domain, and let `xi` be a fixed nonnegative outer- collar cutoff. Under the precise terminal-tube hypotheses in Section 1, the full weighted signed flux obeys

$$\boxed{\mathfrak{X}^{\mathrm{tr}}_{\xi,R}(H,q) \le CL^{2/3}\omega^{1/3}R^{-2/3}(p^{\mathrm{tr}}_{F,J})^{2/3}.} \tag{H.2}$$

The coefficient has the strict frozen rate

$$\lim_{L\to\infty}\frac{1}{L^2}\log\left(L^{2/3}\omega^{1/3}R^{-2/3}\right) = \frac{\rho}{6} - \frac{c_\gamma}{12} = -\frac{4279}{238140000} < 0. \tag{H.3}$$

Thus the pure-transport flux is paid by the corresponding benchmark Version-M functional whenever the terminal tube lies in the frozen scale-`2R` exterior measurement region. In a background with the matching common-shear cubic size, (H.2) is exactly the `alpha=1/3` interaction scale selected by R0.75G. The benchmark pair is not asserted to solve Navier--Stokes.

This does **not** prove the passive advection-diffusion target E.24. When diffusion is restored, the localized identity contains the unknown dissipation itself; the terminal-tube argument alone then becomes circular. The note proves the ballistic benchmark and isolates the diffusion remainder, nothing more.

Frozen geometry and terminal tube

The note uses the following frozen inputs.

input	SHA-256	
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	ti o a re
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5049`	si n d
`research/r075f_modal_phase_integration_identity.md`	`f7a72ebfe0471e18c0d5d44bd3123491d7cd47a79293d1860c5d023c13acf440`	ci b
`research/r075g_signed_flux_gain_threshold.md`	`f2b3424dddb7eee5938200c3433cd012a1820564e43ad446e0c88d8dfa39ff41`	su g

Retain

$$R=\exp\Big(-\frac{\rho}{4}L^2\Big),\qquad \omega=\exp\Big(-\frac{c_\gamma}{4}L^2\Big),\qquad \rho=\frac{9}{10000},\quad c_\gamma=\frac{8}{3969}. \tag{H.4}$$

Let `[s,t_2]` be the full time interval and let the frozen cutoff satisfy

$$0\le \eta_R\le 1, qquad \eta_R(s)=0, qquad \eta_R(t_2)=1, qquad \eta_R'\ge 0. \tag{H.5}$$

Fix a terminal interval

$$J=[t_2-\delta_R,t_2],\qquad c_-R^3\le \delta_R\le c_+R^3, \tag{H.6}$$

contained in the region where `eta_R=1`. Let `0<=xi<=1`, and denote its spatial support by `Omega_0`. Suppose there is a measurable enlarged tube `Omega_+` such that

$$\Omega_0-\big(q(t_2)-q(t)\big)e_2\subset \Omega_+\quad (t\in J),\qquad |\Omega_+|\le C_V L^2R^3. \tag{H.7}$$

The subtraction in (H.7) is on the periodic domain using one fixed lift; no seam crossing is allowed in this terminal passage. For the frozen outer collar, (H.7) follows from its `O(R)` padding whenever

$$|q(t_2)-q(t)|\le c_qR\quad (t\in J). \tag{H.8}$$

In particular, `|q'|<=C_q R^(-2)` and (H.6), with a sufficiently small fixed `c_+`, imply (H.8). No lower speed is required for the theorem.

Finally assume the scale-`2R` exterior weight satisfies

$$W_{2R}\ge \omega\quad \text{on } J\times\Omega_+. \tag{H.9}$$

These hypotheses are stated explicitly because the result is a local terminal-tube theorem, not a whole-annulus or all-winding assertion.

Exact weighted transport identity

Define

$$E_\xi(t):=\int \xi(x)|H(t,x)|^2\,dx. \tag{H.10}$$

Multiplication of (H.1) by `xi H` and periodic integration by parts gives

$$\frac{1}{2}E_\xi'(t) = \frac{1}{2}\int q'(t)\partial_2\xi|H|^2\,dx. \tag{H.11}$$

Set

$$\mathcal{T}^{\mathrm{tr}}_{\xi,\eta} := \frac{1}{2}\int_s^{t_2}\int \eta_R q'(t)\partial_2\xi|H|^2\,dxdt. \tag{H.12}$$

Using (H.5) and integrating (H.11) in time yields

$$\mathcal{T}^{\mathrm{tr}}_{\xi,\eta} = \frac{1}{2}E_\xi(t_2) - \frac{1}{2}\int_s^{t_2}\eta_R'(t)E_\xi(t)\,dt. \tag{H.13}$$

Both `eta_R'` and `E_xi` are nonnegative, so

$$\boxed{[\mathcal{T}^{\mathrm{tr}}_{\xi,\eta}]_+ \le \frac{1}{2}E_\xi(t_2).} \tag{H.14}$$

This is the full-window signed cancellation. It does not estimate the absolute flux and introduces no block-count factor.

Terminal persistence and cubic payment

The characteristic formula is

$$H(t,x) = H\big(t_2, x + (q(t_2) - q(t))e_2\big). \tag{H.15}$$

By (H.7), a change of variables gives, for every `t in J`,

$$\int_{\Omega_+}|H(t,x)|^2\,dx \ge \int_{\Omega_0}|H(t_2,x)|^2\,dx \ge E_\xi(t_2). \tag{H.16}$$

Consequently

$$\int_J\int_{\Omega_+}|H|^2 \ge \delta_R E_\xi(t_2). \tag{H.17}$$

Spacetime Holder on `J times Omega_+` gives

$$\int_J \int_{\Omega_+} |H|^2 \le (\delta_R |\Omega_+|)^{1/3} \Big(\int_J \int_{\Omega_+} |H|^3 \Big)^{2/3}. \tag{H.18}$$

Combining (H.17)--(H.18),

$$\boxed{E_\xi(t_2) \le \delta_R^{-2/3} |\Omega_+|^{1/3} \Big(\int_J \int_{\Omega_+} |H|^3 \Big)^{2/3}.} \tag{H.19}$$

No endpoint pointwise lower bound, frequency truncation, or observability constant is used.

Exact `R^(1/3)` gain

Define the terminal-tube cubic atom

$$p^{\mathrm{tr}}_{F,J} := R^{-2} \omega \int_J \int_{\Omega_+} |H|^3. \tag{H.20}$$

Let $P_R^{M,\mathrm{tr}}$ denote the same nonnegative Version-M measurement formula evaluated on this pure-transport benchmark. This notation does not assert that the benchmark pair is a Navier--Stokes solution. By (H.9), the atom is bounded by a constant multiple of its scale-`2R` exterior velocity row and hence

$$p^{\mathrm{tr}}_{F,J} \le C P_R^{M,\mathrm{tr}}. \tag{H.21}$$

Normalize the positive flux exactly as in R0.75E:

$$\mathfrak{X}^{\mathrm{tr}}_{\xi,R} := \frac{\omega}{R} [\mathcal{T}^{\mathrm{tr}}_{\xi,\eta}]_+. \tag{H.22}$$

Equations (H.6), (H.7), (H.14), (H.19), and (H.20) give

$$\begin{aligned} \mathfrak{X}^{\mathrm{tr}}_{\xi,R} &\le C \frac{\omega}{R} (R^3)^{-2/3} (L^2 R^3)^{1/3} (R^2 \omega^{-1} p^{\mathrm{tr}}_{F,J})^{2/3} \\ &\le C L^{2/3} \omega^{1/3} R^{-2/3} (p^{\mathrm{tr}}_{F,J})^{2/3}. \end{aligned} \tag{H.23}$$

This proves (H.2). Using (H.4), its coefficient has rate

$$-\frac{c_\gamma}{12} + \frac{\rho}{6} = -\frac{4279}{238140000}, \tag{H.24}$$

which proves (H.3) and, with (H.21),

$$\boxed{\mathfrak{X}^{\mathrm{tr}}_{\xi,R} \le C(P_R^{M,\mathrm{tr}})^{2/3} \quad \text{for all sufficiently large } L.} \tag{H.25}$$

If the background atom also has the frozen matching lower scale `p_b >= c L^2 omega R^(-3)`, then

$$L^{2/3}\omega^{1/3}R^{-2/3} \le CR^{1/3}p_b^{1/3}. \tag{H.26}$$

Thus (H.23) realizes the R0.75G `alpha=1/3` scale in this exact benchmark.

Why diffusion is not covered

For the actual passive equation

$$(\partial_t + b\partial_2 - \Delta_{23})F = 0, \tag{H.27}$$

the frozen localized identity instead gives

$$\begin{aligned} \mathcal{T}_{\xi,\eta}(F,b) = \frac{1}{2}\int \xi|F(t_2)|^2 + \int_s^{t_2}\int \eta_R\xi|\nabla_{23}F|^2 \\ - \frac{1}{2}\int_s^{t_2}\int [\eta_R'\xi + \eta_R\Delta_{23}\xi]|F|^2. \end{aligned} \tag{H.28}$$

The first and last rows can be handled by terminal persistence and the already-paid cutoff estimate. The middle row is precisely the unknown outer-collar accumulated dissipation. Bounding the flux through (H.28) would therefore assume the quantity needed to close E.24; this is the R0.75F circularity in physical-space form.

The characteristic identity (H.15) also fails after diffusion. A valid extension must replace it with information that is independent of the target dissipation, for example a Feynman--Kac occupation estimate with a separately paid initial/source row, a resolvent estimate for the signed cutoff functional, or a plateau/transition decomposition with a genuinely small commutator. None is proved here.

Status and minimum next proposition

Proved: the full-window signed pure-transport identity H.13--H.14, terminal-tube persistence H.16--H.19, the exact payment estimate H.23, the favorable frozen rate H.24, and the conditional identification with the R0.75G `alpha=1/3` scale H.26.

Open: any analogue of H.16 or H.23 for the arbitrary diffusing frozen passive field that does not reuse the unknown dissipation; shear-transition bands; periodic recrossing beyond the terminal tube; E.24; complete-clock extraction; fixed deletion; suitable-weak transfer; and all regularity or singularity conclusions.

The minimum next proposition is a one-terminal-tube Feynman--Kac or resolvent estimate that retains the signed cutoff functional and produces some gain `R^alpha` with

$$\alpha > \frac{27163}{107163} \tag{H.29}$$

without placing the target dissipation on its right-hand side. No simulation or numerical fit is used. **NOT CLAY**.

Result and exact boundary

R0.75H obtains the required `R^(1/3)` gain for one pure-transport passage by reducing the full signed flux to a terminal endpoint. The present note separates two issues that must not be conflated:

1. payment of the flux on one `O(R^3)` block; and 2. aggregation over the `O(R^(-1))` blocks in the full `O(R^2)` clock.

The first issue is completely insensitive to diffusion. On every short block `J_j`, the frozen pointwise bounds and spacetime Holder give

$$\boxed{\mathfrak{X}_j \le CL^{2/3}\omega^{1/3}R^{-2/3}p_j^{2/3}.} \tag{I.1}$$

No equation for the passive field is used in this estimate. It therefore holds, in particular, for every smooth solution of the frozen advection--diffusion equation.

For a collection of blocks, define the cubic participation count

$$N_{\mathrm{eff}} := \frac{\left(\sum_j p_j^{2/3}\right)^3}{\left(\sum_j p_j\right)^2}. \tag{I.2}$$

Then the exact aggregation loss is `N_eff^(1/3)`. If

$$N_{\mathrm{eff}} \le CR^{-\theta}, \qquad \theta < \theta_* := \frac{8558}{35721} \approx 0.2395789592, \tag{I.3}$$

the selected-block flux is paid at the target `P^(2/3)` scale for all sufficiently large `L`. In contrast, uniform payment across all `N asymp R^(-1)` blocks has a strictly adverse exponential rate. Thus R0.75I reduces the **absolute block-summation route** from "diffusion on one passage" to a precise occupation/recrossing statement: prove (I.3), or obtain signed cancellation that is at least as strong. Small participation is sufficient, not necessary; a zero-flux sector can have maximal participation.

This note does **not** prove (I.3) for the frozen passive solution, does not close E.24, and proves no complete-clock or regularity statement.

Frozen short-block geometry

The immediately used frozen inputs are

input	SHA-256
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`

input	SHA-256
`research/r075c_background_shear_packing_false_positive.md`	`1f72f3c9d9d348f86188206690ce714df28aed661a9192c7b53bc1e5921f2f89`
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5049
`research/r075g_signed_flux_gain_threshold.md`	`f2b3424dddb7eee5938200c3433cd012a1820564e43ad446e0c88d8dfa39ff41
`research/r075h_single_pass_transport_flux_closure.md`	`849379bea9cf22e0d892ac11ac05bb3b3bc2967a1735753dbc4a6ffc7bb7d7b9`

Retain

$$R=\exp\Big(-\frac{\rho}{4}L^2\Big),\qquad \omega=\exp\Big(-\frac{c_\gamma}{4}L^2\Big),\qquad \rho=\frac{9}{10000},\quad c_\gamma=\frac{8}{3969}. \tag{I.4}$$

The smooth frozen collar cutoff and common shear obey

$$|\partial_2\xi|\le C_\xi R^{-1},\qquad |\operatorname{supp}\partial_2\xi|\le C_\Omega L^2R^3, qquad |b|\le C_bR^{-2}, \tag{I.5}$$

and `0<=eta_R<=1`. Let `J_1,...,J_N` be pairwise disjoint time intervals with

$$|J_j|\le C_JR^3. \tag{I.6}$$

Write

$$\Omega_\partial:=\operatorname{supp}\partial_2\xi,\qquad Q_j:=J_j\times\Omega_\partial. \tag{I.7}$$

Equations (I.5)--(I.6) imply

$$|Q_j|\le CL^2R^6. \tag{I.8}$$

The result below only needs these measure and pointwise bounds. In particular, it does not use characteristics, a heat kernel, a frequency cutoff, or a sign for `b`.

One-block estimate

For an arbitrary real measurable field `F` with finite cubic integral, set

$$
\begin{aligned}
\mathcal{T}_j(F,b) &:= \frac{1}{2}\int_{J_j}\!\!\int \eta_R b\,\partial_2\xi\,|F|^2,\\
\mathfrak{X}_j(F,b) &:= \frac{\omega}{R}[\mathcal{T}_j(F,b)]_+,\\
p_j(F) &:= R^{-2}\omega\int_{Q_j}|F|^3.
\end{aligned}
\tag{I.9}
$$

The pointwise bounds give

$$
|\mathcal{T}_j| \le CR^{-3}\int_{Q_j}|F|^2. \tag{I.10}
$$

Spacetime Holder on `Q_j`, followed by (I.8), yields

$$
\int_{Q_j}|F|^2 \le |Q_j|^{1/3}\left(\int_{Q_j}|F|^3\right)^{2/3} \le CL^{2/3}R^2\left(\int_{Q_j}|F|^3\right)^{2/3}. \tag{I.11}
$$

Substitution of

$$
\int_{Q_j}|F|^3 = R^2\omega^{-1}p_j \tag{I.12}
$$

into (I.10)--(I.11), followed by multiplication by `omega/R`, gives

$$
\begin{aligned}
\mathfrak{X}_j &\le C\frac{\omega}{R}R^{-3}L^{2/3}R^2(R^2\omega^{-1}p_j)^{2/3}\\
&= CL^{2/3}\omega^{1/3}R^{-2/3}p_j^{2/3}.
\end{aligned}
\tag{I.13}
$$

This proves (I.1). The coefficient has the strict rate

$$
\lim_{L\to\infty}\frac{1}{L^2}\log\left(L^{2/3}\omega^{1/3}R^{-2/3}\right) = \frac{\rho}{6}-\frac{c_\gamma}{12} = -\frac{4279}{238140000} < 0. \tag{I.14}
$$

Consequently one short exterior block is paid uniformly for large `L` after using the domination `p_j<=p_F<=C P_R^M` in (I.18). Since no PDE identity entered (I.10)--(I.13), physical diffusion cannot invalidate this conclusion.

Exact participation count

Let `A` be any finite collection of the disjoint blocks and define

$$p_A := \sum_{j\in A} p_j, \qquad \mathfrak{X}_A := \frac{\omega}{R}\left[\sum_{j\in A} \mathcal{T}_j\right]_+. \tag{I.15}$$

If `p_A>0`, define `N_eff(A)` by (I.2), with the sums restricted to `A`; set `N_eff(A)=0` when `p_A=0`. Positivity and Holder for finite sums give

$$1 \le N_{\mathrm{eff}}(A) \le |A| \quad (p_A > 0). \tag{I.16}$$

The lower equality occurs when only one block carries payment. The upper equality occurs when all `p_j`, `j in A`, are equal and positive. This definition records the actual cubic distribution and is sharper than simply counting every block intersecting the clock.

Using `[sum_j a_j]_+<=sum_j|a_j|`, (I.13), and (I.2),

$$\boxed{\mathfrak{X}_A \le CL^{2/3}\omega^{1/3}R^{-2/3}N_{\mathrm{eff}}(A)^{1/3}p_A^{2/3}.} \tag{I.17}$$

For blocks inside the frozen exterior cylinder, disjointness and the Version-M cubic row imply

$$p_A \le p_F \le CP_R^M. \tag{I.18}$$

Thus a dynamical estimate

$$N_{\mathrm{eff}}(A) \le C_N R^{-\theta} \tag{I.19}$$

would give

$$\mathfrak{X}_A \le CL^{2/3}\omega^{1/3}R^{-(2+\theta)/3}(P_R^M)^{2/3}. \tag{I.20}$$

The exponential rate of the coefficient in (I.20) is

$$\frac{\rho(2+\theta)-c_\gamma}{12}. \tag{I.21}$$

It is strictly negative exactly when

$$\boxed{\theta < \frac{c_\gamma}{\rho} - 2 = \frac{8558}{35721} = \theta_*.} \tag{I.22}$$

The inequality must be strict: at equality the exponential rate is zero and the remaining factor `L^(2/3)` is unbounded. Equivalently, if the active time fraction is written `R^beta` so that `theta=1-beta`, then

$$\beta > 1 - \theta_* = \frac{27163}{35721} \approx 0.7604210408, \tag{I.23}$$

which agrees exactly with the residence threshold in R0.75G.

The full-block obstruction

The frozen clock has length `O(R^2)`. A partition into `O(R^3)` blocks contains

$$N \asymp R^{-1} \tag{I.24}$$

blocks. Uniformly spread cubic payment has `N_eff=N`, hence `theta=1`. The coefficient in (I.20) then becomes

$$L^{2/3}\omega^{1/3}R^{-1}, \tag{I.25}$$

whose exponential rate is

$$\frac{\rho}{4} - \frac{c_\gamma}{12} = \frac{27163}{476280000} > 0. \tag{I.26}$$

Therefore independent absolute estimates on all blocks reproduce the known adverse block-count loss. The favorable one-block estimate cannot be summed naively over the whole clock.

A high-participation zero-flux diagnostic

The participation count is not itself an obstruction to E.24. Let `F=f_0(t,x_3)` be independent of `x_2`. Since `b` and `F` are also independent of `x_2`, periodic integration gives, on every block,

$$\mathcal{T}_j = \frac{1}{2}\int_{J_j}\int_{\mathbb{T}_{x_1}\times\mathbb{T}_{x_3}} \eta_R b|f_0|^2 \left(\int_{\mathbb{T}_{x_2}} \partial_2\xi\, dx_2\right) dx_1 dx_3 dt = 0. \tag{I.27}$$

A temporally persistent nonzero zero mode can nevertheless have comparable positive `p_j` on all blocks, hence `N_eff asymp R^(-1)`. Thus a large passive-cubic participation count can be a false positive for the absolute route, just as R0.75C found for the background cubic count. Equation (I.19) is only a sufficient route. Failure of (I.19) neither disproves E.24 nor removes the signed cross-mode alternative.

What remains dynamical

The algebraic theorem reduces the next step to a falsifiable dichotomy.

1. Prove the sufficient participation estimate (I.19) with some `theta<theta_*` from shear transport, diffusion, collar geometry, and Version-M rows already present; or 2. prove signed inter-block cancellation that dominates the right side of (I.17); or 3. construct an admissible frozen passive sequence with `N_eff >= R^(-theta_*+o(1))` and uncompensated positive flux.

Diffusion enters only at this stage, through occupation and recrossing across many blocks. It is not an obstruction to (I.1). A future Feynman--Kac, Davies--Gaffney, or resolvent argument is useful only if it controls the multi-block participation or signed aggregation; a one-block localization estimate alone would not close E.24.

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Status boundary

Proved: the arbitrary-field one-block estimate I.10--I.14, the exact participation identity and bounds I.15--I.17, the conditional threshold I.19--I.23, the adverse full-uniform-block rate I.24--I.26, and the high-participation zero-flux diagnostic I.27.

Not used: the passive PDE, characteristics, heat-kernel estimates, frequency localization, or numerical simulation.

Open: the required bound on `N_eff` or an equivalent signed cancellation for the frozen diffusing field; shear-transition bands; periodic recrossing; E.24; complete-clock extraction; fixed deletion; suitable-weak transfer; and every regularity or singularity conclusion. **NOT CLAY**.

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Result and exact boundary

R0.75H leaves open a Feynman--Kac or resolvent treatment of the signed diffusive collar flux, and R0.75I shows that one short block is not the difficulty. The present note tests the most direct adjoint construction.

Let

$$\mathcal{L} := \partial_t + b(t,x_3)\partial_2 - \Delta_{23}, \qquad \mathcal{L}^* := -\partial_t - b(t,x_3)\partial_2 - \Delta_{23}. \tag{J.1}$$

For the exact flux source

$$a(t,x) := \eta_R(t) b(t,x_3)\partial_2\xi(x), \tag{J.2}$$

consider the zero-terminal adjoint problem

$$\mathcal{L}^*\psi = a, \qquad \psi(t_2) = 0. \tag{J.3}$$

The source has zero `(x_2,x_3)` mean for every fixed `(t,x_1)`. Hence so does `psi`. Therefore, unless `a` vanishes identically, the exact adjoint weight cannot be nonnegative:

$$\boxed{a \not\equiv 0 \quad\Longrightarrow\quad \psi \text{ changes sign.}} \tag{J.4}$$

For a real passive solution `mathcal L F=0`, exact duality gives

$$\boxed{\mathcal{T}_{\xi,\eta}(F,b) = \frac{1}{2}\int \psi(s)|F(s)|^2 - \int_s^{t_2}\int \psi|\nabla_{23}F|^2.} \tag{J.5}$$

Thus the negative part of `psi` places an uncontrolled positive dissipation term on the upper-bound side. Adding a constant to make the weight nonnegative is not free: the exact global energy identity cancels that constant when all terms are retained, while dropping the favorable dissipation creates a boundary surcharge equal to that constant times the global energy drop.

Consequently an exact zero-terminal adjoint inversion of the signed source does not by itself improve R0.75F/H. A viable adjoint argument must instead construct a **nonnegative majorant** whose initial/source boundary row is independently paid. This note proves the obstruction and the required replacement architecture; it does not construct the paid majorant or close E.24.

Frozen setting

The immediately used frozen inputs are

input	SHA-256	
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5049`	p e a si
`research/r075f_modal_phase_integration_identity.md`	`f7a72ebfe0471e18c0d5d44bd3123491d7cd47a79293d1860c5d023c13acf440`	a ci b
`research/r075h_single_pass_transport_flux_closure.md`	`849379bea9cf22e0d892ac11ac05bb3b3bc2967a1735753dbc4a6ffc7bb7d7b9`	p tr e m a d re
`research/r075i_diffusion_safe_block_participation.md`	`c851169 0dc52988b3f3715e589379c72dae0a892dcabd8d0ca218dddbe0fd3a7`	d sa b b m b

Work on `[s,t_2] times T^3`. The passive field is independent of `x_1`, the shear `b=b(t,x_3)` is independent of `x_1,x_2`, and all functions are smooth and periodic. The cutoff `xi` may depend on all spatial variables. The signed flux is

$$\mathcal{T}_{\xi,\eta}(F,b) := \frac{1}{2}\int_s^{t_2}\int_{\mathbb{T}^3} a(t,x)|F(t,x_2,x_3)|^2\, dx dt. \tag{J.6}$$

The note is an exact smooth identity. No backward well-posedness for the passive equation is assumed: (J.3) is an ordinary backward adjoint problem, equivalently a forward parabolic problem after reversing time.

Zero-mean forcing and sign change

For every fixed `(t,x_1)`, periodicity and independence of `b` from `x_2` give

$$\begin{aligned}\int_{\mathbb{T}^2_{23}} a(t,x_1,x_2,x_3)\,dx_2dx_3 &= \eta_R(t)\int_{\mathbb{T}_{x_3}} b(t,x_3)\left(\int_{\mathbb{T}_{x_2}}\partial_2\xi\,dx_2\right)dx_3 \\ &= 0.\end{aligned} \tag{J.7}$$

Integrating (J.3) over `(x_2,x_3)` removes both the periodic Laplacian and the divergence-free shear drift. Hence

$$-\frac{d}{dt}\int_{\mathbb{T}^2_{23}}\psi(t,x)\,dx_2dx_3 = 0. \tag{J.8}$$

The terminal condition gives

$$\boxed{\int_{\mathbb{T}^2_{23}}\psi(t,x_1,x_2,x_3)\,dx_2dx_3 = 0 \quad \text{for every } (t,x_1).} \tag{J.9}$$

If `psi>=0`, (J.9) and continuity force `psi=0`. Equation (J.3) would then force `a=0`. This proves (J.4). More precisely, whenever the exact adjoint solution is nonzero at a time slice, it has both a positive and a negative spatial part on that slice.

This conclusion uses the physical derivative source. Replacing `a` by `|a|` or `a_+` changes the equation and loses the signed cancellation that the route was meant to exploit.

Exact duality and reappearance of dissipation

Set

$$g := |F|^2. \tag{J.10}$$

Since `mathcal L F=0` and `F` is real,

$$\mathcal{L}g = -2|\nabla_{23}F|^2. \tag{J.11}$$

Periodic integration by parts gives the general duality formula

$$\begin{aligned}\int_s^{t_2}\int g\,\mathcal{L}^*\phi = \int_{\mathbb{T}^3}\phi(s)g(s) - \int_{\mathbb{T}^3}\phi(t_2)g(t_2) \\ + \int_s^{t_2}\int \phi\,\mathcal{L}g.\end{aligned} \tag{J.12}$$

Substituting `phi=psi`, (J.3), (J.6), and (J.11) yields (J.5). Decomposing `psi=psi_+-psi_-` gives the valid but nonclosing upper bound

$$\mathcal{T}_{\xi,\eta} \le \frac{1}{2}\int \psi_+(s)|F(s)|^2 + \int_s^{t_2}\int \psi_-|\nabla_{23}F|^2. \tag{J.13}$$

The second row is a positive weighted portion of the passive dissipation. No Version-M cubic payment for it is proved here. Thus the exact adjoint representation has moved, rather than removed, the H.28 obstruction.

Why a positive constant shift is not free

Let

$$E(t) := \int_{\mathbb{T}^3} |F(t)|^2, \qquad D := \int_s^{t_2}\int_{\mathbb{T}^3} |\nabla_{23}F|^2. \tag{J.14}$$

The global passive energy identity is

$$E(s) - E(t_2) = 2D. \tag{J.15}$$

Choose any constant `C>=||psi_-||_infinity` and set `phi=psi+C>=0`. Since `mathcal L^*C=0`, the source remains exactly `a`, but the terminal value becomes `phi(t_2)=C`. Formula (J.12) now gives

$$\begin{aligned}\mathcal{T}_{\xi,\eta} = \frac{1}{2}\int (\psi(s)+C)|F(s)|^2 - \frac{C}{2}E(t_2) \\ - \int_s^{t_2}\int (\psi+C)|\nabla_{23}F|^2.\end{aligned} \tag{J.16}$$

The constant contribution to the exact right-hand side is

$$\frac{C}{2}\big(E(s)-E(t_2)\big) - CD = 0 \tag{J.17}$$

by (J.15). Thus the shift produces no new exact information. If one uses `phi>=0` and discards the last nonpositive row in (J.16), the resulting upper bound pays the surcharge

$$\frac{C}{2}\big(E(s)-E(t_2)\big) = CD. \tag{J.18}$$

That is a global energy-drop/dissipation row. It is not the desired local cubic payment and cannot be assumed away. The same cancellation occurs for any homogeneous adjoint correction once its own boundary and dissipation terms are retained.

The admissible replacement: a paid positive majorant

The obstruction does not rule out every adjoint method. Suppose one can construct a nonnegative `Phi` such that

$$a \le \mathcal{L}^*\Phi, \qquad \Phi \ge 0, \qquad \Phi(t_2) \ge 0. \tag{J.19}$$

Since `g>=0`, (J.11)--(J.12) imply

$$\begin{aligned} \mathcal{T}_{\xi,\eta} &\le \frac{1}{2}\int_s^{t_2}\int g\,\mathcal{L}^*\Phi \\ &\le \frac{1}{2}\int_{\mathbb{T}^3} \Phi(s)|F(s)|^2. \end{aligned} \tag{J.20}$$

The terminal and dissipation terms are now favorable. The price is the initial occupation/source row on the right. Taking `Phi` to be the zero-terminal adjoint solution driven by `a_+` is the canonical Feynman--Kac majorant; the parabolic maximum principle makes it nonnegative. However, no estimate of its initial row by `C(P_R^M)^(2/3)` with the required positive `R` gain is proved here.

Thus the next valid proposition is precise: build a majorant satisfying (J.19) whose boundary row in (J.20) is paid by existing Version-M atoms, including the plateau/transition and periodic-copy geometry. Exact inversion of the signed mean-zero source is not a substitute for that estimate.

Status boundary

Proved: zero mean of the physical flux source J.7; zero mean and forced sign change of its exact zero-terminal adjoint J.8--J.9; the dual identity J.12--J.13; exact cancellation and global-energy surcharge under a constant positivity shift J.14--J.18; and the sufficient positive-majorant architecture J.19--J.20.

Not proved: a paid positive majorant, an occupation estimate with the required `R^alpha` gain, transition-band control, periodic recrossing, E.24, complete-clock extraction, fixed deletion, suitable-weak transfer, or any regularity or singularity conclusion.

This is a structural route-pruning theorem for the frozen smooth passive problem. It is not a no-go theorem for all resolvent or Feynman--Kac methods, uses no simulation or numerical fit, and makes no novelty or priority claim. **NOT CLAY**.

Result and exact boundary

R0.75J identifies a nonnegative adjoint majorant as the admissible replacement for exact inversion of the sign-changing collar source. The present note tests whether the resulting initial boundary row can be paid directly by the local spacetime cubic atom used in R0.75I.

The answer is negative for every fixed nontrivial nonnegative entrance weight, even in a constant-shear one-dimensional subfamily. There are smooth real passive solutions `F_k` for which

$$\int_0^T\int |F_k|^3 = O(k^{-2}), \qquad \int \Phi(0)|F_k(0)|^2 \longrightarrow \frac{A^2}{2}\int \Phi(0) > 0. \tag{K.1}$$

Consequently no frequency-independent constant can bound the positive majorant's entrance row by the two-thirds power of that spacetime cubic mass.

For the explicit physical source `a=cos x_2`, the actual signed flux of the same sequence is exactly zero. Thus the obstruction is a loss in the positive-majorant proof architecture, not a counterexample to E.24. A successful adjoint route must preserve signed/frequency cancellation, adapt the test to the passive frequency content, or introduce an independently controlled entrance-trace/frequency atom. None of those repairs is proved here.

Frozen setting

The immediately used frozen inputs are

input	SHA-256	
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5049`	si a ta
`research/r075i_diffusion_safe_block_participation.md`	`c8511690dc52988b3f3715e589379c72dae0a892dcabd8d0ca218dddbe0fd3a7`	lo sp cu
`research/r075j_mean_zero_adjoint_flux_obstruction.md`	`960e3cbc18ac8207253a8802da215b3eac07a714ddbcc7209985f27a00c9ff4d`	p m b a

Work on `[0,T] times T`, where `T>0` is fixed and the circle is represented by `[0,2pi]`. The unused `x_1,x_3` variables can be restored as normalized constant factors. Take

$$\mathcal{L} = \partial_t + \partial_2 - \partial_2^2, \qquad \mathcal{L}^* = -\partial_t - \partial_2 - \partial_2^2. \tag{K.2}$$

This is the frozen passive operator with smooth constant shear `b=1`. Choose `eta=1` and `xi(x_2)=sin x_2`. The physical signed source is

$$a = b\partial_2\xi = \cos x_2. \tag{K.3}$$

Every construction below is smooth and periodic. It is an exact passive subfamily, not a Navier--Stokes solution.

A canonical smooth positive majorant

Define

$$q(x_2) := 1 + \cos x_2 \geq a(x_2), \qquad q \geq 0, \tag{K.4}$$

and let

$$\mathcal{L}^*\Phi = q, \qquad \Phi(T) = 0. \tag{K.5}$$

After reversing time, semigroup positivity gives `Phi>=0`. Because the operator has constant coefficients and `q` contains only Fourier modes `0,+/-1`, the entrance weight `Phi(0)` contains only those same modes. Integrating (K.5) in space gives

$$-\frac{d}{dt}\int_0^{2\pi}\Phi(t,x_2)\,dx_2=2\pi,\qquad \boxed{\int_0^{2\pi}\Phi(0,x_2)\,dx_2=2\pi T.} \tag{K.6}$$

Thus this admissible smooth majorant has a strictly positive zero Fourier mode at the entrance time. J.20 gives the valid upper bound

$$\mathcal{T}(F)\le\frac{1}{2}\int_0^{2\pi}\Phi(0,x_2)|F(0,x_2)|^2\,dx_2. \tag{K.7}$$

Exact high-frequency passive family

For every integer `k>=1` and amplitude `A>0`, set

$$F_k(t,x_2):=Ae^{-k^2t}\cos\bigl(k(x_2-t)\bigr). \tag{K.8}$$

The phase translation cancels the shear drift and the amplitude cancels the Laplacian:

$$\boxed{\mathcal{L}F_k=0.} \tag{K.9}$$

At the entrance time,

$$|F_k(0,x_2)|^2=\frac{A^2}{2}\bigl(1+\cos(2kx_2)\bigr). \tag{K.10}$$

Since `Phi(0)` has only frequencies `0,+/-1`, Fourier orthogonality and (K.6) give, for every integer `k>=1`,

$$\boxed{B_k:=\frac{1}{2}\int_0^{2\pi}\Phi(0)|F_k(0)|^2=\frac{A^2\pi T}{2}.} \tag{K.11}$$

The boundary payment is independent of `k`.

By contrast, `int_0^(2pi)|cos(kx)|^3 dx=8/3`, so the exact spacetime cubic mass is

$$\begin{aligned}M_k&:=\int_0^T\int_0^{2\pi}|F_k|^3\,dx_2dt\\&=\frac{8A^3}{9k^2}\bigl(1-e^{-3k^2T}\bigr)\le\frac{8A^3}{9k^2}.\end{aligned} \tag{K.12}$$

Therefore

$$\boxed{\frac{B_k}{M_k^{2/3}} \geq \frac{\pi T}{2}\left(\frac{9}{8}\right)^{2/3} k^{4/3} \longrightarrow \infty.} \tag{K.13}$$

For fixed `R,omega,T`, replacing `M_k` by the R0.75I atom `p_k=R^(-2)omega M_k` changes only the fixed prefactor. Hence there is no constant independent of `k` for an estimate of the form

$$B_k \leq C(R,\omega,T)p_k^{2/3}. \tag{K.14}$$

This is an entrance-trace failure: rapid diffusion makes the spacetime cubic mass small while the initial quadratic trace remains fixed.

The actual signed flux is zero

The physical signed flux for (K.3) and (K.8) is

$$\mathcal{T}_k := \frac{1}{2}\int_0^T\int_0^{2\pi} \cos x_2 \left|F_k(t,x_2)\right|^2 dx_2 dt. \tag{K.15}$$

For every time, `|F_k|^2` has only frequencies `0,+/-2k`. Since no integer `k>=1` satisfies `2k=1`, orthogonality gives

$$\boxed{\mathcal{T}_k = 0.} \tag{K.16}$$

The positive majorant has replaced the exact difference-frequency cancellation by a positive zero mode. Its boundary row is therefore arbitrarily larger than the cubic payment even though the target signed flux vanishes exactly. This prevents using (K.13) as an E.24 counterexample.

General fixed-weight trace lemma

The mechanism is not special to (K.5). Let `W` be any fixed continuous nonnegative entrance weight on the circle with `int W>0`. Then

$$\begin{aligned}\frac{1}{2}\int_0^{2\pi} W(x)|F_k(0,x)|^2\,dx = \frac{A^2}{4}\int_0^{2\pi} W(x)\,dx \\ +\frac{A^2}{4}\int_0^{2\pi} W(x)\cos(2kx)\,dx.\end{aligned} \tag{K.17}$$

The Riemann--Lebesgue lemma makes the second row tend to zero. Combining this positive limit with (K.12) proves that no fixed nontrivial nonnegative entrance weight, independent of the passive frequency, can be paid uniformly by `M_k^(2/3)` on this family.

This statement does not rule out an `F`-dependent adjoint test, a signed kernel representation, frequency-localized cancellation, or payment by a separate trace/frequency atom. It only closes the tempting route

$$\text{fixed positive majorant} \quad + \quad \text{local spacetime cubic atom alone.} \tag{K.18}$$

Consequence for the research route

The next admissible proposition must retain information destroyed by source positivity. There are three logically distinct options:

1. keep the signed adjoint kernel and control its negative dissipation row by frequency/phase cancellation; 2. build an `F`-dependent positive test whose entrance zero mode decays with the active frequency and whose dependence can be audited without circularity; or 3. add a genuinely available entrance-trace or frequency-weighted atom to the Version-M ledger and prove that its coefficient is affordable.

R0.75K proves none of these options. It does prove that the raw Feynman--Kac positivity route cannot simply invoke the existing local spacetime cubic atom after (J.20).

Status boundary

Proved: the smooth positive majorant K.4--K.7; the exact real passive family K.8--K.10; its frequency-independent boundary row K.11; the exact cubic mass and divergent trace ratio K.12--K.14; the zero actual signed flux K.15--K.16; and the general fixed-weight version K.17.

Not proved: failure of every adjoint/resolvent argument, failure of an `F`-dependent test, failure of payment by any full Version-M row, a counterexample to E.24, transition-band or periodic-copy control, complete-clock extraction, fixed deletion, suitable-weak transfer, or any regularity or singularity conclusion. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75K shows that a fixed positive adjoint entrance weight loses the high-frequency cancellation of the signed collar source. The present note returns to the physical signed flux and asks whether diffusion itself creates a measurable gain before any positive majorant is introduced.

For a constant shear and one real horizontal harmonic of frequency `k`, the answer is yes. If `k^2 T>=1`, then

$$\boxed{|\mathcal{T}_{k,\eta}| \le C_*|B|V_\xi k^{-2/3} M_k^{2/3},} \tag{L.1}$$

where `B` is the constant shear, `V_xi=int|partial_2 xi|`, and `M_k` is the full spacetime cubic mass of the passive harmonic. The exponent `k^(-2/3)` is exact for the elementary conversion used here: the flux has a `k^(-2)` diffusive time integral, whereas `M_k^(2/3)` has size `k^(-4/3)`.

This is a genuine bound for the physical signed source, unlike the positive-majorant boundary row in R0.75K. It is nevertheless only a constant-shear, single-real-harmonic benchmark with a full-torus cubic mass. It does not sum arbitrary cross modes, localize the cubic mass to the frozen spherical collar, treat `x_3`-dependent shear, or prove E.24.

Frozen inputs and exact subfamily

The immediately used frozen inputs are

input	SHA-256
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5
`research/r075g_signed_flux_gain_threshold.md`	`f2b3424dddb7eee5938200c3433cd012a1820564e43ad446e0c88d8dfa39
`research/r075k_positive_majorant_high_frequency_trace_loss.md`	`9282fb30eb7517853759fb835579220e0da763974d5543e2fb260ec8ca6da

Work on `[0,T] times T`, with the circle represented by `[0,2pi]`. Let `B` be a real constant and set

$$\mathcal{L}_B := \partial_t + B\partial_2 - \partial_2^2. \tag{L.2}$$

Let `xi` be any smooth periodic real cutoff and let `eta` be any measurable time weight satisfying `0<=eta<=1`. Define

$$V_\xi := \int_0^{2\pi} |\partial_2\xi(x_2)|\, dx_2. \tag{L.3}$$

For an integer `k>=1` and amplitude `A>0`, put

$$F_k(t,x_2) := Ae^{-k^2t}\cos\big(k(x_2 - Bt)\big). \tag{L.4}$$

The phase and amplitude cancellations give

$$\boxed{\mathcal{L}_B F_k = 0.} \tag{L.5}$$

This is a smooth real passive solution. The unused variables can be restored as normalized constant factors, but the result is not asserted for a nonconstant `x_3` shear.

Exact diagonal cancellation and diffusive flux bound

The physical signed cutoff flux is

$$\mathcal{T}_{k,\eta} := \frac{1}{2}\int_0^T\int_0^{2\pi} \eta(t)B\partial_2\xi(x_2)|F_k(t,x_2)|^2\, dx_2 dt. \tag{L.6}$$

Using

$$|F_k|^2 = \frac{A^2 e^{-2k^2 t}}{2}\left[1 + \cos\bigl(2k(x_2 - Bt)\bigr)\right], \tag{L.7}$$

the constant horizontal mode makes no contribution because `int partial_2 xi=0`. Hence

$$\mathcal{T}_{k,\eta} = \frac{A^2 B}{4} \int_0^T \eta(t) e^{-2k^2 t} \left(\int_0^{2\pi} \partial_2 \xi(x_2) \cos\bigl(2k(x_2 - Bt)\bigr)\, dx_2 \right) dt. \tag{L.8}$$

Taking the absolute value only after this exact diagonal cancellation gives

$$\begin{aligned} |\mathcal{T}_{k,\eta}| &\le \frac{A^2 |B| V_\xi}{4} \int_0^T e^{-2k^2 t}\, dt \\ &= \frac{A^2 |B| V_\xi}{8k^2}\bigl(1 - e^{-2k^2 T}\bigr) \\ &\le \frac{A^2 |B| V_\xi}{8k^2}. \end{aligned} \tag{L.9}$$

The `k^(-2)` factor is the ordinary horizontal heat decay of this exact mode. No enhanced-dissipation theorem, resolvent estimate, or stochastic representation is used.

Conversion to the spacetime cubic payment

The exact full-torus cubic mass is

$$\begin{aligned} M_k &:= \int_0^T \int_0^{2\pi} |F_k|^3\, dx_2 dt \\ &= \frac{8A^3}{9k^2}\bigl(1 - e^{-3k^2 T}\bigr). \end{aligned} \tag{L.10}$$

Assume

$$k^2 T \ge 1. \tag{L.11}$$

Then `1-e^(-3k^2T)>=1-e^(-3)`, and (L.10) gives

$$A^2 \le \left(\frac{9}{8(1 - e^{-3})} \right)^{2/3} k^{4/3} M_k^{2/3}. \tag{L.12}$$

Substituting (L.12) into (L.9) proves (L.1), with

$$C_* := \frac{1}{8}\left(\frac{9}{8(1-e^{-3})}\right)^{2/3}. \tag{L.13}$$

Both sides scale like `A^2`; the gain cannot be created by passive-amplitude normalization. It comes from comparing the heat time integral with the two-thirds power of the cubic time integral.

Target normalization and exponent diagnostic

For fixed positive `R,omega`, define the full-torus diagnostic atom and normalized flux

$$p_k^{\mathrm{tor}} := R^{-2}\omega M_k, \qquad \mathfrak{X}_k^{\mathrm{tor}} := \frac{\omega}{R}[\mathcal{T}_{k,\eta}]_+. \tag{L.14}$$

Equation (L.1) becomes

$$\boxed{\mathfrak{X}_k^{\mathrm{tor}} \le C_*|B|V_\xi R^{1/3}\omega^{1/3}k^{-2/3}(p_k^{\mathrm{tor}})^{2/3}.} \tag{L.15}$$

This is not yet the frozen Version-M estimate: `p_k^tor` uses the whole circle, and `|B|V_xi` has not been converted into the frozen background atom. It does isolate the exact frequency factor that survives the target normalization.

If a future paid high-frequency decomposition has `k>=R^(-kappa)`, then

$$k^{-2/3} \le R^{2\kappa/3}. \tag{L.16}$$

Purely at the exponent-diagnostic level, this factor exceeds the R0.75G threshold `R^(alpha_*)` when

$$\boxed{\kappa > \frac{3}{2}\alpha_* = \frac{27163}{71442} \approx 0.3802105204.} \tag{L.17}$$

This implication is conditional on paying every other coefficient and on replacing the full-torus cubic mass by frozen local atoms. It is not a proof of G.1.

What fails beyond one harmonic

The proof uses three properties simultaneously:

1. one real `+/-k` pair produces only the diagonal zero mode and the difference frequencies `+/-2k` in `|F_k|^2`; 2. periodicity kills the zero mode before an absolute value is taken; and 3. both members of the pair have the same exact heat factor `e^(-k^2t)`.

For a general real passive field, `|F|^2` contains all differences `m-n`. Even for constant shear, summing pairwise time-decay factors against the local cubic payment requires a nontrivial convolution estimate. For the frozen `b(t,x_3)`, vertical diffusion and `x_3`-dependent modal phases add another layer. Localizing the full-torus cubic mass to the spherical collar can also reintroduce frequency leakage.

The next valid target is therefore a paid high/low difference-frequency split: use diffusive time decay and cutoff Fourier tails on the high part, and reserve pathwise/resolvent or geometric information for the low part. No such summation or low-frequency estimate is proved here.

Status boundary

Proved: the exact constant-shear passive family L.2--L.5; diagonal cancellation before absolute values L.6--L.8; the `k^(-2)` flux bound L.9; the exact cubic mass and `k^(-2/3)` payment gain L.10--L.13; the normalized diagnostic L.14--L.15; and the conditional frequency threshold L.16--L.17.

Not proved: a multimode convolution bound, localization of `p_k^tor` to the frozen collar, payment of `|B|V_xi` by Version-M, nonconstant-shear enhanced dissipation, the low difference-frequency sector, E.24, complete-clock extraction, fixed deletion, suitable-weak transfer, or any regularity or singularity conclusion. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75L obtains a `k^(-2/3)` physical signed-flux gain for one real constant-shear harmonic. The present note proves that the same exponent survives arbitrary finite interference inside one dyadic horizontal frequency packet.

Let the initial passive field have Fourier support `K<=|n|<=2K`, where `K>=1`, and assume `K^2T>=1`. If

$$d_\ell := \frac{1}{2\pi}\int_0^{2\pi} \partial_2\xi(x_2)e^{-i\ell x_2}\,dx_2, \qquad \mathcal{W}_\xi := \sum_{\ell\in\mathbb{Z}}|d_\ell|, \tag{M.1}$$

then the physical signed flux obeys

$$\boxed{|\mathcal{T}_{K,\eta}| \le e(2\pi)^{1/3}|B|\mathcal{W}_\xi K^{-2/3}M_K^{2/3}.} \tag{M.2}$$

Here `M_K` is the full-torus spacetime cubic mass. The proof has two independent pieces: a Schur estimate gives `K^(-2)` against the initial quadratic energy, and a short-time heat lower bound converts that energy to `K^(4/3)M_K^(2/3)`. No pair count appears because the absolutely summable cutoff Fourier coefficients absorb all within-packet interference.

This is a strict extension of the R0.75L benchmark, but it remains a constant-shear, full-torus, single-dyadic-packet result. The scale of `W_xi`, localization to the spherical collar, interactions between different dyadic packets, nonconstant `x_3` shear, and E.24 are open.

Frozen inputs and packet solution

The immediately used frozen inputs are

input	SHA-256
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c50
`research/r075g_signed_flux_gain_threshold.md`	`f2b3424dddb7eee5938200c3433cd012a1820564e43ad446e0c88d8dfa39ff

input	SHA-256
`research/r075l_single_harmonic_diffusive_signed_flux_gain.md`	`52e25b2fdf1a224609c9e8fafa1c041b7f09a361f75f4b3e44ebcdddb756cdf5`

Work on `[0,T] times T` with

$$\mathcal{L}_B = \partial_t + B\partial_2 - \partial_2^2, \qquad B \in \mathbb{R}. \tag{M.3}$$

Let `Lambda_K={n in Z: K<=|n|<=2K}` and take a finite real-admissible initial packet

$$F_0(x_2) = \sum_{n\in\Lambda_K} c_n e^{inx_2}, \qquad c_{-n} = \overline{c_n}. \tag{M.4}$$

Its exact passive evolution is

$$F(t,x_2) = \sum_{n\in\Lambda_K} c_n e^{-n^2 t} e^{in(x_2-Bt)}, \qquad \mathcal{L}_B F = 0. \tag{M.5}$$

Let `xi` be smooth and periodic, let `0<=eta<=1`, and define

$$\mathcal{T}_{K,\eta} := \frac{1}{2}\int_0^T \int_0^{2\pi} \eta(t) B\partial_2\xi(x_2) |F(t,x_2)|^2 \, dx_2 dt. \tag{M.6}$$

Smoothness implies `W_xi<infinity`. Section 4 records a quantitative Sobolev bound requiring only the first two derivatives of `xi` in `L^2`.

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Exact modal kernel and Schur estimate

Expanding `|F|^2`, using the coefficients in (M.1), and noting `d_0=0` gives

$$\mathcal{T}_{K,\eta} = \pi B \sum_{n,m\in\Lambda_K} d_{m-n} c_n \overline{c_m} \int_0^T \eta(t) e^{-(n^2+m^2)t} e^{-i(n-m)Bt} \, dt. \tag{M.7}$$

The diagonal `n=m` vanishes before any absolute value is taken. For every remaining pair,

$$\left| \int_0^T \eta(t) e^{-(n^2+m^2)t} e^{-i(n-m)Bt} \, dt \right| \le \frac{1}{n^2+m^2} \le \frac{1}{2K^2}. \tag{M.8}$$

For the nonnegative matrix

$$A_{nm} := \frac{|d_{m-n}|}{n^2+m^2}, \qquad n,m \in \Lambda_K, \tag{M.9}$$

both row and column sums are bounded by

$$\sup_n \sum_m A_{nm} \le \frac{\mathcal{W}_\xi}{2K^2}. \tag{M.10}$$

Schur's test and Parseval therefore give

$$\begin{aligned} |\mathcal{T}_{K,\eta}| &\le \frac{\pi |B| \mathcal{W}_\xi}{2K^2} \sum_{n \in \Lambda_K} |c_n|^2 \\ &= \frac{|B| \mathcal{W}_\xi}{4K^2} E_0, \qquad E_0 := \int_0^{2\pi} |F_0|^2. \end{aligned} \tag{M.11}$$

Thus within-packet mode count does not appear. If one discards the Fourier tail and uses only `sup|d_l|`, the corresponding row sum can lose one power of `K`; absolute summability is the precise anti-aggregation input here.

Short-time conversion from energy to cubic mass

Set

$$M_K := \int_0^T \int_0^{2\pi} |F(t,x_2)|^3 \, dx_2 dt. \tag{M.12}$$

For `0<=t<=1/(8K^2)`, the upper packet edge `|n|<=2K` gives

$$\begin{aligned} \|F(t)\|_2^2 &= 2\pi \sum_{n \in \Lambda_K} |c_n|^2 e^{-2n^2 t} \\ &\ge e^{-8K^2 t} E_0 \ge e^{-1} E_0. \end{aligned} \tag{M.13}$$

On a circle of measure `2pi`, Holder gives

$$\|F(t)\|_3^3 \ge (2\pi)^{-1/2} \|F(t)\|_2^3 \ge (2\pi)^{-1/2} e^{-3/2} E_0^{3/2}. \tag{M.14}$$

The assumption `K^2T>=1` ensures that the entire interval `[0,1/(8K^2)]` lies inside `[0,T]`. Integration yields

$$M_K \ge \frac{e^{-3/2}}{8(2\pi)^{1/2}} K^{-2} E_0^{3/2}. \tag{M.15}$$

Equivalently,

$$E_0 \le 4e(2\pi)^{1/3}K^{4/3}M_K^{2/3}. \tag{M.16}$$

Combining (M.11) and (M.16) proves (M.2). The passive-amplitude degree is two on both sides, and no inverse heat flow or entrance-trace estimate is assumed.

The cutoff Wiener row

The exact anti-aggregation quantity in (M.2) is the Wiener norm of `partial_2 xi`, not merely its pointwise derivative bound. Cauchy--Schwarz in Fourier space gives

$$\begin{aligned}\mathcal{W}_\xi &\le \left(\sum_{\ell\in\mathbb{Z}}(1+\ell^2)|d_\ell|^2\right)^{1/2}\left(\sum_{\ell\in\mathbb{Z}}(1+\ell^2)^{-1}\right)^{1/2}\\ &\le C\left(\|\partial_2\xi\|_{L^2}+\|\partial_2^2\xi\|_{L^2}\right).\end{aligned} \tag{M.17}$$

This shows that no derivative beyond the second is logically required in the one-dimensional packet theorem. It does not yet give the calibrated `R,L` size of the `x_1,x_3`-dependent frozen collar cutoff after averaging, nor show that the resulting coefficient is affordable.

Target normalization and remaining split

Define the full-torus diagnostic atom

$$p_K^{\mathrm{tor}} := R^{-2}\omega M_K, \qquad \mathfrak{X}_K^{\mathrm{tor}} := \frac{\omega}{R}[\mathcal{T}_{K,\eta}]_+. \tag{M.18}$$

Then (M.2) gives

$$\boxed{\mathfrak{X}_K^{\mathrm{tor}} \le e(2\pi)^{1/3}|B|\mathcal{W}_\xi R^{1/3}\omega^{1/3}K^{-2/3}(p_K^{\mathrm{tor}})^{2/3}.} \tag{M.19}$$

The same conditional exponent diagnostic as R0.75L follows: a paid packet threshold `K>=R^(-kappa)` produces the factor `R^(2kappa/3)`, whose exponent exceeds the R0.75G requirement only when

$$\kappa > \frac{27163}{71442}. \tag{M.20}$$

The unresolved task is no longer interference inside one dyadic packet. It is to control the sum over packet pairs, quantify the frozen cutoff Wiener row, localize `M_K` to existing collar atoms, and handle the low difference-frequency sector and nonconstant shear without circularity.

Status boundary

Proved: the exact finite-packet solution M.3--M.6; modal kernel and diagonal cancellation M.7; the Schur/Wiener energy bound M.8--M.11; the short-time cubic lower bound and energy conversion M.12--M.16; the second-derivative Wiener estimate M.17; and the normalized dyadic-packet diagnostic M.18--M.20.

Not proved: an inter-packet summation, the calibrated frozen-collar size of `W_xi`, replacement of `p_K^tor` by existing local Version-M atoms, nonconstant-shear control, the low difference-frequency sector, E.24, complete-clock extraction, fixed deletion, suitable-weak transfer, or any regularity or singularity conclusion. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75M compresses all interference inside one horizontal dyadic packet into the Wiener norm of the cutoff derivative. The physical R0.75E cutoff is three-dimensional, radial in the central lift, and averaged in `x_1` before it meets the horizontal modes. The present note computes the scale of that averaged Fourier row.

Retain the frozen calibration

$$p = \frac{32}{63}, \qquad a = pL, \qquad r = aR, \qquad 0 < R \le 1, qquad a \ge a_0. \tag{N.1}$$

For a fixed smooth radial profile supported in an `O(R)` enlargement of the outer collar, let `d_l(x_3)` be the horizontal Fourier coefficient of the `x_1`-averaged derivative. Then

$$\boxed{\sum_{\ell\in\mathbb{Z}} \|d_\ell\|_{L^\infty_{x_3}} \le C_\vartheta a \le C_\vartheta L.} \tag{N.2}$$

Thus the pointwise derivative size `O(R^(-1))` does not survive as a negative power of `R` in the exact cross-mode Wiener row. If the cutoff is also averaged in `x_3`, the corresponding one-dimensional row satisfies

$$\boxed{\sum_{\ell\in\mathbb{Z}} |D_\ell| \le C_\vartheta R a^2 \le C_\vartheta L^2 R.} \tag{N.3}$$

The proof is a rescaling plus a low/high Fourier-sample split. It avoids the crude periodic `H^1` estimate from R0.75M, whose second-derivative term would lose `R^(-1/2)` after the same rescaling.

This is a geometric coefficient theorem. It does not extend the R0.75M time kernel to `x_3`-dependent shear or vertical diffusion, localize the full-torus cubic mass, sum dyadic packets, or prove E.24.

Frozen inputs and a canonical collar representative

The immediately used frozen inputs are

input	SHA-256
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`

input	SHA-256
`research/r075c_background_shear_packing_false_positive.md`	`1f72f3c9d9d348f86188206690ce714df28aed661a9192c7b53bc1e5921f2f89`
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5049
`research/r075m_dyadic_packet_diffusive_flux_gain.md`	`13434bbc15eabecd5a695eceef01a7d63415e96511b14c29cc8abcd1297c7bf7`

Fix `delta>0` and a nonnegative `vartheta in C_c^infinity((-delta,delta))`. It may be chosen equal to one on the normalized radial set supporting the complementary R0.75B clock piece. Consequently the periodic lift of

$$\xi_{a,R}(x) := \vartheta\left(\frac{|x|-r}{R}\right) = \vartheta\left(\frac{|x|}{R} - a\right) \tag{N.4}$$

is an admissible canonical representative of the outer-collar cover. Take `a>=max(2delta,1)` and `(a+delta)R<pi/2`. Its Euclidean support then lies in the central torus chart, so periodization introduces no overlap in any integral below. Derivatives of order `j` cost at most `C_jR^(-j)`.

For the R0.75E convention, define

$$\begin{aligned} \Xi_\ell(x_3) &:= \int_{-\pi}^{\pi} \frac{1}{2\pi} \int_{-\pi}^{\pi} \xi_{a,R}(x_1,x_2,x_3) e^{-i\ell x_2}\, dx_2 dx_1, \\ d_\ell(x_3) &:= \frac{1}{2\pi} \int_{-\pi}^{\pi} \partial_2\left(\int_{-\pi}^{\pi} \xi_{a,R}\, dx_1\right) e^{-i\ell x_2}\, dx_2 = i\ell \Xi_\ell(x_3). \end{aligned} \tag{N.5}$$

In particular `d_0=0` exactly. Formula (N.2) is stronger than `sup_(x_3) sum_l |d_l(x_3)|<=CL`; the supremum is taken separately for each Fourier coefficient before summation.

A scale-correct Fourier sampling lemma

Use the real-line Fourier transform

$$\widehat{h}(\zeta) := \int_{\mathbb{R}} h(y) e^{-i\zeta y}\, dy. \tag{N.6}$$

If a family `h_z` is compactly supported, belongs to `W^(2,1)(R)`, and

$$\sup_z \left(\|h_z\|_{L^1} + \|h_z''\|_{L^1}\right) \le A, \tag{N.7}$$

then for every `nu>=1` and `0<R<=1`,

$$R^{\nu}\sum_{\ell\in\mathbb{Z}}\sup_{z}|\widehat{h}_z(\ell R)|\le CR^{\nu-1}A. \tag{N.8}$$

Indeed, for `|ell|<=R^(-1)` use `|hat h_z(ell R)|<=||h_z||_1`; there are `O(R^(-1))` such integers. For `|ell|>R^(-1)`, two integrations by parts give

$$|\widehat{h}_z(\ell R)|\le \frac{\|h_z''\|_{L^1}}{\ell^2R^2},\qquad \sum_{|\ell|>R^{-1}}\ell^{-2}\le CR. \tag{N.9}$$

The two ranges both contribute at most `CR^(nu-1)A`. No comparison of a discrete sum with an unsigned Riemann integral is assumed.

The `x_1`-averaged row

Write `x_3=Rz`, `x_2=Ry`, and set

$$G_{a,z}(y):=\int_{\mathbb{R}}\vartheta\Big(\sqrt{u^2+y^2+z^2}-a\Big)\,du,\qquad h_{a,z}:=G'_{a,z}. \tag{N.10}$$

Central support and the change of variables `x_1=Ru` give

$$\int_{-\pi}^{\pi}\xi_{a,R}(x_1,Ry,Rz)\,dx_1=RG_{a,z}(y),\qquad d_\ell(Rz)=\frac{R}{2\pi}\widehat{h}_{a,z}(\ell R). \tag{N.11}$$

If the slice meets the radial support, then `|z|<=a+delta`. Its scaled two-dimensional support is

$$A_{a,z}:=\Big\{(u,y):\Big|\sqrt{u^2+y^2+z^2}-a\Big|<\delta\Big\},\qquad |A_{a,z}|\le 4\pi a\delta. \tag{N.12}$$

For `|z|<=a-delta`, this is the exact difference of two disk areas. In the tangency band `a-delta<|z|<=a+delta`, it is bounded by the same outer disk difference. Since the radial variable on the support is at least `a-delta>=a/2`, the first and third `y` derivatives of the smooth radial profile are uniformly bounded. Differentiation under the integral and Fubini therefore give

$$\sup_{z}\Big(\|h_{a,z}\|_{L^1_y}+\|h''_{a,z}\|_{L^1_y}\Big)\le C_\vartheta a. \tag{N.13}$$

Apply (N.8) with `nu=1` to (N.11). This proves (N.2), including spherical tangencies and every horizontal Fourier mode.

The fully averaged row

Define

$$
\begin{aligned}
\bar{\xi}_{a,R}(x_2) &:= \int_{\mathbb{T}_{x_1}} \int_{\mathbb{T}_{x_3}} \xi_{a,R}(x_1,x_2,x_3)\,dx_1dx_3,\\
D_\ell &:= \frac{1}{2\pi}\int_{-\pi}^{\pi} \partial_2\bar{\xi}_{a,R}(x_2)e^{-i\ell x_2}\,dx_2.
\end{aligned}
\tag{N.14}
$$

With

$$
G_a(y) := \int_{\mathbb{R}^2} \vartheta\left(\sqrt{u_1^2+y^2+u_3^2}-a\right)du_1du_3, \qquad h_a := G_a', \tag{N.15}
$$

scaling gives

$$
\bar{\xi}_{a,R}(Ry) = R^2G_a(y), \qquad D_\ell = \frac{R^2}{2\pi}\widehat{h}_a(\ell R). \tag{N.16}
$$

The scaled three-dimensional shell has volume at most `C_delta a^2`. The same derivative and Fubini argument gives `||h_a||_1+||h_a''||_1<=C_vartheta a^2`. Applying (N.8) with `nu=2` proves (N.3).

Frequency diagnostic and remaining analytic gate

At the upper heat threshold isolated in R0.75D, `K>=R^(-3/2)` implies

$$
\boxed{\left(\sum_\ell \|d_\ell\|_\infty\right)K^{-2/3} \le C_\vartheta LR, \qquad \left(\sum_\ell |D_\ell|\right)K^{-2/3} \le C_\vartheta L^2R^2.} \tag{N.17}
$$

Hence neither within-packet mode count nor radial-collar Fourier geometry forces a negative `R` power in the high-frequency coefficient. What remains is genuinely dynamical and local: extend the R0.75M time-kernel argument through vertical diffusion and nonconstant `b(t,x_3)`, replace the full-torus cubic mass by buffered collar atoms without backward heat amplification, and sum packet interactions including low differences.

Status boundary

Proved: a canonical radial cutoff compatible with the frozen outer collar; the scale-correct Fourier sampling lemma N.6--N.9; the uniform `x_1`-averaged coefficient row N.10--N.13; the fully averaged row N.14--N.16; and the high-frequency coefficient diagnostic N.17.

Not proved: that the previously unspecified cutoff must have this bound without choosing the canonical radial representative; a two-dimensional or nonconstant-shear version of the R0.75M semigroup estimate; local cubic payment; inter-packet or low-difference control; E.24; complete-clock extraction; fixed deletion; suitable-weak transfer; or any regularity or singularity conclusion. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75M proves a `K^(-2/3)` signed-flux gain for one horizontal dyadic packet with no vertical variable. R0.75N proves that the physical `x_1`-averaged radial-collar derivative has an `x_3`-uniform Wiener row of size `O(L)`. The present note joins those two inputs without suppressing vertical diffusion.

For constant shear `B`, arbitrary vertical frequencies preserve the quadratic-energy estimate

$$\boxed{|\mathcal{T}_{K,\eta}^{(2)}| \le \frac{|B|\mathcal{W}_\infty}{4K^2}E_0, \qquad \mathcal{W}_\infty := \sum_{\ell\in\mathbb{Z}} \|d_\ell\|_{L^\infty_{x_3}}.} \tag{O.1}$$

If the total initial Fourier support additionally obeys `n^2+j^2<=4K^2` and `K^2T>=1`, then the full two-dimensional spacetime cubic mass `M_K^(2)` controls the initial energy and

$$\boxed{|\mathcal{T}_{K,\eta}^{(2)}| \le e(2\pi)^{2/3}|B|\mathcal{W}_\infty K^{-2/3}(M_K^{(2)})^{2/3}.} \tag{O.2}$$

For the canonical radial collar from R0.75N, `mathcal W_infty<=C_vartheta L`. At the frozen threshold `K>=R^(-3/2)` and for a constant plateau shear `|B|<=C_B R^(-2)`, the normalized coefficient is

$$C_{\vartheta,B}L\,\omega^{1/3}R^{-2/3} = C_{\vartheta,B}L\exp\left(-\frac{4279}{238140000}L^2\right), \tag{O.3}$$

and therefore tends to zero. This gives the same two-thirds homogeneity for the packet's own full-`T^2` diagnostic atom under constant shear.

This does **not** localize the cubic mass to the physical collar, sum different packets, treat low horizontal differences, or handle the actual nonconstant `b(t,x_3)`. It is a constant-shear benchmark, not E.24.

Frozen inputs and two-dimensional packet

The immediately used frozen inputs are

input	SHA-256	role
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5049`	exact `x_1`-averaged flux
`research/r075g_signed_flux_gain_threshold.md`	`f2b3424dddb7eee5938200c3433cd012a1820564e43ad446e0c88d8dfa39ff41`	calibrated gain and shear size
`research/r075m_dyadic_packet_diffusive_flux_gain.md`	`13434bbc15eabecd5a695eceef01a7d63415e96511b14c29cc8abcd1297c7bf7`	one-dimensional packet argument
`research/r075n_radial_collar_averaged_wiener_row.md`	`ba59a4df399d8580b35d8dbb3f0758f9d2ffcc7f97f1147e5804c428f3740318`	canonical collar Wiener row

Work on `[0,T] times T^2_(x_2,x_3)` with

$$\mathcal{L}_B^{(2)} := \partial_t + B\partial_2 - \Delta_{23}, \qquad B\in\mathbb{R}. \tag{O.4}$$

For an integer `K>=1`, let the horizontal Fourier support satisfy `K<=|n|<=2K`. Initially allow

$$F_0(x_2,x_3)=\sum_{K\le|n|\le 2K} f_n^0(x_3)e^{inx_2},\qquad f_n^0\in L^2(\mathbb{T}),\tag{O.5}$$

with finitely many nonzero `n`; smooth finite Fourier data may be used throughout and the estimate then extends by density. Its exact evolution is

$$f_n(t)=e^{-n^2t}e^{-inBt}e^{t\partial_3^2}f_n^0,\qquad F(t)=\sum_n f_n(t,x_3)e^{inx_2}.\tag{O.6}$$

Let

$$H(x_2,x_3):=\int_{-\pi}^{\pi}\xi(x_1,x_2,x_3)\,dx_1,\qquad d_\ell(x_3):=\frac{1}{2\pi}\int_{-\pi}^{\pi}\partial_2H\,e^{-i\ell x_2}\,dx_2.\tag{O.7}$$

Periodicity gives `d_0=0`. Assume `mathcal W_infty<infinity`, take `0<=eta<=1`, and define the physical `x_1`-integrated signed flux

$$\mathcal{T}^{(2)}_{K,\eta}:=\frac{1}{2}\int_0^T\int_{\mathbb{T}^3}\eta(t)B\partial_2\xi(x)|F(t,x_2,x_3)|^2\,dxdt.\tag{O.8}$$

Vertical-semigroup Schur estimate

Horizontal Fourier expansion of (O.8) gives the exact identity

$$\mathcal{T}^{(2)}_{K,\eta}=\pi B\,\mathrm{Re}\sum_{n,m}\int_0^T\eta(t)\int_{\mathbb{T}}d_{m-n}(x_3)f_n(t,x_3)\overline{f_m(t,x_3)}\,dx_3dt.\tag{O.9}$$

The diagonal vanishes before absolute values. Put `a_n=||f_n^0||_(L^2_(x_3))`. The vertical heat semigroup is an `L^2` contraction, so for every pair

$$\begin{aligned}&\int_0^T\eta(t)\left|\int_{\mathbb{T}}d_{m-n}f_n\overline{f_m}\,dx_3\right|dt\\&\quad\le\|d_{m-n}\|_\infty a_na_m\int_0^T e^{-(n^2+m^2)t}\,dt\\&\quad\le\frac{\|d_{m-n}\|_\infty}{n^2+m^2}a_na_m.\end{aligned}\tag{O.10}$$

The nonnegative matrix

$$A_{nm}:=\frac{\|d_{m-n}\|_\infty}{n^2+m^2}\tag{O.11}$$

has both row and column sums at most `mathcal W_infty/(2K^2)`. Schur's test and horizontal Parseval give

$$
\begin{aligned}
|\mathcal{T}^{(2)}_{K,\eta}| &\le \frac{\pi|B|\mathcal{W}_\infty}{2K^2}\sum_n \|f^0_n\|_2^2 \\
&= \frac{|B|\mathcal{W}_\infty}{4K^2}E_0, \qquad E_0 := \int_{\mathbb{T}^2} |F_0|^2.
\end{aligned} \tag{O.12}
$$

This proves (O.1). No upper vertical-frequency bound was used: vertical diffusion can only reduce the `L^2` factors in the flux estimate. An upper bound becomes necessary only in the reverse comparison from later cubic mass back to the entrance energy.

Total-frequency cap and cubic conversion

Now restrict to the finite real-admissible packet

$$
\Gamma_K := \{(n,j)\in\mathbb{Z}^2 : K \le |n| \le 2K, \quad n^2+j^2 \le 4K^2\}, \tag{O.13}
$$

and take

$$
F_0(x_2,x_3) = \sum_{(n,j)\in\Gamma_K} c_{n,j}e^{i(nx_2+jx_3)}, \qquad c_{-n,-j} = \overline{c_{n,j}}. \tag{O.14}
$$

Set

$$
M^{(2)}_K := \int_0^T \int_{\mathbb{T}^2} |F(t)|^3. \tag{O.15}
$$

For `0<=t<=1/(8K^2)`, the total upper-frequency cap gives

$$
\begin{aligned}
\|F(t)\|_2^2 &= (2\pi)^2 \sum_{(n,j)\in\Gamma_K} |c_{n,j}|^2 e^{-2(n^2+j^2)t} \\
&\ge e^{-8K^2 t}E_0 \ge e^{-1}E_0.
\end{aligned} \tag{O.16}
$$

Holder on the torus of measure `(2pi)^2` yields `||F(t)||_3^3>=(2pi)^(-1)||F(t)||_2^3`. If `K^2T>=1`, integration on the same short interval gives

$$
M^{(2)}_K \ge \frac{e^{-3/2}}{16\pi}K^{-2}E_0^{3/2}, \qquad E_0 \le e(16\pi)^{2/3}K^{4/3}(M^{(2)}_K)^{2/3}. \tag{O.17}
$$

Combining (O.12) and (O.17) leaves `(16pi)^(2/3)/4=(2pi)^(2/3)` and proves (O.2). The vertical packet cardinality does not occur; it is absorbed by Parseval and the torus Holder inequality.

Canonical collar and frozen normalization

For the R0.75N canonical radial cutoff,

$$\mathcal{W}_\infty = \sum_\ell \|d_\ell\|_\infty \le C_\vartheta a \le C_\vartheta L. \tag{O.18}$$

Define the full-`T^2` diagnostic atom and normalized flux by

$$p^{\mathrm{tor}}_{K,23} := R^{-2}\omega M^{(2)}_K, \qquad \mathfrak{X}^{\mathrm{tor}}_{K,23} := \frac{\omega}{R}[\mathcal{T}^{(2)}_{K,\eta}]_+. \tag{O.19}$$

Then (O.2) and (O.18) imply

$$\boxed{\mathfrak{X}^{\mathrm{tor}}_{K,23} \le C_\vartheta |B| L R^{1/3}\omega^{1/3} K^{-2/3} (p^{\mathrm{tor}}_{K,23})^{2/3}.} \tag{O.20}$$

If `K>=R^(-kappa)` and `|B|<=C_BR^(-2)`, its scale coefficient is

$$C_{\vartheta,B} L \omega^{1/3} R^{(2\kappa-5)/3}. \tag{O.21}$$

Under the frozen calibration `R=exp(-rho L^2/4)`, `omega=exp(-c_gamma L^2/4)`, this tends to zero precisely when

$$\boxed{\kappa > \frac{1}{2}\left(5 - \frac{c_\gamma}{\rho}\right) = \frac{98605}{71442} \approx 1.3802105204.} \tag{O.22}$$

At equality the exponential rate vanishes while the factor `L` grows, so strict inequality is required. The frozen choice `kappa=3/2` is safely above threshold and gives

$$L\omega^{1/3} R^{-2/3} = L \exp\left[\left(\frac{\rho}{6} - \frac{c_\gamma}{12}\right) L^2\right] = L \exp\left(-\frac{4279}{238140000} L^2\right) \longrightarrow 0. \tag{O.23}$$

Consequently, for sufficiently large `L`, one full-`T^2` packet obeying the total-frequency cap and frozen high-frequency threshold satisfies

$$\boxed{\mathfrak{X}^{\mathrm{tor}}_{K,23} \le C_{\vartheta,B} (p^{\mathrm{tor}}_{K,23})^{2/3}.} \tag{O.24}$$

This is the first certified row in the current route that simultaneously retains horizontal interference, vertical diffusion, the physical radial-collar Fourier row, the calibrated `R^(-2)` shear size, and a bounded coefficient against its own full-torus two-thirds atom. This is not yet the Version-M payment; the full-torus and constant-shear restrictions remain decisive.

Status boundary

Proved: the arbitrary-vertical-frequency energy estimate O.4--O.12; the total-frequency-capped cubic conversion O.13--O.17; the canonical radial-collar insertion O.18--O.20; the exact paid-frequency threshold O.21--O.22; and the frozen `kappa=3/2` closure O.23--O.24.

Not proved: a replacement of the full-`T^2` cubic atom by the buffered physical-collar atom; stability under nonconstant `b(t,x_3)`; inter-packet summation; low horizontal difference control; removal of the total upper- frequency cap; E.24; complete-clock extraction; fixed deletion; suitable-weak transfer; or any regularity or singularity conclusion. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75O controls the full-time signed flux of one constant-shear packet by its entrance energy, but pays that energy with a full-`T^2` cubic mass. The present note gives a sufficient condition that replaces that global mass by a genuine three-dimensional radial-collar atom.

Choose the canonical R0.75N radial profile with a fixed plateau and a smooth cutoff `phi_0` inside its transverse projection. Let

$$E_{\rm in} := \int_{\mathbb{T}^2} \phi_0 |F_0|^2, \qquad E_0 := \int_{\mathbb{T}^2} |F_0|^2, \qquad E_{\rm in} \geq \mu E_0, \qquad 0 < \mu \leq 1. \tag{P.1}$$

For a real constant-shear packet with

$$K \leq |n| \leq 2K, qquadn^2 + j^2 \leq 4K^2, qquadK \geq R^{-3/2}, qquad|B| \leq C_B R^{-2}, \tag{P.2}$$

the cutoff transported by `B` retains at least half of the entrance fraction for a time `tau=c_0 mu K^(-2)`. The three-dimensional plateau fibres then give the local cubic lower bound

$$\boxed{M_{K,\rm col} \geq c_* a^{-1} \mu^{5/2} K^{-2} E_0^{3/2}, \qquad a = pL.} \tag{P.3}$$

Combining this with R0.75O yields

$$\boxed{\mathfrak{X}_{K,\rm col} \leq C|B| a^{5/3} \mu^{-5/3} R^{1/3} \omega^{1/3} K^{-2/3} p_{K,\rm col}^{2/3}.} \tag{P.4}$$

Here `p_(K,col)=R^(-2)omega M_(K,col)` is supported inside the actual radial-collar payment region. At the frozen `B,K` scales, the coefficient is bounded for large `L` whenever

$$\boxed{\mu \geq c_\mu R^\sigma, \qquad 0 \leq \sigma < \sigma_* := \frac{1}{5}\left(\frac{c_\gamma}{\rho} - 2\right) = \frac{8558}{178605} \approx 0.0479157918.} \tag{P.5}$$

Under (P.5), the packet's signed flux is paid by the existing Version-M row whenever the packet is an actual coordinate component of the same velocity measured by that row, and the collar is aligned as specified below. A Fourier projection of a larger component is not covered. This is a conditional realized-subclass closure, not an arbitrary-packet result. A spatially spread packet can have an entrance fraction of order `L^2R^2`, far below (P.5), while its signed flux may still be small by localization or cancellation. The low-concentration branch therefore requires a spatially localized signed-kernel estimate; it cannot be settled by the global entrance energy in R0.75O.

The theorem remains constant-shear, single-packet, and total-frequency capped. It does not prove E.24 for an arbitrary passive field.

Frozen inputs and canonical plateau geometry

The immediately used frozen inputs are

input	SHA-256	role
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	outer-collar payment and scale-`2R` weight
`research/r075i_diffusion_safe_block_participation.md`	`c8511690dc52988b3f3715e589379c72dae0a892dcabd8d0ca218dddbe0fd3a7`	short-block payment scale and participation boundary
`research/r075n_radial_collar_averaged_wiener_row.md`	`ba59a4df399d8580b35d8dbb3f0758f9d2ffcc7f97f1147e5804c428f3740318`	selectable radial cutoff and Wiener row
`research/r075o_vertical_diffusion_packet_gain.md`	`3efb39d2624cf5b5a0e7f348f6cde2ef2416eca900f1aa3ecc90a6ad734849a9`	constant-shear vertical-diffusion packet estimate

Retain

$$p = \frac{32}{63}, \qquad a = pL, qquadr = aR, qquadR = e^{-\rho L^2/4}, \qquad \omega = e^{-c_\gamma L^2/4}. \tag{P.6}$$

Translate the time origin to the left endpoint of the full signed-flux window and assume this window is contained in the frozen scale-`2R` exterior measurement interval `I_(2R)`. Use the Version-M translated spatial coordinates, and assume that the canonical plateau shell below is contained in the selected outer-collar cover. In particular, the short interval `[0,tau]` constructed below is measured by the same Version-M cubic row. This is a ledger-alignment hypothesis. It does not assert that an arbitrary constant-shear packet realizes the frozen inversion-paired zero-trajectory family.

For the final payment statement P.31, impose the additional field-realization hypothesis: after the common coordinate translation, `F` is an actual coordinate component of the same smooth velocity `v_R` to which `P_R^M` is applied throughout the aligned tube. Thus `|F|<=|v_R|` pointwise there. In particular, `F` is not a Littlewood--Paley or Fourier projection of a larger velocity component. Statements P.3--P.30 do not use this realization hypothesis.

Choose the admissible canonical profile from R0.75N so that, for a fixed `delta_0>0`,

$$\vartheta(s) = 1 \quad (|s| \le \delta_0), \qquad \xi_{a,R}(x) = \vartheta(|x|/R - a). \tag{P.7}$$

Assume `a>=4delta_0` and `(a+delta)R<pi/2`, as in the central-chart construction. Define its plateau shell

$$\mathcal{S}^{\mathrm{plat}}_{a,R} := \{x \in \mathbb{T}^3 : ||x|/R - a| \le \delta_0\}. \tag{P.8}$$

Let `y=(x_2,x_3)` and choose `phi_0 in C_c^infinity(T^2)` with

$$\begin{aligned} &0 \le \phi_0 \le 1, \qquad \operatorname{supp}\phi_0 \subset \{|y| \le (a - 3\delta_0)R\}, \\ &\|\Delta_y \phi_0\|_\infty \le C_\phi R^{-2}, \qquad V_\phi := |\operatorname{supp}\phi_0| \le \pi a^2 R^2. \end{aligned} \tag{P.9}$$

For every `|y|<= (a-2delta_0)R`, the `x_1`-fibre of the plateau shell has length at least `4delta_0R`. Indeed, with `q=|y|/R`, its exact length is

$$\ell_a(q) = 2R\left[\sqrt{(a+\delta_0)^2 - q^2} - \sqrt{(a-\delta_0)^2 - q^2}\right] \geq 4\delta_0 R. \tag{P.10}$$

The difference in brackets is increasing in `q` and equals `2delta_0` at `q=0`. This is a lower fibre estimate on the chosen plateau, not an upper estimate for the whole shell.

Packet, moving cutoff, and local persistence

Let

$$\Gamma_K = \{(n,j) \in \mathbb{Z}^2 : K \leq |n| \leq 2K,\ n^2 + j^2 \leq 4K^2\}, \tag{P.11}$$

and take the real finite datum

$$F_0(y) = \sum_{(n,j)\in\Gamma_K} c_{n,j} e^{i(nx_2 + jx_3)}, \qquad c_{-n,-j} = \overline{c_{n,j}}. \tag{P.12}$$

Its constant-shear evolution is

$$F(t,y) = \sum_{(n,j)\in\Gamma_K} c_{n,j} e^{-(n^2+j^2)t} e^{i(n(x_2 - Bt) + jx_3)}. \tag{P.13}$$

Transport the entrance cutoff by the same shear,

$$\phi_t(x_2, x_3) := \phi_0(x_2 - Bt, x_3), \qquad \partial_t \phi_t + B\partial_2 \phi_t = 0, \tag{P.14}$$

and put

$$E_\phi(t) := \int_{\mathbb{T}^2} \phi_t |F(t)|^2. \tag{P.15}$$

The exact local energy identity is

$$E_\phi'(t) = \int_{\mathbb{T}^2} \Delta_y \phi_t |F|^2 - 2\int_{\mathbb{T}^2} \phi_t |\nabla_y F|^2. \tag{P.16}$$

The total-frequency cap is preserved and gives

$$\|\nabla_y F(t)\|_2^2 \le 4K^2\|F(t)\|_2^2 \le 4K^2 E_0. \tag{P.17}$$

Since `K>=R^(-3/2)` and `R<=1`, one has `R^(-2)<=K^2`. Hence

$$E_\phi'(t) \ge -(8 + C_\phi)K^2 E_0. \tag{P.18}$$

Choose a fixed `c_0>0` satisfying

$$c_0 \le \min\left\{\frac{1}{2(8+C_\phi)}, \frac{\delta_0}{C_B}, 1\right\}, \qquad \tau := c_0 \mu K^{-2}. \tag{P.19}$$

If the available time window obeys `K^2T>=1`, then `tau<=T`. Moreover, `|B|tau<=delta_0R`, so `supp phi_t` stays inside `{|y|<=(a-2delta_0)R}` for `0<=t<=tau`. Integrating (P.18) and using (P.1) gives

$$\boxed{E_\phi(t) \ge \frac{\mu}{2}E_0 \quad (0 \le t \le \tau).} \tag{P.20}$$

Physical-collar cubic lower bound

Define the local three-dimensional cubic mass

$$M_{K,\mathrm{col}} := \int_0^\tau \int_{\mathcal{S}_{a,R}^{\mathrm{plat}}} |F(t,x_2,x_3)|^3 \, dx dt. \tag{P.21}$$

The field is independent of `x_1`. Equations (P.10) and (P.20), followed by Holder on `supp phi_t`, give for every `0<=t<=tau`

$$\begin{aligned}\int_{\mathcal{S}_{a,R}^{\mathrm{plat}}} |F(t)|^3 &\ge 4\delta_0 R \int_{\operatorname{supp}\phi_t} |F(t,y)|^3 \, dy \\ &\ge 4\delta_0 R\, V_\phi^{-1/2} E_\phi(t)^{3/2} \\ &\ge \frac{4\delta_0}{\sqrt{\pi}\, a}\left(\frac{\mu}{2}E_0\right)^{3/2}.\end{aligned} \tag{P.22}$$

Integration over `tau=c_0mu K^(-2)` proves (P.3) with the explicit constant

$$c_* := \frac{\sqrt{2}\,\delta_0 c_0}{\sqrt{\pi}}. \tag{P.23}$$

Equivalently,

$$E_0 \le c_*^{-2/3} a^{2/3} \mu^{-5/3} K^{4/3} M_{K,\mathrm{col}}^{2/3}. \tag{P.24}$$

No backward heat estimate or spectral observability constant is used. The power `mu^(5/2)` consists of `mu^(3/2)` from the persistent local energy and one further `mu` from the certified time length.

Flux closure and exact concentration threshold

Let `mathcal T_(K,eta)^(2)` be the full-window physical signed flux from R0.75O. Its arbitrary-vertical-frequency energy row and the canonical R0.75N Wiener estimate give

$$|\mathcal{T}_{K,\eta}^{(2)}| \le \frac{|B|\mathcal{W}_\infty}{4K^2} E_0, \qquad \mathcal{W}_\infty \le C_\vartheta a. \tag{P.25}$$

Substitution of (P.24) proves

$$|\mathcal{T}_{K,\eta}^{(2)}| \le C|B| a^{5/3} \mu^{-5/3} K^{-2/3} M_{K,\mathrm{col}}^{2/3}. \tag{P.26}$$

Define

$$p_{K,\mathrm{col}} := R^{-2}\omega M_{K,\mathrm{col}}, \qquad \mathfrak{X}_{K,\mathrm{col}} := \frac{\omega}{R}[\mathcal{T}_{K,\eta}^{(2)}]_+. \tag{P.27}$$

This gives (P.4). At `|B|<=C_BR^(-2)` and `K>=R^(-3/2)`,

$$\mathfrak{X}_{K,\mathrm{col}} \le C L^{5/3} \mu^{-5/3} R^{-2/3} \omega^{1/3} p_{K,\mathrm{col}}^{2/3}. \tag{P.28}$$

If `mu>=c_mu R^sigma`, the exponential rate of the coefficient in (P.28) is

$$\frac{\rho}{6} - \frac{c_\gamma}{12} + \frac{5\sigma\rho}{12}. \tag{P.29}$$

It is strictly negative exactly for (P.5), because

$$\frac{1}{5}\left(\frac{c_\gamma}{\rho} - 2\right) = \frac{1}{5}\left(\frac{80000}{35721} - 2\right) = \frac{8558}{178605}. \tag{P.30}$$

At equality the exponential rate vanishes and the factor `L^(5/3)` is unbounded, so the strict endpoint is excluded. Under the ledger-alignment hypotheses above, the plateau shell lies in the scale-`2R` exterior measurement region where the frozen weight is at least `omega`. Since `F` is the actual coordinate component of that same velocity, `|F|<=|v_R|` pointwise. Nonnegativity of the exterior cubic row therefore gives

$$p_{K,\mathrm{col}} \le C P_R^M, \qquad \boxed{\mathfrak{X}_{K,\mathrm{col}} \le C (P_R^M)^{2/3}} \tag{P.31}$$

for all sufficiently large `L` under (P.5).

The remaining low-concentration branch

For a packet whose energy is spatially spread at torus scale, the fraction captured by `phi_0` can be of order `a^2R^2`, hence of order `L^2R^2`. This is much smaller than `R^sigma` for every `sigma<sigma_*`. Such a packet need not have a large signed collar flux: the same spatial geometry and difference-frequency cancellation discarded by the global energy `E_0` may make the flux small.

Therefore failure of (P.5) is not a counterexample. It shows that the next estimate must localize the signed R0.75O kernel to the moving collar, or split near and far entrance data while preserving the horizontal packet cancellation. Replacing `E_0` by a positive fixed entrance majorant would repeat the R0.75K trace loss.

Status boundary

Proved: canonical plateau fibres P.7--P.10; moving-cutoff local energy and persistence P.14--P.20; the physical-collar cubic lower bound P.21--P.24; the conditional signed-flux estimate P.25--P.28; the exact entrance-concentration threshold P.29--P.30; and conditional Version-M ledger payment P.31 under the stated space-time alignment hypothesis.

Not proved: the entrance concentration P.5 for every packet; the low-concentration complement; a spatially localized signed heat kernel; nonconstant shear; inter-packet summation; low differences; removal of the total upper-frequency cap; arbitrary-field E.24; complete-clock extraction; fixed deletion; suitable-weak transfer; or any regularity or singularity conclusion. P.31 is not proved for a Fourier projection of a larger velocity component. No arbitrary constant-shear packet is claimed to realize the frozen inversion-paired zero-trajectory family. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75L proves a favorable diffusive factor for one real constant-shear harmonic but pays it with a full-circle cubic mass. R0.75P replaces the full-torus mass by a physical radial-collar atom when a quantified fraction of the entrance energy lies in a central transverse disk. The present note closes the complementary benchmark in which the field is one harmonic and is independent of `x_1,x_3`.

Let

$$F_k(t,x_2) = A e^{-k^2 t} \cos(k(x_2 - Bt)), \qquad A > 0, \quad k \in \mathbb{N}, \tag{Q.1}$$

and use the canonical radial plateau shell of radius `aR` and fixed thickness. If

$$k^2 T \ge 1, \qquad kaR \ge 2\pi, \qquad k \ge R^{-3/2}, \qquad |B| \le C_B R^{-2}, \qquad a \ge 4\delta_0, \qquad (a+\delta)R < \frac{\pi}{2}, \tag{Q.2}$$

then its physical-collar cubic mass satisfies

$$\boxed{M_{k,\mathrm{col}} \ge c_{\mathrm{box}}\delta_0 a^2 R^3 k^{-2} A^3, \qquad c_{\mathrm{box}} := \frac{2(1-e^{-3})}{9\pi}.} \tag{Q.3}$$

For every measurable time weight `0<=eta<=1`, the signed physical cutoff flux defined below is consequently bounded by

$$\boxed{|\mathcal{T}^{(3)}_{k,\eta}| \le C_\vartheta \delta_0^{-2/3}|B|a^{2/3}k^{-2/3}M_{k,\mathrm{col}}^{2/3}.} \tag{Q.4}$$

After the frozen normalization this gives

$$\boxed{\mathfrak{X}_{k,\mathrm{col}} \le CL^{2/3}R^{-2/3}\omega^{1/3}p_{k,\mathrm{col}}^{2/3} \le Cp_{k,\mathrm{col}}^{2/3}} \tag{Q.5}$$

for all sufficiently large `L`. The coefficient tends to zero at the exact rate

$$L^{2/3}\exp\left(-\frac{4279}{238140000}L^2\right). \tag{Q.6}$$

No entrance-concentration hypothesis is used. Indeed, the entrance fraction for this spatially spread harmonic can be at most of order `a^2R^2`, far below the R0.75P threshold. This closes the physical-collar localization left open by R0.75L for this one subfamily.

The theorem remains one real harmonic, constant shear, independent of `x_1,x_3`, and total-field rather than packet-projection based. It does not handle arbitrary low-concentration packets, vertical structure, interference, nonconstant shear, or E.24.

Frozen inputs and radial geometry

The immediately used frozen inputs are

input	SHA-256	role
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	outer-collar weight and frozen scales
`research/r075l_single_harmonic_diffusive_signed_flux_gain.md`	`52e25b2fdf1a224609c9e8fafa1c041b7f09a361f75f4b3e44ebcdddb756cdf5`	exact real harmonic and diagonal cancellation
`research/r075n_radial_collar_averaged_wiener_row.md`	`ba59a4df399d8580b35d8dbb3f0758f9d2ffcc7f97f1147e5804c428f3740318`	canonical radial cutoff
`research/r075p_buffered_collar_entrance_concentration.md`	`8df38e54514d82102cd3e568e89ec1db93913da3ceac52f1371d77fd79c1b7a6`	plateau fibres and actual-component ledger boundary

Retain

$$a = pL, \qquad p = \frac{32}{63}, \qquad R = e^{-\rho L^2/4}, \qquad \omega = e^{-c_\gamma L^2/4}, \qquad \rho = \frac{9}{10000}, \qquad c_\gamma = \frac{8}{3969}. \tag{Q.7}$$

Choose `0<delta_0<delta` and `0<=vartheta<=1`, and let the central-chart formula below denote its periodic lift:

$$\xi_{a,R}(x)=\vartheta(|x|/R-a),\qquad \vartheta=1\quad\text{on }[-\delta_0,\delta_0],\qquad \operatorname{supp}\vartheta\subset(-\delta,\delta). \tag{Q.8}$$

Assume `a>=4delta_0` and `(a+delta)R<pi/2`. Its plateau shell is

$$\mathcal{S}^{\mathrm{plat}}_{a,R}:=\{x\in\mathbb{T}^3:||x|/R-a|\le\delta_0\}. \tag{Q.9}$$

The derivative is supported in a shell of volume at most `C_delta a^2R^3`, while `|partial_2 xi|<=C_vartheta R^(-1)`. Therefore

$$V_{\xi,3}:=\int_{\mathbb{T}^3}|\partial_2\xi_{a,R}|\,dx\le C_\vartheta a^2R^2. \tag{Q.10}$$

This direct `L^1` row is sufficient for one harmonic. No Wiener summation is needed.

Exact signed-flux bound

The field Q.1 obeys

$$(\partial_t+B\partial_2-\partial_2^2)F_k=0. \tag{Q.11}$$

For any measurable `0<=eta<=1`, define the three-dimensional physical flux

$$\mathcal{T}^{(3)}_{k,\eta}:=\frac{1}{2}\int_0^T\int_{\mathbb{T}^3}\eta(t)B\partial_2\xi_{a,R}(x)|F_k(t,x_2)|^2\,dxdt. \tag{Q.12}$$

The square identity is

$$|F_k|^2=\frac{A^2e^{-2k^2t}}{2}[1+\cos(2k(x_2-Bt))]. \tag{Q.13}$$

The constant row vanishes because the integral of `partial_2 xi` over the torus is zero. Taking absolute values only after that cancellation and using Q.10 gives

$$\begin{aligned}|\mathcal{T}^{(3)}_{k,\eta}|&\le\frac{A^2|B|V_{\xi,3}}{4}\int_0^T e^{-2k^2t}\,dt\\&\le\frac{A^2|B|V_{\xi,3}}{8k^2}\le C_\vartheta A^2|B|a^2R^2k^{-2}.\end{aligned} \tag{Q.14}$$

The bound uses ordinary heat decay and the exact zero row. It is not an enhanced-dissipation estimate.

A phase-uniform physical-collar lower bound

For

$$\mathcal{R}_{a,R} := \{(x_2,x_3) : |x_2| \le aR/4,\ |x_3| \le aR/4\}, \tag{Q.15}$$

one has `|(x_2,x_3)|<=aR/(2sqrt(2))<=(a-2delta_0)R`. R0.75P's exact fibre formula therefore gives

$$\left|\{x_1 : (x_1,x_2,x_3) \in \mathcal{S}^{\mathrm{plat}}_{a,R}\}\right| \ge 4\delta_0 R \quad ((x_2,x_3) \in \mathcal{R}_{a,R}). \tag{Q.16}$$

The function `|cos z|^3` has period `pi` and

$$\int_0^\pi |\cos z|^3\, dz = \frac{4}{3}. \tag{Q.17}$$

An interval of length `ell` contains `floor(k ell/pi)` complete periods of `|cos(kx-phi)|^3`, starting from its own left endpoint. If `k ell/pi>=1`, then `floor(k ell/pi)>=(k ell/pi)/2`. With `ell=aR/2` and Q.2, this gives, uniformly in the phase `phi`,

$$\int_{-aR/4}^{aR/4} |\cos(kx_2-\phi)|^3\, dx_2 \ge \frac{aR}{3\pi}. \tag{Q.18}$$

Combining Q.16, Q.18, and the `x_3` interval of length `aR/2` yields

$$\int_{\mathcal{S}^{\mathrm{plat}}_{a,R}} |\cos(k(x_2-Bt))|^3\, dx \ge \frac{2\delta_0 a^2}{3\pi} R^3 \tag{Q.19}$$

for every time, with no restriction on the translated phase.

Define

$$M_{k,\mathrm{col}} := \int_0^T \int_{\mathcal{S}^{\mathrm{plat}}_{a,R}} |F_k|^3\, dx dt. \tag{Q.20}$$

Since `k^2T>=1`, Q.19 gives

$$\begin{aligned} M_{k,\mathrm{col}} &\ge \frac{2\delta_0 a^2 R^3}{3\pi} A^3 \int_0^T e^{-3k^2 t}\, dt \\ &\ge \frac{2(1-e^{-3})}{9\pi} \delta_0 a^2 R^3 k^{-2} A^3. \end{aligned} \tag{Q.21}$$

This proves Q.3 and implies

$$A^2 \le c_{\rm box}^{-2/3}\delta_0^{-2/3}a^{-4/3}R^{-2}k^{4/3}M_{k,\rm col}^{2/3}. \tag{Q.22}$$

Substitution in Q.14 cancels the full `R^2` and proves Q.4.

Frozen normalization and Version-M inclusion

Set

$$p_{k,\rm col} := R^{-2}\omega M_{k,\rm col}, \qquad \mathfrak{X}_{k,\rm col} := \frac{\omega}{R}[\mathcal{T}_{k,\eta}^{(3)}]_+. \tag{Q.23}$$

Equation Q.4 becomes

$$\mathfrak{X}_{k,\rm col} \le C|B|a^{2/3}R^{1/3}\omega^{1/3}k^{-2/3}p_{k,\rm col}^{2/3}. \tag{Q.24}$$

At the frozen `B,k` bounds this coefficient is at most

$$CL^{2/3}R^{-2/3}\omega^{1/3} = CL^{2/3}\exp\left(-\frac{4279}{238140000}L^2\right) \longrightarrow 0. \tag{Q.25}$$

This proves Q.5--Q.6 against the harmonic's own physical-collar atom.

For the final Version-M payment only, impose the same ledger-alignment and realized-subclass conditions as R0.75P. Translate the time origin to the left endpoint of the flux window, require `[0,T]` to lie in the same scale-`2R` exterior measurement interval, and require the plateau tube to lie in an exterior row whose weight is at least `omega`. After the common coordinate translation, require `F_k` to be an actual coordinate component of the same smooth velocity `v_R` measured by `P_R^M` throughout that tube. Then `|F_k|<=|v_R|` pointwise and nonnegativity of the exterior cubic row gives

$$p_{k,\rm col} \le CP_R^M, \qquad \boxed{\mathfrak{X}_{k,\rm col} \le C(P_R^M)^{2/3}} \tag{Q.26}$$

for sufficiently large `L`. This is not asserted for a harmonic projection of a larger velocity component, and it does not assert arbitrary zero-trajectory realization.

Why this result belongs to the low-entrance branch

Let `phi_0` be any R0.75P entrance cutoff. At `t=0`,

$$E_0 = 2\pi^2A^2, \qquad E_{\rm in} \le A^2|\operatorname{supp}\phi_0| \le \pi a^2R^2A^2, \qquad \frac{E_{\rm in}}{E_0} \le \frac{a^2R^2}{2\pi}. \tag{Q.27}$$

For every fixed `0<=sigma<2`,

$$\frac{a^2R^2}{R^\sigma}=p^2L^2R^{2-\sigma}\longrightarrow 0. \tag{Q.28}$$

In particular this can lie far below the P threshold `sigma_*=8558/178605`. Q.26 is therefore an independent cancellation and spatial-spreading mechanism, not a hidden use of P's entrance hypothesis.

Status boundary

Proved: the radial derivative `L^1` row Q.8--Q.10; exact diagonal cancellation and flux bound Q.11--Q.14; the phase-uniform rectangular subcollar estimate Q.15--Q.21; the physical-collar cubic conversion Q.22--Q.25; conditional actual-component Version-M payment Q.26; and the low-entrance diagnostic Q.27--Q.28.

Not proved: the same result for a Fourier or Littlewood--Paley projection of a larger field; two or more horizontal harmonics; arbitrary vertical structure; a general low-entrance packet; nonconstant shear; inter-packet or low-difference summation; removal of the total upper- frequency cap in the broader route; arbitrary-field E.24; complete-clock extraction; fixed deletion; suitable-weak transfer; or any regularity or singularity conclusion. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75Q pays the signed radial-cutoff flux of one spatially spread shear harmonic with that harmonic's cubic mass on the physical plateau shell. The present note tests whether the same plateau atom can pay an arbitrary high-frequency packet. It cannot.

Fix an integer `m>=1` and the canonical radial cutoff from R0.75Q. For every sufficiently large frozen parameter `L`, the construction below gives a real trigonometric polynomial `G_K`, supported in the horizontal frequency band

$$\{j\in\mathbb{Z}:K\le|j|\le 2K\}, \tag{R.1}$$

and an exact global smooth Navier--Stokes shear solution

$$u_K(t,x)=(0,B,F_K(t,x_2)),\quad F_K(t,x_2)=e^{t\partial_2^2}G_K(x_2-Bt). \tag{R.2}$$

Here `B=-bR^(-2)` for a fixed sufficiently small `b>0` and the observation time is `T=K^(-2)`. With `eta=1`, define

$$\begin{aligned}\mathcal{T}_K&:=\frac{1}{2}\int_0^T\int_{\mathbb{T}^3}B\,\partial_2\xi_{a,R}(x)|F_K(t,x_2)|^2\,dxdt,\\ M_{K,\mathrm{plat}}&:=\int_0^T\int_{\mathcal{S}_{a,R}^{\mathrm{plat}}}|F_K(t,x_2)|^3\,dxdt.\end{aligned} \tag{R.3}$$

There are constants depending only on `m` and the fixed cutoff such that

$$\boxed{\mathcal{T}_K \geq c_m|B|aRA^2n^{-1}K^{-2}, \quad M_{K,\mathrm{plat}} \leq C_mA^3a^2R^3K^{-2}(nR)^{-6m}.} \tag{R.4}$$

After the frozen normalization

$$p_{K,\mathrm{plat}} := R^{-2}\omega M_{K,\mathrm{plat}}, \quad \mathfrak{X}_K := \frac{\omega}{R}[\mathcal{T}_K]_+, \tag{R.5}$$

the quotient grows at least as

$$\boxed{\frac{\mathfrak{X}_K}{p_{K,\mathrm{plat}}^{2/3}} \geq c_mbL^{-1/3}R^{-2m-1/6}\omega^{1/3} = c_mbL^{-1/3}e^{\kappa_mL^2},} \tag{R.6}$$

where

$$\kappa_m := \frac{(2m+1/6)\rho}{4} - \frac{c_\gamma}{12} > 0. \tag{R.7}$$

For the smallest choice `m=1`, the exact rate is

$$\kappa_1 = \frac{304373}{952560000} > 0, \quad R^{-13/6}\omega^{1/3} = R^{-304373/214326}. \tag{R.8}$$

Consequently there is no constant independent of `L` in the plateau-only multimode estimate

$$\mathfrak{X}_K \leq Cp_{K,\mathrm{plat}}^{2/3}. \tag{R.9}$$

This is a negative result about that specific attempted extension of Q. The complete Version-M payment sees exterior rows beyond the plateau and also sees the background component `B`. Thus R.9 is not a counterexample to E.24 or to Version-M. In addition, this note does not prove that its constant background belongs to the frozen mean-zero, inversion-paired Version-M subclass. It does not concern singularity formation.

Frozen geometry and the exact outer-cap identity

Retain

$$a = pL, \quad p = \frac{32}{63}, \quad R = e^{-\rho L^2/4}, \quad \omega = e^{-c_\gamma L^2/4}, \quad \rho = \frac{9}{10000}, \quad c_\gamma = \frac{8}{3969}. \tag{R.10}$$

The immediately used frozen inputs are

input	SHA-256	role
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	frozen `a,R,omega` scales and shear bound
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5049`	definition and open status of E.24
`research/r075n_radial_collar_averaged_wiener_row.md`	`ba59a4df399d8580b35d8dbb3f0758f9d2ffcc7f97f1147e5804c428f3740318`	canonical radial cutoff geometry
`research/r075q_spatially_spread_harmonic_collar_payment.md`	`9d7058fd7fbc61136967227507e47b0e866c7a4eeafebae198ab05a23645ed9c`	one-harmonic plateau payment being tested

Let `0<delta_0<delta`, and let

$$\xi_{a,R}(x) = \vartheta(|x|/R - a), \quad 0 \le \vartheta \le 1, \quad \vartheta = 1 \text{ on } [-\delta_0, \delta_0], \quad \operatorname{supp} \vartheta \subset (-\delta, \delta). \tag{R.11}$$

As before, assume `a>=4delta_0` and `(a+delta)R<pi/2`, so the support lies in one Euclidean chart of the torus. Integrate the cutoff derivative over the two coordinates that do not occur in `F_K`:

$$D_R(y) := \int_{\mathbb{T}^2_{x_1,x_3}} \partial_2 \xi_{a,R}(x_1, y, x_3)\, dx_1 dx_3. \tag{R.12}$$

The cross-sectional cutoff mass is

$$\Xi_R(y) = 2\pi \int_{|y|}^{\infty} \vartheta(\varrho/R - a)\varrho\, d\varrho. \tag{R.13}$$

Differentiating its moving lower endpoint gives the exact identity

$$\boxed{D_R(y) = \Xi_R'(y) = -2\pi y \vartheta(|y|/R - a).} \tag{R.14}$$

In particular, `D_R<0` on the positive cap and `D_R>0` on the negative cap. This identity is stronger than the absolute `L^1` estimate used in Q.

Continuity and `vartheta(delta_0)=1` provide numbers

$$s_* \in (\delta_0, \delta), \quad h > 0, \quad c_\vartheta > 0 \tag{R.15}$$

such that

$$\delta_0 < s_* - 3h < s_* + 3h < \delta, \quad \vartheta(s) \ge c_\vartheta \quad (|s - s_*| \le 3h). \tag{R.16}$$

Set `y_0=(a+s_*)R`. The interval

$$I_+ := [y_0 - 2hR, y_0 + 2hR] \tag{R.17}$$

lies in the positive cutoff cap but outside the `x_2` projection of the plateau shell if `h` is decreased once more. On this interval,

$$-D_R(y) \geq c_\vartheta aR. \tag{R.18}$$

An explicit high-band packet

Let `K` be the smallest multiple of `16m` not below `R^(-3/2)`, and put

$$n = \frac{K}{16m}, \quad q = \frac{3K}{2}, \quad T = K^{-2}. \tag{R.19}$$

For all sufficiently large `L`,

$$R^{-3/2} \leq K \leq 2R^{-3/2}, \quad n \asymp_m R^{-3/2}, \quad nR \longrightarrow \infty. \tag{R.20}$$

Use the normalized Dirichlet kernel

$$d_n(z) = \frac{1}{2n+1} \sum_{r=-n}^{n} e^{irz} = \frac{\sin((n+1/2)z)}{(2n+1)\sin(z/2)} \tag{R.21}$$

and define the real packet

$$G_K(y) = A\, d_n(y-y_0)^{2m} \cos(q(y-y_0)), \quad A > 0. \tag{R.22}$$

The Fourier support of `d_n^(2m)` lies in `[-2mn,2mn]=[-K/8,K/8]`. Multiplication by the real carrier shifts this support by `+q` and `-q`. Therefore

$$\operatorname{supp} \widehat{G}_K \subset \left\{ j : \frac{11K}{8} \leq |j| \leq \frac{13K}{8} \right\}, \tag{R.23}$$

which proves R.1 without a projection remainder.

The elementary Dirichlet bound

$$|d_n(z)| \leq C \min\left\{1, \frac{1}{n \operatorname{dist}(z, 2\pi\mathbb{Z})}\right\} \tag{R.24}$$

and a fixed interval of length comparable to `K^(-1)` on which both factors in R.22 are bounded below show that

$$c_m A^2 n^{-1} \le \|G_K\|^2_{L^2(\mathbb{T})} \le C_m A^2 n^{-1}. \tag{R.25}$$

For every fixed `r>0`, R.24 also gives the tail bounds

$$\begin{gathered}\sup_{\operatorname{dist}(y,y_0)\ge rR} |G_K(y)| \le C_{m,r} A(nR)^{-2m}, \\ \int_{\operatorname{dist}(y,y_0)\ge rR} |G_K(y)|^2 dy \le C_{m,r} A^2 n^{-1} (nR)^{1-4m}.\end{gathered} \tag{R.26}$$

The relative tail in the second line tends to zero for every `m>=1`.

Exact Navier--Stokes evolution

Choose

$$0 < b \le b_* := \min\{C_B, h/8, (s_* - \delta_0)/16\}, \quad B = -bR^{-2}. \tag{R.27}$$

Since `K>=R^(-3/2)`, the drift over R.19 satisfies

$$|B|T \le bR. \tag{R.28}$$

Define `F_K` and `u_K` by R.2. Direct differentiation gives

$$(\partial_t + B\partial_2 - \partial_2^2)F_K = 0. \tag{R.29}$$

Moreover `div u_K=0` and

$$\nabla\cdot u_K = 0, \quad (u_K\cdot\nabla)u_K = (0,0,B\partial_2 F_K).$$

Consequently

$$\partial_t u_K + (u_K\cdot\nabla)u_K - \Delta u_K = 0 \tag{R.30}$$

with constant pressure. Thus the example is an exact unforced smooth three-dimensional Navier--Stokes solution, not a passive-scalar surrogate or a numerical simulation.

The support row R.23 is preserved by the heat semigroup. Hence, for `0<=t<=T`,

$$\|F_K(t)\|_2^2 \geq e^{-8}\|G_K\|_2^2 \geq c_m A^2 n^{-1}. \tag{R.31}$$

The constant `8` is deliberately coarse; `|j|<=2K` and `t<=K^(-2)` are sufficient.

Short-time localization and the positive flux

For two subsets of the circle separated by distance `d`, the one-dimensional heat semigroup satisfies the elementary Gaussian off-diagonal estimates

$$\|1_E e^{t\partial_2^2} 1_H\|_{2\to 2} \leq e^{-d^2/(4t)}, \quad \sup_{w\in H}\int_{\mathrm{dist}(z,H)\geq d} H_t(z-w)\,dz \leq Ce^{-d^2/(8t)}. \tag{R.32}$$

They follow directly from the periodized Gaussian heat kernel; only fixed changes in the numerical constants are used below.

Split `G_K` into the part within distance `hR/2` of `y_0` and its complement. Equations R.26, R.28, and R.32 imply, uniformly for `0<=t<=T`,

$$\int_{I_+} |F_K(t,y)|^2 dy \geq c_m A^2 n^{-1} \tag{R.33}$$

once `L` is large. Indeed, the square root of the relative initial tail is `O((nR)^((1-4m)/2))`, while the heat leakage is `O(exp(-c(KR)^2))`; both tend to zero. Explicitly,

$$\|1_{I_+}F_K(t)\|_2 \geq \|F_K(t)\|_2 - \|1_{I_+^c}F_K(t)\|_2,$$

so the global lower bound minus the vanishing leakage has the required direction. The same argument, with a larger separation, gives

$$\int_{\{y<0:\, D_R(y)>0\}} |F_K(t,y)|^2 dy = o_m(A^2 n^{-1}). \tag{R.34}$$

Because `B<0`, the integrand `BD_R|F_K|^2` is nonnegative for `y>0` and nonpositive for `y<0`. Use R.18 and R.33 for the positive contribution and R.34 for the only adverse contribution. For sufficiently large `L`,

$$\begin{aligned} \mathcal{T}_K &= \frac{B}{2}\int_0^T\int_{\mathbb{T}} D_R(y)|F_K(t,y)|^2 dydt \\ &\geq c_m|B|aR\int_0^T A^2 n^{-1} dt \geq c_m|B|aRA^2n^{-1}K^{-2}. \end{aligned} \tag{R.35}$$

This is the first inequality in R.4.

What the plateau atom misses

If `(x_1,y,x_3)` belongs to the plateau shell, then

$$|y| \le (a+\delta_0)R. \tag{R.36}$$

Its circular distance from `y_0` is therefore at least `(s_*-delta_0)R`. R.27 keeps the transported center a fixed multiple of `R` away from that projection throughout `[0,T]`.

Split the heat convolution into initial points near and far from `y_0`. The near part is controlled by the Gaussian tail in R.32; the far part is controlled in `L^infinity` by R.26 and heat-semigroup contraction. Thus

$$\sup_{\substack{0\le t\le T\\ |y|\le (a+\delta_0)R}} |F_K(t,y)| \le C_m A\left[(nR)^{-2m} + e^{-c(KR)^2}\right] \le C_m A(nR)^{-2m} \tag{R.37}$$

for large `L`. The plateau shell has volume at most `C a^2R^3`. Multiplying R.37 cubed by this volume and by `T=K^(-2)` proves

$$M_{K,\mathrm{plat}} \le C_m A^3 a^2 R^3 K^{-2} (nR)^{-6m}, \tag{R.38}$$

the second inequality in R.4.

Frozen exponent and failure of the plateau-only estimate

Equations R.35 and R.38 give

$$\frac{\mathcal{T}_K}{M_{K,\mathrm{plat}}^{2/3}} \ge c_m |B| a^{-1/3} R^{-1} n^{-1} K^{-2/3} (nR)^{4m}. \tag{R.39}$$

The normalization R.5 multiplies this quotient by `R^(1/3)omega^(1/3)`. Substituting `|B|=bR^(-2)` and R.20 yields

$$\frac{\mathfrak{X}_K}{p_{K,\mathrm{plat}}^{2/3}} \ge c_m b a^{-1/3} R^{-2m-1/6} \omega^{1/3}. \tag{R.40}$$

Since `a=(32/63)L`, its polynomial factor is `C L^(-1/3)`. Finally,

$$R^{-2m-1/6}\omega^{1/3} = \exp\left[\left(\frac{(2m+1/6)\rho}{4} - \frac{c_\gamma}{12}\right)L^2\right]. \tag{R.41}$$

At `m=1`, exact rational arithmetic gives R.8. Because `kappa_1>0` and `kappa_m` increases with `m`, R.6 diverges for every fixed `m>=1`. This proves R.9.

The amplitude `A` cancels from the quotient. The obstruction is therefore geometric and spectral, not an artifact of a large-amplitude limit.

Consequence for the route

The one-harmonic Q argument used phase-uniform spatial spreading across the plateau fibres. R shows that band limitation alone does not retain that property for a multimode field: coherent modes can concentrate at scale `n^(-1)<<R` in an outer cap and leave only an algebraic tail on the plateau.

Any valid continuation of the route must therefore supply at least one of the following ingredients:

1. a payment atom covering the full signed-flux cap rather than only the plateau; 2. a quantitative spreading or thickness hypothesis that rules out the packet R.22; 3. a signed multimode cancellation that is not reduced to the plateau's absolute cubic mass; or 4. an independent mechanism that transfers outer-cap mass into an already counted Version-M row.

This narrows the next positive target. It does not show that all localized or signed methods fail.

Status boundary

Proved: the exact cross-sectional identity R.14; the explicit real high-band support R.23; the Dirichlet concentration and tail rows R.25--R.26; exact global smooth Navier--Stokes realization R.29--R.30; short-time outer-cap persistence R.31--R.34; the positive signed-flux lower bound R.35; the plateau cubic upper bound R.38; and the divergent normalized quotient R.40--R.41.

Ruled out: a uniform extension of Q to arbitrary high-band packets when the only payment is the cubic mass of the canonical plateau shell.

Not ruled out or proved: a payment using the full cutoff support; the complete nonnegative exterior rows already present in Version-M; a spectral or physical spreading hypothesis; arbitrary nonconstant shear; nonlinear mode transfer; the arbitrary-field estimate E.24; complete-clock extraction; fixed deletion; suitable-weak transfer; or any regularity or singularity conclusion. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75Q pays one spatially spread real shear harmonic on a physical radial collar under the high-frequency assumptions `k^2 T>=1`, `kaR>=2pi`, and `k>=R^(-3/2)`. R0.75R shows why the same plateau-only argument cannot be extended to an arbitrary concentrated high-band packet. The present note returns to the one-harmonic class and removes all three frequency restrictions by using the complete frozen time window.

Set

$$T_R := t_2 - s_R = 4R^2 \tag{S.1}$$

and translate `s_R` to time zero. Let `eta` be the frozen nondecreasing time cutoff, so `0<=eta<=1`, `eta(0)=0`, and its total variation is at most one. For every `A>0`, integer `k>=1`, phase `phi in R`, and constant `B in R`, define

$$F_k(t,x_2) = Ae^{-k^2 t}\cos\bigl(k(x_2 - Bt) - \phi\bigr),\quad 0 \le t \le T_R. \tag{S.2}$$

For the canonical radial cutoff and its plateau shell, put

$$\begin{aligned}\mathcal{T}_{k,R} &:= \frac{1}{2}\int_0^{T_R}\int_{\mathbb{T}^3}\eta(t)B\,\partial_2\xi_{a,R}|F_k|^2\,dxdt,\\ M_{k,R}^{\mathrm{plat}} &:= \int_0^{T_R}\int_{\mathcal{S}_{a,R}^{\mathrm{plat}}}|F_k|^3\,dxdt.\end{aligned} \tag{S.3}$$

There is a constant depending only on the fixed radial profile and plateau width such that, for all sufficiently large `L`,

$$\boxed{|\mathcal{T}_{k,R}| \le Ca^{2/3}R^{-1/3}\big(M_{k,R}^{\mathrm{plat}}\big)^{2/3}} \tag{S.4}$$

simultaneously for every `k`, `B`, `phi`, and `A`. In particular, no lower or upper bound on the constant shear is needed for this estimate.

With the frozen normalization

$$p_{k,R}^{\mathrm{plat}} := R^{-2}\omega M_{k,R}^{\mathrm{plat}},\quad \mathfrak{X}_{k,R} := \frac{\omega}{R}[\mathcal{T}_{k,R}]_+, \tag{S.5}$$

equation S.4 becomes

$$\boxed{\mathfrak{X}_{k,R} \le Ca^{2/3}\omega^{1/3}\big(p_{k,R}^{\mathrm{plat}}\big)^{2/3}.} \tag{S.6}$$

Since `a=pL` and `omega=exp[-(c_gamma/4)L^2]`, its coefficient has the strict rate

$$\lim_{L\to\infty}\frac{1}{L^2}\log\big(a^{2/3}\omega^{1/3}\big) = -\frac{c_\gamma}{12} < 0. \tag{S.7}$$

Thus the full frozen clock pays the physical-collar flux of every single real horizontal harmonic, including all frequencies omitted by Q. The proof is not a multimode estimate. It neither contradicts R nor controls interference between distinct harmonics, a nonconstant shear, or the arbitrary-field target E.24.

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Frozen geometry and exact scalar reduction

Retain

$$a = pL,\quad p = \frac{32}{63},\quad R = e^{-\rho L^2/4},\quad \omega = e^{-c_\gamma L^2/4},\quad \rho = \frac{9}{10000},\quad c_\gamma = \frac{8}{3969}. \tag{S.8}$$

The immediately used frozen inputs are

input	SHA-256	role
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	complete time window and monotone cutoff

input	SHA-256	role
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5049`	physical signed flux and E.24
`research/r075q_spatially_spread_harmonic_collar_payment.md`	`9d7058fd7fbc61136967227507e47b0e866c7a4eeafebae198ab05a23645ed9c`	phase-uniform high-frequency plateau mass
`research/r075r_outer_cap_spectral_concentration_obstruction.md`	`e5eba5b262a8e140eaa149b6d914f355f2f3c636ec1e74cf85515f1c38fd32f3`	exact radial cross section and multimode boundary

Choose fixed `0<delta_0<delta` and a smooth profile `0<=vartheta<=1` such that

$$\xi_{a,R}(x)=\vartheta(|x|/R-a),\quad \vartheta=1\text{ on }[-\delta_0,\delta_0],\quad \operatorname{supp}\vartheta\subset(-\delta,\delta). \tag{S.9}$$

Assume `a>=4delta_0` and `(a+delta)R<pi/2`. Its plateau shell is

$$\mathcal{S}^{\mathrm{plat}}_{a,R}:=\{x\in\mathbb{T}^3:||x|/R-a|\le\delta_0\}. \tag{S.10}$$

The exact cross-sectional derivative from R0.75R is

$$D_R(y):=\int_{\mathbb{T}^2_{x_1,x_3}}\partial_2\xi_{a,R}(x_1,y,x_3)\,dx_1dx_3=-2\pi y\vartheta(|y|/R-a). \tag{S.11}$$

It is odd. Define its sine coefficient

$$S_{k,R}:=\int_{-\pi}^{\pi}D_R(y)\sin(2ky)\,dy. \tag{S.12}$$

The constant and cosine rows vanish by oddness. Expanding the square in S.2 therefore gives the exact one-dimensional identity

$$\boxed{\mathcal{T}_{k,R}=\frac{A^2BS_{k,R}}{4}\int_0^{T_R}\eta(t)e^{-2k^2t}\sin(2\phi+2kBt)\,dt.} \tag{S.13}$$

This preserves both signs until the scalar oscillatory integral has been identified.

Three bounds for the radial sine coefficient

Put

$$q:=kR,\quad \varepsilon:=kaR=aq. \tag{S.14}$$

After `y=Rz`, equation S.12 becomes

$$S_{k,R} = -2\pi R^2 \int z\vartheta(|z|-a)\sin(2qz)\,dz. \tag{S.15}$$

The integrand is supported on two intervals of fixed length centered at `+/-a`. Consequently

$$\|z\vartheta(|z|-a)\|_{L^1} \le Ca, \quad \|\partial_z^N[z\vartheta(|z|-a)]\|_{L^1} \le C_N a \quad (N \ge 1). \tag{S.16}$$

The elementary bound `|sin(2qz)|<=2q|z|`, the first inequality in S.16, and `N` integrations by parts give, respectively,

$$\boxed{|S_{k,R}| \le C_N a R^2 \min\{\varepsilon, 1, q^{-N}\}.} \tag{S.17}$$

The three entries have different roles: `epsilon` resolves low-frequency oddness, `1` is the direct cross-sectional `L^1` size, and `q^(-N)` pays frequencies above the inverse collar thickness.

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Two elementary phase--mass lemmas

Let

$$d(\psi) := \operatorname{dist}\big(\psi, \pi/2 + \pi\mathbb{Z}\big), \quad Q_\varepsilon(\psi) := \min\{1, \varepsilon + d(\psi)\}. \tag{S.18}$$

The rectangular subcollar used in Q contains

$$|x_2| \le aR/4, \quad |x_3| \le aR/4, \quad |\{x_1 : (x_1, x_2, x_3) \in \mathcal{S}^{\mathrm{plat}}_{a,R}\}| \ge 4\delta_0 R. \tag{S.19}$$

For `0<epsilon<=2pi`, a one-dimensional node estimate yields

$$\int_{-1/4}^{1/4} |\cos(\varepsilon z - \psi)|^3\,dz \ge c\,Q_\varepsilon(\psi)^3. \tag{S.20}$$

Indeed, if `epsilon+d(psi)` is small, translate the nearest cosine node and use `|sin r|>=2|r|/pi` on `|r|<=pi/2`; the integral of `|epsilon z-r_0|^3` is bounded below by a constant times `(epsilon+|r_0|)^3`. If `epsilon+d(psi)` is bounded below, compactness on one phase period gives a positive lower bound. Equations S.19--S.20 imply

$$\int_{\mathcal{S}^{\mathrm{plat}}_{a,R}} |\cos(kx_2 - \psi)|^3\,dx \ge c\,a^2 R^3 Q_\varepsilon(\psi)^3. \tag{S.21}$$

The second lemma couples a moving node to the signed time phase. If `0<epsilon<=2pi`, `0<=lambda<=1`, `sigma in R`, and `w(s)=eta(T_Rs)e^{-2lambda s}`, then

$$\boxed{\left|\sigma \int_0^1 w(s)\sin(2\phi+2\sigma s)\,ds\right| \le C\left[\int_0^1 Q_\varepsilon(\phi+\sigma s)^3\,ds\right]^{2/3}.} \tag{S.22}$$

To verify it, denote the integral inside the brackets by `J`. For `|sigma|<=1`, an interval of phase length `|sigma|` has

$$J^{1/3} \ge c\min\{1,\varepsilon+|\sigma|\}. \tag{S.23}$$

Moreover `|sin(2psi)|<=2Q_epsilon(psi)`. Holder then gives

$$|\sigma|\int_0^1 |\sin(2\phi+2\sigma s)|\,ds \le C|\sigma|J^{1/3} \le CJ^{2/3}. \tag{S.24}$$

For `|sigma|>=1`, a phase interval of length at least one gives `J>=c`. The monotonicity of `eta` gives, in fact for every `lambda>=0`,

$$\operatorname{Var}_{[0,1]} w \le \int_0^1 d\eta(T_R s) + 2\lambda\int_0^1 \eta(T_R s)e^{-2\lambda s}\,ds \le 2. \tag{S.25}$$

One integration by parts against the phase then bounds the left side of S.22 by an absolute constant. This proves S.22 in both regimes. The estimate is exactly what prevents a harmonic that initially lies near a cosine node from escaping payment: either the shear is too small to create flux, or it moves the node through enough plateau mass.

Low spatial frequency: node motion pays the clock

Assume first

$$\varepsilon = kaR \le 2\pi. \tag{S.26}$$

For all sufficiently large `L`, `a>=4pi`, and hence

$$\lambda := k^2T_R = 4q^2 \le \frac{16\pi^2}{a^2} \le 1. \tag{S.27}$$

Set `sigma=kBT_R`. Equations S.13 and S.17 give

$$|\mathcal{T}_{k,R}| \le CA^2a^2R^3\left|\sigma\int_0^1 \eta(T_R s)e^{-2\lambda s}\sin(2\phi+2\sigma s)\,ds\right|. \tag{S.28}$$

On the other hand, S.21 and `e^(-3lambda s)>=e^(-3)` give

$$M_{k,R}^{\mathrm{plat}} \geq cA^3a^2R^3T_R \int_0^1 Q_\varepsilon(\phi+\sigma s)^3\,ds. \tag{S.29}$$

Apply S.22 and use `T_R=4R^2`. The amplitude cancels and yields

$$|\mathcal{T}_{k,R}| \leq Ca^{2/3}R^{-1/3}\big(M_{k,R}^{\mathrm{plat}}\big)^{2/3}. \tag{S.30}$$

Thus the complete low-frequency sector is paid without a lower bound on `B` and without assuming a phase-uniform spatial lower bound at each time.

High spatial frequency: phase variation and radial Fourier decay

Now assume

$$\varepsilon = kaR \geq 2\pi. \tag{S.31}$$

The phase-uniform plateau estimate Q.19 gives

$$\int_{\mathcal{S}_{a,R}^{\mathrm{plat}}} |\cos(kx_2-\psi)|^3\,dx \geq ca^2R^3 \quad \text{for every } \psi. \tag{S.32}$$

Consequently

$$M_{k,R}^{\mathrm{plat}} \geq cA^3a^2R^3 \min\{T_R, k^{-2}\}. \tag{S.33}$$

If `B=0`, the flux vanishes. Otherwise let `w(t)=eta(t)e^(-2k^2t)`. Its total variation is at most two, exactly as in S.25. Integration by parts in the phase gives

$$\left| B\int_0^{T_R} w(t)\sin(2\phi+2kBt)\,dt \right| \leq \frac{C}{k}. \tag{S.34}$$

Therefore S.13 implies

$$|\mathcal{T}_{k,R}| \leq CA^2\frac{|S_{k,R}|}{k}. \tag{S.35}$$

It remains only to compare scales. If `q<=1`, then S.31 gives `q>=2pi/a`, and S.17 yields

$$\frac{|S_{k,R}|}{k} \le CaR^3 q^{-1} \le Ca^2 R^3. \tag{S.36}$$

In this range `min{T_R,k^(-2)}>=cR^2`. If `q>=1`, use S.17 with `N=1`:

$$\frac{|S_{k,R}|}{k} \le CaR^3 q^{-2}, \quad \min\{T_R, k^{-2}\} = R^2 q^{-2}. \tag{S.37}$$

Since `q^(-2)<=q^(-4/3)` and `a<=a^2`, equations S.33--S.37 again give

$$|\mathcal{T}_{k,R}| \le Ca^{2/3} R^{-1/3} \big(M_{k,R}^{\mathrm{plat}}\big)^{2/3}. \tag{S.38}$$

Equations S.30 and S.38 prove S.4 for every integer frequency.

Normalization, exact-solution status, and open boundary

Substitution of S.5 into S.4 gives S.6. If, in addition, the harmonic is an actual coordinate component of the same smooth velocity measured by the frozen Version-M ledger throughout the plateau tube, and that tube belongs to an exterior row of weight at least `omega`, then

$$p_{k,R}^{\mathrm{plat}} \le CP_R^M, \quad \mathfrak{X}_{k,R} \le C(P_R^M)^{2/3} \quad (L \ge L_0). \tag{S.39}$$

This last implication is conditional on the same realized-subclass and ledger-alignment requirements as R0.75Q: the whole interval `[0,T_R]` must lie in the same scale-`2R` measurement window, and the entire plateau spacetime tube must lie in an exterior row of weight at least `omega` for the same velocity `v_R`. It is not asserted for a Fourier projection of a larger velocity component.

The scalar family is also embedded in the exact smooth unforced shear

$$u_k(t,x) = (0, B, F_k(t, x_2)), \tag{S.40}$$

because `div u_k=0` and

$$\partial_t F_k + B\partial_2 F_k - \partial_2^2 F_k = 0. \tag{S.41}$$

More explicitly,

$$(u_k \cdot \nabla)u_k = (0,0,B\partial_2 F_k), \quad \Delta u_k = (0,0,\partial_2^2 F_k),$$

Thus S.40 solves three-dimensional Navier--Stokes with constant pressure. Its nonzero constant background has not been shown to belong to the frozen mean-zero, inversion-paired Version-M subclass.

Proved: the exact scalar flux identity S.11--S.13; the three radial coefficient bounds S.15--S.17; the spatial node and moving-phase lemmas S.18--S.25; the low-frequency payment S.26--S.30; the high-frequency payment S.31--S.38; and the full-frequency normalized estimate S.6.

Open: two or more harmonics; interference and packet aggregation; nonconstant `x_3` shear; arbitrary vertical structure; transfer from a harmonic projection to the actual velocity component; arbitrary-field E.24; complete Version-M clock extraction; fixed deletion; suitable-weak transfer; and every regularity or singularity conclusion. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75S closes the complete-clock collar payment for one real horizontal harmonic. R0.75R rules out the corresponding plateau-only estimate for an arbitrary concentrated high-band packet. The first unresolved case between those results is therefore not another frequency endpoint: it is the destructive interference of exactly two harmonics.

This note resolves the spatial part of that problem. Let

$$f(x_2)=A\cos(kx_2-\phi)+C\cos(mx_2-\psi),\quad 0\le C,A,\qquad 1\le m<k\le 2m, \tag{T.1}$$

put `d=k-m`, `ell=aR`, and define

$$\begin{aligned}\delta_\pi &:= \operatorname{dist}(\phi-\psi,\pi+2\pi\mathbb{Z}),\\ q_{d,\ell}^2 &:= \min\{1,(d\ell)^2+\delta_\pi^2\},\\ H_{d,\ell}^2 &:= (A-C)^2+ACq_{d,\ell}^2.\end{aligned} \tag{T.2}$$

There is an absolute constant `C_0>0` and a constant `c_0>0` depending only on the frozen plateau width `delta_0` such that, whenever `maR>=C_0`,

$$\boxed{\int_{\mathcal{S}_{a,R}^{\text{plat}}}|f(x_2)|^3\,dx\ge c_0a^2R^3H_{d,aR}^3.} \tag{T.3}$$

The lower bound is uniform in the carrier frequency, the two amplitudes, and both phases. Its degeneracy is also sharp: when `A=C`, the relative phase is `pi`, and `dell<<1`, the two waves cancel to first order and both sides have size `A^3(dell)^3a^2R^3`.

For the exact diffusive constant-shear pair

$$\begin{aligned}F(t,x_2)&=A_t\cos(kx_2-\phi_t)+C_t\cos(mx_2-\psi_t),\\ A_t&=Ae^{-k^2t},\quad C_t=Ce^{-m^2t},\\ \phi_t&=\phi+kBt,\quad \psi_t=\psi+mBt,\end{aligned} \tag{T.4}$$

the same estimate holds at every time, with

$$\delta_\pi(t)=\operatorname{dist}(\phi-\psi+dBt,\pi+2\pi\mathbb{Z}) \tag{T.5}$$

and `A,C` replaced by `A_t,C_t`. Define the complete-clock plateau mass by

$$M_{k,m,R}^{\mathrm{plat}} := \int_0^{T_R} \int_{\mathcal{S}_{a,R}^{\mathrm{plat}}} |F(t,x_2)|^3 \, dx dt.$$

Then

$$\boxed{M_{k,m,R}^{\mathrm{plat}} \geq c_0 a^2 R^3 \int_0^{T_R} H_{d,aR}(t)^3 \, dt.} \tag{T.6}$$

Equation T.6 is the exact spatial coercivity input needed for the remaining difference-frequency time estimate. This note does **not** yet prove that the complete two-harmonic signed flux is bounded by `(M_(k,m,R)^plat)^(2/3)`.

Frozen inputs and plateau fibre

Retain

$$a = pL, \quad R = e^{-\rho L^2/4}, \quad T_R = 4R^2, \tag{T.7}$$

and the radial profile from R0.75R--S:

$$\xi_{a,R}(x) = \vartheta(|x|/R - a), \quad \vartheta = 1 \text{ on } [-\delta_0, \delta_0], \qquad a \geq 4\delta_0, \tag{T.8}$$

with `(a+delta)R<pi/2`. The immediately used frozen inputs are

input	SHA-256	role
`research/r075e_horizontal_cross_mode_flux_reduction.md`	`99529b8934412775e19a1b70784cddf3e62190b50c42669ccd34135a134c5049`	exact difference-frequency flux
`research/r075m_dyadic_packet_diffusive_flux_gain.md`	`13434bbc15eabecd5a695eceef01a7d63415e96511b14c29cc8abcd1297c7bf7`	dyadic within-packet boundary
`research/r075r_outer_cap_spectral_concentration_obstruction.md`	`e5eba5b262a8e140eaa149b6d914f355f2f3c636ec1e74cf85515f1c38fd32f3`	arbitrary-packet obstruction
`research/r075s_full_frequency_single_harmonic_clock_payment.md`	`d2736eaa43443048bd620567c4acd72024dc4c662320a8aa58af31ccc6047ccd`	one-harmonic complete-clock theorem

For every `|y|<=aR/2`, both radial boundaries of the plateau shell meet the `(x_1,x_3)` fibre. Their squared radii differ by

$$((a+\delta_0)R)^2 - y^2 - (((a-\delta_0)R)^2 - y^2) = 4a\delta_0 R^2. \tag{T.9}$$

Consequently the fibre area is exactly

$$\left|\{(x_1,x_3) : (x_1,y,x_3) \in \mathcal{S}_{a,R}^{\mathrm{plat}}\}\right| = 4\pi a \delta_0 R^2. \tag{T.10}$$

Taking `I_ell=[-ell/2,ell/2]` therefore gives

$$\int_{\mathcal{S}_{a,R}^{\mathrm{plat}}} |f(x_2)|^3\,dx \ge 4\pi a\delta_0 R^2 \int_{I_\ell} |f(y)|^3\,dy. \tag{T.11}$$

No observability theorem is hidden in this geometric reduction.

A slow-envelope sampling lemma

The only analytic issue is cancellation of the two waves. The following elementary two-dimensional lemma keeps the sharp beat defect.

Lemma T.1. There are absolute constants `M_0,c_1>0` with the following property. Let `I=[-1/2,1/2]`, `0<=beta<=1`, `mu>=M_0`, and `alpha,gamma in C`. Set

$$Z(s) = \alpha e^{i\beta s/2} + \gamma e^{-i\beta s/2}. \tag{T.12}$$

Then

$$\int_I |\operatorname{Re}(e^{i\mu s} Z(s))|^2\,ds \ge c_1 \int_I |Z(s)|^2\,ds. \tag{T.13}$$

To prove the lemma without a singular constant as `beta` tends to zero, use the regular basis

$$u_\beta(s) = \cos(\beta s/2), \quad v_\beta(s) = \begin{cases} 2\beta^{-1}\sin(\beta s/2), & \beta > 0, \\ s, & \beta = 0. \end{cases} \tag{T.14}$$

The Gram matrix of `(u_beta,v_beta)` on `I` is continuous and positive definite for `0<=beta<=1`. Compactness of this closed parameter interval therefore gives the uniform inverse bounds

$$\|Z\|_{L^\infty(I)} + \|Z'\|_{L^2(I)} \le C\|Z\|_{L^2(I)}. \tag{T.15}$$

The identity

$$\int_I |\operatorname{Re}(e^{i\mu s}Z)|^2 = \frac{1}{2}\int_I |Z|^2 + \frac{1}{2}\operatorname{Re}\int_I e^{2i\mu s}Z^2 \tag{T.16}$$

and one integration by parts give

$$\left|\int_I e^{2i\mu s}Z^2\right| \le \frac{C}{\mu}\int_I |Z|^2. \tag{T.17}$$

Choosing `M_0` once proves T.13. This is a fixed two-dimensional norm equivalence, not an invocation of a general spectral inequality.

The sharp two-wave `L^2` row

Write

$$d=k-m,\quad c=\frac{k+m}{2},\quad \Delta=\phi-\psi,\quad \Sigma=\phi+\psi. \tag{T.18}$$

Then T.1 has the exact carrier-envelope representation

$$f(y)=\mathrm{Re}\left[e^{i(cy-\Sigma/2)}Z(y)\right],\quad Z(y)=Ae^{i(dy/2-\Delta/2)}+Ce^{-i(dy/2-\Delta/2)}. \tag{T.19}$$

3.1 Unresolved beat: `dell<=1`

Rescale `y=ell s`. Since `cell>=mell`, Lemma T.1 applies once `mell>=C_0`. It yields

$$\int_{I_\ell}|f|^2\,dy\ge c\int_{I_\ell}|Z|^2\,dy. \tag{T.20}$$

The envelope integral is exact:

$$\frac{1}{\ell}\int_{I_\ell}|Z|^2\,dy=A^2+C^2+2AC\,\mathrm{sinc}(d\ell/2)\cos\Delta, \tag{T.21}$$

where `sinc z=sin(z)/z`. With `theta=dist(Delta,pi+2pi Z)`, the last row is

$$(A-C)^2+2AC\left[1-\mathrm{sinc}(d\ell/2)\cos\theta\right]. \tag{T.22}$$

For `0<=dell<=1`, elementary Taylor bounds give

$$1-\mathrm{sinc}(d\ell/2)\cos\theta\ge c\min\{1,(d\ell)^2+\theta^2\}. \tag{T.23}$$

Thus

$$\int_{I_\ell}|f|^2\,dy\ge c\ell H_{d,\ell}^2. \tag{T.24}$$

3.2 Resolved beat: `dell>=1`

Direct integration gives

$$\int_{I_\ell} |f|^2 = \frac{\ell}{2}(A^2+C^2) + \frac{A^2\sin(k\ell)\cos(2\phi)}{2k} + \frac{C^2\sin(m\ell)\cos(2\psi)}{2m} + AC\ell\,\mathrm{sinc}(d\ell/2)\cos\Delta + AC\ell\,\mathrm{sinc}((k+m)\ell/2)\cos\Sigma. \tag{T.25}$$

For `dell>=1`,

$$|\mathrm{sinc}(d\ell/2)| \le 2\sin(1/2) < 1. \tag{T.26}$$

The three remaining boundary-frequency errors in T.25 are at most `C(mell)^(-1)ell(A^2+C^2)`. Increasing `C_0` if necessary gives

$$\int_{I_\ell} |f|^2\,dy \ge c\ell(A^2+C^2) \ge c\ell H_{d,\ell}^2. \tag{T.27}$$

Equations T.24 and T.27 cover all beat frequencies.

Cubic coercivity and the diffusive corollary

Holder's inequality on an interval of length `ell` gives

$$\int_{I_\ell} |f|^3\,dy \ge \ell^{-1/2}\left(\int_{I_\ell} |f|^2\,dy\right)^{3/2} \ge c\ell H_{d,\ell}^3. \tag{T.28}$$

Combining T.28 with the exact fibre T.10 proves T.3.

For T.4, each time slice is again a dyadic pair with nonnegative amplitudes `A_t,C_t` and relative phase T.5. The frequency condition is unchanged. Applying T.3 pointwise in time and integrating proves T.6. Notice that unequal heat rates are retained exactly in `A_t-C_t`; they are not replaced by a common dyadic damping factor.

Exact flux decomposition and the next scalar target

Let

$$J_{n,R} := \int_{-\pi}^{\pi} D_R(y)\sin(ny)\,dy, \quad D_R(y) = -2\pi y\vartheta(|y|/R - a). \tag{T.29}$$

Oddness of `D_R` and the product-to-sum identity give the exact signed-flux decomposition

$$
\begin{aligned}
\mathcal{T}_{k,m,R} = \frac{B}{4}\int_0^{T_R} \eta(t)\left[A_t^2 J_{2k,R}\sin(2\phi_t) + C_t^2 J_{2m,R}\sin(2\psi_t)\right]dt \\
+\frac{B}{2}\int_0^{T_R} \eta(t) A_t C_t\left[J_{d,R}\sin(\phi_t-\psi_t) + J_{k+m,R}\sin(\phi_t+\psi_t)\right]dt.
\end{aligned}
\tag{T.30}
$$

The dangerous term is now explicit: it is the low difference frequency `d`, and its phase is exactly the phase entering `H_(d,aR)(t)`. The next positive theorem must prove a weighted moving-phase estimate of the form

$$
\left|BJ_{d,R}\int_0^{T_R} \eta A_t C_t \sin(\phi-\psi+dBt)\,dt\right| \le Ca^{2/3}R^{-1/3}\left(a^2R^3\int_0^{T_R} H_{d,aR}(t)^3 dt\right)^{2/3},
\tag{T.31}
$$

together with compatible control of the two self frequencies and the sum frequency. T.31 is not assumed or proved here.

Meaning and open boundary

Proved: the exact plateau-fibre identity T.10; the uniform slow-envelope sampling lemma T.13; the sharp unresolved-beat defect T.21--T.24; the resolved-beat estimate T.25--T.27; the local cubic lower bound T.3; its exact diffusive time-slice integration T.6; and the four-frequency flux identity T.30.

What changes: destructive interference of one dyadic pair is no longer an unstructured loss. It is measured by one explicit scalar defect. The defect simultaneously records amplitude mismatch, beat distance `dell`, and distance from the cancelling relative phase. No carrier-frequency or mode-count loss occurs once the collar contains a fixed number of periods of the lower carrier.

Not proved: T.31; complete two-harmonic flux payment; low-carrier pairs with `maR<C_0`; three or more harmonics; arbitrary packets; inter-packet summation; nonconstant or vertically dependent shear; projection from a larger velocity; arbitrary-field E.24; complete Version-M extraction; suitable-weak transfer; regularity; or singularity. The result does not contradict R0.75R, whose concentrating packet uses an increasing number of modes. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75T converts the local cubic mass of one two-harmonic dyadic pair into a sharp scalar beat defect. The open row T.31 asks whether that defect pays the low difference-frequency part of the signed collar flux uniformly in the constant shear speed. This note answers that question affirmatively.

Let

$$
\begin{aligned}
F(t,x_2) &= A_t\cos(kx_2-\phi_t) + C_t\cos(mx_2-\psi_t),\\
A_t &= Ae^{-k^2t},\quad C_t = Ce^{-m^2t},\\
\phi_t &= \phi + kBt,\quad \psi_t = \psi + mBt,
\end{aligned}
\tag{U.1}
$$

where `A,C>=0`, `1<=m<k<=2m`, `d=k-m`, and `maR>=C_0`, with `C_0` the carrier threshold in R0.75T. Set

$$
\mathcal{D}_{k,m,R} := \frac{B}{2}J_{d,R}\int_0^{T_R} \eta_R(t) A_t C_t \sin(\phi_t-\psi_t)\,dt,
\tag{U.2}
$$

where

$$J_{n,R} := \int_{-\pi}^{\pi} D_R(y)\sin(ny)\,dy, \quad D_R(y) = -2\pi y\vartheta(|y|/R - a). \tag{U.3}$$

Then, uniformly in both frequencies, amplitudes, phases, and `B`,

$$\boxed{|\mathcal{D}_{k,m,R}| \le Ca^{2/3}R^{-1/3}\big(M^{\mathrm{plat}}_{k,m,R}\big)^{2/3}.} \tag{U.4}$$

With

$$p^{\mathrm{plat}}_{k,m,R} := R^{-2}\omega M^{\mathrm{plat}}_{k,m,R}, \quad \mathfrak{X}^{\mathrm{diff}}_{k,m,R} := \frac{\omega}{R}[\mathcal{D}_{k,m,R}]_+, \tag{U.5}$$

the normalized bound is

$$\boxed{\mathfrak{X}^{\mathrm{diff}}_{k,m,R} \le Ca^{2/3}\omega^{1/3}\big(p^{\mathrm{plat}}_{k,m,R}\big)^{2/3}.} \tag{U.6}$$

All powers of `R` cancel. The exact logarithmic `L^2` coefficient is again

$$\lim_{L\to\infty} \frac{1}{L^2}\log(a^{2/3}\omega^{1/3}) = -\frac{c_\gamma}{12} = -\frac{2}{11907} < 0. \tag{U.7}$$

This proves the difference-frequency target T.31. It does not yet pay the two self frequencies `2k,2m` and the sum frequency `k+m` as a combined quantity, so it is not a complete two-harmonic flux theorem.

Frozen inputs and the radial row

Retain `T_R=4R^2`, the frozen nondecreasing cutoff with

$$0 \le \eta_R \le 1, \quad \eta_R(0) = 0, \quad |\eta_R'(t)| \le C_\eta R^{-2}, \tag{U.8}$$

after translating the left endpoint of the clock to zero. The immediately used frozen inputs are

input	SHA-256	role
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	complete clock and cutoff onset
`research/r075r_outer_cap_spectral_concentration_obstruction.md`	`e5eba5b262a8e140eaa149b6d914f355f2f3c636ec1e74cf85515f1c38fd32f3`	exact radial cross section
`research/r075s_full_frequency_single_harmonic_clock_payment.md`	`d2736eaa43443048bd620567c4acd72024dc4c662320a8aa58af31ccc6047ccd`	radial sine rows and phase-distance method

input	SHA-256	role
research/r075t_two_harmonic_collar_coercivity.md	822059f8a6248143ff3f36938a2333bee9f909b9166db951e227c426c2e8bc66	sharp dyadic-pair beat defect

Scaling y=Rz in U.3 gives, exactly as in the one-harmonic row,

$$|J_{n,R}| \le CaR^2 \min\{naR, 1\}. \tag{U.9}$$

Both entries imply the single bound needed here:

$$\boxed{\frac{|J_{n,R}|}{n} \le Ca^2R^3 \quad (n \ge 1).} \tag{U.10}$$

Indeed, use the first entry when naR<=1; otherwise n^(-1)<=aR and use the second.

A weighted moving-phase lemma

The time estimate is independent of the carrier frequencies after nondimensionalization.

Lemma U.1. Let zeta be absolutely continuous on [0,4], with

$$0 \le \zeta \le 1, \quad \zeta(0) = 0, \quad |\zeta'| \le C_\eta. \tag{U.11}$$

For every Lambda>=0, sigma in R, and alpha in R, define

$$h(s) := \min\{1, \operatorname{dist}(\alpha + \sigma s, \pi + 2\pi\mathbb{Z})\}. \tag{U.12}$$

Then

$$\boxed{\left|\sigma \int_0^4 \zeta(s) e^{-\Lambda s} \sin(\alpha + \sigma s)\, ds\right| \le C\left(\int_0^4 e^{-3\Lambda s/2} h(s)^3\, ds\right)^{2/3}.} \tag{U.13}$$

The constant depends only on C_eta.

2.1 The phase-distance moment

Put

$$\tau := \begin{cases} 1, & 0 \le \Lambda \le 1, \\ \Lambda^{-1}, & \Lambda \ge 1, \end{cases} \quad r := h(0), \quad q := \min\{1, r + |\sigma|\tau\}. \tag{U.14}$$

On `[0,tau]`, the exponential weight is bounded below by `e^(-3/2)`. The distance to `pi+2pi Z` is a periodic triangular wave. Integrating one affine segment, and splitting only when a corner is crossed, gives

$$\int_0^\tau h(s)^3\,ds \ge c\tau q^3. \tag{U.15}$$

The estimate includes intervals that cross the cancelling phase: for an affine distance decreasing from `r` to zero, its cubic mean is `r^3/4`, so there is no loss at a phase node. Consequently

$$\left(\int_0^4 e^{-3\Lambda s/2}h(s)^3\,ds\right)^{2/3} \ge c\tau^{2/3}q^2. \tag{U.16}$$

2.2 Slow phase on the effective heat interval

Assume `|sigma|tau<=1`. If `Lambda<=1`, then `tau=1`, and on `[0,4]`

$$|\sin(\alpha+\sigma s)| \le Cq. \tag{U.17}$$

Therefore the left side of U.13 is at most `C|sigma|q<=Cq^2`, because `q>=c|sigma|`.

If `Lambda>=1`, U.11 gives `zeta(s)<=C_eta s`. The elementary Laplace moments yield

$$\begin{aligned}&|\sigma|\int_0^4 \zeta(s)e^{-\Lambda s}|\sin(\alpha+\sigma s)|\,ds\\ &\le C|\sigma|\int_0^\infty se^{-\Lambda s}\min\{1,r+|\sigma|s\}\,ds\\ &\le C|\sigma|\tau^2 q \le C\tau q^2 \le C\tau^{2/3}q^2.\end{aligned} \tag{U.18}$$

Together with U.16, this proves the slow-phase case.

2.3 Fast phase on the effective heat interval

Assume `|sigma|tau>=1`; then `q=1`. Put `w(s)=zeta(s)e^(-Lambda s)`. Since `w(0)=0`, one integration by parts gives

$$\left|\sigma\int_0^4 w(s)\sin(\alpha+\sigma s)\,ds\right| \le |w(4)| + \int_0^4 |w'(s)|\,ds. \tag{U.19}$$

If `Lambda<=1`, the right side is bounded by a constant. If `Lambda>=1`, use `zeta(s)<=C_eta s` to obtain

$$|w(4)| + \int_0^4 |w'| \le C\left(\Lambda^{-1} + \Lambda\int_0^\infty se^{-\Lambda s}\,ds\right) \le C\tau. \tag{U.20}$$

Since `0<tau<=1`, both cases are at most `C tau^(2/3)`. Equation U.16 with `q=1` proves the fast-phase case and completes Lemma U.1.

Application to the exact dyadic pair

If `AC=0`, the difference-frequency flux U.2 vanishes and U.4 is immediate. Assume below that `AC>0`.

Scale time by

$$t = R^2 s, \quad \Lambda = (k^2 + m^2)R^2, \quad \sigma = dBR^2, \tag{U.21}$$

and set `zeta(s)=eta_R(R^2s)`. Equation U.8 gives U.11. Let

$$h(s) = \min\{1, \operatorname{dist}(\phi - \psi + \sigma s, \pi + 2\pi\mathbb{Z})\}. \tag{U.22}$$

The R0.75T defect, evaluated at `t=R^2s`, obeys

$$H_{d,aR}(R^2 s)^2 \geq A_{R^2 s} C_{R^2 s} h(s)^2 = ACe^{-\Lambda s} h(s)^2. \tag{U.23}$$

Hence Lemma U.1 implies

$$\begin{aligned} &\left| B \int_0^{T_R} \eta_R(t) A_t C_t \sin(\phi - \psi + dBt)\, dt \right| \\ &\leq \frac{C}{dR^{4/3}} \left(\int_0^{T_R} H_{d,aR}(t)^3\, dt \right)^{2/3}. \end{aligned} \tag{U.24}$$

The amplitude product cancels exactly: the left side is `AC/d` times the left side of U.13, while U.23 contributes `(AC)^(3/2)` inside the cubic integral.

Multiply U.24 by `|J_(d,R)|/2` and use U.10:

$$|\mathcal{D}_{k,m,R}| \leq Ca^2 R^{5/3} \left(\int_0^{T_R} H_{d,aR}(t)^3\, dt \right)^{2/3}. \tag{U.25}$$

R0.75T gives

$$M^{\text{plat}}_{k,m,R} \geq ca^2 R^3 \int_0^{T_R} H_{d,aR}(t)^3\, dt. \tag{U.26}$$

Substitution of U.26 into U.25 proves U.4. Substitution of U.5 then proves U.6, including the exact cancellation of the `R` powers.

Exact-solution and Version-M boundary

The pair U.1 is a solution of

$$\partial_t F + B\partial_2 F - \partial_2^2 F = 0. \tag{U.27}$$

It embeds in the exact smooth unforced shear `u=(0,B,F(t,x_2))` with constant pressure. Its nonzero constant background has not been shown to belong to the frozen mean-zero, inversion-paired Version-M subclass. If the entire complete clock and plateau tube lie in the same scale-`2R` Version-M measurement row of weight at least `omega`, and `F` is an actual component of that same velocity, then

$$p^{\text{plat}}_{k,m,R} \le C P_R^M \tag{U.28}$$

and U.6 pays the difference-frequency component by `C(P_R^M)^(2/3)`. This is conditional on the same realized-subclass and ledger-alignment hypotheses as R0.75S. It is not a statement about a Fourier projection of a larger field, and it does not pay the remaining three frequency rows in the total flux.

Meaning and open boundary

Proved: the uniform radial quotient U.10; the weighted complete-clock moving-phase lemma U.13; its slow/fast proof U.14--U.20; exact scaling and amplitude cancellation U.21--U.24; the difference-frequency payment U.4; and the normalized rate U.6--U.7.

What changes: the most obviously dangerous term in a dyadic pair is the low difference frequency, because its radial Fourier coefficient need not have high-frequency decay. U shows that this term is nevertheless paid by the same beat defect that measures destructive interference in the local cubic mass. Beyond the automatic integer separation `d>=1`, no additional lower bound on `d`, `dR`, `d aR`, or `B` is used.

Open: joint payment of the self frequencies `2k,2m` and sum frequency `k+m`; a complete two-harmonic signed-flux theorem; low carriers with `maR<C_0`; three or more harmonics; arbitrary packets; inter-packet aggregation; nonconstant or vertically dependent shear; arbitrary-field E.24; complete Version-M extraction; fixed deletion; suitable-weak transfer; regularity; and singularity. No novelty or priority claim is made. **NOT CLAY**.

Result and exact boundary

R0.75T proves sharp collar coercivity for one two-harmonic dyadic pair. R0.75U uses that coercivity to pay the low difference-frequency row of the signed flux. Three coupled rows remain: the two self frequencies `2k,2m` and the sum frequency `k+m`. Bounding those rows separately destroys the cancellation measured by the T defect. This note keeps their quadratic combination intact and closes the exact two-harmonic calculation.

Let

$$\begin{aligned} F(t,x_2) &= A_t\cos(kx_2-\phi_t) + C_t\cos(mx_2-\psi_t),\\ A_t &= Ae^{-k^2t},\quad C_t = Ce^{-m^2t},\\ \phi_t &= \phi + kBt,\quad \psi_t = \psi + mBt, \end{aligned} \tag{V.1}$$

where `A,C>=0`, `1<=m<k<=2m`, `d=k-m`, and `maR>=C_0`, with the same carrier threshold as R0.75T. Put `n=k+m` and let

$$\begin{aligned} \mathcal{E}_{k,m,R} := {}& \frac{B}{4}\int_0^{T_R} \eta_R(t)\left[A_t^2 J_{2k,R}\sin(2\phi_t) + C_t^2 J_{2m,R}\sin(2\psi_t)\right]dt\\ &+\frac{B}{2}\int_0^{T_R}\eta_R(t)A_tC_tJ_{n,R}\sin(\phi_t+\psi_t)\,dt. \end{aligned} \tag{V.2}$$

This is exactly the self/sum block left open in U. There is a constant depending only on the frozen radial profile, plateau width, carrier threshold, and time cutoff such that

$$\boxed{|\mathcal{E}_{k,m,R}| \le Ca^{2/3}R^{-1/3}\left(M^{\mathrm{plat}}_{k,m,R}\right)^{2/3}.} \tag{V.3}$$

Combining V.3 with the R0.75U difference-frequency estimate gives the full exact signed collar flux

$$\boxed{|\mathcal{T}_{k,m,R}| \le Ca^{2/3}R^{-1/3}\left(M^{\mathrm{plat}}_{k,m,R}\right)^{2/3}.} \tag{V.4}$$

With

$$p^{\mathrm{plat}}_{k,m,R} = R^{-2}\omega M^{\mathrm{plat}}_{k,m,R}, \quad \mathfrak{X}_{k,m,R} = \frac{\omega}{R}[\mathcal{T}_{k,m,R}]_+, \tag{V.5}$$

equation V.4 becomes

$$\boxed{\mathfrak{X}_{k,m,R} \le Ca^{2/3}\omega^{1/3}\left(p^{\mathrm{plat}}_{k,m,R}\right)^{2/3}.} \tag{V.6}$$

All powers of `R` cancel, and the exact logarithmic `L^2` rate is

$$\lim_{L\to\infty}\frac{1}{L^2}\log(a^{2/3}\omega^{1/3}) = -\frac{2}{11907} < 0. \tag{V.7}$$

This is a complete flux theorem only for the exact high-carrier dyadic pair V.1. It is not an arbitrary two-mode projection theorem and not an arbitrary-field estimate.

Frozen inputs and the four-frequency split

Retain

$$a = pL, \quad R = e^{-\rho L^2/4}, \quad T_R = 4R^2, \quad \ell = aR, \tag{V.8}$$

and the frozen time cutoff

$$0 \le \eta_R \le 1, \quad \eta_R(0) = 0, \quad |\eta_R'(t)| \le C_\eta R^{-2}. \tag{V.9}$$

The immediately used frozen inputs are

input	SHA-256	role
research/r075b_bulk_clock_outer_padding_gate.md	430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a	complete clock and cutoff onset
research/r075r_outer_cap_spectral_concentration_obstruction.md	e5eba5b262a8e140eaa149b6d914f355f2f3c636ec1e74cf85515f1c38fd32f3	exact radial cross section
research/r075t_two_harmonic_collar_coercivity.md	822059f8a6248143ff3f36938a2333bee9f909b9166db951e227c426c2e8bc66	two-wave defect and plateau mass
research/r075u_two_harmonic_difference_frequency_payment.md	f9fb331cf880b20f3b407fe66453bce71517ac1ef2af4fa0863c00325c1022a4	difference-frequency payment

Write

$$J_{r,R}=\int_{-\pi}^{\pi} D_R(y)\sin(ry)\,dy,\quad D_R(y)=-2\pi y\vartheta(|y|/R-a). \tag{V.10}$$

The exact expansion from T is

$$\begin{aligned}\mathcal{T}_{k,m,R}&=\mathcal{E}_{k,m,R}+\mathcal{D}_{k,m,R},\\ \mathcal{D}_{k,m,R}&=\frac{B}{2}J_{d,R}\int_0^{T_R}\eta_R A_t C_t\sin(\phi_t-\psi_t)\,dt.\end{aligned} \tag{V.11}$$

R0.75U already proves V.3 with `mathcal E` replaced by `mathcal D`. It therefore remains to prove V.3 without taking absolute values inside the three-row combination V.2.

Radial quotient and its two-jet

For real `r>0`, define the smooth quotient

$$K_R(r):=\frac{J_{r,R}}{r}. \tag{V.12}$$

The formula in V.10 extends `K_R` smoothly and evenly through `r=0`. After `y=Rz`, differentiation under the integral and integration by parts in `z` give, for `j=0,1,2` and every integer `N>=0`,

$$\boxed{|\partial_r^j K_R(r)|\le C_{j,N}a^{j+2}R^{j+3}(1+rR)^{-N}.} \tag{V.13}$$

Indeed, each `r` derivative supplies one factor `Rz`, the support consists of two fixed-width intervals with `|z|` comparable to `a`, and the frozen profile is smooth. For `rR<=1`, the direct moment bound gives V.13. For `rR>=1`, arbitrary integration by parts gives the last factor.

Set

$$q=nR,\quad \varepsilon=\min\{1,d\ell\},\quad \Lambda_N=C_Na^2R^3(1+q)^{-N}. \tag{V.14}$$

Since `d<=n/3`, all three arguments `n-d,n,n+d` are comparable to `n`. Since `n\ell>=2maR>=2C_0`, also

$$\frac{d}{n} \le C\varepsilon. \tag{V.15}$$

Taylor's theorem, V.13, and the direct bound when `dell>=1` imply

$$\begin{aligned} |K_R(n)| + \left|\frac{K_R(n+d)+K_R(n-d)}{2}\right| &\le \Lambda_N, \\ \left|\frac{K_R(n+d)-K_R(n-d)}{2}\right| &\le \Lambda_N \varepsilon, \\ \left|\frac{K_R(n+d)+K_R(n-d)}{2} - K_R(n)\right| &\le \Lambda_N \varepsilon^2. \end{aligned} \tag{V.16}$$

The same statement holds for

$$L_R(r) := \frac{r^2}{2} K_R(r), \tag{V.17}$$

with `Lambda_N` replaced by `C n^2 Lambda_N`. Terms produced by differentiating `r^2` carry `d/n` or `(d/n)^2` and are absorbed by V.15.

The quadratic cancellation lemma

Let

$$\Delta_t = \phi_t - \psi_t, \quad \delta_t = \mathrm{dist}(\Delta_t, \pi + 2\pi\mathbb{Z}), \tag{V.18}$$

and retain the T defect

$$H(t)^2 = (A_t - C_t)^2 + A_t C_t \min\{1, (d\ell)^2 + \delta_t^2\}. \tag{V.19}$$

Put `u=A_t exp(i Delta_t/2)` and `v=C_t exp(-i Delta_t/2)`. Then

$$|u+v|^2 \le CH(t)^2, \quad \varepsilon(A_t + C_t) \le CH(t). \tag{V.20}$$

Consider any real multiplier triple `G_+,G_0,G_-` satisfying V.16 with size `Lambda`. Its quadratic combination can be written exactly as

$$\begin{aligned} &G_+ A_t^2 e^{i\Delta_t} + 2G_0 A_t C_t + G_- C_t^2 e^{-i\Delta_t} \\ &= \overline{G}\,(u+v)^2 + 2A_t C_t (G_0 - \overline{G}) + G_\Delta (u^2 - v^2), \end{aligned} \tag{V.21}$$

where `overline G=(G_++G_-)/2` and `G_Delta=(G_+-G_-)/2`. Equations V.16 and V.20 give

$$\boxed{|G_+A_t^2e^{i\Delta_t}+2G_0A_tC_t+G_-C_t^2e^{-i\Delta_t}|\le C\Lambda H(t)^2.} \tag{V.22}$$

The last term uses `|u^2-v^2|<=|u+v|(A_t+C_t)`. Thus the first multiplier difference is paired with one factor of the beat defect and the second difference with two factors. This is the cancellation lost by three separate absolute values.

Apply V.22 first to

$$G_+=K_R(2k),\quad G_0=K_R(n),\quad G_-=K_R(2m). \tag{V.23}$$

For the heat row, apply it to `L_R(2k),L_R(n),L_R(2m)`. The actual cross coefficient differs from the central one by

$$(k^2+m^2)K_R(n)-\frac{n^2}{2}K_R(n)=\frac{d^2}{2}K_R(n). \tag{V.24}$$

This remainder is also bounded by `C n^2 Lambda_N H(t)^2`, because `(d/n)^2A_tC_t<=C H(t)^2` by V.15 and V.19. Consequently the cutoff and heat quadratic forms obey

$$|Q_K(t)|\le C\Lambda_N H(t)^2,\quad |Q_L(t)|\le Cn^2\Lambda_N H(t)^2. \tag{V.25}$$

Exact time integration by parts

For a self phase, direct differentiation gives

$$\int_0^{T_R} Bg(t)\sin(2\phi+2kBt)\,dt=-\left[\frac{g(t)\cos(2\phi+2kBt)}{2k}\right]_0^{T_R}+\int_0^{T_R}\frac{g'(t)}{2k}\cos(2\phi+2kBt)\,dt. \tag{V.26}$$

The corresponding identity for the sum phase has denominator `n`. Use V.26 separately on all three rows of V.2. Because `eta_R(0)=0`, the initial boundary vanishes. Factoring the common phase `Sigma_t=phi_t+psi_t` after, not before, the integrations gives the exact identity

$$\begin{aligned}\mathcal{E}_{k,m,R}=&-\frac{\eta_R(T_R)}{4}\operatorname{Re}\big(e^{i\Sigma_{T_R}}Q_K(T_R)\big)\\&+\frac{1}{4}\int_0^{T_R}\eta_R'(t)\operatorname{Re}\big(e^{i\Sigma_t}Q_K(t)\big)\,dt\\&-\frac{1}{4}\int_0^{T_R}\eta_R(t)\operatorname{Re}\big(e^{i\Sigma_t}Q_L(t)\big)\,dt,\end{aligned} \tag{V.27}$$

where

$$\begin{aligned}Q_K(t)&=A_t^2K_R(2k)e^{i\Delta_t}+2A_tC_tK_R(n)+C_t^2K_R(2m)e^{-i\Delta_t},\\Q_L(t)&=2k^2A_t^2K_R(2k)e^{i\Delta_t}+2(k^2+m^2)A_tC_tK_R(n)+2m^2C_t^2K_R(2m)e^{-i\Delta_t}.\end{aligned} \tag{V.28}$$

No derivative of the relative phase occurs in V.27. Integrating after a premature common-phase factorization would create an unnecessary `dB` term and would obscure the exact cancellation.

Complete-clock trace for the beat defect

Set

$$I_H := \int_0^{T_R} H(t)^3\,dt. \tag{V.29}$$

Holder immediately gives

$$\int_0^{T_R} H(t)^2\,dt \le T_R^{1/3} I_H^{2/3} \le C R^{2/3} I_H^{2/3}. \tag{V.30}$$

The endpoint in V.27 needs one further elementary fact.

Lemma V.1. With `q=nR`,

$$\boxed{(1+q)^{-8} H(T_R)^2 \le C R^{-4/3} I_H^{2/3}.} \tag{V.31}$$

To prove it, scale `t=R^2s`, so the clock is `[0,4]`, and write `x=A_(T_R)`, `y=C_(T_R)`. Backwards from the terminal time,

$$A_{R^2 s} \ge x, \quad C_{R^2 s} \ge y. \tag{V.32}$$

For every affine phase and every fixed `epsilon>=0`, direct integration on the affine pieces of the periodic distance function gives

$$\int_0^4 \min\{1, \varepsilon^2 + \operatorname{dist}(\alpha+\sigma s, \pi + 2\pi\mathbb{Z})^2\}^{3/2} ds \ge c \min\{1, \varepsilon^2 + \operatorname{dist}(\alpha + 4\sigma, \pi + 2\pi\mathbb{Z})^2\}^{3/2}. \tag{V.33}$$

For `|sigma|<=1`, this is the cubic mean of finitely many affine pieces; for `|sigma|>=1`, a fixed fraction of a period has distance bounded below. Equations V.19, V.32, and V.33 pay the terminal product-phase term `xy min{1,(dell)^2+delta_(T_R)^2}`.

It remains to pay `|x-y|^2`. Put `epsilon=min{1,dell}`. If `xy=0`, one amplitude vanishes on the whole clock and the other is no smaller backwards, so the required cubic lower bound is immediate. Suppose `xy>0`. If `|x-y|<=4epsilon sqrt(xy)`, that term is already bounded by the product-phase term. Otherwise, the backward amplitude ratio is monotone, and its logarithmic speed is

$$(k^2 - m^2)R^2 = dnR^2. \tag{V.34}$$

On a terminal subinterval of length at least `c(1+q^2)^(-1)` in the `s` variable, the ratio stays at least a fixed fraction of its terminal distance from one and the common heat factor changes by at most a fixed factor. Indeed, when `dell<=1`, the ratio speed divided by `epsilon` is `q/a`; when `dell>=1`, V.34 is at most `q^2/3`. Hence

$$\int_0^4 |A_{R^2 s} - C_{R^2 s}|^3 ds \geq c(1+q^2)^{-1}|x-y|^3. \tag{V.35}$$

The `2/3` power of V.35, together with the product-phase row, proves

$$H(T_R)^2 \leq C(1+q^2)^{2/3}\left(R^{-2} I_H\right)^{2/3}. \tag{V.36}$$

Multiplication by `(1+q)^(-8)` proves V.31.

Payment of the self/sum block

Choose `N=8` in V.14--V.25 and abbreviate `w_q=(1+q)^(-8)`. Equations V.25 and V.27 give

$$\begin{aligned} |\mathcal{E}_{k,m,R}| \leq{}& Ca^2 R^3 w_q H(T_R)^2 \\ &+ Ca^2 R^3 w_q R^{-2} \int_0^{T_R} H(t)^2 dt \\ &+ Ca^2 R^3 w_q n^2 \int_0^{T_R} H(t)^2 dt. \end{aligned} \tag{V.37}$$

Use V.30--V.31 and `q^2w_q<=C`. Every row in V.37 is bounded by

$$Ca^2 R^{5/3} I_H^{2/3}. \tag{V.38}$$

R0.75T gives

$$M^{\mathrm{plat}}_{k,m,R} \geq ca^2 R^3 I_H. \tag{V.39}$$

Substitution of V.39 into V.38 yields

$$a^2 R^{5/3}\left(\frac{M^{\mathrm{plat}}_{k,m,R}}{a^2 R^3}\right)^{2/3} = a^{2/3} R^{-1/3}\left(M^{\mathrm{plat}}_{k,m,R}\right)^{2/3}, \tag{V.40}$$

which proves V.3.

Full flux and normalization

Equation V.11 and the triangle inequality applied only after both coupled blocks have been proved give

$$|\mathcal{T}_{k,m,R}| \le |\mathcal{E}_{k,m,R}| + |\mathcal{D}_{k,m,R}|. \tag{V.41}$$

R0.75U and V.3 therefore prove V.4. Substituting `M=R^2 omega^(-1)p` into `(omega/R)V.4` gives V.6. With the frozen values `a=pL` and `omega=exp[-(c_gamma/4)L^2]`, the polynomial factor from `a` does not affect the `L^2` rate, and `-c_gamma/12=-2/11907`, proving V.7.

Exact-solution and Version-M boundary

The field V.1 solves

$$\partial_t F + B\partial_2 F - \partial_2^2 F = 0 \tag{V.42}$$

and embeds in the exact smooth unforced shear `u=(0,B,F(t,x_2))` with constant pressure. The nonzero constant background has not been shown to belong to the frozen mean-zero, inversion-paired Version-M subclass.

If the complete clock and plateau tube belong to the same scale-`2R` Version-M measurement row of weight at least `omega`, and `F` is an actual component of that same velocity, then

$$p^{\mathrm{plat}}_{k,m,R} \le C P^M_R \tag{V.43}$$

and V.6 pays the full exact two-harmonic collar flux by `C(P_R^M)^(2/3)`. This is conditional on the realized-subclass and ledger-alignment hypotheses. It is not a bound for a Fourier projection of a larger velocity.

What is closed and what remains open

Closed here: the joint self/sum block; together with U, the complete signed collar flux of one exact diffusive two-harmonic dyadic pair with `maR>=C_0`; the scale-free normalization and exact negative coefficient rate.

Open: low-carrier pairs; three or more harmonics; arbitrary dyadic packets; inter-packet aggregation; nonconstant or vertically dependent shear; projection from a larger velocity; arbitrary-field E.24; complete Version-M extraction; fixed deletion; suitable-weak transfer; regularity; singularity.

R0.75V is a finite-dimensional exact-subfamily theorem. It does not prove that the general three-dimensional Navier--Stokes solution is smooth. **NOT CLAY.**

Result and exact boundary

R0.75V proves the complete signed collar-flux estimate for one exact two-harmonic dyadic shear under the carrier condition `maR>=C_0`. The condition enters only through the pointwise-in-time spatial sampling lemma of R0.75T. It is false that the same T defect controls a low-carrier pair: two slowly varying waves can cancel at the centre of the observed interval for amplitude and phase combinations not measured by that defect.

This note does not repair the T defect. Instead it treats the complementary low-carrier sector by one local energy identity and a uniform four-dimensional observation lemma. The exact initial slice with `k=2m`, `C=2A`, `phi=pi/2`, and `psi=-pi/2` is `A sin(2my)-2A sin(my)=-`

`A(my)^3+O(A(my)^5)`. Its T amplitude-mismatch defect is of order `A`, while its observed low-carrier mass vanishes to ninth order in `maR`. This is a counterexample to extending the T pointwise defect, but not to the complete signed-flux estimate below. Let

$$
\begin{aligned}
F(t,x_2) &= A_t\cos(kx_2-\phi_t)+C_t\cos(mx_2-\psi_t),\\
A_t &= Ae^{-k^2t},\quad C_t=Ce^{-m^2t},\\
\phi_t &= \phi+kBt,\quad \psi_t=\psi+mBt,
\end{aligned}
\tag{W.1}
$$

where `A,C>=0`, `1<=m<k<=2m`, and `B` is any real constant. There is a constant depending only on the frozen radial profile, plateau width, time cutoff, and the carrier threshold in R0.75T such that

$$
\mathcal{T}_{k,m,R}:=\frac{1}{2}\int_0^{T_R}\int_{\mathbb{T}^3}\eta_R(t)B\,\partial_2\xi_{a,R}|F|^2\,dxdt,\quad M^{\mathrm{plat}}_{k,m,R}:=\int_0^{T_R}\int_{\mathcal{S}^{\mathrm{plat}}_{a,R}}|F|^3\,dxdt.
$$

Then

$$
\boxed{|\mathcal{T}_{k,m,R}|\le Ca^{2/3}R^{-1/3}\left(M^{\mathrm{plat}}_{k,m,R}\right)^{2/3}}
\tag{W.2}
$$

for every dyadic pair W.1 and all sufficiently large `L`. There is no lower or upper carrier-frequency restriction in W.2.

With

$$
p^{\mathrm{plat}}_{k,m,R}=R^{-2}\omega M^{\mathrm{plat}}_{k,m,R},\quad \mathfrak{X}_{k,m,R}=\frac{\omega}{R}[\mathcal{T}_{k,m,R}]_+,
\tag{W.3}
$$

the normalized estimate is

$$
\boxed{\mathfrak{X}_{k,m,R}\le Ca^{2/3}\omega^{1/3}\left(p^{\mathrm{plat}}_{k,m,R}\right)^{2/3},\quad \lim_{L\to\infty}\frac{1}{L^2}\log(a^{2/3}\omega^{1/3})=-\frac{2}{11907}.}
\tag{W.4}
$$

This is a full-frequency theorem only for the exact pair W.1. It is not a three-mode estimate, a packet theorem, or an estimate for a Fourier projection of a larger velocity.

Frozen inputs and the carrier split

Retain

$$
a=pL,\quad R=e^{-\rho L^2/4},\quad T_R=4R^2,\quad \ell=aR,
\tag{W.5}
$$

and the translated time cutoff

$$0 \le \eta_R \le 1, \quad \eta_R(0) = 0, \quad |\eta_R'(t)| \le C_\eta R^{-2}. \tag{W.6}$$

The immediately used frozen inputs are

input	SHA-256	role
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	complete clock and cutoff onset
`research/r075r_outer_cap_spectral_concentration_obstruction.md`	`e5eba5b262a8e140eaa149b6d914f355f2f3c636ec1e74cf85515f1c38fd32f3`	exact radial cross section
`research/r075t_two_harmonic_collar_coercivity.md`	`822059f8a6248143ff3f36938a2333bee9f909b9166db951e227c426c2e8bc66`	plateau fibre and carrier threshold
`research/r075v_complete_two_harmonic_flux_payment.md`	`6917ff77099b6271b005ca90335df589434a38b0a57001893dcae8b02fd34824`	complete high-carrier theorem

The high-carrier sector

$$maR \ge C_0 \tag{W.7}$$

is exactly R0.75V. It remains to prove W.2 when `maR<C_0`. Since `k<=2m`, both scaled frequencies in that sector obey a fixed upper bound.

Low-carrier scaling

For `maR<C_0`, set

$$s = \frac{t}{R^2}, \quad z = \frac{x_2}{aR}, \quad \alpha = kaR, \quad \beta = maR, \quad v = \frac{BR}{a}. \tag{W.8}$$

Thus `0<beta<alpha<=2C_0`, the time interval becomes `[0,4]`, and

$$\begin{aligned} G(s,z) := F(R^2 s, aRz) = {} & Ae^{-\alpha^2 s/a^2} \cos\big(\alpha(z - vs) - \phi\big) \\ & + Ce^{-\beta^2 s/a^2} \cos\big(\beta(z - vs) - \psi\big). \end{aligned} \tag{W.9}$$

The exact scalar equation becomes

$$\partial_s G + v\partial_z G - a^{-2}\partial_z^2 G = 0. \tag{W.10}$$

Put `I=[-1/2,1/2]` and

$$h(s) = \int_I |G(s,z)|^3 \, dz, \quad H = \int_0^4 h(s)\, ds. \tag{W.11}$$

The proof below bounds the dimensionless signed flux by `CH^(2/3)`.

A confluent spatial observation lemma

Lemma W.1. Fix `K<infinity` and compact intervals `I` and `J`, with `I` having nonempty interior. There is `C=C(K,I,J)` such that every real or complex solution of

$$(\partial_z^2+\alpha^2)(\partial_z^2+\beta^2)g=0,\quad 0\le\alpha,\beta\le K, \tag{W.12}$$

obeys

$$\boxed{\|g\|_{L^\infty(J)}+\|g'\|_{L^\infty(J)}\le C\|g\|_{L^3(I)}.} \tag{W.13}$$

This includes `alpha=beta` and `alpha=beta=0`. In the latter limit the solution space is the cubic-polynomial space, so the lemma retains the worst confluent cancellation, including the explicit cubic node above, rather than assuming separated frequencies.

To prove it, write `X=(g,g',g'',g''')`. Equation W.12 is the first-order system

$$X'=\mathbb{A}_{\alpha,\beta}X,\quad \mathbb{A}_{\alpha,\beta}=\begin{pmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\-\alpha^2\beta^2&0&-(\alpha^2+\beta^2)&0\end{pmatrix}. \tag{W.14}$$

The matrices form a compact bounded family. Gronwall therefore propagates the initial jet uniformly from `I` to `J`. If the initial jet were not controlled by the `L^3(I)` norm, a normalized countersequence would have convergent parameters and initial jets. Continuous dependence for W.14 would give a nonzero limiting solution that vanishes almost everywhere on `I`, hence vanishes identically, a contradiction. This proves W.13.

A frequency-gap-free terminal trace

Lemma W.2. Fix `Lambda<infinity`. If

$$q(s)=\sum_{j=1}^{N}c_je^{\lambda_js},\quad N\le 4,\quad |\operatorname{Re}\lambda_j|\le\Lambda, \tag{W.15}$$

then

$$\boxed{|q(4)|^3\le C_\Lambda\int_0^4|q(s)|^3\,ds.} \tag{W.16}$$

The constant is independent of the imaginary parts of the exponents and of the gaps between them. This is a direct `L^3` consequence of the Turan--Nazarov inequality

$$\sup_{[0,4]} |q| \le e^{4\max_j |\operatorname{Re}\lambda_j|} \left(\frac{4C}{|E|}\right)^{N-1} \sup_E |q| \tag{W.17}$$

for a measurable `E` of positive measure. Take `E` to be the half-measure sublevel set supplied by Chebyshev's inequality. Nazarov's original result allows complex exponents and has no imaginary-frequency or gap factor; an accessible primary restatement is recorded in the source report.

For each fixed `z`, W.9 is an exponential polynomial in `s` with at most four terms and exponents

$$-\alpha^2/a^2 \pm i\alpha v, \quad -\beta^2/a^2 \pm i\beta v. \tag{W.18}$$

Their real parts are uniformly bounded in the low-carrier sector. Applying W.16 pointwise in `z` and integrating over `I` gives

$$\boxed{h(4) \le CH.} \tag{W.19}$$

This is the required right-endpoint trace. It remains uniform as `v` tends to zero or infinity and as the two frequencies coalesce.

Scaled radial primitive

Let the frozen radial profile be `vartheta`, as in R0.75R--V, and define

$$W_a(z) = -2\pi a z \vartheta\big(a(|z|-1)\big), \quad \Xi_a(z) = \int_{-\infty}^{z} W_a(r)\,dr. \tag{W.20}$$

Oddness gives `int W_a=0`, so `Xi_a` is compactly supported. Once `a>=max(1,2delta)`, its support lies in `J=[-3/2,3/2]`, and direct scaling of the two fixed-width transition intervals gives

$$\|W_a\|_{L^1} + \|\Xi_a\|_{L^1} + \|\Xi_a\|_{L^\infty} \le C, \quad \|\Xi_a''\|_{L^1} = \|W_a'\|_{L^1} \le Ca. \tag{W.21}$$

The exact cross-sectional derivative satisfies

$$\int_{-\pi}^{\pi} D_R(y) F(t,y)^2\,dy = aR^2 \int_{\mathbb{R}} W_a(z) G(s,z)^2\,dz. \tag{W.22}$$

With `zeta(s)=eta_R(R^2s)`, equations W.8 and W.22 give the exact flux scaling

$$\boxed{\mathcal{T}_{k,m,R} = \frac{a^2R^3}{2} v \int_0^4 \zeta(s) \int_{\mathbb{R}} W_a(z) G(s,z)^2\,dz ds.} \tag{W.23}$$

The low-carrier local energy identity

Set `Q=G^2`. Equation W.10 gives

$$\partial_s Q + v\partial_z Q - a^{-2}\partial_z^2 Q = -2a^{-2}|\partial_z G|^2. \tag{W.24}$$

Let `E(s)=int Xi_a G^2`. Since `W_a=Xi_a'`, integration by parts in `z` and W.24 yield the exact identity

$$v\int W_a G^2 = E'(s) - a^{-2}\int \Xi_a'' G^2 + 2a^{-2}\int \Xi_a|\partial_z G|^2. \tag{W.25}$$

Multiplying by `zeta`, integrating on `[0,4]`, and using `zeta(0)=0` gives

$$\begin{aligned} v\int_0^4 \zeta\int W_a G^2 = \zeta(4)E(4) - \int_0^4 \zeta' E\,ds \\ -a^{-2}\int_0^4 \zeta\int \Xi_a'' G^2\,dzds \\ +2a^{-2}\int_0^4 \zeta\int \Xi_a|\partial_z G|^2\,dzds. \end{aligned} \tag{W.26}$$

At each time, G satisfies W.12 with `K=2C_0`. Lemma W.1 and W.21 imply

$$|E(s)| + a^{-2}\left|\int \Xi_a'' G^2\right| + a^{-2}\int |\Xi_a||\partial_z G|^2 \le Ch(s)^{2/3}. \tag{W.27}$$

For the terminal row, apply Lemma W.1 first and W.19 second:

$$|E(4)| \le Ch(4)^{2/3} \le CH^{2/3}. \tag{W.28}$$

The remaining rows in W.26 are bounded by W.27, the cutoff bounds, and Holder on the fixed time interval. Therefore

$$\boxed{\left|v\int_0^4 \zeta\int W_a G^2\right| \le CH^{2/3}.} \tag{W.29}$$

No division by `v`, `alpha-beta`, or either frequency occurs. The identity keeps the endpoint, cutoff, heat, and dissipation rows together, including the cancellations that are lost by integrating the four Fourier rows separately.

Return to physical scales

The exact plateau-fibre area from R0.75T is `4pi a delta_0 R^2` for every `|x_2|<=ell/2`. Equations W.8 and W.11 therefore give

$$\boxed{M^{\mathrm{plat}}_{k,m,R} \geq 4\pi\delta_0 a^2 R^5 H.} \tag{W.30}$$

Combining W.23, W.29, and W.30 proves, throughout `maR<C_0`,

$$|\mathcal{T}_{k,m,R}| \leq Ca^2R^3H^{2/3} \leq Ca^{2/3}R^{-1/3}(M^{\mathrm{plat}}_{k,m,R})^{2/3}. \tag{W.31}$$

R0.75V gives the same estimate when `maR>=C_0`; W.7 and W.31 partition all possibilities. This proves W.2. Substitution of W.3 proves W.4, with the same exact rate `-c_gamma/12=-2/11907`.

Exact-solution and Version-M boundary

The pair W.1 satisfies

$$\partial_t F + B\partial_2 F - \partial_2^2 F = 0 \tag{W.32}$$

and embeds in the exact smooth unforced shear `u=(0,B,F(t,x_2))` with constant pressure. The constant background has not been shown to belong to the frozen mean-zero, inversion-paired Version-M subclass.

If the complete clock and plateau tube belong to the same scale-`2R` Version-M measurement row of weight at least `omega`, and `F` is an actual component of that same velocity, then

$$p^{\mathrm{plat}}_{k,m,R} \leq CP^M_R, \quad \mathfrak{X}_{k,m,R} \leq C(P^M_R)^{2/3}. \tag{W.33}$$

This last implication remains conditional on the realized-subclass and ledger-alignment hypotheses. It is not valid merely because W.1 is a Fourier projection of a larger field.

What is closed and what remains open

Closed here: the low-carrier sector for one exact diffusive dyadic pair; the frequency-gap-free spatial and terminal observations needed by the local energy identity; and, after combining with V, the full-frequency two-harmonic signed-flux estimate and its scale-free normalization.

Open: three or more harmonics; arbitrary dyadic packets; inter-packet aggregation; nonconstant or vertically dependent shear; projection from a larger velocity; arbitrary-field E.24; complete Version-M extraction; fixed deletion; suitable-weak transfer; regularity; and singularity.

The proof is analytic. A numerical plot would neither certify the compact ODE lemma nor the Turan--Nazarov trace, so no formal scientific figure or simulation is claimed for this section. No novelty or priority claim is made. R0.75W is not a proof of global regularity. **NOT CLAY.**

Result and exact boundary

R0.75W proves a full-frequency signed collar-flux estimate for one exact two-harmonic dyadic shear. Its low-carrier proof uses only two properties of that pair: a finite-dimensional spatial ODE and a finite-term temporal exponential polynomial. The present note proves that this mechanism is not special to two modes.

Fix an integer `q>=1`. Let

$$1 \le n_1 < n_2 < \cdots < n_q \le 2n_1 \tag{X.1}$$

and set

$$F(t,x_2) = \sum_{j=1}^{q} A_j e^{-n_j^2 t} \cos\big(n_j x_2 - \phi_j - n_j B t\big), \quad A_j \ge 0, \quad B \in \mathbb{R}. \tag{X.2}$$

For the frozen radial collar, plateau, and time cutoff, define

$$\begin{aligned} \mathcal{T}_{\boldsymbol{n},R} &:= \frac{1}{2}\int_0^{4R^2}\int_{\mathbb{T}^3} \eta_R(t) B\, \partial_2 \xi_{a,R} |F|^2 \, dx dt, \\ M^{\mathrm{plat}}_{\boldsymbol{n},R} &:= \int_0^{4R^2}\int_{\mathcal{S}^{\mathrm{plat}}_{a,R}} |F|^3 \, dx dt. \end{aligned} \tag{X.3}$$

Assume the low-carrier condition

$$n_1 a R < C_0. \tag{X.4}$$

There is a constant `C_q`, depending on `q`, `C_0`, and the frozen collar and cutoff profiles, but not on `R`, the frequencies, amplitudes, phases, or `B`, such that

$$\boxed{|\mathcal{T}_{\boldsymbol{n},R}| \le C_q a^{2/3} R^{-1/3} \big(M^{\mathrm{plat}}_{\boldsymbol{n},R}\big)^{2/3}.} \tag{X.5}$$

With

$$p^{\mathrm{plat}}_{\boldsymbol{n},R} = R^{-2}\omega M^{\mathrm{plat}}_{\boldsymbol{n},R}, \quad \mathfrak{X}_{\boldsymbol{n},R} = \frac{\omega}{R}[\mathcal{T}_{\boldsymbol{n},R}]_+, \tag{X.6}$$

equation X.5 becomes

$$\boxed{\mathfrak{X}_{\boldsymbol{n},R} \le C_q a^{2/3}\omega^{1/3}\big(p^{\mathrm{plat}}_{\boldsymbol{n},R}\big)^{2/3}, \quad \lim_{L\to\infty}\frac{1}{L^2}\log(a^{2/3}\omega^{1/3}) = -\frac{2}{11907}.} \tag{X.7}$$

For every fixed `q`, the factor `C_q` has zero logarithmic cost on the frozen `L^2` scale. No uniform control of `C_q` as `q` grows is proved. For `q=2`, X.5 is the low-carrier half of R0.75W. For `q>=3`, the high-carrier sector remains open.

Frozen inputs and scaling

Retain

$$a = pL, \quad R = e^{-\rho L^2/4}, \quad \omega = e^{-c_\gamma L^2/4}, \quad aR = \ell, \tag{X.8}$$

and

$$0 \le \eta_R \le 1, \quad \eta_R(0) = 0, \quad |\eta_R'(t)| \le C_\eta R^{-2}. \tag{X.9}$$

The immediately used frozen inputs are

input	SHA-256	role
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	complete clock and cutoff onset
`research/r075r_outer_cap_spectral_concentration_obstruction.md`	`e5eba5b262a8e140eaa149b6d914f355f2f3c636ec1e74cf85515f1c38fd32f3`	canonical radial cross section and growing-packet obstruction
`research/r075w_full_frequency_two_harmonic_flux_payment.md`	`571b8152e3e5f81becec4dd691488fb5889fac23e94ca7c99bd546399dc320d4`	local energy identity and the two-mode case

In the sector X.4, put

$$s = \frac{t}{R^2}, \quad z = \frac{x_2}{aR}, \quad \alpha_j = n_j aR, \quad v = \frac{BR}{a}. \tag{X.10}$$

Then

$$0 < \alpha_1 < \cdots < \alpha_q \le 2C_0 \tag{X.11}$$

and

$$G(s, z) := F(R^2 s, aRz) = \sum_{j=1}^{q} A_j e^{-\alpha_j^2 s/a^2} \cos\big(\alpha_j(z - vs) - \phi_j\big). \tag{X.12}$$

The exact equation is

$$\partial_s G + v\partial_z G - a^{-2}\partial_z^2 G = 0. \tag{X.13}$$

Let

$$I=[-1/2,1/2],\quad h(s)=\int_I |G(s,z)|^3\,dz,\quad H=\int_0^4 h(s)\,ds. \tag{X.14}$$

Uniform confluent observation in dimension 2q

Lemma X.1. Fix `q`, `K<infinity`, and compact intervals `I,J`, with `I` having nonempty interior. There is `C=C(q,K,I,J)` such that, for every parameter vector `boldsymbol alpha in [0,K]^q`, every real or complex solution of

$$\prod_{j=1}^{q}(\partial_z^2+\alpha_j^2)g=0 \tag{X.15}$$

satisfies

$$\boxed{\|g\|_{L^\infty(J)}+\|g'\|_{L^\infty(J)}\le C\|g\|_{L^3(I)}.} \tag{X.16}$$

Repeated parameters and `alpha_1=...=alpha_q=0` are included. In the fully confluent zero-frequency limit, X.15 is `partial_z^(2q)g=0`, so its solution space is the polynomials of degree at most `2q-1`.

To prove the lemma, expand

$$\prod_{j=1}^{q}(\lambda+\alpha_j^2)=\lambda^q+\sigma_1\lambda^{q-1}+\cdots+\sigma_q \tag{X.17}$$

and write `X=(g,g',...,g^(2q-1))`. Equation X.15 is

$$X'=\mathbb{A}_{\alpha}X, \tag{X.18}$$

where the superdiagonal entries are one and the final row has entries

$$(-\sigma_q,0,-\sigma_{q-1},0,\ldots,-\sigma_1,0). \tag{X.19}$$

The matrices form a compact bounded family because every `sigma_j` is a continuous elementary symmetric polynomial on `[0,K]^q`. It is enough to control the initial jet at one point of `I`. Otherwise there would be parameters `boldsymbol alpha^(r)` and solutions with unit initial jet and `L^3(I)` norm tending to zero. After taking a subsequence, both the parameters and initial jets converge. Continuous dependence for X.18 gives a nonzero limiting solution whose first component vanishes on `I`. Uniqueness for X.18 then forces the whole limiting initial jet to vanish, a contradiction. Uniform propagation of the controlled jet from `I` to `J` proves X.16.

At every fixed time, G satisfies X.15 with `K=2C_0`. Therefore

$$\|G(s)\|_{L^\infty(J)} + \|\partial_z G(s)\|_{L^\infty(J)} \le C_q h(s)^{1/3}. \tag{X.20}$$

No inverse frequency gap occurs.

A 2q-term terminal trace

Lemma X.2. Fix `q` and `Lambda<infinity`. If

$$Q(s) = \sum_{r=1}^{N} c_r e^{\lambda_r s}, \quad N \le 2q, \quad |\operatorname{Re}\lambda_r| \le \Lambda, \tag{X.21}$$

then

$$\boxed{|Q(4)|^3 \le C_{q,\Lambda} \int_0^4 |Q(s)|^3 \, ds.} \tag{X.22}$$

The constant is independent of the imaginary parts and of every exponent gap. Nazarov's measurable-set inequality gives

$$\sup_{[0,4]} |Q| \le e^{4\Lambda} \left(\frac{4C}{|E|} \right)^{N-1} \sup_E |Q|. \tag{X.23}$$

If `I_Q=int_0^4|Q|^3`, then `E={s:|Q(s)|^3<=I_Q/2}` has measure at least two. Substitution in X.23 proves X.22. This also displays an at-most-exponential contribution in `q` from the temporal trace; it does not give a uniform-in-`q` estimate.

For each fixed `z`, X.12 has at most `2q` exponential terms, with exponents

$$-\alpha_j^2/a^2 \pm i\alpha_j v, \quad 1 \le j \le q. \tag{X.24}$$

Their real parts are bounded by `4C_0^2` once `a>=1`. Apply X.22 and integrate over `z in I` to obtain

$$\boxed{h(4) \le C_q H.} \tag{X.25}$$

This remains uniform for `B=0`, arbitrarily large `|B|`, and colliding scaled frequencies.

Radial primitive and local energy identity

Let the frozen radial profile be `vartheta` and set

$$W_a(z) = -2\pi a z\,\vartheta\big(a(|z|-1)\big),\;\; \Xi_a(z) = \int_{-\infty}^{z} W_a(r)\,dr. \tag{X.26}$$

Oddness gives `int W_a=0`, so `Xi_a` is compactly supported. For all sufficiently large `L`, its support lies in `J=[-3/2,3/2]` and

$$\|W_a\|_1 + \|\Xi_a\|_1 + \|\Xi_a\|_\infty \le C,\;\; \|\Xi_a''\|_1 \le Ca. \tag{X.27}$$

The exact cross-sectional calculation and `zeta(s)=eta_R(R^2s)` give

$$\boxed{\mathcal{T}_{n,R} = \frac{a^2R^3}{2} v \int_0^4 \zeta(s) \int_{\mathbb{R}} W_a(z)G(s,z)^2\,dzds.} \tag{X.28}$$

For `Q=G^2` and `E(s)=int Xi_aG^2`, equation X.13 yields

$$\begin{aligned} \partial_s Q + v\partial_z Q - a^{-2}\partial_z^2 Q &= -2a^{-2}|\partial_z G|^2, \\ v\int W_a G^2 &= E'(s) - a^{-2}\int \Xi_a'' G^2 + 2a^{-2}\int \Xi_a |\partial_z G|^2. \end{aligned} \tag{X.29}$$

After multiplication by `zeta` and integration on `[0,4]`, the right side is

$$\zeta(4)E(4) - \int_0^4 \zeta' E\,ds - a^{-2}\int_0^4 \zeta \int \Xi_a'' G^2 + 2a^{-2}\int_0^4 \zeta \int \Xi_a |G_z|^2. \tag{X.30}$$

Use X.20 and X.27 at every time. Since `a>=1`,

$$|E(s)| + a^{-2}\left|\int \Xi_a'' G^2\right| + a^{-2}\int |\Xi_a||G_z|^2 \le C_q h(s)^{2/3}. \tag{X.31}$$

For the endpoint, X.20 and X.25 give

$$|E(4)| \le C_q h(4)^{2/3} \le C_q H^{2/3}. \tag{X.32}$$

The cutoff satisfies `|zeta'|<=C_eta`. Holder on the fixed time interval then bounds every other row of X.30 by `C_qH^(2/3)`. Consequently

$$\boxed{\left| v\int_0^4 \zeta \int W_a G^2 \right| \le C_q H^{2/3}.} \tag{X.33}$$

The proof forms the complete energy identity before taking absolute values. It never divides by `v`, an amplitude, a frequency, or a frequency gap.

Return to physical scales

The exact plateau-fibre area and X.14 give

$$\boxed{M^{\mathrm{plat}}_{\boldsymbol{n},R} \geq 4\pi\delta_0 a^2 R^5 H.} \tag{X.34}$$

Combining X.28, X.33, and X.34 yields

$$\begin{aligned} |\mathcal{T}_{\boldsymbol{n},R}| &\leq C_q a^2 R^3 H^{2/3} \\ &\leq C_q a^{2/3} R^{-1/3} (M^{\mathrm{plat}}_{\boldsymbol{n},R})^{2/3}, \end{aligned} \tag{X.35}$$

which proves X.5. Substitution of X.6 proves X.7.

Dependence on the number of modes

For fixed `q`, `C_q` is independent of `L`; hence it does not alter the strict rate `-2/11907`. The proof deliberately does not hide what happens when `q` grows:

1. the temporal Turan--Nazarov factor in X.23 grows at most exponentially in `q` for the fixed half-measure choice; 2. the compactness proof of Lemma X.1 supplies a finite spatial constant for each `q`, but no quantitative uniform bound in `q`; 3. the outer-cap construction of R0.75R shows that a growing high-band packet can concentrate away from the plateau, so a uniform arbitrary- packet conclusion cannot be inferred from X.35.

Thus X.35 is a fixed-finite-dimensional theorem, not an arbitrary-packet theorem in disguise.

Exact-solution and Version-M boundary

Equation X.2 satisfies

$$\partial_t F + B\partial_2 F - \partial_2^2 F = 0 \tag{X.36}$$

and embeds in the exact smooth unforced shear `u=(0,B,F(t,x_2))` with constant pressure. The nonzero constant background has not been shown to belong to the frozen mean-zero, inversion-paired Version-M subclass.

If the complete clock and plateau tube are in the same scale-`2R` Version-M measurement row with weight at least `omega`, and F is an actual component of that same velocity, then X.7 gives the same conditional `C_q(P_R^M)^(2/3)` consequence as W. This is not valid merely for a Fourier projection of a larger field.

What is closed and what remains open

Closed here: the complete low-carrier signed-flux estimate for every fixed finite real harmonic family in one dyadic band; all frequency collisions; arbitrary phases, amplitudes, and constant shear speed; and the scale-free normalization for fixed `q`.

Open: a quantitative spatial observation constant suitable for `q=q(L)`; the high-carrier sector for three or more modes; arbitrary dyadic packets; inter-packet aggregation; nonconstant or vertically dependent shear; projection from a larger velocity; arbitrary-field E.24; complete Version-M extraction; fixed deletion; suitable-weak transfer; regularity; and singularity.

The proof is analytic. A plot or finite parameter scan would not certify the `2q`-dimensional compactness argument or the continuum exponential-polynomial trace, so no formal scientific figure or simulation is claimed. No completeness, novelty, or priority claim is made. **NOT CLAY.**

Result and exact boundary

R0.75X pays the low-carrier signed collar flux for every fixed finite harmonic family, without a frequency-gap condition. The high-carrier sector for three or more modes remained open. This note closes a quantitatively separated part of that sector.

Fix an integer `q>=1`, let

$$1 \leq n_1 < n_2 < \cdots < n_q \leq 2n_1, \quad \ell = aR, \tag{Y.1}$$

and define the separation of the signed spectrum by

$$\delta_{\boldsymbol n} := \min(\{2n_1\} \cup \{n_j - n_i : 1 \leq i < j \leq q\}). \tag{Y.2}$$

For `q=1`, the second set in Y.2 is empty, so `\delta_{\boldsymbol n}=2n_1`. Assume

$$\boxed{\ell \delta_{\boldsymbol n} \geq 8q.} \tag{Y.3}$$

For `q>=2`, Y.1 and Y.3 imply `n_1\ell>=8q(q-1)`. Thus this is genuinely a high-carrier, increasingly sparse class when `q` grows; it is not a dense-packet hypothesis in another form.

Consider the exact real diffusive shear

$$\begin{gathered} F(t,x_2) = \sum_{j=1}^{q} A_j(t)\cos\bigl(n_j x_2 - \phi_j(t)\bigr), \\ A_j(t) = A_j e^{-n_j^2 t}, \quad \phi_j(t) = \phi_j + n_j B t, \end{gathered} \tag{Y.4}$$

where `A_j>=0`, the phases are arbitrary, and `B` is any real constant. With the frozen complete clock `T_R=4R^2`, put

$$\begin{gathered} \mathcal{T}_{\boldsymbol n,R} := \frac{1}{2}\int_0^{T_R}\int_{\mathbb{T}^3} \eta_R(t) B\,\partial_2\xi_{a,R}|F|^2\,dx dt, \\ M_{\boldsymbol n,R}^{\mathrm{plat}} := \int_0^{T_R}\int_{\mathcal{S}_{a,R}^{\mathrm{plat}}} |F|^3\,dx dt. \end{gathered} \tag{Y.5}$$

Then

$$\boxed{|\mathcal{T}_{\boldsymbol{n},R}| \le Cq^2 a^{2/3} R^{-1/3} \left(M^{\rm plat}_{\boldsymbol{n},R}\right)^{2/3}.} \tag{Y.6}$$

Here `C` depends only on the frozen radial profile, plateau width, and time cutoff. It is independent of `q`, `R`, the admissible frequencies, amplitudes, phases, and `B`; all displayed mode-count dependence is the explicit factor `q^2`.

With

$$p^{\rm plat}_{\boldsymbol{n},R} := R^{-2}\omega M^{\rm plat}_{\boldsymbol{n},R}, \quad \mathfrak{X}_{\boldsymbol{n},R} := \frac{\omega}{R}[\mathcal{T}_{\boldsymbol{n},R}]_+, \tag{Y.7}$$

equation Y.6 gives

$$\boxed{\mathfrak{X}_{\boldsymbol{n},R} \le Cq^2 a^{2/3} \omega^{1/3} \left(p^{\rm plat}_{\boldsymbol{n},R}\right)^{2/3}.} \tag{Y.8}$$

For fixed `q`, and more generally whenever `\log q=o(L^2)`, the exact `L^2`-scale coefficient remains

$$\lim_{L\to\infty} \frac{1}{L^2} \log\left(q^2 a^{2/3} \omega^{1/3}\right) = -\frac{2}{11907}. \tag{Y.9}$$

The theorem does not cover unresolved spectral clusters `\ell\delta_{\boldsymbol n}<8q`. It therefore neither contradicts nor removes the consecutive-frequency outer-cap packet obstruction of R0.75R.

Frozen inputs

Retain

$$a = pL, \quad R = e^{-\rho L^2/4}, \quad \omega = e^{-c_\gamma L^2/4}, \quad T_R = 4R^2, \tag{Y.10}$$

and the frozen radial chart and cutoff conditions

$$0 < \delta_0 < \delta, \quad a \ge 4\delta_0, \quad (a+\delta)R < \frac{\pi}{2}, \quad 0 \le \eta_R \le 1, \quad \eta_R(0) = 0, \quad |\eta_R'(t)| \le C_\eta R^{-2}. \tag{Y.11}$$

The immediately used frozen inputs are

input	SHA-256	role
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	complete clock and cutoff
`research/r075r_outer_cap_spectral_concentration_obstruction.md`	`e5eba5b262a8e140eaa149b6d914f355f2f3c636ec1e74cf85515f1c38fd32f3`	radial fibre and unresolved-packet boundary
`research/r075u_two_harmonic_difference_frequency_payment.md`	`f9fb331cf880b20f3b407fe66453bce71517ac1ef2af4fa0863c00325c1022a4`	radial quotient and complete-clock scaling

input	SHA-256	role
`research/r075x_fixed_finite_mode_low_carrier_payment.md`	`8e0c412528578c15d807b33b64f0996e62a2dabe2ebd58fa297f67c093929763`	complementary low-carrier theorem

No external observability theorem is needed below. The only spatial estimate is a finite Gram-matrix calculation with its constant shown.

Signed-spectrum Gram coercivity

Let

$$I_\ell=[-\ell/2,\ell/2],\quad S(t)^2:=\sum_{j=1}^{q}A_j(t)^2. \tag{Y.12}$$

At a fixed time, write

$$F(t,y)=\sum_{\lambda\in\Lambda}c_\lambda(t)e^{i\lambda y},\quad \Lambda=\{-n_q,\dots,-n_1,n_1,\dots,n_q\}, \tag{Y.13}$$

where

$$c_{n_j}(t)=\frac{A_j(t)}{2}e^{-i\phi_j(t)},\quad c_{-n_j}(t)=\overline{c_{n_j}(t)}. \tag{Y.14}$$

The minimum distance between two distinct members of `\Lambda` is exactly `\delta_{\boldsymbol n}`. Direct integration gives

$$\begin{aligned}\int_{I_\ell}|F(t,y)|^2\,dy=\ell\sum_{\lambda\in\Lambda}|c_\lambda(t)|^2\\+\sum_{\substack{\lambda,\mu\in\Lambda\\ \lambda\neq\mu}}c_\lambda(t)\overline{c_\mu(t)}\frac{2\sin((\lambda-\mu)\ell/2)}{\lambda-\mu}.\end{aligned} \tag{Y.15}$$

Since `|\Lambda|=2q`,

$$\sum_{\lambda\neq\mu}|c_\lambda||c_\mu|\le(2q-1)\sum_{\lambda\in\Lambda}|c_\lambda|^2. \tag{Y.16}$$

Consequently, the absolute value of the off-diagonal row in Y.15 is at most

$$\frac{2(2q-1)}{\delta_{\boldsymbol n}}\sum_{\lambda\in\Lambda}|c_\lambda(t)|^2\le\frac{\ell}{2}\sum_{\lambda\in\Lambda}|c_\lambda(t)|^2, \tag{Y.17}$$

where the final inequality uses Y.3. Because `\sum_{\lambda\in\Lambda}|c_\lambda|^2=S(t)^2/2`, equations Y.15--Y.17 yield

$$\boxed{\int_{I_\ell} |F(t,y)|^2\,dy \ge \frac{\ell}{4} S(t)^2.} \tag{Y.18}$$

Holder's inequality on I_\ell then gives

$$\boxed{\int_{I_\ell} |F(t,y)|^3\,dy \ge \ell^{-1/2}\left(\int_{I_\ell} |F(t,y)|^2\,dy\right)^{3/2} \ge \frac{\ell}{8} S(t)^3.} \tag{Y.19}$$

This is phase-uniform and contains no hidden compactness or inverse-gap constant: the price of separation is stated entirely in Y.3.

A phase-free complete-clock row

The next lemma is the phase-free version of the moving-phase estimate. It is sufficient because Y.19 controls the individual modal amplitudes, rather than only a cancellation defect.

Lemma Y.1. Let \zeta be absolutely continuous on [0,4] and satisfy

$$0 \le \zeta \le 1, \quad \zeta(0) = 0, \quad |\zeta'| \le C_\eta. \tag{Y.20}$$

For every \Lambda_0\ge0, \sigma,\alpha\in\mathbb R,

$$\boxed{\left|\sigma \int_0^4 \zeta(s) e^{-\Lambda_0 s} \sin(\alpha + \sigma s)\,ds\right| \le C\left(\int_0^4 e^{-3\Lambda_0 s/2}\,ds\right)^{2/3}.} \tag{Y.21}$$

The constant depends only on C_\eta.

To prove it, let

$$\tau = \begin{cases} 1, & 0 \le \Lambda_0 \le 1, \\ \Lambda_0^{-1}, & \Lambda_0 \ge 1. \end{cases} \tag{Y.22}$$

The right side of Y.21 is comparable from above and below to \tau^{2/3}.

If |\sigma|\tau\le1 and \Lambda_0\le1, estimate the left side by 4|\sigma|\le4. If |\sigma|\tau\le1 and \Lambda_0\ge1, use \zeta(s)\le C_\eta s to obtain

$$|\sigma| \int_0^4 \zeta(s) e^{-\Lambda_0 s}\,ds \le C|\sigma|\Lambda_0^{-2} \le C\tau \le C\tau^{2/3}. \tag{Y.23}$$

If |\sigma|\tau\ge1, set w(s)=\zeta(s)e^{-\Lambda_0s} and integrate the sine once:

$$\left|\sigma\int_0^4 w(s)\sin(\alpha+\sigma s)\,ds\right| \le |w(4)| + \int_0^4 |w'(s)|\,ds. \tag{Y.24}$$

For $\Lambda_0\le1$, the last row is bounded by a constant. For $\Lambda_0\ge1$, Y.20 and $\zeta(s)\le C_\eta s$ give

$$|w(4)| + \int_0^4 |w'(s)|\,ds \le C\Lambda_0^{-1} = C\tau \le C\tau^{2/3}. \tag{Y.25}$$

Equations Y.22--Y.25 prove the lemma. Notice that no division by σ occurs in the slow-phase case and no phase-distance lower bound is used.

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Exact modal expansion and row payment

Use the frozen odd radial kernel

$$D_R(y) = -2\pi y\vartheta(|y|/R-a),\quad J_{r,R} := \int_{-\pi}^{\pi} D_R(y)\sin(ry)\,dy. \tag{Y.26}$$

The exact radial quotient from R0.75U is

$$\boxed{\frac{|J_{r,R}|}{r} \le Ca^2R^3 \quad (r\ge1).} \tag{Y.27}$$

Oddness of D_R and the product-to-sum identity expand Y.5 into exactly q^2 nonconstant sine rows:

$$\begin{aligned}\mathcal{T}_{n,R} = {} & \frac{B}{4}\sum_{j=1}^{q} J_{2n_j,R}\int_0^{T_R} \eta_R A_j(t)^2\sin(2\phi_j(t))\,dt \\ & +\frac{B}{2}\sum_{1\le i<j\le q} J_{n_j-n_i,R}\int_0^{T_R} \eta_R A_i(t)A_j(t)\sin(\phi_j(t)-\phi_i(t))\,dt \\ & +\frac{B}{2}\sum_{1\le i<j\le q} J_{n_i+n_j,R}\int_0^{T_R} \eta_R A_i(t)A_j(t)\sin(\phi_i(t)+\phi_j(t))\,dt.\end{aligned} \tag{Y.28}$$

The count is $q+2\binom q2=q^2$: one self row per mode and one difference plus one sum row per unordered pair.

Every row has the form

$$BJ_{r,R}\int_0^{T_R} \eta_R(t)P_0e^{-\lambda t}\sin(\alpha+rBt)\,dt, \tag{Y.29}$$

up to a factor $1/4$ or $1/2$, where r is a positive integer, $P(t)=P_0e^{-\lambda t}$ is either $A_j(t)^2$ or $A_i(t)A_j(t)$, and λ is respectively $2n_j^2$ or $n_i^2+n_j^2$.

Scale `t=R^2s` and put

$$\Lambda_0 = \lambda R^2, \quad \sigma = rBR^2, \quad \zeta(s) = \eta_R(R^2 s). \tag{Y.30}$$

Lemma Y.1 gives the exact physical-time estimate

$$\boxed{\left| B \int_0^{T_R} \eta_R(t) P(t) \sin(\alpha + rBt)\, dt \right| \le \frac{C}{rR^{4/3}} \left(\int_0^{T_R} P(t)^{3/2}\, dt \right)^{2/3}.} \tag{Y.31}$$

Indeed, the left side after scaling is `P_0/r` times the left side of Y.21, while

$$\left(\int_0^{T_R} P(t)^{3/2}\, dt \right)^{2/3} = R^{4/3} P_0 \left(\int_0^4 e^{-3\Lambda_0 s/2}\, ds \right)^{2/3}. \tag{Y.32}$$

For every self or cross row,

$$P(t)^{3/2} \le S(t)^3. \tag{Y.33}$$

Multiplying Y.31 by Y.27 and summing the exactly `q^2` rows of Y.28 therefore yields

$$|\mathcal{T}_{n,R}| \le Cq^2 a^2 R^{5/3} \left(\int_0^{T_R} S(t)^3\, dt \right)^{2/3}. \tag{Y.34}$$

Absolute values are safe here because the strong-separation observation Y.19 controls every modal product. This is precisely the step that is not available for an unresolved cluster.

Plateau mass and scale conversion

The frozen radial fibre inequality is

$$\int_{S_{a,R}^{\mathrm{plat}}} |F(t, x_2)|^3\, dx \ge 4\pi a \delta_0 R^2 \int_{I_\ell} |F(t, y)|^3\, dy. \tag{Y.35}$$

Equations Y.19 and Y.35 imply

$$\boxed{M_{n,R}^{\mathrm{plat}} \ge \frac{\pi \delta_0}{2} a^2 R^3 \int_0^{T_R} S(t)^3\, dt.} \tag{Y.36}$$

Substitute Y.36 into Y.34:

$$
\begin{aligned}
|\mathcal{T}_{\boldsymbol{n},R}| &\le Cq^2a^2R^{5/3}\left(\frac{M^{\mathrm{plat}}_{\boldsymbol{n},R}}{a^2R^3}\right)^{2/3} \\
&= Cq^2a^{2/3}R^{-1/3}\left(M^{\mathrm{plat}}_{\boldsymbol{n},R}\right)^{2/3}.
\end{aligned}
\tag{Y.37}
$$

This proves Y.6. Substituting $M=R^2\omega^{-1}p$ into (ω/R) times Y.6 proves Y.8. The frozen identity $-c_\gamma/12=-2/11907$, together with $\log q=o(L^2)$ and $\log a=o(L^2)$, proves Y.9.

Exact-solution and Version-M boundary

The field Y.4 satisfies

$$
\partial_t F + B\partial_2 F - \partial_2^2 F = 0 \tag{Y.38}
$$

and embeds in the exact smooth unforced shear $u=(0,B,F(t,x_2))$ with constant pressure. The nonzero constant background has not been shown to belong to the frozen mean-zero, inversion-paired Version-M subclass.

If the complete clock and plateau tube are in the same scale-$2R$ Version-M measurement row with weight at least ω, and F is an actual component of that same velocity, then Y.8 gives the conditional payment

$$
\mathfrak{X}_{\boldsymbol{n},R} \le Cq^2(P_R^M)^{2/3}. \tag{Y.39}
$$

This is not valid merely for a Fourier projection of a larger velocity.

What is closed and what remains open

Closed here: the complete signed collar-flux estimate for every strongly separated finite harmonic family in one dyadic band; all self-, sum-, and difference-frequency rows; arbitrary amplitudes, phases, and constant shear speed; an explicit q^2 payment factor; and the exact negative normalized rate whenever $\log q=o(L^2)$.

Open: unresolved high-carrier clusters with $\ell\delta_{\boldsymbol n}<8q$; removal or weakening of the strong separation hypothesis; arbitrary dyadic packets; inter-packet aggregation; nonconstant or vertically dependent shear; projection from a larger velocity; arbitrary-field E.24; complete Version-M extraction; fixed deletion; suitable-weak transfer; regularity; and singularity.

R0.75Y does not control the consecutive-frequency packet in R0.75R, whose signed-spectrum separation is far below Y.3 on the frozen collar scale. The proof is analytic, and no simulation or formal scientific figure is needed. No completeness, novelty, or priority claim is made. **NOT CLAY.**

Result and exact boundary

R0.75X pays every fixed finite family in the low-carrier sector. R0.75Y pays every strongly separated family in the high-carrier sector. This note identifies the exact part of parameter space left between them and tests the most immediate proposed recursion: demodulate each unresolved cluster and apply the low-carrier theorem to its envelope.

Fix an integer $q>=1$, let

$$1 \le n_1 < n_2 < \cdots < n_q \le 2n_1, \quad \ell = aR, \tag{Z.1}$$

and retain the exact diffusive shear

$$F(t,y) = \sum_{j=1}^{q} A_j e^{-n_j^2 t} \cos\big(n_j(y-Bt) - \phi_j\big). \tag{Z.2}$$

For fixed q, choose the R0.75X threshold $C_0=8q$. Then every family falls into exactly one of the following three sectors:

$$\begin{array}{ll} \text{X-sector:} & n_1\ell < 8q, \\ \text{Y-sector:} & n_1\ell \ge 8q \text{ and } (n_{j+1}-n_j)\ell \ge 8q \quad (1 \le j < q), \\ \text{Z-sector:} & n_1\ell \ge 8q \text{ and at least one } (n_{j+1}-n_j)\ell < 8q. \end{array} \tag{Z.3}$$

The X-sector is paid by R0.75X. The Y-sector satisfies the signed-spectrum condition of R0.75Y and is paid there. Thus the Z-sector is the complete remaining high-carrier region for this fixed-q family, rather than a selected example of it.

Cut the ordered frequencies at every adjacent gap at least $8q/\ell$. This gives a unique partition into maximal consecutive clusters. Each non-singleton cluster $C=\{r,...,s\}$ has carrier $N=n_r$ and offsets $d_j=n_j-N$ satisfying

$$0 = d_r < d_{r+1} < \cdots < d_s, \quad d_s\ell < 8q(s-r) \le 8q(q-1), \quad d_s \le N. \tag{Z.4}$$

Its analytic signal has the exact carrier-envelope representation

$$\begin{aligned} H_C(t,y) &:= \sum_{j=r}^{s} A_j e^{-n_j^2 t} e^{i(n_j(y-Bt)-\phi_j)} \\ &= e^{-N^2 t} e^{iN(y-Bt)} Z_C(t,y), \\ Z_C(t,y) &:= \sum_{j=r}^{s} A_j e^{-(2Nd_j+d_j^2)t} e^{i(d_j(y-Bt)-\phi_j)}, \quad F_C = \operatorname{Re} H_C. \end{aligned} \tag{Z.5}$$

The envelope is spatially low-band for fixed q, by Z.4, but it does not satisfy the low-carrier equation used in R0.75X. Instead,

$$\boxed{\partial_t Z_C + B\partial_y Z_C - \partial_y^2 Z_C - 2iN\partial_y Z_C = 0.} \tag{Z.6}$$

If $Q_C=|Z_C|^2$ and $J_C=\operatorname{Im}(\overline{Z_C}\partial_y Z_C)$, then

$$\boxed{\partial_t Q_C + B\partial_y Q_C - \partial_y^2 Q_C = -2|\partial_y Z_C|^2 - 4NJ_C.} \tag{Z.7}$$

The carrier current is globally favorable because the offsets are nonnegative, but it has no fixed sign locally. Moreover no constant independent of N can control $N|J_C|$ pointwise by $Q_C+|\partial_y Z_C|^2$, even for a two-term envelope. Consequently the R0.75X local-energy argument cannot be recursively applied to `Z_C` after taking the new current term in absolute value. This is a no-go result for that proof strategy, not a counterexample to the desired cluster flux estimate.

The next admissible route must keep the one-sided offset spectrum and the sign of the current, or estimate the density and carrier blocks jointly before local absolute values are taken. No full Z-sector flux payment is claimed here.

Frozen inputs

The immediately used inputs are

input	SHA-256	role
`research/r075b_bulk_clock_outer_padding_gate.md`	`430feb9efc151e2b968dd1b7f785a19dc0e38416270ff8ed0275cfc6429b1a5a`	frozen complete clock and cutoff
`research/r075r_outer_cap_spectral_concentration_obstruction.md`	`e5eba5b262a8e140eaa149b6d914f355f2f3c636ec1e74cf85515f1c38fd32f3`	unresolved packet boundary
`research/r075x_fixed_finite_mode_low_carrier_payment.md`	`8e0c412528578c15d807b33b64f0996e62a2dabe2ebd58fa297f67c093929763`	every fixed-`q` low-carrier family
`research/r075y_strongly_separated_multimode_flux_payment.md`	`74790f910b596c86b204291d997ef723cabbc85d14a89e3fe900814fcd88b0a6`	strongly separated high-carrier family

The frozen geometric and temporal assumptions remain

$$a = pL,\quad R = e^{-\rho L^2/4},\quad \omega = e^{-c_\gamma L^2/4},\quad T_R = 4R^2,\quad \ell = aR. \tag{Z.8}$$

This note uses only exact finite Fourier algebra. It imports no external observability, unique-continuation, or cluster theorem.

Exhaustive fixed-q parameter partition

R0.75X allows any fixed low-carrier threshold `C_0`. Taking `C_0=8q` for the present fixed `q` proves the X-sector row of Z.3.

Suppose now that `n_1 ell>=8q`. The minimum gap of the signed spectrum is

$$\delta_n = \min\left(2n_1, \min_{1\le j<q}(n_{j+1}-n_j)\right). \tag{Z.9}$$

Indeed the smallest positive-positive gap is adjacent, the smallest negative-negative gap is the same, and the smallest positive-negative gap is `2n_1`. If every adjacent gap obeys the Y-sector row of Z.3, then

$$2n_1\ell \ge 16q,\quad \ell \min_{1\le j<q}(n_{j+1}-n_j) \ge 8q,\quad \ell\delta_n \ge 8q. \tag{Z.10}$$

This is exactly the hypothesis Y.3. If an adjacent gap fails it, the family lies in the Z-sector. The three rows of Z.3 are disjoint and cover all possibilities, so the partition is exhaustive.

For the cluster partition, place a cut between `n_j` and `n_{j+1}` exactly when

$$(n_{j+1}-n_j)\ell \geq 8q. \tag{Z.11}$$

Maximality makes the consecutive blocks unique. In a block `C={r,...,s}`, every internal gap is strictly less than `8q/ell`; hence

$$(n_s-n_r)\ell=\sum_{j=r}^{s-1}(n_{j+1}-n_j)\ell<8q(s-r). \tag{Z.12}$$

The dyadic band in Z.1 also gives

$$n_s-N \leq 2n_1-N \leq n_1 \leq N, \tag{Z.13}$$

which completes Z.4. A Z-sector family has at least one non-singleton cluster, while the all-singleton partition is exactly the Y-sector within `n_1 ell>=8q`.

Exact carrier-envelope equation

Fix one non-singleton cluster and abbreviate `Z=Z_C`, `H=H_C`. The `j`th envelope mode is

$$Z_j(t,y)=A_je^{-(2Nd_j+d_j^2)t}e^{i(d_j(y-Bt)-\phi_j)}. \tag{Z.14}$$

Direct differentiation gives

$$\begin{aligned} \partial_t Z_j &= (-2Nd_j-d_j^2-iBd_j)Z_j, \\ B\partial_y Z_j &= iBd_jZ_j, \\ -\partial_y^2 Z_j &= d_j^2 Z_j, \\ -2iN\partial_y Z_j &= 2Nd_jZ_j. \end{aligned} \tag{Z.15}$$

The four rows sum to zero term by term, proving Z.6. Conversely, multiplication by `e^{-N^2t}e^{iN(y-Bt)}` restores

$$\partial_t H+B\partial_y H-\partial_y^2 H=0. \tag{Z.16}$$

Thus demodulation has not produced the real advection-diffusion equation X.13. It has produced a complex equation with an `N`-dependent imaginary first derivative. The real decay rates

$$\lambda_j=2Nd_j+d_j^2 \tag{Z.17}$$

also remain unbounded as the carrier tends to infinity whenever `d_j>0`. Spatial boundedness of the offsets alone therefore does not put the envelope into the compact coefficient family used by R0.75X.

Exact square and flux split for one cluster

The real cluster component is F_C=(H_C+\overline{H_C})/2. Therefore

$$\boxed{F_C^2 = \frac{1}{2} e^{-2N^2 t} |Z_C|^2 + \frac{1}{2} e^{-2N^2 t} \operatorname{Re}\left(e^{2iN(y-Bt)} Z_C^2\right).} \tag{Z.18}$$

Use the frozen odd radial kernel

$$D_R(y) = -2\pi y \vartheta(|y|/R - a), \quad \mathcal{T}_C := \frac{1}{2} \int_0^{T_R} \eta_R(t) B \int_{-\pi}^{\pi} D_R(y) F_C(t,y)^2 \, dy dt. \tag{Z.19}$$

Substitution of Z.18 gives the exact decomposition

$$\boxed{\mathcal{T}_C = \mathcal{T}_C^{\mathrm{den}} + \mathcal{T}_C^{\mathrm{car}},} \tag{Z.20}$$

where

$$\begin{aligned} \mathcal{T}_C^{\mathrm{den}} &:= \frac{1}{4} \int_0^{T_R} \eta_R B e^{-2N^2 t} \int_{-\pi}^{\pi} D_R |Z_C|^2 \, dy dt, \\ \mathcal{T}_C^{\mathrm{car}} &:= \frac{1}{4} \operatorname{Re} \int_0^{T_R} \eta_R B e^{-2N^2 t} \int_{-\pi}^{\pi} D_R e^{2iN(y-Bt)} Z_C^2 \, dy dt. \end{aligned} \tag{Z.21}$$

The density block contains only offset differences. The carrier block contains the self and offset-sum rows translated by `2N`. This split is exact for the square of one cluster. For the square of the full family, cross-cluster products must still be added; Z.20 does not pay or discard them.

Local carrier current

Set

$$Q = |Z|^2, \quad J = \operatorname{Im}(\overline{Z} \, \partial_y Z). \tag{Z.22}$$

From Z.6,

$$\partial_t Z = -B \partial_y Z + \partial_y^2 Z + 2iN \partial_y Z. \tag{Z.23}$$

Taking twice the real part after multiplication by `overline Z`, and using

$$2\,\mathrm{Re}(\overline{Z}\,\partial_y^2 Z) = \partial_y^2|Z|^2 - 2|\partial_y Z|^2, \quad 4N\,\mathrm{Re}(i\overline{Z}\,\partial_y Z) = -4NJ, \tag{Z.24}$$

proves Z.7.

The same term is visible in the unmodulated gradient:

$$|\partial_y H|^2 = e^{-2N^2 t}\left(N^2 Q + |\partial_y Z|^2 + 2NJ\right). \tag{Z.25}$$

There is no contradiction with the ordinary energy identity for `H`; Z.7 and Z.25 are the same dissipation written in modulated variables.

The current is favorable only after full-period integration. Orthogonality of the distinct nonnegative integer offsets yields

$$\boxed{\int_{-\pi}^{\pi} J(t,y)\,dy = 2\pi \sum_{j=r}^{s} d_j A_j^2 e^{-2(2Nd_j + d_j^2)t} \geq 0.} \tag{Z.26}$$

Consequently,

$$\frac{d}{dt}\int_{-\pi}^{\pi} Q\,dy = -2\int_{-\pi}^{\pi} |\partial_y Z|^2\,dy - 4N\int_{-\pi}^{\pi} J\,dy \leq 0. \tag{Z.27}$$

The collar argument is local in `y`, so Z.26 cannot simply be substituted for the current inside a weighted interval.

Exact failure of carrier-uniform pointwise absolute payment

Consider the two-term envelope

$$Z^{(N)}(y) = 2 - e^{iy}, \tag{Z.28}$$

which is the time-zero envelope of frequencies `N` and `N+1`, with a phase change on the second mode. At the centre of the frozen interval, `y=0`,

$$Z^{(N)}(0) = 1, \quad \partial_y Z^{(N)}(0) = -i, \quad Q(0) = 1, \quad J(0) = -1. \tag{Z.29}$$

If a constant `C` independent of `N` satisfied

$$2N|J(y)| \leq C\big(Q(y) + |\partial_y Z(y)|^2\big) \tag{Z.30}$$

for every admissible carrier-envelope pair, Z.29 would imply `N<=C` for every positive integer `N`, which is impossible. More sharply, the modulated dissipation density at this point is

$$|\partial_y Z^{(N)}(0)|^2 + 2NJ(0) = 1 - 2N < 0. \tag{Z.31}$$

For example, with `q=2` and `ell=1`, the frequencies `(N,N+1)` lie in an unresolved high-carrier cluster for every `N>=16`.

This calculation rules out only a carrier-uniform pointwise estimate of the absolute current. It does not rule out a signed weighted estimate, a Hardy-space argument using the nonnegative offset spectrum, a cancellation between Z.21's two blocks, or the final desired collar-flux inequality.

Route decision and open obligations

The exhaustive reduction Z.3 changes the next question. There is no longer an unspecified `q>=3` high-carrier region: after X and Y, it is precisely the high-carrier family containing at least one maximal block of nearby modes. Z.5--Z.7 show that this block is a low-band complex envelope with a large carrier current, not another instance of the real low-carrier equation.

The next proof attempt must establish at least one of the following before claiming cluster payment:

1. a localized signed-current inequality that exploits the one-sided offset spectrum without an `N` loss;
2. a joint multiplier identity for `\mathcal T_C^{\rm den}+\mathcal T_C^{\rm car}` in which the large current is never estimated separately; or
3. a finite-dimensional cluster observable that controls cross-cluster products and remains uniform in the carrier.

Closed here: an exhaustive fixed-`q` X/Y/Z parameter partition; the unique maximal cluster decomposition; the exact carrier-envelope equation; the exact density/carrier split of one cluster square; the local and global carrier-current identities; and a concrete carrier-uniform pointwise no-go example.

Open: every full Z-sector collar-flux estimate; a localized signed current estimate; joint density/carrier cancellation; cross-cluster aggregation; arbitrary growing packets; nonconstant or vertically dependent shear; projection from a larger velocity; arbitrary-field E.24; complete Version-M extraction; fixed deletion; suitable-weak transfer; regularity; and singularity.

The field Z.2 remains an exact smooth unforced shear component, subject to the same constant-background and Version-M boundary recorded in R0.75Y. No simulation or formal scientific figure is needed for these exact identities. No completeness, novelty, or priority claim is made. **NOT CLAY.**

Result and exact boundary

R0.75Z leaves one possible cluster route: exploit the nonnegative offset spectrum to keep the carrier current signed instead of estimating its absolute value. The full-period current is nonnegative, but the collar multiplier is local. This note tests the sign after inserting the actual frozen radial primitive and the complete-clock cutoff.

Let the frozen radial profile satisfy

$$0 \le \vartheta \le 1, \quad \vartheta = 1 \text{ on } [-\delta_0, \delta_0], \quad \operatorname{supp}\vartheta \subset (-\delta, \delta), \quad 0 < \delta_0 < \delta, \tag{A.1}$$

and define

$$W_a(z) = -2\pi a z \vartheta\big(a(|z| - 1)\big), \quad \Xi_a(z) = \int_{-\infty}^{z} W_a(r)\, dr. \tag{A.2}$$

Assume

$$a \geq \max(24, 2\delta), \quad 0 < \ell = aR \leq \frac{1}{11}. \tag{A.3}$$

Set `q=2` and choose the integer carrier and scaled frequencies

$$N = \left\lceil \frac{16}{\ell} \right\rceil, \quad \alpha = N\ell, \quad \beta = \ell. \tag{A.4}$$

Then

$$16 \leq \alpha < 16 + \beta < 17, \quad N\ell \geq 8q, \quad ((N+1) - N)\ell = \beta < 8q. \tag{A.5}$$

Thus `(N,N+1)` is an actual unresolved high-carrier cluster. On the scaled complete clock `0<=s<=4`, take `v=1`, amplitudes `(2,1)`, and phases `(0,pi)`. After removal of the first carrier, its exact envelope is

$$Z(s,z) = 2 - r(s)e^{i\beta(z-s)}, \quad r(s) = e^{-\mu s}, \quad \mu = \frac{2\alpha\beta + \beta^2}{a^2}. \tag{A.6}$$

Let

$$J(s,z) = \operatorname{Im}(\overline{Z}\,\partial_z Z). \tag{A.7}$$

For every `s in [0,4]` and every `z` in the support of `Xi_a`,

$$\boxed{J(s,z) \leq -\frac{9}{16}\beta < 0,} \tag{A.8}$$

and, more strongly,

$$\boxed{|\partial_z Z(s,z)|^2 + 2\alpha J(s,z) \leq -\alpha\beta < 0.} \tag{A.9}$$

The primitive `Xi_a` is nonnegative, nonzero, and has a uniform mass lower bound. Hence every nonnegative complete-clock cutoff `zeta` with `int_0^4 zeta>0` satisfies

$$\boxed{\int_0^4 \zeta(s)e^{-2\alpha^2 s/a^2} \int_{\mathbb{R}} \Xi_a(z)J(s,z)\,dzds < 0.} \tag{A.10}$$

The same strict negativity holds with `J` replaced by the left side of A.9. Thus one-sided offset spectrum does not give a nonnegative localized current, even after the actual collar primitive, common carrier heat factor, complete time window, and nonnegative cutoff are inserted.

This is not a counterexample to the two-mode collar-flux estimate: R0.75W already pays that estimate. It rules out only the strategy of discarding the localized carrier-current row by sign. In this fixed-amplitude example the scaled current contribution is at most `C ell/a^2=C R/a`, so it may still be perturbative or cancel against the carrier-density rows. No cluster-payment or regularity claim is made.

Frozen inputs

input	SHA-256	role
`research/r075r_outer_cap_spectral_concentration_obstruction.md`	`e5eba5b262a8e140eaa149b6d914f355f2f3c636ec1e74cf85515f1c38fd32f3`	frozen radial profile and exact cross section
`research/r075w_full_frequency_two_harmonic_flux_payment.md`	`571b8152e3e5f81becec4dd691488fb5889fac23e94ca7c99bd546399dc320d4`	scaled primitive and already-paid two-mode class
`research/r075z_unresolved_cluster_carrier_current_gate.md`	`30d2811e8747aa2b40b4787e6f169af19d1381b66fc84610327da221168f3d97`	exhaustive cluster sector and carrier-current identity

Retain

$$s=\frac{t}{R^2},\quad z=\frac{y}{aR},\quad v=\frac{BR}{a},\quad \ell=aR. \tag{A.11}$$

The real cluster field corresponding to A.6 is

$$\begin{aligned}G(s,z)=2e^{-\alpha^2 s/a^2}&\cos\bigl(\alpha(z-s)\bigr)\\ -e^{-(\alpha+\beta)^2 s/a^2}&\cos\bigl((\alpha+\beta)(z-s)\bigr).\end{aligned} \tag{A.12}$$

It is the scaled form of the exact smooth diffusive shear with physical frequencies `(N,N+1)`, positive amplitudes `(2,1)`, second phase `pi`, and constant speed `B=a/R`.

Positivity and mass of the frozen primitive

The function `W_a` is odd. On the negative half-line it is nonnegative; on the positive half-line it is nonpositive. Its total integral vanishes. Therefore

$$\Xi_a(z)\ge 0\quad (z\in\mathbb{R}). \tag{A.13}$$

The support condition in A.1 gives

$$\operatorname{supp}\Xi_a\subset\left[-1-\frac{\delta}{a},1+\frac{\delta}{a}\right]\subset\left[-\frac{3}{2},\frac{3}{2}\right]. \tag{A.14}$$

On

$$I_- = \left[-1-\frac{\delta_0}{a}, -1+\frac{\delta_0}{a}\right], \tag{A.15}$$

one has `vartheta=1` and `|z|>=1/2`, so

$$W_a(z) \geq \pi a, \quad \int_{I_-} W_a(z)\,dz \geq 2\pi\delta_0. \tag{A.16}$$

Between the two transition lobes, `Xi_a` equals the full mass of the negative lobe. The interval `[-1+delta/a,1-delta/a]` has length at least one. Consequently,

$$\boxed{\int_{\mathbb{R}} \Xi_a(z)\,dz \geq 2\pi\delta_0 > 0.} \tag{A.17}$$

The earlier frozen estimate also gives

$$\|\Xi_a\|_{L^1} + \|\Xi_a\|_{L^\infty} \leq C_\vartheta. \tag{A.18}$$

Uniform phase and damping bounds

From A.3--A.5,

$$0 < \beta \leq \frac{1}{11}, \quad 16 \leq \alpha < 17, \quad 0 \leq 4\mu < \frac{140}{11\cdot 24^2} < \frac{1}{4}. \tag{A.19}$$

Thus, throughout the complete clock,

$$\frac{3}{4} < e^{-1/4} \leq r(s) \leq 1. \tag{A.20}$$

For `z` in the support of `Xi_a`, A.14 gives

$$|\beta(z-s)| \leq \beta\left(\frac{3}{2}+4\right) \leq \frac{1}{2}. \tag{A.21}$$

The elementary inequality `cos x>=1-x^2/2` now yields

$$\cos\big(\beta(z-s)\big) \geq \frac{7}{8}. \tag{A.22}$$

Exact negative localized current

Differentiating A.6 gives

$$\partial_z Z = -i\beta r e^{i\beta(z-s)}. \tag{A.23}$$

Therefore

$$\begin{aligned} J &= \operatorname{Im}\left[(2 - re^{-i\beta(z-s)})(-i\beta r e^{i\beta(z-s)})\right] \\ &= \beta r\,(r - 2\cos(\beta(z-s))). \end{aligned} \tag{A.24}$$

Equations A.20 and A.22 imply

$$r - 2\cos(\beta(z-s)) \leq 1 - \frac{7}{4} = -\frac{3}{4}, \tag{A.25}$$

and hence

$$J \leq -\frac{3}{4}\beta r \leq -\frac{9}{16}\beta. \tag{A.26}$$

This proves A.8. Also

$$|\partial_z Z|^2 = \beta^2 r^2 \leq \beta^2 \leq \frac{1}{8}\alpha\beta. \tag{A.27}$$

Combining A.26 and A.27 gives

$$|\partial_z Z|^2 + 2\alpha J \leq \frac{1}{8}\alpha\beta - \frac{9}{8}\alpha\beta = -\alpha\beta, \tag{A.28}$$

which proves A.9.

Complete-clock insertion

Let

$$\zeta : [0,4] \longrightarrow [0,1], \quad \zeta \geq 0, \quad \int_0^4 \zeta(s)\,ds > 0. \tag{A.29}$$

No monotonicity is needed for the sign. Every nonzero frozen cutoff of this class is admissible. Because every remaining factor is nonnegative, A.8 and A.17 give

$$\begin{aligned} &\int_0^4 \zeta e^{-2\alpha^2 s/a^2} \int_{\mathbb{R}} \Xi_a J \,dzds \\ &\quad \leq -\frac{9\pi\delta_0}{8}\beta \int_0^4 \zeta(s) e^{-2\alpha^2 s/a^2}\,ds < 0. \end{aligned} \tag{A.30}$$

This proves A.10. The same calculation using A.9 gives

$$\begin{aligned} &\int_0^4 \zeta e^{-2\alpha^2 s/a^2} \int_{\mathbb{R}} \Xi_a \left(|\partial_z Z|^2 + 2\alpha J\right) dzds \\ &\quad \leq -2\pi\delta_0\alpha\beta \int_0^4 \zeta(s) e^{-2\alpha^2 s/a^2}\,ds < 0. \end{aligned} \tag{A.31}$$

There is no conflict with the full gradient identity

$$|i\alpha Z + \partial_z Z|^2 = \alpha^2|Z|^2 + |\partial_z Z|^2 + 2\alpha J \geq 0. \tag{A.32}$$

The positive carrier-density term `alpha^2|Z|^2` was not included in A.31. It is one of the rows that a joint multiplier argument must retain.

Size and claim boundary

In scaled variables the envelope density equation contains the current as

$$\partial_s|Z|^2 + v\partial_z|Z|^2 - a^{-2}\partial_z^2|Z|^2 = -2a^{-2}|\partial_z Z|^2 - 4\frac{\alpha}{a^2}J. \tag{A.33}$$

For the fixed amplitudes in A.6, A.18 and the direct upper bound `|J|<=3 beta` give

$$\left|4\frac{\alpha}{a^2}\int_0^4 \zeta e^{-2\alpha^2 s/a^2} \int \Xi_a J\right| \leq C\frac{\alpha\beta}{a^2} \leq C\frac{\ell}{a^2} = C\frac{R}{a}. \tag{A.34}$$

Thus the example proves a strict sign obstruction but not a large-error obstruction. It leaves open a perturbative estimate, an inequality with a localized boundary error, and cancellation with the carrier block. The fact that the same two-mode family is fully paid by R0.75W confirms that a joint mechanism exists in this test class.

Closed here: failure of localized current nonnegativity for an exact integer-frequency high-carrier cluster, with the actual frozen primitive, carrier heat factor, complete clock, and every nonnegative nonzero cutoff; uniform negativity of the current-correction density; and its exact fixed-amplitude upper scale `C  R/a`.

Open: a quantitative localized current estimate for arbitrary cluster coefficients; joint density/carrier-block payment; every full Z-sector collar-flux estimate; cross-cluster aggregation; growing packets; nonconstant or vertically dependent shear; projection from a larger velocity; arbitrary-field E.24; complete Version-M extraction; fixed deletion; suitable-weak transfer; regularity; and singularity.

The proof is analytic. No simulation or formal scientific figure is needed. No completeness, novelty, or priority claim is made. **NOT CLAY.**

A / 冻结证据

Step 52 主文、primary-source boundary、双实现证书与 fail-closed QA

Step 52 主文 · primary audit · primary-source boundary · fixtures JSON · expected JSON · certificate JSON · Python report · Ruby independent audit · certificate QA · Python script · Ruby script · QA script

同步 reader PDF · 上一大里程碑累计回顾（截止 R0.75W） · W recap PDF

Certificate：Python 15/15、Ruby 15/15、A.1--A.34、34/34 tags 与 displays，3 个 Python hash seeds 及完整 regeneration 字节稳定；两套实现分别拒绝 86/86 定向 mutations，unknown mutations 均 fail closed。完整冻结 ledger 为 12/12。有限检查不代替 continuum identities；本节无正式图、simulation、DNS 或 DGX。

NAV / 相邻研究节点

Z 的局部载频问题与 A 的严格负号障碍

Z：cluster normal form and current gate · A：actual primitive, complete clock, and strict localized negativity · 后续边界 →

NEXT / 后续未授权、未读取

general clustered-sector payment remains OPEN

本站在 R0.76A Step 52 停止。A 只证明实际 primitive 与完整时钟上的 localized current/correction row 严格负号，排除凭单边 offset spectrum 丢弃该行或宣称非负。W 的 exact q=2 通量支付保持有效；perturbative estimate、localized boundary error、nonlocal signed estimate、joint density/carrier cancellation、一般簇支付、跨簇聚合、E.24、complete Version-M extraction、regularity 与 singularity 仍开放。后续工作未授权、未读取、未公开。